TFPT — Changelog

All changes and new results, by date

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This file is the canonical, dated record of every change to the TFPT theory development: new verification modules (`vN`), status changes, editorial passes and infrastructure. It is maintained *in the same change* as the work it records, and it is also imported at the end of the `introduction` document. Module numbers `vN` refer to `verification/vN_*.py`; claim IDs refer to rows of `verification/status_ledger.csv`.

2026-07-21 (XXIII) (`v511` — WP5e-δ_2 of `CELEST.SEAM.01` "the full-tensor ledger" promoted as `CELEST.WP5E.DELTA2.01` (intermediate verdict, exact): the named escape route of the `v508` exchange no-go executed on the FULL adjoint tensor structure of $\mathfrak{e}_8$ — the COLLAPSE IS CONFIRMED full-tensorially AND arity-crossing (the INNERNESS THEOREM: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ semisimple with NO $\mathfrak{u}(1)$ (class-0 roots 40+12, root span rank 8), h in the Cartan OF $\mathfrak{g}_0$ $\Rightarrow$ every invariant tensor carries total charge $0 \bmod 4$: the bilinear Hom table is nonzero ONLY at $j' = -j$ (dims 2/1/1/1) and ALL 15 non-neutral trilinear sector triples have $\mathrm{Hom} = 0$), BUT ONE CUBIC DOOR OPENS: of the neutral trilinears ($\{0,0,0\}: 3 = 1$ sym $+\, 2$ antisym, $\{0,1,3\}: 2$, $\{0,2,2\}: 2$, $\{1,1,2\}: 1$, $\{2,3,3\}: 1$; all cross-sector ones pure brackets) the UNIQUE totally symmetric survivor is the $\mathfrak{su}(4)$ d-symbol ($\mathfrak{so}(10)$ has no cubic Casimir, $\mathrm{Inv}(S^3\,45) = 0$ protects T_5), whose exchange quartic $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$ (TWO independent routes, Killing propagator $2h^\vee = 60$ from the roots) has $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ — the `v508` master kill (product theorem) does NOT extend to cubic vertices; the pairing stays OBSTRUCTED in the charge reading with the genuinely WEAKER certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\text{fix}}) = 72 - 8 = 64$ ($M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ rank 3, augmented 4), and becomes EXACTLY solvable relaxed: $A_{\text{fix}} = -u + 8v + 2w + 1920\,Q_{dd}$ (residual 0); [C] fences $64 = \dim \mathfrak{g}_1 = 2^6$, $1920 = |W(D_5)| = 8 \cdot 240$; controls: $SO(16)/D_8$ has NO symmetric cubic and no A_3 block (E_8/μ_4 doubly special), false $\mathfrak{g}_0$/sector assignments separate; the δ_1 burden SHARPENED to the $\psi = 64$ slice or the selection-rule relaxation — and the δ_1 exploration run itself stands UNDECIDED at exploration level (strict holomorphic q^0 reading REFUTED at the $\mathbb{Z}_2$/Eguchi–Hanson anchor = method boundary, not a kill; contact term = the MODULAR COMPLETION of the AB data, a Harvey–Moore τ-integral with forced leading (T_5, T_3) ratio 4:3; massive coset levels $n \geq 2$ mandatory); Wolfram extension $463 \to 470$ (verified engine run); *NO marker moves — WP5e proper stays [O], `SEAM.EQUIV.01` stays [O]*)

- **`v511_celestial_wp5e_delta2_tensor_ledger.py` (`CELEST.WP5E.DELTA2.01`, 41 checks, ~4 s; integer-scaled weight multisets + Kostant/Weyl alternating sums + sympy/Fraction, no floats; deterministic).** D0/D1 structure: 240 roots, class split $(52, 64, 60, 64)$; $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ SEMISIMPLE without $\mathfrak{u}(1)$ (40+12 class-0 roots, no mixed root, root span rank 8); sectors $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$ as multiplicity-free minuscule Weyl-orbit products, each irreducible (64/60/64); the grading is INNER ($h = (2, 2, 2, 2, 2; 0^4)$

in the $\mathfrak{g}_0$ Cartan). D3/D4 Hom tables: bilinear nonzero ONLY at $j' = -j$ (dims 2/1/1/1; all 12 charged pairs zero — the inner-torus theorem at arity 2); trilinear: ALL 15 non-neutral triples Hom = 0 (arity 3), neutral dims $(3, 2, 2, 1, 1)$ with all cross-sector trilinears ANTISYMMETRIC in their repeated legs — the unique symmetric survivor is the $\mathfrak{su}(4)$ d-symbol ($\mathrm{Inv}(S^3\,45) = 0$, $\mathrm{Inv}(S^3\mathfrak{a}_3) = 1$). D5 the new channel: $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$ by Killing-propagator contraction AND the hand route $|x^2 - (\mathrm{tr}\,x^2/4)\mathbf{1}|^2/60$; $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$; brackets/tadpoles/$d{\times}f$ die exactly; no T_5 analogue. D6 decision: strict reading unreachable (rank 2 vs 3); generous reading STILL unsolvable (rank 3 vs 4) with the weaker certificate $\psi(A_{\mathrm{fix}}) = 64$; relaxed reading EXACTLY solvable with $c_d = 1920 = 32 \times 60$. D7 controls: Killing control = `v508` replication (A_{fix}, E_{13}, E_{22}, $E_{00} = 225E_{22}$); $SO(16)/D_8$ bilinears $(1, 0, 1)$, trilinears $(1, 0, 1, 0)$, $\mathrm{Inv}(S^3\mathrm{adj}_{D_8}) = 0$; false $\mathfrak{g}_0 = \mathfrak{d}_5{+}\mathfrak{a}_2{+}\mathfrak{u}(1)$: $(5, 2, 2) \neq (2, 1, 1)$, $\mathrm{Inv}(S^3\mathfrak{g}_0) = 6 \neq 1$; false sector content $(16_s, \bar{4})$ kills every pairing. Discovery provenance: `experiments/tfpt-discovery/celestial_seam_wp5e_delta2_full_tensor_ledger_probe.py` (41/41), same day.

- **The δ_1 fence (exploration-level, NOT promoted — verdict UNDECIDED).** The twisted BCOV contact-term probe (`celestial_seam_wp5e_delta1_bcov_contact_probe.py`, 27/27, stays sandbox; v-numbered on demand) contributes its four exact sharpenings as fence text: (i) the strict holomorphic q^0 reading is REFUTED at the $\mathbb{Z}_2$/Eguchi–Hanson anchor (method boundary, not a kill); (ii) the forced leading (T_5, T_3) ratio is 4:3 (sector 2 : sectors 1+3); (iii) the $SL(2, \mathbb{Z})$ orbit decomposition $15 = 12 + 3$ identifies the contact term as the MODULAR COMPLETION of the AB data (a Harvey–Moore-type τ-integral), not a free number; (iv) massive coset levels $n \geq 2$ are mandatory. The remaining δ_1 target is the τ-integral, evaluated against the δ_2-sharpened criterion ($\psi = 64$ slice or selection-rule relaxation).
- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.DELTA2.01` (**[E]** Hom tables/innerness/Q_{dd}/certificate; **[C]** the number fences; **[O]** arity ≥ 4, cubic GS term, δ_1); `CELEST.SEAM.01` updated (delta-2 executed via v511 — the full-tensor collapse is confirmed (innerness), but the cubic d-symbol channel opens T_3 with the weaker certificate $\psi = 64$; delta-1 stands UNDECIDED at exploration level). Papers: `tfpt_research_contracts` WP5e bullet gains the executed δ_2 stage + the δ_1 exploration fence, WP5e-proper targets renamed (τ-integral, cubic GS term, ε_1, back-reaction); `tfpt_3` celestial section extended to the FOURTEEN-step story (new Step 14 "the full-tensor ledger: the collapse is real, and one cubic door opens") with the "What remains" block updated. Wolfram: seven mirrors (innerness count + $\mathfrak{g}_0$ structure; the bilinear and trilinear Hom tables by exact Kostant alternating sums; the unique symmetric survivor; the $SO(16)$/false-$\mathfrak{g}_0$/ false-sector controls; Q_{dd} by two routes; the ψ certificate + the relaxed solution $(-1, 8, 2; 1920)$) verified with the active engine the same day ($463 \to 470$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). Registry + `run_all` (505 modules, suite green: `ALL CHECKS PASSED`). No gate closes, no marker moves — WP5e proper stays **[O]**.

2026-07-21 (XXII) (P2 consolidation — the axiomatics reduction state written as ONE coherent statement across the papers (editorial, NO new claims, NO marker moves): the compiler's dimensionless inputs reduce to *four marks* (derived from P1-side topology, v216), *ONE square-modulus datum* ($\tau = i$ — equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; `v491/v499/v506/v507/v510`) and π, plus the [C] identifications R1/R2 of `v499`; `AX.P2.01` stays a DECLARED axiom and `QGEO.REALIZE.01` stays [C]/[O] — the consolidation describes the *reduction state*, not an abolition)

- **What is consolidated (and why now).** The P2 state of affairs was spread over five module rounds: the partition corollary (`v491`: $4 = 2{+}1{+}1$ unique $\Rightarrow g_{\mathrm{car}} = e_2 = 5$), the weight typing (`v499`: Deligne + Birkhoff–Grothendieck + Mehta–Seshadri force $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$, residuals R1/R2 **[C]**), the residual convergence (`v503`: QGEO-R1 $\equiv$ P2-R1), the clock rigidity and bit origin (`v506/v507`: one alignment bit, order fermionically forced, deck class derived) and the freedom theorems (`v510`: the position half is topology, the edge class excluded). The honest consolidated statement, now in one place: *the compiler's dimensionless inputs reduce to four marks (derived from P1-side topology), one square-modulus datum ($\tau = i$, equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle), and π* — plus the **[C]** identifications R1/R2 of `v499`. No new claims; conservative wording; no "axiom abolished" language anywhere.

- **Papers + ledger + website.** `tfpt_1_architecture_e8`: new keybox "The current reduction state of P2" next to the foundational-postulates keybox, with `v491/v499/v506/v507/v510` citations. `tfpt_3`: mirror bullet "The current reduction state" in the honest-status section. `introduction`: the anchor paragraph gains the consolidated one-sentence state and the status card row $a = (1, 1, 2)$ now names the reduction state. Ledger: SYNTHESIS NOTE appended to `AX.P2.01` ("reduction state 2026-07-21: four marks + one square-modulus datum ($\tau = i$) + π"), marker *Axiom unchanged.* Website: `papers.ts` mirrors of the changed sections. No gate closes, no marker moves.

2026-07-21 (XXI) (`v510` — the freedom attack on the `v506/v507` alignment bit promoted as `SEAM.BIT.FREEDOM.01`: the POSITION half of the bit is TOPOLOGY — the collar deck is a covering deck (the $\mathbb{Z}_2$ seam of the Möbius double), and covering decks act FREELY; the DIHEDRAL THEOREM (on the seam circle the free involution is UNIQUE = the antipode: 17 involutions in D_{16}, exactly 1 free; $\mathrm{Aut}(C_{16}) = D_{16}$ complete census, brute force $8!$ at $N{=}8$; quotient topology circle/antipode = CIRCLE $\chi{=}0$ vs circle/reflection = INTERVAL $\chi{=}1$ — the 6π Gauss–Bonnet budget needs closed circles, so freeness IS the established P1 datum); NONSPLIT $\iff$ FREE over the complete 17-involution census in exact $\mathrm{Cl}(16)$ (free $\Rightarrow \tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16} = (-1)^F$ nonsplit with 2 roots = the $\mathbb{Z}_8$ tower; ALL 16 reflections $\Rightarrow U^2$ = scalar $\{2^7, 2^8\}$, split, 0 roots); the REAL-STRUCTURE classification (deck-commuting families = exactly two; $\mu_4 \to$ equatorial, deck FREE; silver $\to$ real axis THROUGH the deck poles, NOT free — the edge class is the restriction of the BRANCHED pillowcase involution; hexagonal $\to$ NO mark-fixing real structure, RP-incompatible); HARMONIC + FREE $\Rightarrow \mu_4$ exactly ($b = \pm i$); honest counterwitnesses ("nonsplit $\Rightarrow$ clock" is FALSE: $\{0, 1, 8, 9\}$ and $\{\pm 1, \pm(3{+}4i)/5\}$ with $j = 148176/25 \neq 1728$); the alignment bit REDUCES from "harmonic AND central" to the square-modulus datum $\tau = i$ alone; Wolfram extension $457 \to 463$ (verified engine run); *NO marker moves — `QGEO.REALIZE.01` stays [C]/[O], `AX.P2.01` stays an axiom, P2-R1 halved*)

- **`v510_seam_bit_freedom.py` (`SEAM.BIT.FREEDOM.01`, 30 checks, <5 s; sympy + exact $\mathrm{Cl}(16)$ Fractions, no floats; deterministic).** Part A (dihedral/fermionic): the involution census in D_N on the refined $2N$-point seam circle ($N = 16$: 17 involutions, EXACTLY ONE free = the antipodal shift by 8; all 16 reflections have exactly 2 circle fixed points — sites for even axis, edge midpoints for odd; robustness $N = 8, 12$; odd control $N = 9$: ZERO free — an even carrier count is needed, consistent with $16 = 2^{g_{\mathrm{car}}-1}$); the census is COMPLETE because $\mathrm{Aut}(C_{16}) = D_{16}$ exactly (brute force over all $8! = 40320$ permutations at $N = 8$, propagation at $N = 16$); quotient topology: circle/antipode = CIRCLE ($\chi = 0$, closed), circle/reflection = INTERVAL ($\chi = 1$, two boundary corners) — a fixed-point deck would break the $6\pi = 3 \times 2\pi$ closed-circle count, so "deck free on the seam circle" is exactly the established "the seam is a closed circle", not a new premise; the complete Fock census: ALL 17 involutions implemented in exact $\mathrm{Cl}(16)$ — NONSPLIT $\iff$ FREE with no exception, root dichotomy 2 vs 0; the chain: free $\Rightarrow$ shift 8 $\Rightarrow$ crossing (all 28 invariant 4-sets) $\Rightarrow U^2 = (-1)^F \Rightarrow$ central class $\Rightarrow$ the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control (the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION: the silver arrangement mistakes the mark-creating branch $\mathbb{Z}_2$ for the seam deck). Part B (Möbius/RP): the cylinder-double-cover deck is free on all 120 model points and restricts to the antipode on the $\mathbb{Z}_2$ seam fibre; one-sidedness TYPES the $\mathbb{Z}_2$ as the free antipode (quotient $\mathbb{RP}^2$ = the $|\mathbb{Z}_2|$ of $c_3 = 1/(|\mathbb{Z}_2|2\pi\chi)$); the deck-commuting real structures are EXACTLY two families ($\mu\bar{z}$: fix line through the poles, NOT free; $\mu/\bar{z}$, $\mu > 0$: fix circle $|z| = \sqrt{\mu}$, FREE; $\mu < 0$: empty); the per-configuration table (μ_4 equatorial/free, silver real-axis/not free, generic-unit free without harmonicity, hexagonal RP-incompatible); harmonic + free solves EXACTLY to $b = \pm i$ (on free circles the silver family "harmonic AND edge" is EMPTY — harmonicity becomes sufficient); honest counterwitnesses: the discrete $\{0, 1, 8, 9\}$ and the continuous $\{\pm 1, \pm(3+4i)/5\}$ (free, central, $j = 148176/25$, no clock) — the Fock dichotomy measures the CLASS, not the modulus. Premise typing: (P-i) deck = Möbius covering deck ⇒ free [standard topology; the one-sidedness strand `origin_theory/v456`]; (P-ii) marks on the Θ-fix seam circle = existing REALIZE **[C]** content (`QGEO.REALIZE.02` hypotheses, `v264`) — NO new input. Audit finding documented: "circle-freeness" was nowhere typed before (only mark-freeness, `v216`); `v507` F1.1 explicitly admitted the reflection reading of the edge class — `v510` closes that door (a reflection is not a covering deck). The **[C]** rest, verbatim: *the alignment bit reduces from "harmonic AND central" to the square-modulus datum $\tau = i$ alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input.* Discovery provenance: `experiments/tfpt-discovery/qgeo_realize_deck_freedom_dihedral_probe.py` (14/14) and `qgeo_realize_rp_fixcircle_moebius_probe.py` (16/16), same day.

- **Ledger + papers + Wolfram + website.** Ledger: new row `SEAM.BIT.FREEDOM.01` (**[E]** dihedral theorem + complete Fock census + real-structure classification + harmonic-free collapse + counterwitnesses; **[C]** the square-modulus datum); TEXT precisions (markers unchanged) in `SEAM.CLOCK.RIGIDITY.01`, `SEAM.BIT.ORIGIN.01`, `QGEO.REALIZE.01` (bit → square-modulus datum; edge class topologically excluded) and `P2.TYPING.01` (R1 halved). Papers: `tfpt_3` celestial/QGEO passage extended ("the position half is topology"), `tfpt_research_contracts` QGEO kill-test + QGEO.MARKS.03 + celestial-contract passages extended, one-sentence precisions in `introduction`, `tfpt_1`, `tfpt_2`, `origin_theory` where `v506`/`v507` are cited. Wolfram: six mirrors (dihedral census + quotient topology; Aut completeness incl. the 8! brute force; the Cl(16) dichotomy census over all 17 involutions on 256×256 Jordan–Wigner gammas; the real-structure table; harmonic + free ⇒ $\pm i$; the counterwitnesses) verified with the active engine the same day (457 → 463, `ALL WOLFRAM EXTENSION CHECKS PASSED`; base file re-verified 116/116). Registry + `run_all` (504 modules, suite green: `ALL CHECKS PASSED`). No gate closes, no marker moves.

2026-07-21 (XX) (v509 — WP5e-ε_2 of `CELEST.SEAM.01` "the CPS level-from-flux dial" promoted as `CELEST.WP5E.EPS2.01`, verdict B: the Costello–Paquette–Sharma Burns-holography mechanism ("level = flux quantum × Dynkin index") executed on the lockstep spheres — the CPS SKELETON exact (S^3 period $(2\pi i)^2 N$ with pullback density $-\sin 2\theta$, exceptional-sphere Kähler flux $2\pi N$, boundary level magnitude $2N$ by the symplectic-boson contraction, $T(\text{vector}) = 1$); the PAIRING MATRIX pinned, not postulated (ch_2 ledger + integrality + unimodularity + effectivity $\Rightarrow P_{mi} = \delta_{mi}$ by complete enumeration: rows $8/6/8$, 48 unimodular, exactly 2 effective = identity + diagram flip), fluxes $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ with uniformly ONE quantum per charged current; "level = total flux" KILLED by the lockstep test itself (F_i unequal = flavour multiplicity), the per-current reading $(1, 1, 1)$ anchored by the lattice current count $(240, 0, 0, 0)$ and embedding index 1; ord-4-vs-level-1 RESOLVED (the clock order counts fractional flux sectors: $\mu = 16 = \mathrm{ord}^2$, Lagrangian μ_4 glue, KLM $16 \to 1$) with the NEW sector-counter dial $\#\mathrm{prim}((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ — exactly ONE sector iff $k = 1$; the LOCKSTEP THEOREM lifts the v505 "one scale $\Rightarrow$ one level" to a THEOREM of clock invariance ($|\Pi_j| = \sqrt{2}\,t$ equal, monodromy = pure phase i, $\det(A{-}1) = -4$: no invariant flux vector; NEW falsifier $Z^4{-}Z$: $0/24$ orderings lockstep); honest limit: uniform doubling keeps the lockstep (equality, not the value); controls $k{=}2/SO(16)/A_2$ separate honestly; Wolfram extension $450 \to 457$ (verified engine run); *NO marker moves — WP5e stays [O] (the CPS dictionary on $\mathbb{PT}/\mathbb{Z}_4$ typed [C], the type-I B-model back-reaction [O]), `SEAM.EQUIV.01` stays [O]*)

- **`v509_celestial_wp5e_eps2_level_from_flux.py` (`CELEST.WP5E.EPS2.01`, 28 checks, < 1 s; sympy/Fraction exact, no floats; deterministic).** Target: derive the boundary level $k = 1$ from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL **130**, 061602; `arXiv:2306.00940`). *CPS skeleton replicated exactly*: the S^3 period of the brane back-reaction 3-form is $(2\pi i)^2 N$ (their eq. 3.33; large-gauge invariance quantises N), the exceptional-sphere Kähler flux is $2\pi N$ exactly ($K = |u|^2 + N \log |u|^2$, the flat part dies at $t \to 0$), and the boundary level magnitude is $2N$ via the symplectic-boson double contraction ($N = 1, 2, 3$; their eq. 9.4), Dynkin normalisation $T(\mathrm{so}_8 \text{ vector}) = 1$, $T(\mathrm{adj}) = 6$. *Pairing matrix pinned*: the `v505` ch_2 ledger ($f(m) = -(C^{-1})_{mm}/2 = (-\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8})$) + integrality + unimodularity + effectivity force $P_{mi} = c_1(T_m).[\Sigma_i] = \delta_{mi}$ by COMPLETE enumeration (row candidates $8/6/8$; 48 unimodular solutions of $PC^{-1}P^T = C^{-1}$; with $P \geq 0$ exactly 2: identity + A_3 diagram flip); fluxes $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$, total $188 = 248 - 60$, flip-invariant; flux per charged root current uniformly ONE quantum through exactly one sphere. *The level formula, honest*: "level = total open-string flux" is KILLED by the lockstep test itself ($k_i = (64, 60, 64)$ unequal — F_i is the FLAVOUR multiplicity, the brane matter content); what works: k = flux per adjoint current $= F_i / \dim \mathfrak{g}_i = (1, 1, 1)$, anchored non-circularly by the lattice current count $(240, 0, 0, 0)$ for $k = 1..4$ (`v505` S4.8) and the embedding index $j(D_5 {\subset} E_8) = j(A_3 {\subset} E_8) = 1$ ($60 = 2h^\vee$). *Ord-4-vs-level-1 resolved*: the clock order counts FRACTIONAL flux sectors, not the level ($\mu = \det(D_5)\det(A_3) = 16 = \mathrm{ord}^2$; the μ_4 glue diagonal isotropic + Lagrangian; KLM $16 \to 16/4^2 = 1$, zero residual fractional flux); NEW DIAL: $\#\mathrm{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ from the affine marks — exactly ONE sector iff $k = 1$ (independent of the CPS transplantation). *The lockstep theorem* (the lifting of `v505` S4.10): clock invariance $P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ sharply, the four branch points are ONE μ_4 orbit, $|\Pi_j| = \sqrt{2}\,t$ EQUAL on all three spheres, the monodromy is the pure phase i and $\det(A{-}1) = -4 \neq 0$ (no invariant flux vector) — $k_1 = k_2 = k_3$ IS A THEOREM of clock invariance; NEGATIVE PROBE (new falsifier): the clock-forbidden family $Z^4{-}Z$ (smooth, disc -27, branch points $\{0, 1, \omega, \omega^2\}$ not a μ_4 orbit): over ALL 24 orderings ZERO chains with three equal period moduli (1 vs $\sqrt{3}$); HONEST LIMIT: uniform flux doubling keeps the lockstep — the theorem proves EQUALITY, not the value (1 from the counters and the CPS minimal quantum $N = 1$). *Negative controls*: $k = 2$ (current count 0, residual 360, prefactor $-31/48$, #prim 3 — while the lockstep dial honestly does NOT fire); $SO(16)$ (fluxes $(0, 60, 0)$: spheres 1/3 FLUX-DARK, defect $-30 \neq -78$, condensation stuck at $4 = \det \mathrm{Cartan}(D_8)$); A_2 (det $3 \neq 4$, glue diagonal $23/24 \notin \mathbb{Z}$, $\mu = 12$ no perfect square: NO Lagrangian subgroup, Coxeter $3 \neq 4$). Typed honestly: the CPS dictionary itself on $\mathbb{PT}/\mathbb{Z}_4$ stays **[C]** (their geometry is the $\mathbb{C}^2$ blow-up with one (-1)-sphere and $\mathfrak{so}(8)$ at negative level — not the A_3 ALE); the type-I B-model back-reaction stays **[O]** (the same fence as `v502/v505`).

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.EPS2.01` (**[E]** CPS skeleton/pairing/lockstep/sector arithmetic + **[C]** CPS dictionary + **[O]** back-reaction); `CELEST.SEAM.01` updated ("ε_2 executed via v509: the one-level statement is now a THEOREM (lockstep from clock invariance, falsifier $Z^4 - Z$), level = 1 carries the new sector-counter dial $\#\mathrm{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$; the $\mathbb{PT}/\mathbb{Z}_4$ back-reaction stays **[O]**"); the `v505` "one scale ⇒ one level" sentence LIFTED to theorem status in ledger and prose (text move, no marker move). Papers: `tfpt_research_contracts` WP5e section extended by the ε_2 block (done, verdict B; roadmap now $\delta_1/\delta_2/\varepsilon_1$ + back-reaction); `tfpt_3` celestial section becomes a THIRTEEN-step story with Step 13 "one level is a theorem; the sector counter pins $k = 1$" and the "What remains" paragraph updated. Wolfram: seven mirrors (the CPS period integrals; the $2N$ contraction + Dynkin normalisation; the pairing enumeration $48 \to 2$; the fluxes $(64, 60, 64)$; the #prim table $(1, 3, 5, 10, 15, 27)$; the lockstep periods + the 0/24 negative probe; the $SO(16)/A_2$ controls) verified with the active engine the same day ($450 \to 457$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `VerificationDag`: `v509` on the QFT/celestial node; registry + `run_all` (503 modules, suite green). No gate closes, no marker moves — WP5e proper stays **[O]**.

2026-07-21 (XIX) (`e8-ladder-bed v2` — the 2DCS extension executed with an honest preregistered `kill_2d` verdict at the DETECTOR level: the known-answer map recovers 8/8 rungs incl. m_7/m_8 via F3 difference coordinates (MC $p = 2\times10^{-4}$), but the 2D control battery breaks (scramble 27%, jitter 18% at the destructive 30% level) — the v1 1D results (BCVO 8/8, $CoNb_2O_6$ 6/8) are untouched; scorecard row updated, status/stage unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)

- **`experiments/e8-ladder-bed` v2 (2DCS): amendment preregistered, run, and killed by its own controls.** Trigger: arXiv:2512.16829 (Watanabe, Trebst, Hickey) — a THEORY-only paper (four-kink/ED localized-limit spectra at $B_y = 0$–$3\,\mathrm{T}$, not the E8 point; no measurements); as of 2026-07-21 *no experimental 2DCS measurement of $CoNb_2O_6$ exists anywhere* (web-checked). `amendment_v2` was frozen in `hypotheses/e8_ladder_bed_v1.yaml` BEFORE any 2D recovery run (detector, catalog F1/F2/F3, gates, null model, verdict criteria); `amendment_v2_1` discloses a null-region correction (add `n_matched` to the false-alarm region) found on synthetic diagnostics, timing disclosed. Runs (`results/results2d.json`, deterministic): known-answer KA2D 8/8 rungs with m_7/m_8 recovered ONLY via F3 difference coordinates ($\chi^2/\mathrm{dof} = 1.30$, MC $p = 2.0 \times 10^{-4}$, 0/5000) — the mechanism that pulls m_7/m_8 inside a low-pass window works; the NEG2D localized-limit map (the paper's own non-E8 closed forms) is correctly rejected ($p = 0.029$); offset and wrong-ladder controls rejected. BUT `scramble2d` fires at 54/200 = 27% and `jitter2d` shows no dose-response decay (29/21/17/18%; 18% at the destructive 30% level) ⇒ preregistered verdict `kill_2d`: permuting the detection axis preserves the pump-axis marginal (which carries the full 1D ladder) and the 120-entry catalog is dense enough that rung coverage follows from the marginal alone — the v2 detector does not test the 2D *pairing*; failure kills the detector, not TFPT. The $CoNb_2O_6$ leg stays 6/8 ($p = 0.0038$) per the preregistered `m7_m8_upgrade_rule`; a future v3 would need coincidence-based coverage and a sparser catalog, preregistered BEFORE real data exists.
- **Surfaces synced (scorecard row updated; counters unchanged, 120 rows).** Scorecard row "E8 mass-ladder blind recovery bed": `data_value` extended ("CoNb2O6 6/8

unchanged; 2DCS extension killed at detector level (scramble 27%, jitter 18% @30%); KA 8/8 incl. m7/m8 via F3, p=2e-4; no experimental 2DCS data exists (arXiv:2512.16829 theory-only)") and `status_note` extended (amendment v2/v2.1 preregistered+disclosed; real-data leg blocked; v3 criteria) via `build_evidence_scorecard.py`; status/stage UNCHANGED (`consistent/not_applicable`, `analog_positive_control` basket — the v1 detector validation stands). Experiment README v2 section, hypotheses YAML amendments, `experiments/next.txt` note; data-request plan on record (Armitage lab/JHU, Wang lab/Ames; 2D peak table at $B \parallel b$, 4.7–5.5 T, sub-Kelvin). No claim; firewall intact.

2026-07-21 (XVIII) (`v508` — WP5e-γ of `CELEST.SEAM.01` "the sphere-axion pairing check" promoted as `CELEST.WP5E.GAMMA.01`, an honest rigid NEGATIVE result: the VERTEX-SPACE THEOREM — the $W(D_5)\times W(A_3)$-invariant quadratics on the 8-dim glue Cartan are EXACTLY $\mathrm{span}\{s_5, s_3\}$ (dim 2) and the quartics EXACTLY $\mathrm{span}\{P_1, P_2, P_3, T_5, T_3\}$ (dim 5), with the PRODUCT THEOREM as the master kill (any two invariant quadratics multiply into $\mathrm{span}\{P_1, P_2, P_3\}$ — zero T_5/T_3 content, so Φ_{T_3} annihilates EVERY exchange image while $\Phi_{T_3}(A_{\mathrm{fix}}) = 32 \neq 0$); the strict two-index $\mathbb{Z}_4$ rule COLLAPSES (the sphere axions have no invariant bilinear Cartan vertex at all), the twist-insertion channels give a rank-2 exchange matrix with $\mathrm{rank}([M \mid A_{\mathrm{fix}}]) = 3$ and the annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\mathrm{fix}}) = (0, 32, 72)$; side discovery $K^{(0)} = -15\, K^{(2)}$; naturalness DISSOLVES (ch$_2$-natural and AB-weight couplings both certify $(0, 0)$ vs required $(32, 72)$ — scale-independent); controls: $SO(16)$ has no sphere partners AND uncancelled T_5, D_8 no T_3 structure; the O-fence is SHARPENED: the exchange sub-branch is dead, the $32\, T_3$ residual MUST be carried by twisted BCOV contact terms (δ_1 binary: coefficient exactly $(9, -30, -15, 0, 32)$, computed not fitted; escape δ_2: the full-tensor ledger); Wolfram extension $444 \to 450$ (verified engine run); *NO marker moves — `CELEST.SEAM.01` stays Open, the slot bijection untouched, the preregistered level-kill still does not fire, `SEAM.EQUIV.01` stays [O]*)

- **`v508_celestial_wp5e_gamma_pairing.py` (`CELEST.WP5E.GAMMA.01`, 27 checks, $\sim$4 s; sympy/Fraction exact, no floats; deterministic).** Question: can the three twisted sphere axions η_m (Coxeter eigencharacters $\{i, -1, -i\}$ on $H^2(\mathrm{ALE})$, `v505` S5.8) cancel the rigid $32\, T_3$ residual of the Atiyah–Bott ledger ($A_{\mathrm{fix}} = Q^{(1)}/2 + Q^{(2)}/4 + Q^{(3)}/2 = 9P_1 - 30P_2 - 15P_3 + 32\, T_3$, `v505` S3.7) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? *No, with a certificate. Vertex-space theorem*: exact Weyl nullspace arithmetic over all bidegrees gives quadratics dim 2 (bidegrees $(1, 0, 1)$) and quartics dim 5 (bidegrees $(2, 0, 1, 0, 2)$) — the vertex basis is COMPLETE; *product theorem* (the master kill): $(as_5 + bs_3)(cs_5 + ds_3) = ac\, P_1 + (ad + bc)P_2 + bd\, P_3$ identically — every exchange image (any axion count, any rule, any couplings) is annihilated by Φ_{T_3}, and $\Phi_{T_3}(A_{\mathrm{fix}}) = 32 \neq 0$. *Selection enumeration*: the strict two-index reading collapses ($C_{j,j'} \neq 0$ only for $j' = -j$, forcing $m = 0$: 16 raw triples, 4 nonzero, all $m = 0$); twist insertion: $\eta_1 \leftrightarrow K^{(3)}$, $\eta_2 \leftrightarrow K^{(2)}$, $\eta_3 \leftrightarrow K^{(1)}$, $\eta_0 \leftrightarrow K^{(0)}$; channels $E_{13} = (16, -96, 144, 0, 0)$, $E_{22} = (16, 32, 16, 0, 0)$, $E_{00} = 225\, E_{22}$; SIDE DISCOVERY: $K^{(0)} = -15\, K^{(2)}$ (the even-sector quadratics are PARALLEL — this collapse drives the Φ_P obstruction). *Unsolvability with certificate (the core)*: M is 5×2, rank 2, kernel 0; $\mathrm{rank}([M \mid A_{\mathrm{fix}}]) = 3$; annihilator certificate $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\mathrm{fix}}) = (0, 32, 72)$ — two independent obstructions ($32\, T_3$ structural, rule-independent; $\Phi_P = 72$ selection-rule-driven); $\ker M = 0$ forces the unique non-disturbing coupling $c = 0$ (the sphere axions are RIGIDLY SILENT); $\Phi_P(Q^{(0)}) = 0$ (the sector-0 Okubo mechanism undisturbed); relaxed control: dropping charge conservation reaches the P-block ($\det = -64$) but $T_5 = T_3 = 0$ persists. *Naturalness resolution*: ch$_2$-natural couplings $(9/64, 1/4)$ give $(25/4, -11/2, 97/4, 0, 0)$ and AB-weight couplings $(1/2, 1/4)$ give $(12, -40, 76, 0, 0)$ — BOTH with certificate $(0, 0)$ vs required $(32, 72)$: the failure is scale- and coupling-INDEPENDENT (no "unnatural factor"). *Negative controls*:

invariant-only = `v505` S3.9 replicated + sharpened; flipped rule $m{+}j{-}j'$: Φ_{T_3} kill flip-invariant, Φ_P rule-dependent; $SO(16)$: odd classes EMPTY and AB sum $(15,-18,-15,-4,40)$ — T_5 does not even cancel (E_8 doubly special); D_8: $16\,T_8{+}12\,s_8^2$, no T_3 structure — the question cannot even be posed. Number fences **[C]** only: $32=(h,h){+}(h',h')=20{+}12$, $72=2\tilde\lambda^2_{e_8}$.

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.GAMMA.01` (**[E]** vertex space/selection/certificate + **[C]** fences + **[O]** contact terms); `CELEST.SEAM.01` stays Open with the WP5e targets UPDATED ("the exchange sub-branch is closed by v508; the named remaining targets are the twisted BCOV contact terms (δ_1, binary test) and the full-tensor ledger (δ_2)"). Papers: `tfpt_research_contracts` WP5e section extended by the γ block (γ done — negative result with certificate; the $\delta_1/\delta_2/\varepsilon_1/\varepsilon_2$ roadmap with success/kill criteria); `tfpt_3` celestial section becomes a TWELVE-step story with Step 12 "the exchange no-go and the contact-term criterion" and the "What remains" paragraph updated. Wolfram: six mirrors (the vertex-space dims 2/5 by generator-wise nullspace arithmetic; the twisted quadratics with $K^{(0)}=-15\,K^{(2)}$; the product theorem, symbolic; the rank-3 certificate $(0,32,72)$ with channels and naturalness images; the $SO(16)$ AB sum $(15,-18,-15,-4,40)$; the D_8 quartic $16\,T_8{+}12\,s_8^2$) verified with the active engine the same day ($444\to450$, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `VerificationDag`: `v508` on the QFT/celestial node; registry + `run_all` (502 modules, suite green). No gate closes, no marker moves — the O-fence is only sharpened.

2026-07-21 (XVII) (`v507` — the tautology attack on the `v506` alignment bit, executed and refuted, promoted as `SEAM.BIT.ORIGIN.01`: the ORIGIN THEOREM — marks = fixed points of the hyperelliptic involution = the 4 half-periods $\Lambda/2\Lambda$ of the seam double, and EVERY mark-free collar deck IS a half-period translation (in all three stabiliser types $(4,3,3)/(8,5,3)/(12,3,3)$ the free involutions are exactly the three σ_c: the CLASS of the deck is forced, its POSITION is not); the CORE SOLVE (symbolic λ): a marks-preserving root of σ_c exists $\iff$ $\lambda\in\{-1,2,\frac{1}{2}\}$ (one per c; intrinsic: the deck pairs separate harmonically), union = the harmonic orbit $\iff$ $j=1728$ $\iff$ $\tau=i$; the EQUIVALENCE TETRAD "Uhr existiert $\iff$ ($\tau{=}i$ AND deck = CM-fixed half-period) $\iff$ (Stab = D_4 AND deck central) $\iff$ deck pairs harmonic $\iff$ NS Fock lift NONSPLIT"; the FERMION DICHOTOMY (the new physical face): central $\tilde U^2=256\,\gamma_1\cdots\gamma_{16}\Rightarrow U^2=(-1)^F$ nonsplit vs edge $\tilde U_e^2=+2^7\Rightarrow U^2=+1$ SPLIT (2 vs 0 roots — the bit IS a Fidkowski–Kitaev-type extension class, distinguished but not derived); the silver set = the in-family misalignment witness; Wolfram extension $438\to444$ (verified engine run); *NO marker moves — the bit stays [C], `SEAM.CLOCK.RIGIDITY.01/QGEO.REALIZE.01/P2.TYPING.01` text precised only*)

- **`v507_seam_bit_origin.py` (`SEAM.BIT.ORIGIN.01`, 26 checks, ~2.6 s; Part A sympy-exact torus/Möbius arithmetic + Part B exact $\mathrm{Cl}(16)$ Fractions; deterministic).** Conjecture under attack: marks and deck both descend from the SAME torus structure (the seam double $T^2_\tau\to S^2$, `v214`/`v260`), so the common origin would force the central alignment automatically and the `v506` bit would be a tautology. *Origin theorem (die Herkunft)*: the 4 marks are the fixed points of $w\mapsto -w$ on $\mathbb{C}/(\mathbb{Z}{+}\tau\mathbb{Z})$ = the half-periods $\Lambda/2\Lambda$ (a $\frac{1}{3}$-coefficient never qualifies); t_c descends iff $2c\in\Lambda$ — exactly the three half-period translations (control $c=\frac{1}{3}$ fails); the descended maps are the EXPLICIT rational involutions $\sigma_i(x)=e_i+(e_i{-}e_j)(e_i{-}e_k)/(x{-}e_i)$ with the exact curve-lift identity $\prod(\sigma_i{-}e_l)=\psi_i^2\prod(x{-}e_l)$, ψ_i RATIONAL (the $\wp$-addition formula as pure algebra), closing to Klein V_4; CONVERSELY the freely-acting involutions in ALL three stabiliser types (generic V_4 $(4,3,3)$, harmonic D_4 $(8,5,3)$, hexagonal A_4 $(12,3,3)$) are EXACTLY the three σ_c $\Rightarrow$ every mark-free collar deck (`v216`) IS a half-period translation. *Core solve (symbolic λ)*: σ_c admits a marks-preserving PGL_2 root $\iff$ $\lambda=-1$ (σ_1), 2 (σ_2), $\frac{1}{2}$ (σ_3) — per half-period exactly ONE solution, intrinsically "the deck pairs separate harmonically" (pair cross-ratio -1); generic modulus: V_4 abelian, ALL σ_c central yet clockless ("central alone is empty" — the bit is the CONJUNCTION $D_4 \wedge$ central); at $\tau = i$ the stabiliser jumps to D_4 with σ_1 central (2 roots = the clock pair), σ_2/σ_3 edge-type (0 roots); the `v506` seam deck lands on the central $\sigma_1 = -1/x$ under all 8 mark bijections; CM dictionary: mult-by-i fixes exactly $c^* = (1+\tau)/2$ (mult-by-ρ 3-cycles all three), the clock lift needs the CM unit i ($\psi = 2i/(x+1)^2$, t^2+1 irreducible over $\mathbb{Q}$) while half-period decks lift rationally. *Equivalence tetrad + circularity audit*: clock exists $\iff$ (τ=i AND deck = the CM-fixed half-period) $\iff$ (Stab = D_4 AND deck central) $\iff$ deck pairs harmonic $\iff$ NS Fock lift nonsplit; the `v493` converse independently re-verified — two independently computed converses of ONE equivalence, no circle. *Fermion dichotomy (Part B)*: central arrangement (shift by 8): $\tilde{U}^2 = 256\,\gamma_1\cdots\gamma_{16} \Rightarrow U^2 = (-1)^F$, nonsplit $\mathbb{Z}_4$, 2 roots in the 32-element dihedral lift (the $\mathbb{Z}_8$ tower); edge arrangement (reflection $j \mapsto 4-j$, fixed sites = non-marks = the silver $\{0, \infty\}$ between neighbour marks): the implementer EXISTS but $\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1$, the extension SPLITS, 0 roots, no $\mathbb{Z}_8$; one-particle $(\lambda^2+1)^8$ vs $(\lambda^2-1)^8$; census: all 28 shift-invariant 4-sets cross (central type), all 21 mark-free reflection-invariant sets do not; robust at 8 sites; R-sector edge control splits too — the bit IS a physical extension class (Fidkowski–Kitaev-type): distinguished, but not derived (WHICH class nature realises stays the input). *Silver placement*: $\{\pm 1, \pm(3+2\sqrt{2})\}$ = "$\tau = i$ with deck = a non-CM-fixed half-period translation" (all 8 bijections $\to \sigma_2/\sigma_3$, never σ_1) — the in-family witness: TAUTOLOGY REFUTED. *The one [C] rest (verbatim)*: the alignment bit remains a carrier datum; "no tautology — the origin forces the CLASS of the deck (half-period translation), not its POSITION". No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row `SEAM.BIT.ORIGIN.01` (**[E]** origin theorem/core solve/tetrad/fermion dichotomy/silver placement + **[C]** the bit); TEXT precisions at *unchanged* markers: `SEAM.CLOCK.RIGIDITY.01` ("the bit is now fully characterised as a tetrad … and carries a physical face: the non-split NS Fock extension class — v507"), `QGEO.REALIZE.01` ("the class of the deck IS derived (half-period translation, v507); only its position stays the input", stays **[C]**/**[O]**), `P2.TYPING.01` (origin cross-reference on the common R1 bit). Papers: `tfpt_3` celestial Step-1/2 environment + QGEO paragraph, contracts marks/kill-test paragraph, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates where `v506` is cited. Wolfram: six mirrors (half-period descent $2c \in \Lambda$ with the rational σ_i lift identity; the core solve $\lambda \in \{-1, 2, \frac{1}{2}\}$; the stabiliser counts $(4,3,3)/(8,5,3)/(12,3,3)$ via the complete 24-triple enumeration; the CM fixed point c^* + the lemniscatic $\psi = 2i/(x+1)^2$; the Cl(16) dichotomy $256\,\gamma$-monomial vs scalar 2^7 on explicit 256×256 Jordan–Wigner matrices + the 2-vs-0 root census in the 32-element dihedral lift; the silver bijection census) verified with the active engine the same day (438 $\to$ 444, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the `v493` audit direction and the 28/21 arrangement census stay Python-side (flagged). `VerificationDag`: `v507` on the P2/carrier and QGEO nodes; registry + `run_all` (501 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XVI) (`v506` — seam clock rigidity, Route B of the `v503` classification, promoted as `SEAM.CLOCK.RIGIDITY.01`: TWO exact theorems reduce the order-4 clock input to ONE alignment bit — Theorem A (Möbius): the deck $z \mapsto -z$ has EXACTLY TWO square roots $z \mapsto \pm iz$ in $\mathrm{PGL}_2(\mathbb{C})$, both *automatically* order 4, and a marks-preserving root exists $\iff$ the deck is the CENTRAL involution of the mark-D_4 (counterexample $\{\pm 1, \pm(3+2\sqrt{2})\}$: harmonic with full D_4 but deck non-central $\Rightarrow$ 0 roots — harmonicity alone insufficient); Theorem B (Fock, the core): the NS deck implementation is FORCED to order 4 — $\tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16}$, i.e. $U^2 = (-1)^F$ EXACTLY, a nonsplit $\mathbb{Z}_4$, with the $\mathbb{Z}_8$ tower $V^2 \propto U$, $V^4 \propto (-1)^F$ ($8 = 2|\mu_4|$ fermionically explained) and the R-sector control split ($U_R^2 = +1$); the one [C] rest: "the collar deck is the central involution of the mark-D_4" — the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment; Wolfram extension $434 \to 438$ (verified engine run); *NO marker moves — `QGEO.REALIZE.01` stays [C]/[O] (text precised), `P2.TYPING.01` R1 mapped onto the same bit*)

- **`v506_seam_clock_rigidity.py` (`SEAM.CLOCK.RIGIDITY.01`, 34 checks, ∼3.5 s; Part A sympy-exact Möbius/divisor arithmetic + Part B exact $\mathrm{Cl}(16)$ Fractions; deterministic).** `v503` localised the one remaining identification input ("why does the raw seam carry the order-4 clock", QGEO-R1 ≡ P2-R1); `v506` reduces it. *Theorem A (Möbius side, complete)*: Cayley–Hamilton + the complete case analysis give $g^2 = \text{deck} \Rightarrow \operatorname{tr} g \neq 0$ forces g diagonal with $(a/d)^2 = -1$, and $\operatorname{tr} g = 0$ forces $\det g = 0$ (impossible) — exactly two roots $z \mapsto \pm iz$, both projective order 4, mutually inverse ($\mathrm{Aut}(\mathbb{Z}_4)$), conjugation-covariant; marks filter: a deck-invariant deck-free 4-set $\{u, -u, w, -w\}$ admits a marks-preserving root $\iff$ $w = \pm iu$ $\iff$ marks = one μ_4 orbit $\Rightarrow$ cross-ratio class $\{2, \frac{1}{2}, -1\}$, $j = 1728$, $\tau = i$, free 4-cycle, deck = clock^2, only the scale free; group form: $|\mathrm{Stab}(\mu_4)| = 8 = 2|\mu_4|$ with order profile $\{1{:}1, 2{:}5, 4{:}2\} = D_4$, center = $\{\mathrm{id}, \mathrm{deck}\}$, exactly 2 order-4 elements = the clocks — "deck central in the mark-D_4" $\iff$ "clock exists"; the counterexample $\{\pm 1, \pm(3+2\sqrt{2})\}$ is deck-free AND harmonic ($j = 1728$, cross-ratio $\frac{1}{2}$) with full D_4, but the deck sits *non*-centrally (the central involution crosses the $\pm$ pairs) $\Rightarrow$ 0 roots; the RP filter passes BOTH roots (the $\mathrm{Aut}(\mathbb{Z}_4)$ ambiguity is irreducible, `v492` NC3). *B3/B4 (n-arithmetic + controls)*: free orbit $\Rightarrow n \mid 4$; transitivity $\Rightarrow n = 4$; faithfulness kills $n \geq 5$ (kernel argument through S_4); $n = 3$ fixes a mark; $n = 6$ dies independently at the cusp degree 3 (`v117`); the deck is modulus-blind (`v503` B5 exact); $n = 8$ impossible on 4 marks; hexagonal stabiliser A_4 (0 order-4 elements, squares only orders $\{1, 3\}$); generic stabiliser V_4 (exponent 2); stabiliser trichotomy $V_4/D_4/A_4$ with order-4 counts $(0, 2, 0)$. *Theorem B (Fock side, the core result)*: S^2 has $b_1 = 0 \Rightarrow$ unique spin structure $\Rightarrow$ NS/bounding on every circle (established P1 datum); one-particle: $D^2 = -1$ (charpoly $(\lambda^2{+}1)^8$), quarter-shift $C^2 = D$, $C^8 = 1$ (charpoly $(\lambda^4{+}1)^4$, ζ_8 spectrum); Fock (exact $\mathrm{Cl}(16)$): $\tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8})$ implements the deck on all 16 generators and $\tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16}$ $\Rightarrow U^2 = (-1)^F$ EXACTLY; $(cU)^2 = c^2(-1)^F$ is *never* 1 (center of $\mathrm{Cl}(16)$ = scalars): the $\mathbb{Z}_4$ lift is FORCED, $1 \to \mathbb{Z}_2^{(-1)^F} \to \mathbb{Z}_4 \to \mathbb{Z}_2^{\mathrm{deck}} \to 1$ nonsplit; the quarter-shift lift V (unique even block implementer) has $V^2 \propto \tilde{U}$, $V^4 \propto (-1)^F$, projective order $8 = 2|\mu_4|$ — the `v492` $\mathbb{Z}_8$ spin bridge realised on the 16-Majorana Fock space; R control $U_R^2 = +1$ (split — the forcing rests on NS = established geometry); 16-fold way $\theta_v = 1$ at $\nu = 16$; robustness: the forcing already holds at 8 sites ($16 =$ the carrier count $2^{g_{\mathrm{car}}-1}$, not a tuning). *The one [C] rest (verbatim)*: "the collar deck is the central involution of the mark-D_4" ($\iff$ square-root existence on the mark data) — ONE alignment bit; the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment. No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row `SEAM.CLOCK.RIGIDITY.01` ([**E**] both theorems + [**C**] the alignment bit); TEXT precisions at *unchanged* markers: `QGEO.REALIZE.01` ("the rest is now ONE bit: the central position of the deck in the mark-D_4; order/uniqueness/orbit/$\tau{=}i$ follow — v506", stays [**C**]/[**O**]), `P2.TYPING.01` (R1 mapped onto the same bit, fermionic halving), `QGEO.EMERGE.LIGHT.01` context sharpened.

Papers: `tfpt_3` celestial Step 1 (fermionic $U^2 = (-1)^F$ sentence) + Step 2 (the one-bit reduction), contracts QGEO kill-test paragraph + marks paragraph + CELEST hypothesis note, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates at the `v499/v503` citation sites. Wolfram: four mirrors (the complete square-root solve; D_4 center + order-4 counts + the silver counterexample; the Cl(16) identity $\tilde{U}^2 = 256\,\gamma_1\cdots\gamma_{16}$ on explicit $256{\times}256$ Jordan–Wigner matrices + the split R control; the n-arithmetic + generic kill) verified with the active engine the same day ($434 \to 438$, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the 24-triple stabiliser enumeration and the A_4 profile stay Python-side (flagged). `VerificationDag`: `v506` on the P2/carrier node; registry + `run_all` (500 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XV) (`v505` — WP5e-β of `CELEST.SEAM.01` "the equivariant anomaly ledger on twistor space" promoted as `CELEST.WP5E.BETA.01`: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop twistor inflow is EXACT — denominators $(2,4,2)$ with Dedekind sum $\frac{5}{4} = (|\mathbb{Z}_4|^2{-}1)/12$, equivariant characters $(248,0,-8,0)$ by two routes, invariant Okubo $36\langle x,x\rangle^2$ with the twisted sectors NOT reducing and the AB sum leaving the RIGID residual $32\,T_3$, the index bridge $f(m) = \mathrm{ch}_2(T_m)$ exact with glue defect -78 both routes, the level dials $k=1$ geometric (current count $240/0/0/0$), and the honest E3 REFUTATION: a_0 fills the GRAVITON slot $O(2)$, NOT the axion slot $O(-2)$ — the three sphere classes fill the three twisted axion slots instead; the preregistered kill ("inflow demands a level $\neq 1$") does NOT fire on the equivariant skeleton; WP5e: α (CFT side, `v502`) + β (equivariant twistor skeleton, `v505`) executed, the global BCOV quantisation stays [O]; Wolfram extension $429 \to 434$ (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v505_celestial_wp5e_beta_equivariant_ledger.py` (`CELEST.WP5E.BETA.01`, 47 checks, ∼1.3 s; sympy/Fraction exact, no floats; deterministic).** The twistor side of WP5e pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) on $\mathbb{C}^2/\mathbb{Z}_4$ with the E_8 μ_4-glue monodromy. *AB ledger*: $\det(1-g_j^{-1}) = (2,4,2)$; $\sum_j 1/\det_j = \frac{5}{4} = (|\mathbb{Z}_4|^2{-}1)/12$; $\mathrm{tr}_{g_j}(\mathrm{Ad}\,E_8) = (248,0,-8,0)$ two routes; invariant average 60 = carrier; Frobenius 61568; $\mathrm{tr}_{g_2} = -8 = 120-128$. *Okubo per sector (honest sharpening)*: the invariant sector is exactly Okubo-quadratic ($Q^{(0)} = 36\langle x,x\rangle^2$, $36 = \tilde{\lambda}^2_{e_8}$ — `v495` re-derived through the glue grading); the twisted sectors do NOT reduce (T_5: $(-8,32,-8)$, T_3: $(48,-64,48)$); the AB sum cancels the D_5 quartic exactly and leaves the rigid $32\,T_3$ (the only quartic-free weighting is the $\mathbb{Z}_4$ projector; the graded GS exchange is rank-obstructed in sectors 1–3); heterotic identities (Tr_{120}, Tr_{128}, $\mathrm{Tr}_{E_8} = 9(\mathrm{tr}^2)^2$) exact. *Index bridge*: $f(m) = \frac{1}{4}\sum_j (i^{jm}{-}1)/\det_j = (0,-\frac{3}{8},-\frac{1}{2},-\frac{3}{8}) = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2$ EXACT (E1 = E2 arithmetic); glue ch_2 defect -78 both routes (classes $(-24,-30,-24)$); Killing sums $(320,320,240,320)$, total $1200 = 2h^\vee(h,h)$. $k=1$ *geometric*: lattice current count 240 at $k{=}1$, exactly 0 at $k=2,3,4$; embedding residual $47k^2{+}219k{-}266 = (0,360,814,1362)$; integrality alone fixes nothing (honest, as `v502`); one scale ⇒ one level (`v493` periods). *E3 refutation (prominent)*: $a_0 \in O(8)$ fills the GRAVITON slot $O(2)$, NOT the BSS axion slot $O(-2)$ ($X = -2 \neq +2 = Y$; mismatch $4 = |\mu_4|$ as a **[C]** coincidence) — "the theory brings its GS axion as a_0" is REFUTED; instead the three H^2(ALE) classes carry exactly the Coxeter eigencharacters $\{i,-1,-i\}$ = the three twisted slots (bijection); slot count $1{+}3 = 4 = |\mu_4|$ soft. *Negative controls*: $\mathrm{diag}(i,i)$ breaks the ledger four ways ($\sum 1/\det = \frac{1}{4}$, ω-weight -1, 5 generators); $SO(16)$ glue: characters $(120,0,+120,0)$, defect $-30 \neq -78$, bulk Okubo fails (independent 4th Casimir); $k=2$: current count 0, c-prefactor $-31/48 \neq -\frac{1}{3}$. **[C]** dictionaries ($\mathrm{CS}(\rho)$ mod 1 = discriminant weight; $32 = (h,h){+}(h',h')$; $-30 = -h^\vee$; Ω-contraction). **[O]** the global BCOV/KS quantisation, non-fixed-point contributions, the $O(-2)$ bulk axion, CPS level-from-flux, the pairing $32\,T_3 \leftrightarrow$ sphere axions. VERDICT B; the kill test does not fire on the equivariant skeleton.

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5E.BETA.01`; `CELEST.SEAM.01` updated to "WP5e: α (CFT side, `v502`) + β (equivariant twistor skeleton, `v505`) executed; the global BCOV quantisation stays **[O]** — with the rigid $32\,T_3$ twisted residual and the sphere-axion pairing as the named remaining targets" at *unchanged* Open status. Papers: contracts WP5e bullet → "α + β done" with the a_0 refutation as an honest finding (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *eleven*-step story (new Step 11 "the equivariant anomaly ledger on twistor space", "what remains" = the global BCOV quantisation with named targets). Wolfram: five mirrors (AB denominators + characters two routes; the index bridge + -78 defect; the per-sector Okubo sums over the 240 glue roots with the $32\,T_3$ residual; the lattice current counts + embedding residuals; the axion-slot weight arithmetic + twisted-slot bijection) verified with the active engine the same day (429 → 434, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `VerificationDag` qft node gains `v505`; registry + `run_all` (499 modules at this step; 500 with `v506`, suite green). No gate closes, no marker moves — `SEAM.EQUIV.01` stays [**O**].

2026-07-21 (XIV) (`v504` — WP5d-β of `CELEST.SEAM.01` "split + strong additivity from the lattice" promoted as `CELEST.WP5DB.01`: the two remaining KLM legs of complete rationality witnessed for the orbifold prescription — strong additivity ALGEBRAICALLY EXACT with the shared boundary Majorana ($\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$**, GF2 spans full, rank** $32/32$**; disjoint exactly index** 2 **= the `v501`** $\ln 2$ **bit), the entropic touching defect BOUNDED** $< \ln 2$ **with the Ising** $\frac{1}{4}$**-exponent approach (**$p = 0.2444$**) while the preregistered** $U(1)$**/Dirac control DIVERGES (Klich–Levitov pinned, infinite index), the split property at the elliptic-nome rate** $\pi K(1-x)/K(x)$ **(**1.3–2.0%**) with EXACT orbifold inheritance (**$C \to -C$ **+ the** $\Lambda^2 C$ **compound — Longo heredity), and Pimsner–Popa** $E(a) - a/2 = PaP/2$ **identically (**$\lambda = \frac{1}{2} = 1/[F{:}F_{\mathrm{even}}]$**,** $\lambda_{E_4} = \frac{1}{4} = 1/\mu$**,** $e^{\Delta_\infty} = 2 = 1/\lambda_{\mathrm{PP}}$ **— two independent index routes); with `v501` ALL THREE KLM ingredients carry lattice witnesses; the continuum uplift stays [O] — "finite-group orbifolds of completely rational nets are completely rational" is XU'S THEOREM, cited not claimed; Wolfram extension** $424 \to 429$ **(verified engine run); `SEAM.EQUIV.01` stays [O], `CELEST.SEAM.01` stays Open, no marker moves)**

- **`v504_celestial_wp5d_klm_completeness.py` (`CELEST.WP5DB.01`, 37 checks, $\sim$12 s; numpy/scipy Gaussian machinery ED-validated + exact GF2/integer algebra; deterministic).** Complete rationality (Kawahigashi–Longo–Müger) = split + strong additivity + finite two-interval μ-index; `v501` (WP5d-α) witnessed the index, `v504` witnesses the other two legs on the same critical Majorana seam edge, with the $U(1)$/Dirac number-conserving subnet as the preregistered negative control (the current net, where strong additivity famously *fails* in the continuum — Buchholz–Schulz–Mirbach). *Strong additivity, exact algebra*: "touching = shared boundary Majorana" gives $\mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ EXACTLY (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32); disjoint intervals give exactly HALF (rank 16/32, index 2 — the missing sector odd $\otimes$ odd is the `v501` $\ln 2$ bit, localised at the split point); the neutral $U(1)$ algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). *Entropic witness*: the fermionic touching MI carries the pure UV point-gap law ($(c/3)\ln 2$ increments, 0.115525, dev 1.2×10^{-6} — present for every net, no discriminator); the honest discriminator is *bounded vs divergent*: the $\mathbb{Z}_2$/orbifold deficit rises $0.4045 \to 0.5899$ ($N = 64 \to 4096$), stays $< \ln 2$, and $(\ln 2 - \Delta_2) \sim N^{-p}$ with $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (Ising disorder exponent); split-point scan reflection-symmetric to 3.9×10^{-14}; ED vN same direction; gap continuity to the `v501` $x \to 1$ endpoint; honest note: "defect $\to 0$" would be *false* at the sharp lattice split — bounded $\Leftrightarrow$ finite index (Longo–Xu) is the correct statement; the $U(1)$ control bursts $\ln 2$ from $L = 128$ and grows as $\frac{1}{2}\ln \mathrm{Var}\, Q_A$

(Klich–Levitov slope 0.10134 vs $1/\pi^2 = 0.10132$). *Split/nuclearity*: coupling-block σ_k fall exponentially at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0% (rates 2.29/3.24/4.82/5.81 at $d = 16/64/256/512$); $\sigma_1 \sim x^{0.5044}$ (vs $\frac{1}{2}$); trace norm summable; dimerised control exponential. *Orbifold inheritance exact*: P_A flips $C \to -C$ (σ_k identical, 8.3×10^{-17}) and the even-bilinear coupling Gram is the second compound $\Lambda^2 C$ (singular values $\{\sigma_a\sigma_b\}$, 2.8×10^{-14}) — the same ladder at doubled rate (Longo heredity, **[C]**). *Pimsner–Popa, exact*: $E(a) - a/2 = PaP/2$ identically; attainment in integers (all 16384 (even, odd) monomial pairs at $N = 8$: 8192 extremal with $M^2 = 4$, $\operatorname{Tr} M = 0$, $PMP = -M$, 0 violations; 2048/2048 at $N = 12$; pencil $\min \lambda = 0.5000000000$); $\lambda_{E_4} = \frac{1}{4} = 1/\mu$; index consistency $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\mathrm{PP}}$ and $e^{2\Delta_\infty} = 4 = \mu$; $U(1)$: $\lambda = 1/(m{+}1) \to 0$. *Controls + honesty*: dimerised phase trivial (drift 3.6×10^{-15}); wrong prescription $\{1, P_{AB}\}$ deficit exactly 0; $\mathbb{Z}_2$ finite-size honesty (14.9% below $\ln 2$ at $N = 4096$ — witness, not proof). VERDICT: with `v501` all THREE KLM ingredients of complete rationality are witnessed on the lattice; the continuum theorem for finite-group orbifolds is *Xu's* (cited, not claimed), the concrete seam quotient net and the condensed $(E_8)_1$ net stay WP5e/Costello–Li.

- **Ledger + papers + Wolfram + website.** Ledger: new row `CELEST.WP5DB.01`; `CELEST.SEAM.01` updated to "WP5d: α (μ-index) + β (split + strong additivity) executed via `v501`/`v504` — all three KLM ingredients witnessed on the lattice; continuum uplift stays **[O]** (Xu's theorem cited)" at *unchanged* Open status. Papers: contracts WP5d bullet → "both stages done" with the β-stage numbers + Xu fence (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *ten*-step story (new Step 10 "the KLM completion on the lattice", "what remains" = WP5e twistor inflow only); `tfpt_4` keystone note updated. Wolfram: five mirrors (GF2 span theorem; Pimsner–Popa identity + attainment on explicit gamma matrices; two-interval group $\lambda_{E_4} = \frac{1}{4}$ + index consistency; the Λ^2/Minors compound with Cauchy–Binet and Minors$[-C, 2]$ = Minors$[C, 2]$; the $U(1)$ dephasing $\lambda = 1/(m{+}1)$ family) verified with the active engine the same day (424 → 429, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the lattice curves stay Python-only (flagged). `VerificationDag` qft node gains `v504`; registry + `run_all` (498 modules, suite green). No gate closes, no marker moves — `SEAM.EQUIV.01` stays **[O]**.

2026-07-21 (XIII) (`v503` — the `v215` deferred emergence test (lever 7) executed *light* and promoted as `QGEO.EMERGE.LIGHT.01`: NO dynamical selection of the square modulus — the free field's global $\det'$ winner is the *hexagonal* torus, $\tau{=}i$ is a saddle, criticality at the square is Palais symmetry; the square is pinned by *rigidity* alone (the raw DtN is mod-4 block-diagonal *iff* $\tau{=}i$, an isolated zero); and the residual *converges*: QGEO-R1 ≡ P2-R1 = one common rest, "why does the raw seam carry the order-4 clock"; Wolfram extension 420 → 424 (verified engine run); *NO marker moves — `QGEO.REALIZE.01` stays [C]/[O], `AX.P2.01` stays a declared axiom*)

- **`v503_qgeo_emergence_light.py` (`QGEO.EMERGE.LIGHT.01`, 17 checks; mpmath 40 digits + `v280`-style lattice DtN + sympy exact).** The `v215` lever 7 ("the decisive kill is deferred: compute the raw seam-collar DtN *without* inputting the marks and check whether four square marks *emerge*") executed in its first honest *light* version. *No dynamical selection*: $\log F = \log(\tau_2|\eta|^4)$ (Osgood–Phillips–Sarnak) is exactly S-invariant (dev $< 10^{-35}$), so $t{=}1$ is symmetry-forced critical on the rectangular family $\tau = it$; there $g''(1) = -1.298 < 0$ (strict maximum) but the τ_1-direction has $h''(0) = +0.298 > 0$: $\tau{=}i$ is a *saddle* of the full moduli space; the only critical points are the CM points ($E_2^*(i) = E_2^*(\rho) = 0$ at 40 digits, generic $0.163 \neq 0$); the *global* $\det'$ winner at fixed area is the *hexagonal* torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$, diff 0.021169); criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). *Sharp rigidity*: the DtN mod-4 off-character indicator (the `v215`-K3 lever as a *modulus function*) vanishes to 8×10^{-15} exactly at $\tau{=}i$ and rises strictly monotonically both ways (min 8.4×10^{-2} away) — an isolated zero; $R^2 = \mathbb{Z}_2$ exact as permutation matrices (the `v492` $\mathbb{Z}_8$ bridge on the lattice); $j(it) \geq 1728$ with equality only at $t{=}1$. *The $\mathbb{Z}_2$ chain is modulus-blind*: 4 marks + $[\mathbb{Z}_2, \Delta_t] = 0$ on *every* torus (`v216` confirmed without mark input); pillowcase $\det{}' = \frac{1}{2}$ torus $\det{}'$ exactly ($\pm$-pair structure) — only the order-4 clock distinguishes $\tau{=}i$; $\mathrm{Aut}(E_\rho) = \mathbb{Z}_6$ has *no* order-4 element (the dynamical winner cannot carry the clock); given the clock the modulus freezes automatically ($j = 1728$ identically, `v493` mini). *Residual convergence (the core finding)*: the order-4 element on H^1 is a Coxeter element of $W(A_3)$ (char poly $\lambda^3{+}\lambda^2{+}\lambda{+}1$, order $4 = |\mu_4| = N_{\mathrm{fam}}{+}1$) = the P2 cusp-class carrier datum of `v499` R1 $\Rightarrow$ *QGEO-R1 $\equiv$ P2-R1, one common residual*; chain: marks derived (`v216`), square derived *given* the clock (`v493`), clock = carrier input [[**C**]], rest = its dynamical realisation [**O**]; route classification (`v347` analogue): B-rigidity in the sharpened form "RP seam data $\Rightarrow$ order-4 clock" is the live route. `v215` lever 7 thereby moves from "deferred, not executable" to "light version executed: rigidity, not dynamical selection" — documented in `v503`, `v215` itself unchanged.

- **Ledger + papers + Wolfram + website.** Ledger: new row `QGEO.EMERGE.LIGHT.01`; `QGEO.REALIZE.01` text *precised* (rest = the order-4 clock, identical with the P2 typing rest R1; square + mark count derived) at *unchanged* marker [**C**]/[**O**]; cross-reference added in `P2.TYPING.01` (convergence theorem). Papers: contracts kill-test paragraph (lever-7 light result + the residual-convergence sentence, both with `v503`), `tfpt_3` Step 2 (light-converse sentence), `origin_theory` necessity paragraph, `introduction/tfpt_1/tfpt_2` one-residual cross-references. Wolfram: four mirrors (CM-point identities $E_2(i) = 3/\pi$, $E_2(\rho) = 2\sqrt{3}/\pi$ $\Rightarrow$ $E_2^* = 0$ exact, j/DedekindEta special values; the no-dynamical-selection witnesses; the pillowcase halving as an exact algebraic identity; clock $\Rightarrow$ square + Coxeter) verified with the active engine the same day ($420 \to 424$, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the lattice DtN curves stay Python-only (flagged). `VerificationDag` masses node gains `v503`; registry + `run_all` (497 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XII) (`v502` — WP5e-α of `CELEST.SEAM.01` "prefactor and level bookkeeping" promoted as `CELEST.WP5E.ALPHA.01`: the $q^{-1/3}$ prefactor of E_4/η^8 IS exact μ_4 vacuum-energy bookkeeping (the clock is inner $\Rightarrow$ shift orbifold $\theta = 0^8$, common $-c/24 = -\frac{1}{3}$; discriminant form = Casimir via spectral flow AND 16-Majorana free fermions), and $k = 1$ is forced THREE independent ways (current condition; conformal embedding; central charge = the prefactor itself; plus the WP5b singular-vector dial $31124 - 27000 = 4124$); honest sharpening: glue-h integrality holds for ALL k and fixes nothing; the BCOV/Costello–Li twistor inflow stays [O] — WP5e is subdivided (α done, WP5e proper open); Wolfram extension $416 \to 420$ (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v502_celestial_wp5e_prefactor_level.py` (`CELEST.WP5E.ALPHA.01`, 33 checks, exact sympy/Fraction, no floats).** The two exactly computable consistency anchors of the WP5e twistor uplift established. *Part A — the prefactor*: Hurwitz-ζ exact ($\zeta(-1,\theta) = -B_2(\theta)/2$; $E_b(\theta) = -\frac{1}{24} + \theta(1-\theta)/4$; Ising shift $\frac{1}{16}$ per Majorana); the clock is *inner* ($h = (2,2,2,2,2;0^4)$ and $h' = (0^5;1,1,1,-3)$ both read the class mod 4 on all 240 roots; $(h,h) = 20$, $(h',h') = 12$, sum 32) $\Rightarrow$ twist $\theta = 0^8$ in all four sectors — a *shift* orbifold, NOT a rotation orbifold — so $E_{\mathrm{osc}} = -\frac{1}{3} = -c/24$ is sector-independent ($c = 8 =$ rank $=$ Sugawara 248/31; one common η^8); sector ground exponents $-\frac{1}{3} + (0,1,1,1)$; *discriminant form = Casimir energy* two ways (spectral flow $j^2(h,h)/32$ mod 1; free fermions exact: $D_5 = 10$ Majorana $\to \frac{5}{8}$, $A_3 = D_3 = 6 \to \frac{3}{8}$, glue diagonal $= 16 = 2^{g_{\mathrm{car}}-1}$ = the WP5d seam carrier $\to 1$); the rotation reading *fails* (deck $\frac{3}{16} \neq \frac{3}{8}$ — "clock $\neq$ deck" at the Casimir level); the four sector characters sum exactly to $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$. *Part B — the level*: $k = 1$ forced three independent ways — (i) the current condition $h(J) = k = 1$ (adjoint closure $60{+}64{+}60{+}64 = 248$ at spin 1); (ii) the conformal embedding ($47k^2{+}219k{-}266 = 47(k{-}1)(k{+}266/47) = 0$; D_8: $128k(1{-}k) = 0$); (iii) $c = 8 \Leftrightarrow 240(k{-}1) = 0$ (= the prefactor); plus the `v498` singular-vector dial ($\|(E^{\theta}_{-1})^m 0\|^2 =$

$m!\,k(k-1)\cdots(k-m+1)$, singular vector at grade $k+1$; $31124 - 27000 = 4124$ at k=1 only, $\langle s|s\rangle = +4$ at k=2). *Honest sharpening (mandatory)*: glue-h integrality $h(J^j; k) = k\,(0, 1, 1, 1)$ is integer for ALL $k = 1..8$ — the originally conjectured integrality route fixes *nothing*; the three other dials carry. *Negative controls*: $D_8/SO(16)_1$ (same $c = 8$, SAME $q^{-1/3}$ prefactor, but $h = (0, \frac{1}{2}, 1, 1)$: the discriminator is glue-h integrality, not the prefactor; $\Theta_{D_8} + \Theta_s = E_4$); $\mathbb{Z}_2$ rotation ($E = +\frac{1}{6}$, $h = \frac{1}{2}$); μ_4 rotation (sector-dependent $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$ — the common prefactor breaks); wrong $k \in \{2, 3, 4\}$ fails all five dials coherently (residuals 360/814/1362; $\langle s|s\rangle = 4/12/24$; $31124 \neq 4124$). [[**C**]] Kac generation for $k \geq 3$ ($k = 1, 2$ machine-anchored by WP5b/WP5c). [**O**] the BCOV/Costello–Li twistor inflow itself (Costello's one-loop axion $\lambda_g/(8\pi\sqrt{3})$ on $\mathbb{PT}$, the Costello–Paquette–Sharma level-from-flux mechanism) — v502 anchors the *CFT side* of WP5e only; the KILL branch ("inflow demands a level $\neq 1$") is not triggered on the CFT side, its twistor side is untested.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5e bullet *subdivided* — "WP5e-α (done) [**E**]/[**C**]: the CFT-side prefactor + level pinning" with the core numbers and `v502_celestial_wp5e_prefactor_level.py`; "WP5e proper [**O**]: the twistor inflow"; summit header, honest-success paragraph and status roster updated ("WP5a–WP5d plus WP5e-α landed; WP5e proper the remaining open piece"). `tfpt_3` celestial section: new Step 9 (prefactor + level, one paragraph); `tfpt_4` keystone note updated. Ledger: new row `CELEST.WP5E.ALPHA.01`; `CELEST.SEAM.01` status → "WP5e: alpha stage (CFT-side prefactor + level pinning) executed via v502; the BCOV/twistor inflow itself stays [**O**]". Wolfram: four exact mirrors (Hurwitz-ζ vacuum energies; discriminant = Casimir with the deck failure; the k-fixing equations incl. the "fixes nothing" integrality table; the character sums + D_8 control) verified with the active engine the same day (416 → 420, `ALL WOLFRAM EXTENSION CHECKS PASSED`). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 424/424 after (XIII)) updated; `VerificationDag` qft node gains `v502`; registry + `run_all` (497 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (XI) (v501 — WP5d-α of CELEST.SEAM.01 "the two-interval index from the lattice" promoted as CELEST.WP5D.01: the KLM Haag-duality discriminator measured entropically on the seam edge — the 16-layer carrier saturates the $\ln 2$ offset at machine precision, $\mu_{\text{gauged}} = 4 =$ the v490 census, and the condensation arithmetic $16 \to 4 \to 1$ closes with $\mu = 1$; the preregistered KILL ("μ-offset $\neq 0$ after condensation") does NOT fire; contract now "WP1–WP5d executed; *WP5e the single remaining milestone*"; Wolfram extension $412 \to 416$ (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)

- **`v501_celestial_wp5d_two_interval_index.py` (`CELEST.WP5D.01`, 39 checks, numpy/scipy Gaussian machinery + exact Fractions).** The WP5d roadmap discriminator — $(E_8)_1 \Rightarrow \mu$=1 (Haag-dual two-interval net, one sector) vs $SO(16)_1 \Rightarrow \mu$=4 — executed at the α (lattice-index) stage on the `v367` seam edge (16 critical Majorana layers, $c_- = 8$, `v450`-style ring; Peschel-exact, every Gaussian formula ED-validated to $\leq 1.5 \times 10^{-15}$; the superselection identity $[\rho, P_{AB}] = 0$ machine-exact). *The $\mu = 1$ reference*: the fermionic (fixed-spin-structure) two-interval mutual information is *extensive* — c fit 0.5000, $\max|F(x)| = 5.97 \times 10^{-5}$ at N=512, falling monotonically to 1.3×10^{-6} at N=1024 (Casini–Huerta extensivity as a lattice witness). *The μ-offset*: Rényi-2 exactly $\Delta_2 = -\ln[(1+R)/2]$; the 16-layer seam carrier saturates $\ln 2$ at machine precision ($|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$ at N=512; $R \sim N^{-0.500}$); index reading $[F : F_{\text{even}}] = 2.000000$, $\mu_{\text{gauged}} = 4.000000 =$ the `v490` parity census (*two independent lattice witnesses*); per-layer gauging costs $16 \ln 2$ (the offset measures ln[index], not a blob). *Haag-duality indicator*: fermionic $S(E) = S(E')$ to 8.2×10^{-12}; the orbifold *breaks* complementarity ($D(E) - D(E') = 0.427$ at a=64); the complementary-pair sum is exchange-symmetric and $\leq \ln 4 = \ln \mu(SO(16)_1)$ with max 1.2637 at the self-dual $x = \frac{1}{2}$ — a *double-limit* statement (the first naive fixed-x check was wrong and was replaced — honestly typed in the module). *Condensation arithmetic anchored at both measured ends*: $\det\mathrm{Cartan}(D_5)\cdot\det\mathrm{Cartan}(A_3) = 16$; KLM/Longo–Rehren $16/4^2 = 1$ and $16/2^2 = 4 \to 4/2^2 = 1$ (both routes exact); $\Sigma d_i^2 = (4,4,1)$; $\theta_v = 1$ exactly at $\nu = 2c_- = 16$, rivals $\neq 1$ — so $\mu = 1$ after condensation ([[**C**]] the condensed net is interacting, the `v490` fence): KILL not triggered. *Negative controls*: ν=1 single Majorana (offset non-removable, $\theta_v = e^{i\pi/8} \neq 1$ — the discriminator has teeth); trivial phase (all $\leq 10^{-7}$); wrong sector sum $\{1, P_{AB}\}$ (offset exactly 0). [**O**] WP5d-β (continuum Haag duality / local normality / split of the quotient net; runs as exploration) and WP5e (Costello–Li) stay open.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5d bullet → "α stage done" with the core numbers and `v501_celestial_wp5d_two_interval_index.py`; status line and honest-success paragraph now "four of five milestones landed; WP5e (twistor uplift) is the remaining open piece". `tfpt_3` celestial section: new Step 8 (one paragraph, conservative); `tfpt_4` keystone note updated to WP1–WP5d. Ledger: new row `CELEST.WP5D.01` (fine types Formal + Lattice/Numerical + [[**C**]] dictionary); `CELEST.SEAM.01` status → "WP1–WP5d executed; WP5e [**O**] (single remaining milestone)". Wolfram: four exact mirrors (Cartan determinants; KLM/Longo–Rehren quotients; Σd^2 + index-reading arithmetic; 16-fold-way θ_v) verified with the active engine the same day (412 → 416, `ALL WOLFRAM EXTENSION CHECKS PASSED`); the lattice curves stay Python-only (flagged). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 416/416) updated; `VerificationDag` qft node gains `v501`; registry + `run_all` (495 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (X) (`v500` — WP5c of `CELEST.SEAM.01` "the GNS limit state" promoted as `CELEST.WP5C.01`: the quasi-free family ω_w exists, is positive, stabilises exactly at the WP5a threshold, and its limit carries the null ideal in its GNS kernel — the full 9361-block exact level-2 Gram has rank 4124 exactly with kernel $27000 = V(2\theta)$ weight by weight; $|s\rangle$ IS the GNS zero vector (the WP5b S4.10 tension resolved at the state level); a CCR obstruction proves the family formulation *necessary*; preregistered SUCCESS criterion met, KILL not triggered; Wolfram extension $408 \to 412$ (verified engine run); `SEAM.EQUIV.01` stays [O], no marker moves)

- **`v500_celestial_wp5c_gns_limit_state.py` (`CELEST.WP5C.01`, 35 checks, exact integer/Fraction, no floats, deterministic).** WP5a (`v497`) made the boundary limit precise, WP5b (`v498`) built the deleting operator; WP5c constructs the *state*. ω_w is quasi-free: loop sector = the affine k=1 vacuum n-point functions via the *machine-determined* compact anti-involution $\theta(e_\alpha) = -e_{-\alpha}$ (the unique sign that is an anti-automorphism on all 61504 basis pairs; contravariance exact); radial sector = oscillator pairings x^{wr} (the exact Gibbs regulator, x-adic). *Convergence sharp*: $\omega_w = \omega_\infty \bmod x^{N+1}$ for all $w \geq N{+}1$ (N=8), not yet at $w = N$ — the WP5a threshold $w = n{+}1$ holds *identically at the state level*; positivity of every ω_w checked. *The kernel (core, fully machine)*: the complete exact level-2 Gram — 31124 monomials in 9361 weight blocks (profile 164:1, 37:240, 7:2160, 1:6960) — has level-1 rank 248 positive definite (Cartan block = the E_8 Cartan matrix, det 1), level-2 rank 4124 *exactly* (the preregistered target check), kernel 27000, every block PSD; rank per Weyl orbit $(0,0,1,8,44)$; kernel = $V(2\theta)$ *weight by weight* (Freudenthal cross-check on all 9361 weights); highest-weight norms $(3844, 49, 2, 0)$ at $(0, \omega, \theta, 2\theta)$; direct ideal checks $F_i|s\rangle$, $F_jF_i|s\rangle$ norm 0. *μ_4-equivariance*: $\omega_w \circ \mathrm{clock} = \omega_w$ for all w; the clock descends to GNS; level-2 rank split by clock class $(1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at q^2 — the two-routes identity at the *state* level. *Resolution of WP5b S4.10*: $|s\rangle$ *is* the zero vector of $\mathrm{GNS}(\omega_\infty)$ — the twisted two-C_1-mode floor (6 quarters = $q^{3/2}$) concerns only pre-limit objects (GNS has no state there, 6 mod 4 $\neq 0$). *Negative controls*: k=2 ($\langle s|s\rangle = +4$, dominant blocks full rank — no ideal in the kernel); k=0 (level-1 Gram $\equiv 0$ — no 248 layer without the central extension); D_8 (rank $2076 = 7380 - 5304$, kernel = $V(2\theta_{D_8})$ PSD, but $h = (0, \frac{1}{2}, 1, 1)$: one block of four — one-block completeness stays E_8/μ_4-specific); wrong family $E'_w = mw{+}r$

(the 248 layer erased, $710955 \neq 248$); no damping ($897266 \neq 248$); *CCR obstruction*: at fixed radial pairing $c > 0$ positivity forces $\omega(bb^*) \geq c$ — no w-uniform damping state exists, the family formulation is *necessary* and the family *exists*: the preregistered KILL is probed and not triggered. **[C]** quasi-free extension to the full $*$-algebra; Kac positivity beyond level 2. **[O]** local normality, net structure/Haag duality (WP5d), Costello–Li (WP5e).

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5c bullet → done with the core numbers (state-level threshold $w = N{+}1$, the full 9361-block Gram rank 4124/kernel 27000 weight by weight, the clock-class split, the CCR necessity of the family), citing `v500_celestial_wp5c_gns_limit_state.py`; the WP5b handover marked answered. `tfpt_3` celestial section: new Step 7 (one paragraph, conservative). Ledger: new row `CELEST.WP5C.01` (fine type Formal/Identity **[E]** + **[C]** identifications + **[O]** net questions). Wolfram: four exact mirrors (level-2 block census from the root data; rank/kernel arithmetic with $\dim V(\omega) = 3875$ by Weyl; the glue-frame clock-class split $(1036, 1024, 1040, 1024)$; threshold + level-dial + D_8 census) verified with the active engine the same day ($408 \to 412$); the exact Gram/PSD computation stays Python-side (flagged). `VerificationDag` qft node gains `v500`; registry + `run_all` (494 modules at this step, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

2026-07-21 (IX) (`v499` — the P2 weight-typing postulate HARDENED: the anchor $a = (1, 1, 2)$ typed as the Birkhoff–Grothendieck splitting exponents of the Deligne canonical extension, so integrality/positivity/sum-4 become theorems given the [E] anchors; the `v491` check-9 residual sharpened, the tfpt_2 "balanced" selection typed as stability; Wolfram extension $403 \to 408$ (verified engine run); *no gate closes, no marker moves — `AX.P2.01` stays a declared axiom, `QGEO.SYM` stays [O]*)

- **`v499_p2_weights_deligne_bg.py` (`P2.TYPING.01`, 23 checks, sympy exact, no floats).** `v491` reduced P2 to "3 positive integers summing to 4"; its residual was the *typing* (why integers, why positive, why sum 4). Now typed: given the **[E]** anchors (4 marks, `v216`; rank $3 = \#\text{marks} - 1$, `v137`/`v228`; cusp-class local monodromy $\lambda^3 - 1$ *and* exact irreducibility $|\langle U, M_0\rangle| = 24 = |S_4|$, $\sum_g |\chi|^2 = 24$, `v117`; unitarity (U)), the Deligne canonical extension E of the flavor connection satisfies: $\deg E = -4$ *as a theorem* (residue exponents $\{0, \frac{1}{3}, \frac{2}{3}\}$ the unique lift to $[0, 1)$, trace 1 per mark; par-deg 0; ∞ apparent, monodromy product $= \mathbf{1}$ exactly); $h^0(E) = 0$ *as a theorem* (Mehta–Seshadri stability; all 324 section-saturation cases; BG dictionary $h^0{=}0 \Leftrightarrow a_i \geq 1$); integrality = Birkhoff–Grothendieck. Hence $\{a\} = \{1, 1, 2\}$ unique and $g_{\text{car}} = e_2 = 5$. *Selector typed*: the Schur–Horn permutohedron has integer points exactly $\{(2, 2, 0), (2, 1, 1)\}$ (the two options of the tfpt_2 "Global computation" theorem) and $h^0{=}0$ kills $\{0, 2, 2\}$ — the "balanced/minimal-variance" heuristic is now stability/positivity. *The h^1 discrepancy, real and resolved*: $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$, so "generations $= H^1$ of the rank-3 bundle" is FALSE (negative control: forcing it demands $\deg -6$ and loses uniqueness, $e_2 \in \{9, 11, 12\}$ never 5); correct two-object typing: $L = \mathcal{O}(-D)$ carries the generations ($h^1(L) = 3 = N_{\text{fam}}$, the `v137` basis $= H^0(\Omega^1(\log D))$), E carries them as its *fiber*. *Biswas explicit* (independent witness): the weight denominator $3 = N_{\text{fam}}$ forces the $\mathbb{Z}_3$ cusp cover $y^3 = (z{-}1)(z{-}i)(z{+}1)^2(z{+}i)^2$ (genus 2; NOT the pillowcase double, genus 1, $j = 1728$ — the deck side); $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$; the regular-representation model lands on $\{0, 2, 2\}$ = the *unstable* Schur–Horn companion ($h^0 = 1$): integrality and the sum rule come *out* of the correspondence, stability is the selector. *Negative controls*: no positivity $\to 4/31$ solutions; $\deg -3/-5$ breaks; n-mark scan $e_2 = (n{-}2)(n{+}1)/2 = 5$ ONLY at $n = 4$ (the value, not the uniqueness, is the $n{=}4$ content); MS-rational $e_2 = \frac{16}{3}, \frac{44}{9}$. *Honest residuals* (P2 stays an axiom): R1 module identification **[C]** (`QGEO.SYM` territory), R2 (U) **[C]** (`v33`/`v40`), R3 analytic monodromy (Numerical, `v117` check 4); non-circular (neither `v23`'s splitting input nor `v115`'s $\operatorname{diag} A_0 = a/4$ input used); the `v491` direction-reversal note carries over. Ledger: new row `P2.TYPING.01`; the `v491`/`P2.PARTITION.01` row records "typing residual sharpened by `v499`" (`v491` stays active, no supersede). Registry + `run_all` (493 modules, suite green).

- **Wolfram second path + papers + website.** Five exact mirrors added to the extension file `tfpt_readouts_extension.wl` (partition enumeration under the full typing + naive-typing control + h^1 discrepancy; Deligne residue-trace degree with the $A_0^\star$ witness char poly; Schur–Horn lattice points; Biswas $\mathbb{Z}_3$-cover degree arithmetic incl. deck double $j = 1728$; the n-mark formula), verified with the active engine the same day ($403 \to 408$, `ALL WOLFRAM EXTENSION CHECKS PASSED`); `wolfram/README.md` + `GATE.WOLFRAM.02` + `tfpt_safeguards` Layer-6 count + website counters (116/116+408/408). Papers (one sentence each, conservative): `introduction` anchor paragraph, `tfpt_1` $(5, 3)$ keybox, `tfpt_2` H2 winbox *and* the "Global computation" theorem (the balanced selection typed as stability), `origin_theory` $(5, 3)$ keybox — all citing `v499_p2_weights_deligne_bg.py`; no status card changes. Website: `VerificationDag` P2 node gains `v499`; generated indices via build; no marker moved anywhere.

2026-07-21 (VIII) (Editorial: dedicated celestial-route section in Paper 3 + website mirror — the scattered celestial sentences consolidated into one coherent six-step section; *no new scripts, no status changes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`tfpt_3_e8_audit_bootstrap`: new section "The celestial route: from the μ_4 clock to the $(E_8)_1$ boundary shadow".** The celestial-contract material that had accumulated as one long paragraph inside "The Möbius origin of the irreducible π" is consolidated into a dedicated section (placed after the bootstrap audit, before "Honest status"), telling the executed WP1–WP5b story in six typed steps: (1) setup — the E_8 μ_4-glue as a flat $\mathbb{Z}_4$ monodromy on the A_3 ALE space, clock^2 = deck (spin $\mathbb{Z}_8$, $8 = 2|\mu_4|$) **[E]** (`v492`); (2) deformation — $XY = Z^4+a_0$ pure scale, $j = 1728$ frozen, clock = Picard–Lefschetz/Coxeter monodromy **[E]** (`v493`); (3) consistency honestly demoted — $(\kappa/c_3)^2 = 12$ exact but the alignment format passes 8/8: alignment survives, selectivity does not **[E]**/**[C]** (`v495`); (4) type mismatch + boundary limit — the $(E_8)_1$ character is not a jet-grading conformal block (growth $n^{2/3}$ vs $n^{1/2}$) but the μ_4 period sum 248 holds exactly: a boundary-limit shadow of `SEAM.EQUIV.01`/MMST shape **[E]** (`v496`); (5) the null ideal quantitative — stabilisation threshold $w = n{+}1$, $27000 = (h^\vee)^3$ derived, quotient = sector sum = 4124, $SO(16)_1$ contrast **[E]** (`v497`); (6) the deleting operator explicit — $|s\rangle = (E^\theta_{-1})^2|0\rangle$, $J^a_1|s\rangle = 0$ on all 248 (tally 190/57/1), level dial $2(1-k)$, one-block closure E_8/μ_4-specific **[E]** (`v498`); plus the WP5c–e roadmap and the honest status line (the route does *not* close `SEAM.EQUIV.01`; the GNS/limit question stays **[O]**). All pre-existing script citations (`v492`–`v498`) are carried into the new section; the Möbius section now points to it with one sentence. `tfpt_research_contracts`: a reader's note at the head of the `CELEST.SEAM.01` contract cross-references the new Paper-3 section (the contract stays the full technical reference). `tfpt_4_frontier`: one sentence in the metric-sector keybox records the celestial route as a third, quantitative approach to the keystone (not closing it).
- **Website mirror.** `papers.ts`: the Paper-3 section list gains the celestial-route section (six-step summary in the existing style) and a "Celestial route" highlight; the contracts mirror gains the same cross-reference sentence. `glossary.ts`: three new entries (celestial route, ALE space, null ideal) in the established technical style. Generated indices via build; no status component changes (no marker moved anywhere).

2026-07-21 (VII) (CELEST.SEAM.01 WP5b executed and promoted: `v498` — "the singular vector as an operator", the deleting object of the `v497` null ideal constructed explicitly and machine-verified singular; contract now WP1–WP4 + WP5a + WP5b done, WP5c–e [O] (roadmap); Wolfram extension $398 \to 403$ **(verified engine run);** ***no gate closes, no marker moves, `SEAM.EQUIV.01` untouched*****)**

- **`v498_celestial_wp5b_singular_vector.py` (`CELEST.WP5B.01`, 53 checks, exact integer/Fraction, deterministic, no floats).** Success on the preregistered WP5b criterion: the deleting object exists and is explicit. *Construction*: $|s\rangle = (E^{\theta}_{-1})^2|0\rangle$ (weight 2θ, level $2 = 8$ quarter units $= q^2$, an *integer* level) in a machine-built Chevalley/Frenkel–Kac basis — bimultiplicative 2-cocycle with asymmetry verified on all 57600 ordered root pairs, the $[e_\alpha, e_{-\alpha}]$ sign *forced* by the Jacobi identity (SGN $= -1$, not chosen), the invariant form κ derived ($\kappa(\theta^\vee, \theta^\vee) = 2$), Jacobi exact on all 57600 $\mathfrak{sl}_2$-slice triples + 6000 seeded + all 496 θ-adjacent, all $N_{\alpha\beta} = \pm 1$. *Singularity (machine)*: $J^a_1|s\rangle = 0$ for all 248 basis elements (0 violations), with the case classification 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$ via the central-term cancellation — the *only* case that sees the level k); plus $J^a_2|s\rangle = 0$, $E^a_0|s\rangle = 0$, exact weight, Shapovalov norm 0; the affine PBW engine unit-tested on all 61504 basis pairs. *Level dial*: the F^θ_1 coefficient is $2(1-k)$ — $2/0/-2$ at $k = 0/1/2$: without the central extension the deletion operator does not exist. *μ_4 data*: glue class $j(\theta) = 1$ (machine-built h-adapted chamber, $\langle \theta, h\rangle = 5$, height $29 = h^\vee - 1$), clock phase $i^{2j} = -1$, class$(2\theta) = 2$ sheet-even; the deleting θ-$\mathfrak{sl}_2$ crosses the sheet-odd classes $(1, 3)$ while its square is sheet-even; cross-table θ-grading $\times$ glue class consistent (rows $(1, 56, 126, 56, 1)$, columns $(52, 64, 60, 64)$); 8 quarters $= q^2$ via the per-period dictionary. *Level-2 match*: weight 2θ has multiplicity 1 in the level-2 Fock, so $U(\mathfrak{g})|s\rangle$ is *the* $27000 = (h^\vee)^3$; the direct $\mathfrak{g}_0$-orbit BFS matches the Freudenthal multiplicities of $V(2\theta)$ exactly through depth 4 ([**C**] Weyl complete reducibility beyond); quotient $31124 - 27000 = 4124$; lattice reading $(2\theta, 2\theta)/2 = 4 > 2$: $|s\rangle$ *is* the kernel generator of Fock $\to E_4/\eta^8$. *Negative controls (honest separation)*: the level-1 current state $E^\theta_{-1}|0\rangle$ (one witness, $\to -k|0\rangle$ — the 248-current state that must remain) and 4 generic level-2 states (14–33 witnesses) are *not* singular; at k=2 the *cube* is (generic Kac $(E^\theta)^{k+1}$ mechanics, honestly typed); $SO(16)_1$ has the same singular vector but keeps three extra level-1 primaries ($h = \frac{1}{2}$ breaks one-block fusion; $5304 \neq 14^3$) vs E_8 (comarks all ≥ 2, glue weights $(0, 1, 1, 1)$ integer) — the *one-block closure* is the E_8/μ_4-specific part. *Honest* [**O**]: in the twisted quarter-slot moding two sector-C_1 modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-period dictionary (`v496`), not the per-slot identification, carries $|s\rangle$ to q^2: exactly the WP5c GNS/limit-state question; WP5c–e stay open; `SEAM.EQUIV.01` untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: the WP5b bullet moved to *done* with the key numbers (case tally 190/57/1; level dial $2(1-k)$; $j(\theta) = 1$, clock phase -1; orbit BFS through depth 4; the D_8 contrast) and the twisted-slot tension as the explicit WP5c handover; honest status line and K4 paragraph updated ("WP1–WP4 + WP5a + WP5b executed; WP5c–e [**O**]"). `tfpt_3_e8_audit_bootstrap`: the celestial-contract paragraph extended by the WP5b sentence. Ledger: `CELEST.WP5B.01` (new) + `CELEST.SEAM.01` ("WP1–WP4 + WP5a + WP5b executed; WP5c–e [**O**] (roadmap)"). Wolfram: 5 exact `v498` mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day ($398 \to 403$, `ALL WOLFRAM EXTENSION CHECKS PASSED`); `wolfram/README.md` + `GATE.WOLFRAM.02` counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 403/403. Website: `papers.ts` contract mirror ("WP1–WP4 + WP5a + WP5b done, WP5c–e roadmap"), DAG `qft` node + `v498`, generated indices via build. Suite now 492 scripts, all green; `SEAM.EQUIV.01` stays [**O**].

2026-07-21 (VI) (CELEST.SEAM.01 WP5a executed and promoted: `v497` — "the null ideal from the limit", the `v496` boundary-limit shadow made a precise coefficientwise limit and the level-2 null ideal derived from root data; contract WP5 subdivided, now WP1–WP4 + WP5a done, WP5b–e [O] (roadmap); Wolfram extension 394 → 398 (verified engine run); *no gate closes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`v497_celestial_wp5a_null_ideal.py` (`CELEST.WP5A.01`, 34 checks, exact integer/Fraction, no floats).** Three results make the WP4 limit reading precise. *Stabilisation theorem*: the one-parameter family χ_w of graded Fock characters on the chiral jet generators (loop energy m, radial number r; $E_w = m + wr$ in quarter units $u = q^{1/4}$) contains the chiral jet grading as its w=2 member (generator counts $64, 120, 128, 180, 192, 240, 256, 300$ per degree), and its u^n coefficient equals the quarter-moded loop Fock for *all* $w \geq n$+1, strictly larger for every $w \leq n$ (verified $n \leq 8$, $w \leq 10$) — the `v496` "boundary-limit shadow" is a precise *coefficientwise* limit $\varepsilon = 1/w \to 0$ with explicit threshold $w = n$+1, not a slice (convention clarified: 897266 = the quarter-moded loop Fock at u^4 = *integer* level 1). *The null ideal derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give $\mathrm{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ with residual exactly zero (independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; $C_2 = (124, 96)$, Dynkin indices 6750/750; level-1 integrability ($\langle 2\theta, \theta^\vee\rangle = 4$, $\langle \omega, \theta^\vee\rangle = 2$) leaves the vacuum as the only level-1 primary (Kac, used only through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ exactly. *Two routes, one number*: the μ_4 theta-split sector sum at $q^2 = (1036, 1024, 1040, 1024)$ gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space (q^3: $34752 = \chi_3$). Negative controls: $SO(16)_1$ through the same pipeline gives *four* components ($5304 + 1820 + 135 + 1$), $5304 \neq 14^3$ (the $h^{\vee\,3}$ identity is not generic), quotient $2076 = \Theta_{D_8}/\eta^8$ at q^2 (the recipe validated on a second algebra), block weights $(0, \frac{1}{2}, 1, 1)$ break one-block fusion (vs E_8's integer $(0, 1, 1, 1)$); only $P = 4 = |\mu_4|$ periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor ($64 \neq 60$). *Honest* **[O]**: the limit does *not* generate the truncation — the singular vector as an *operator* in the limit net (WP5b), the GNS limit state (WP5c) and local normality/Haag duality (WP5d) stay open; the celestial route now has the same two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; `SEAM.EQUIV.01` untouched.
- **Contract + surfaces synced.** `tfpt_research_contracts`: WP5 subdivided — WP5a moved to *done* with the key numbers (stabilisation threshold $w = n$+1; $27000 = (h^\vee)^3$ derived; quotient = sector sum = 4124; $SO(16)_1$ contrast), WP5b–e added as roadmap bullets with success/kill criteria (WP5b singular-vector operator in the quarter-moded current Fock; WP5c GNS limit state, kernel $\supseteq$ ideal; WP5d local net/Haag duality via the KLM route; WP5e BCOV/Costello–Li twistor uplift, partition function = E_4/η^8 incl. the $q^{-1/3}$ prefactor); K4 paragraph and the honest-success status line updated. Ledger: `CELEST.WP5A.01` (new) + `CELEST.SEAM.01` ("WP1–WP4 + WP5a executed; WP5b–e **[O]** (roadmap)"). Wolfram: 4 exact `v497` mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day (394 → 398, `ALL WOLFRAM EXTENSION CHECKS PASSED`); `wolfram/README.md` + `GATE.WOLFRAM.02` counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 398/398. Website: `papers.ts` contract mirror ("WP5a done, WP5b–e roadmap"), DAG `qft` node + `v497`, generated indices via build. Suite now 491 scripts, all green; `SEAM.EQUIV.01` stays **[O]**.

2026-07-21 (V) (`comb-meta-limit` re-run with the PG.06b FULL Vela recovery ingested — the stale single-obs PG.06b reference (noise-peak $F_0 = 11.19275\,\mathrm{Hz}$) corrected in the channel definitions; the new X-ray channel enters *censored* (no absolute ε), both aggregate limits numerically unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)

- **`experiments/comb-meta-limit` channel update (methodology untouched).** The Vela X-ray channel (`channels.py`) now ingests the PG.06b FULL reduction (`pg06b_full_vela.json` in `pulsar-glitch-recovery/results`: 665 obs → 491 usable → 119 segment-coherent $\nu(t)$ points) instead of the superseded single-obs `pg06b_vela.json`; the stale code note quoting the noise-peak "detection" $F_0 = 11.19275\,\mathrm{Hz}$ is corrected on record (true Vela $F_0 = 11.1861692\,\mathrm{Hz}$), and the channel key is fixed from the mislabel `PG.07` to `PG.06b`. Censoring handled exactly by the existing Tier-B rule (nothing invented): the comb observable is a *linear* $\delta\nu(\tau)$ residual (μHz, not a fractional ln-flux modulation), both legs sit below the 2.8-period gate (2019: 2.17, 2021: 1.38), and the surrogate-calibrated injection never reaches ε_{90} below $\varepsilon = 1$ ($\varepsilon_{50} = 0.55$ in 2019 only) ⇒ no absolute per-channel ε-UL exists for the DL/HKSJ combination; the channel stays enumerated-only ("no constraint"), carrying the normalised context amplitude $\sqrt{2g} = 0.40$ (leg gains 0.046/0.114, min $p = 0.014$ — the 2019 audit-level chance feature: Bonferroni $\times 10 = 0.14$, shuffle $p = 0.40$).
- **Full re-run (seed 0, deterministic): both limits numerically unchanged.** Boundary/horizon-scoped: `data_limited` (still no horizon channel with an absolute ε); all-channel universal-DSI 95% UL $\varepsilon < 0.11954$ ($\approx 12.0\%$; HKSJ $t_2 = 2.92$, $\rho = 0.3$-inflated 0.137; 6.91× the predicted $\varepsilon = 0.0173$), still driven by the surface log-flux channels A1/A4/A5 (A4 GRB the anchor, per-channel UL 0.108); now 10 channels enumerated / 3 usable. Injection self-consistency unchanged (inject 5% → recover 4.8%, UL 5.5%; coverage 0.92/150 trials). Surfaces synced: experiment README (tier table, ingest list, PG.06b FULL bullet), `results/results.json`, scorecard row `data_value`/source via `build_evidence_scorecard.py` (status stays `data_limited`; counters unchanged, 120 rows), `experiments/README.md` catalog row. No claim; firewall intact.

2026-07-21 (IV) (Wolfram engine reactivated — the deferred second-path mirrors VERIFIED: base 116/116 + extension 394/394 (378 → 394, the 16 mirrors accumulated since 2026-07-14 now counted); `GATE.WOLFRAM.01/02` and all counter surfaces synced; *no gate closes, no marker moves*)

- **Verified engine run 2026-07-21 (Wolfram Engine 14.3, license reactivated).** `wolframscript -file verification/wolfram/tfpt_readouts.wl` → `116 passed, 0 failed`, `ALL WOLFRAM CHECKS PASSED`; `tfpt_readouts_extension.wl` → `394 passed, 0 failed`, `ALL WOLFRAM EXTENSION CHECKS PASSED`. The 16 exact mirrors added while the engine was unactivated all enter the counted total (378 → 394): `v479` Kronheimer bridge (2), `v491` partition corollary (3), `v493` celestial WP2 clock deformation (4), `v495` celestial WP3 Green–Schwarz alignment (3), `v496` celestial WP4 character shadow (4).
- **Counter surfaces synced.** `verification/wolfram/README.md` (current-state block + the five deferral notes resolved; `v484/v485/v486/v487/v488/v492` stay mirrorable-but-unmirrored, Python-exact, so they are *not* counted); ledger `GATE.WOLFRAM.01` (re-verified 2026-07-21) + `GATE.WOLFRAM.02` (394/394, UPDATE note); `tfpt_safeguards` Layer-6 count + website `papers.ts` mirrors (116/116 + 394/394); `v479` registry/`run_all` deferral note resolved. No numeric claim changes; the second computational path is simply verified-current again.

2026-07-21 (III) (CELEST.SEAM.01 WP3 + WP4 executed and promoted: `v495` Green–Schwarz alignment (verdict B, honestly non-selective) + `v496` character-vs-S-algebra type mismatch (verdict B(ii), boundary-limit shadow); contract now WP1–WP4 done, WP5 [O]; *no gate closes, no marker moves, `SEAM.EQUIV.01` untouched*)

- **`v495_celestial_wp3_gs_alignment.py` (`CELEST.WP3.01`, 25 checks, exact Fraction/sympy).** The pre-registered WP3 "alignment test only": the Okubo coefficient $\mathrm{Tr}_{\mathrm{adj}}X^4/(\mathrm{Tr}_{\mathrm{adj}}X^2)^2 = 5/(2(\dim\mathfrak{g}+2))$ is *derived* as an exact polynomial identity for all 8 algebras on Costello's list (sl_2, sl_3, g_2, so_8, f_4, e_6, e_7, e_8; $\mathrm{e}_6/\mathrm{e}_7$ carved out of the `v492` E_8 root set; negative control: sl_4 fails, 5/32 vs 3/32); the closed form $\tilde{\lambda}^2 = 10h^{\vee 2}/(\dim+2) = h^\vee+6$ holds exactly across the Deligne series; E_8: $\tilde{\lambda} = 6$ (unit-trace) resp. $\lambda_{\mathrm{fund}} = 1/10$ (adjoint-trace) exact, so $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\mathrm{fam}}$ resp. 1/300 — with κ/c_3 itself *irrational* ($2\sqrt{3}$). Byproduct: the printed $\lambda^2_{\mathrm{so}_8} = 3/2$ in Costello's appendix A is a factor-2 slip (his own weight content gives 3). **Look-elsewhere (the hard fence):** the alignment format passes 8/8 across the whole list (zero selectivity), $\tilde{\lambda}$-integrality 2/8 (shared with sl_3), only g_{car}=5 | $h^\vee$ is e_8-selective (1/8), the isolating conjunction is post hoc. Kill test K3 does *not* fire but is scope-demoted: *alignment survives; selectivity does not* — compatibility, not e_8-selective evidence. Verdict B.
- **`v496_celestial_wp4_character_shadow.py` (`CELEST.WP4.01`, 25 checks, exact integer/Fraction/sympy).** WP4, the type mismatch head-on: the $(E_8)_1$ character $E_4/\eta^8 = (1, 248, 4124, 34752, 213126, \dots)$ is *not* a conformal block of the glue-equivariant $E_8[\mathbb{C}^2]$ S-algebra in its own jet grading — localised three ways: (a) spin $1 - d/2$ unbounded below and 248 never a tower dimension ($d \leq 100$); (b) growth $n^{2/3}$ vs $n^{1/2}$ (quadratic cumulative count $31s^2$+$92s$+60, $31 = k$+$h^\vee$; f_n/χ_n strictly increasing $n = 1..12$); (c) the level-2 null ideal deletes exactly $27000 = 30^3 = (h^\vee)^3$ out of $\mathrm{Sym}^2(248) = 1$+3875+27000 — no jet analogue. *But* the boundary/period reading holds exactly at the current stratum (zero modes = 60 = C_0 currents, jet slice cycles $(60, 64, 60, 64)$, one μ_4 period sums to 248, glue weights $(0, 1, 1, 1)$ integer; free loop Fock 897266 $\gg$ 248, so the truncation must come from a limit) — the same MMST scaling-limit shape as `SEAM.EQUIV.01` (`v336`/`v449`). K4 fires only against the exact-block reading; the constructive limit question passes to WP5. Verdict B(ii).
- **Contract + surfaces synced.** `tfpt_research_contracts` WP3/WP4 bullets moved to *done* with the look-elsewhere demotion in the text, K3/K4 marked evaluated, status line now "WP1–WP4 executed (verdicts B, B, B, B(ii)), WP5 [**O**]"; one conservative sentence added to the `tfpt_3` celestial paragraph; ledger rows `CELEST.WP3.01` + `CELEST.WP4.01` (new) and `CELEST.SEAM.01` updated; `v495`/`v496` Wolfram mirrors added to `tfpt_readouts_extension.wl` (3 + 4 exact checks; count with the next verified engine run — engine still unactivated); website `papers.ts` contract highlight "WP1–WP4 done" + DAG `qft` node extended. Suite now 490 scripts, all green; `SEAM.EQUIV.01` stays [**O**].

2026-07-21 (II) (PG.06b FULL: the complete NICER Vela archive reduced — the first X-ray $\nu(t)$ leg of the recovery-comb search, verdict `data_limited`; the old PG.06b "pipeline proven" fold corrected on record; *experiments level only — no verification module, no gate closes, no marker moves*)

- **`experiments/pulsar-glitch-recovery` PG.06b FULL executed (`tfpt-pulsar vela-full`).** The full NICER Vela-pulsar (PSR B0833-45) L2 archive is reduced, with the manifest frozen *before* the bulk download (`manifest_full_reduction.json`: all 665 ObsIDs, preregistered 2019/2021/2024 glitch windows, 20-GB abort gate): 665/665 obs downloaded (2.6 GB, raw gitignored) → 491 usable → 240 per-obs detections → **119 segment-coherent $\nu(t)$ points** (108 fully coherent), median $\sigma_\nu \approx 0.6\,\mu$Hz; barycentring validated against PINT to 3.4 μs — the *first X-ray $\nu(t)$ probe* of the search program. Sanity gate PASS: $\Delta\nu/\nu$ 2019 $= 2.39 \times 10^{-6}$ vs 2.5012×10^{-6} published, 2021 $= 1.272 \times 10^{-6}$ vs 1.26×10^{-6}; the 2024 glitch is not reducible (only ∼150-s snapshots, 9.7 ks — documented).

- **Frozen-kernel comb ($\omega = 2.5827$, $\varepsilon_{\mathrm{pred}} = 0.0173$): verdict `data_limited` (range- and amplitude-limited).** Reach 2019: 2.17 comb periods (< the 2.8-period gate; beats the PG.08 public Vela reach 1.83), 2021: 1.38; the 2019 raw $p = 0.014$ has the kernel as smallest-p battery member BUT Bonferroni $\times 10 = 0.14$ and shuffle $p = 0.395 \Rightarrow$ chance-level by the preregistered escalation rule (audit flag, NOT a candidate); 2021 $p = 0.731$; $\ln \tau$ stack $p = 0.049$/shuffle 0.307. Surrogate-calibrated injection at the real sampling and noise: $\varepsilon_{50} \approx 0.55$ (2019 only), ε_{90} never reached below $\varepsilon = 1$, 0% power at the predicted $\varepsilon = 0.0173$ — a weaker amplitude probe than the PG.08 radio residuals, but it independently reproduces the PG.06 range-blindness on real data. Second off-kernel flag: $\lambda = e$ in the 2021 leg (raw $p = 0.002$, Bonferroni 0.018; the same non-kernel periodicity class as PG.08 J0742−2822 — flagged, not claimed). The sub-day early window that would clear the gate is absent from the archive (first usable obs 3.4 d / 7 d post-glitch); next decider: <3-d post-glitch X-ray observations of a future Vela glitch (eXTP class).
- **Correction on record.** The old PG.06b single-obs "pipeline proven" detection at $F_0 = 11.19275\,\mathrm{Hz}$ ($H = 18.4$) was a noise peak; the true Vela frequency at that epoch is $F_0 = 11.1861692\,\mathrm{Hz}$ ($H = 99.7$, consistent with the phase-connected PuMA ephemeris). The pipeline remains proven — by the corrected fold. Scorecard row updated to "recovery comb on the full NICER Vela X-ray nu(t) (PG.06b FULL)" (status stays `data_limited`; counters unchanged, 120 rows: 49 consistent / 9 tension / 26 null / 33 data-limited / 3 parked); `experiments/README.md` catalog + §1 prose, pulsar README, `experiments/next.txt` synced. No claim; firewall intact.

2026-07-21 (the two 2026-07-20 prediction-of-record rows promoted to the suite: v494 — α_s running + μ-distortion branch discriminator machine-checked, ledger rows COSMO.RUNNING.01/COSMO.MUDIST.01, prediction surfaces synced; *no gate closes, no marker moves*)

- **`v494_cosmo_running_mu_record.py` (23 checks, all green).** The α_s-running and μ-distortion scorecard rows of 2026-07-20 (IV) get their load-bearing module so the new paper/website statements are machine-backed. **[E]** leading-order identities (sympy exact): $\alpha_s = \mathrm{d}n_s/\mathrm{d}\ln k = -2/N_\star^2 = -r/6$, $\beta_s = -4/N_\star^3$ (NOT $-8/N^3$); LO band $N_\star \in [50, 60]$: $[-8.0, -5.6] \times 10^{-4}$. **[E]** exact slow roll on the Einstein-frame Starobinsky potential (third-order Lyth–Riotto + finite-difference cross-check < 1%): $\alpha_s(51.4) = -7.09 \times 10^{-4}$, $\beta_s(51.4) = -2.70 \times 10^{-5}$. **[C]** data: Planck 2018 (-0.0045 ± 0.0067) $+0.57\sigma$; record leg P-ACT-LB ($+0.0062 \pm 0.0052$) -1.33σ (watch channel: positive central value; kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch). **[C]** μ-distortion (Chluba window, +21% calibrated on the ΛCDM anchor): sharp branch $N_\star$=51.4 $\mu = 1.6 \times 10^{-8}$, profiled record $N_\star$=56.1 $\mu = 1.94 \times 10^{-8}$, band 1.5–2.3 $\times 10^{-8}$; branch split $\Delta\mu = 3.0 \times 10^{-9} \Rightarrow 3\sigma$ $A_s/N_\star$ branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; PIXIE baseline 9×10^{-8} does not reach; kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$). **[E]** comb blindness (honest S15): the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is structurally invisible in the μ window ($|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage-2050). Fences: $N_\star$ stays an external reheating input (`GATE.NSTAR.01`); the frozen band $[50, 60]$ stays the surface of record (v84/v86). Ledger rows `COSMO.RUNNING.01` and `COSMO.MUDIST.01` added.
- **Prediction surfaces synced (the missing mirror of 2026-07-20 (IV)).** `predictions.txt` gains the two rows (α_s **[C]** with the -1.33σ P-ACT-LB pull and the positive-running kill; μ-distortion **[C]** data-limited with the Voyage-2050 branch split); website `lib/predictions.ts` gains the matching cards `alpha-s-running` and `mu-distortion` (confrontation + experiment blocks pointing at `experiments/tfpt-discovery`); the cosmology DAG node lists `v494` and states the running/μ readouts.
- **Paper bodies.** tfpt_1 (Inflation interface) gains the running consistency-relation

paragraph ($\alpha_s = -2/N_\star^2 = -r/6$, exact slow roll, pulls, positive-running kill) with `\veri{v494_cosmo_running_mu_record.py}`; tfpt_2 (inflation-pivot keybox) gains the μ-distortion branch-decider sentence (sharp 1.6×10^{-8} vs profiled 2.0×10^{-8}, 3σ at Voyage-2050) with the same `\veri`. No marker of any existing claim changes. (v494 is numerical/quadrature, Python-only by the suite convention, flagged in the docstring; `GATE.WOLFRAM.01/02` untouched.)

2026-07-20 (VIII) (E8 mass-ladder blind-recovery bed EXECUTED: verdict `validated` at the detector level — the Cartan–PF ladder pipeline blindly recovers the published Zamolodchikov-E8 spectra of $BaCo_2V_2O_8$ (8/8) and $CoNb_2O_6$ (6/8 in-window) and rejects every control class; *experiments level only — confirms Zamolodchikov E8, not TFPT; no verification module, no gate closes, no marker moves*)

- **`experiments/e8-ladder-bed/` executed (preregistered `hypotheses/e8_ladder_bed_v1.yaml`).** The blind ladder detector (assignment DP + scale fit + 5000-ladder Monte-Carlo null; the same detector class TFPT uses on search surfaces, e.g. `hfqpo-ladder`) on the only public *known-answer* mass-ladder dataset: $BaCo_2V_2O_8$ (Zou+ PRL **127**, 077201; INS at the 4.7 T 1D QCP, Fig. 3a extraction QA-validated, 17 peaks blind) recovers **8/8 rungs**, $\chi^2/\mathrm{dof} = 0.35$, MC $p = 0.0056$ (singles-only run $p = 0.0004$; σ-variant robust); $CoNb_2O_6$ (Amelin+ PRB **102**, 104431; verbatim THz values) recovers **6/8 in-window rungs**, $\chi^2/\mathrm{dof} = 0.52$, $p = 0.0038$ (m_7/m_8 lie outside the 0.6-THz low-pass window; the broad 0.43-THz (m_1+m_2)-continuum onset is correctly left unassigned). Cartan–PF self-test: the sorted PF eigenvector of $2I - A(E_8)$ equals the Zamolodchikov ladder to 1.5×10^{-15} ($m_2/m_1 = \varphi$ to 2.2×10^{-16}; lowest Cartan eigenvalue $2 - 2\cos(\pi/30)$ to 4.5×10^{-14}).
- **Control battery — every class rejected.** Scramble 0.5%/1.5% false-positive rate (200 log-gap permutations, near-duplicate accounting); offset $\pm\{0.3, 0.5\}m_1$: 0/8 detections; jitter dose-response $\leq 5\%$ at the destructive 30% level; wrong ladders on the real data (harmonic 1..8, A_3-PF, D_8-PF, Airy confined-spinon) all beaten by E8; the synthetic Airy bed (the true zero-field physics of these compounds) yields **0** valid rungs. The one disclosed v1.1 criteria amendment (jitter as dose-response, scramble near-duplicates, χ^2-gated wrong-ladder comparator) is fully documented in the hypotheses YAML with all v1 numbers preserved; the recoveries themselves were never touched.
- **Integration (firewall intact).** Scorecard row "E8 mass-ladder blind recovery bed" moves `parked` → `consistent` (stage `not_applicable`, tier `analog_positive_control`, leakage `detector_control` — the executed-analog basket of `qc-recovery-kernel` and `qgeo-eit-soff`); counters 120 rows: 49 consistent / 3 parked; `experiments/README.md` catalog row + stats (generator); website `predictions.ts EXPERIMENTS_AUDIT` counters; `experiments/next.txt`. This validates the *ladder-detector pipeline* — it confirms Zamolodchikov E8, *not* the TFPT axioms: no [**E**], no `\veri`, no ledger row.

2026-07-20 (VII) (v493 — WP2 of `CELEST.SEAM.01` executed: the clock-invariant deformation $XY{=}Z^4{+}a_0$ is a PURE SCALE with the $\tau{=}i$ pillowcase frozen, the μ_4 clock IS the Picard–Lefschetz/Coxeter monodromy of the seam deformation, and the BHS deformed-algebra pattern transfers $\mathbb{Z}_2 \to \mathbb{Z}_4$ with no sector leak; *verdict B, contract updated to WP2 done, no gate closes, no marker moves, `SEAM.EQUIV.01` and the v216 residual stay [O]*)

- **`v493_celestial_wp2_clock_deformation.py` (`CELEST.WP2.01`).** WP2 of the celestial contract, sympy exact (47 checks), promoted from the `experiments/tfpt-discovery/` probe `celestial_seam_wp2_clock_deformation_probe.py`. *Geometry:* $P(iZ){=}P(Z)$ forces $a_3{=}a_2{=}a_1{=}0$ sharply and two-sided ($\prod_k (Z - i^k w) = Z^4 - w^4$); $XY{=}Z^4{+}a_0$ is smooth iff $a_0{\neq}0$ (disc $= 256\, a_0^3$); the branch points $Z^4{=}{-a_0}$ are one free μ_4 orbit in *every* fibre with cross-ratio 2;

the binary-quartic invariants of $w^2{=}Z^4{+}a_0$ are $I{=}12a_0$, $J{=}0$ *identically*, so $j{=}1728$ and the $\tau{=}i$ pillowcase shape (`v168/v214`) is *frozen* for all a_0 — a_0 is a pure scale, no shape modulus survives clock-invariance. *Periods/monodromy:* $\Pi = t(i{-}1)(1, i, i^2)$, all three vanishing spheres carry the same volume $\sqrt{2}\,t$ (lockstep); the clock acts on H_2 as a *Coxeter element* of $W(A_3)$ (char $x^3{+}x^2{+}x{+}1$, eigenvalues $\{i, -1, -i\}$ = the A_3 exponents, $h(A_3){=}4{=}|\mu_4|{=}N_{\mathrm{fam}}{+}1$), fixes no cycle ($\det(A{-}1){=}{-}4$), and *is* the Picard–Lefschetz monodromy of the family ($a_0 \to e^{2\pi i}a_0$ gives the 4-cycle); the surviving direction is the weight-1 χ_1-Fourier diagonal $(1, i, i^2)$, *not* the naive invariant diagonal; the versal weights $(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)$ are the regular μ_4 representation. *Algebra (BHS $\mathbb{Z}_2 \to \mathbb{Z}_4$, `arXiv:2305.09451`):* mode dictionary $\{w[m, n] : m{\equiv}n \bmod 4\}$; the fibre bracket $-4\,\mathrm{Nambu}(XY{-}Z^4{-}a_0)$ anchored at the $a_0{=}0$ orbifold ($\{X, Y\}{=}16Z^3$, Jacobi + Casimir exact, uniform clock-degree -1); S-algebra corrections exactly *linear* in a_0 at the $\mathbb{Z}_4$ wrap, conserving the μ_4 grade — no sector leak, the `v492` equivariant sector deforms consistently; $a_0 \leftrightarrow$ BHS's weight-0 c^2 slot; matrix factorisations deform along all orbit splittings, the clock fixing exactly the antipodal 2+2 sector (= the EH/$\mathbb{Z}_2$ middle node, 4=2×2). *Negative controls:* $a_1Z \Rightarrow j{=}0$ (hexagonal); pure a_2Z^2 stays singular (A_1); $a_2{\neq}0 \Rightarrow j{=}1556068/81{\neq}1728$; explicit grading leak — doubly sharp. *Verdict B* via three named **[C]** identifications (-4Nambu = the $k{=}4$ CCA bracket; period = root difference; seam-scale reading via clock^2=deck); the `v216` residual is *typed*, not moved (given the order-4 clock the square is automatic — a relocation of the same input, no new derivation).

- **Contract updated to WP2 done.** `tfpt_research_contracts` `CELEST.SEAM.01` section: WP2 bullet "finite, next" → *done (verdict B)* with the two sharpenings (clock = Coxeter/Picard–Lefschetz monodromy; surviving direction = χ_1-diagonal, a_0 pure scale); kill-test paragraph K1 "survived by WP1 and WP2"; status line `CELEST.WP2.01` **[E]**/**[C]** executed, WP3–WP5 stay **[O]**. Ledger rows `CELEST.WP2.01` (new) + `CELEST.SEAM.01` (WP2 done) + `GATE.WOLFRAM.02` (four exact `v493` mirrors added to `tfpt_readouts_extension.wl`, counted with the next verified engine run — engine unactivated, same convention as `v491`); registry + `run_all` (487 modules, suite green); website mirrors (`papers.ts` contract section, generated script index). `SEAM.EQUIV.01` stays **[O]**; no marker moves anywhere. Exact (sympy).

2026-07-20 (VI) (HFQPO "third tooth": the preregistered archival RXTE PCA scan has RUN — a well-powered null with limits; scorecard row S1.e `data_limited`→`null`, no marker moves)

- **H3 stage 2 executed (`experiments/hfqpo-ladder/results/archival/`).** The preregistered archival scan (`hypotheses/hfqpo_v1.yaml`, `archival_design`) ran end-to-end on public HEASARC data: 77 ObsIDs / 4.42 GB across all four 3:2-pair sources (GRO J1655−40, XTE J1550−564, GRS 1915+105, H1743−322). Sanity gate PASS (11/12 published pair lines reproduced, e.g. 450.3 Hz at 5.37% rms 4.4σ, 280.1 Hz at 3.71% 9.9σ); injection calibration PASS in all four sources (90%-recovery sensitivities 5.11/2.68/0.89/1.65% rms at the tooth frequencies).
- **Blind-scan verdict: NULL with limits.** Neither the geometric tooth $\nu_3 = 1.5\,\nu_u$ (661.5/414/252/363 Hz) nor the integer control line $4\nu_0$ (588/368/224/322.7 Hz) is detected in any source ($\sim 0\sigma$ single-trial; trials correction $N{=}8$ as preregistered). 3σ rms upper limits (tooth, primary bands): 3.06/1.61/0.53/1.26% — in every source at or below the strength of the detected upper pair line. The ladder reading gains no support but is not killed (the `harmonic_hit` branch did not fire either; the anti-kernel record 92=184/2, 34/68 stands); GR parametric resonance stays the favorite. Caveats: GRS 1915+105 scanned in a proxy state (the 113/168 Hz epochs were never published at ObsID level); H1743's program is 80146 (YAML said 80138); GRO hard-band sensitivity 5.1%.
- **Integration.** Scorecard row S1.e (hfqpo-ladder) `data_limited`→`null` (120 rows: 26 null / 33 data_limited), next decider eXTP-class sensitivity below the 0.5–3.1% rms limits; HFQPO search-target paragraph updated in `tfpt_horizon_readouts` (+ the `papers.ts` mirror); web-

site `predictions.ts` (`hfqpo-tooth` entry + `EXPERIMENTS_AUDIT` counters); `predictions.txt`; `experiments/README.md`; `experiments/next.txt`. No `verification/` module, no ledger row, no marker move (the search target stays **[O]**).

2026-07-20 (V) (v492 + the new research contract CELEST.SEAM.01 — the celestial-holographic route: the E_8 μ_4-glue is the flat $\mathbb{Z}_4$ monodromy of the equivariant celestial chiral algebra on the A_3 ALE space, with the exact correction clock^2=deck (spin bridge $\mathbb{Z}_8$, 8=2$|\mu_4|$); *verdict B, no gate closes, no marker moves, SEAM.EQUIV.01 stays [O]*)

- **`v492_celestial_z4_orbifold.py` (CELEST.WP1.01).** WP1 of the new contract, sympy exact (36 checks): the `v128` μ_4-glue grading of E_8 (240 = 52+64+60+64, additive on all 6720 root pairs) is *inner* — $h = (2,2,2,2,2;0^4)$ reads the glue class mod 4 on all 240 roots, so the glue is $\mathrm{Ad}(\exp(2\pi i h/4))$, an order-4 inner torus automorphism = a flat $\mathbb{Z}_4$ monodromy in the Kronheimer–Nakajima sense; the A_3-side detector $h' = (0^5;1,1,1,-3)$ reads the same class = the diagonal $(1,1)$ of the discriminant group $\mathbb{Z}_4{\times}\mathbb{Z}_4$ (the `v92`/`v125` Lagrangian glue). $\mathbb{C}^2/\mathbb{Z}_4$ is verified as the A_3 singularity $XY{=}Z^4$ (Hilbert series $(1{+}t^4)/((1{-}t^2)(1{-}t^4))$, Molien exact); the glue-equivariant SDYM(E_8) sector *closes* with graded dimensions $60(d{+}1)/64(d{+}1)$ — possible only because $\dim\mathfrak{g}_j = (60,64,60,64)$ — with zero modes = the carrier $D_5{\oplus}A_3$+Cartan = 60 and density 1/4 = $1/|\mathbb{Z}_4|$; the four glue-sector characters sum exactly to $\chi_{(E_8)_1} = 1{+}248q{+}4124q^2{+}34752q^3$ (`v377`), with sector weights = the discriminant form $(5x^2{+}3y^2)/8$ and integer glue-diagonal $h = (0,1,1,1)$ (= locality of the $(E_8)_1$ extension). *Critical correction (verdict B):* the A_3 deck acts on the celestial sphere as $z \mapsto -z$ (order 2), *not* as the order-4 clock; the clock is the $U(2)$ phase $\mathrm{diag}(i,1)$ (normalising the deck, det = i rotating the holomorphic symplectic form by the μ_4 generator), with the exact spin bridge (spin clock)^2 = deck ($\mathbb{Z}_8$, $8 = 2|\mu_4|$, the $c_3 = 1/(8\pi)$ winding integer). Clock-invariance selects the 1-parameter A_3 deformation $XY{=}Z^4{+}a_0$ whose four branch points are one μ_4 orbit with cross-ratio 2 (`v168`/`v214` pillowcase). Negative controls: the false action $\mathrm{diag}(i,i)$ killed three ways, the false glue breaks additivity on 3112/6720 pairs, rigidity = $\mathrm{Aut}(\mathbb{Z}_4)$.
- **New research contract `CELEST.SEAM.01`** in `tfpt_research_contracts` (own section, the fourth contract alongside $U_{\mathrm{wall}}/G_{\mathrm{metric}}$/`CONTRACT.F.01`): hypothesis (seam = glue-equivariant $\mathbb{Z}_4$-orbifold sector of the CCA on the A_3 ALE; twistorial bulk = the *self-dual* sector only), typing/non-circularity (the continuum $(E_8)_1$ existence — the `SEAM.EQUIV.01` target — is *not* an admissible input), work packages WP1 (done, verdict B) through WP5 (second continuum route, Costello–Li grade), pre-registered kill tests K1–K4 (K1 survived by WP1), and an honest success estimate (full success low, partial findings high). Ledger rows `CELEST.WP1.01` + `CELEST.SEAM.01`; website mirror (`papers.ts` section, script index). `SEAM.EQUIV.01` stays **[O]**; no marker moves. Exact (sympy).

2026-07-20 (IV) (evidence scorecard: five new typed rows — the α_s running prediction of record, the μ-distortion $A_s/N_\star$ branch discriminator, and three parked analog beds (bosonic-E8 thermal Hall, E8 mass-ladder recovery, Efimov DSI tuning); *experiments level only — no verification module, no gate closes, no marker moves*)

- **α_s running (prediction of record, consistent).** The Starobinsky branch fixes the running of the spectral index: $\alpha_s = -2/N_\star^2 = -7.6\times10^{-4}$ at $N_\star$=51.4 (exact slow roll -7.09×10^{-4}; band $N_\star \in [50,60]$: $[-8.0,-5.6]\times10^{-4}$) and $\beta_s = -4/N_\star^3$ (leading order; NOT $-8/N^3$) — the identity $\alpha_s = -r/6$ holds only at leading order (∼7% off exact). Data: Planck 2018 -0.0045 ± 0.0067 ($+0.57\sigma$), P-ACT $+0.0060 \pm 0.0055$ (-1.22σ), record leg P-ACT-LB $+0.0062 \pm 0.0052$ (-1.33σ, Calabrese et al. 2025 JCAP); β_s (Planck run+runrun) 0.010 ± 0.013 (-0.77σ). Forecast: LiteBIRD+CMB-S4 $\sigma(\alpha_s) \approx 4.9\times10^{-3}$ (0.14σ); CMB-S4+DESI+Euclid statistical $\sim 3.3\times10^{-4}$

(2.2σ, systematics-limited). Same `N_star_reheating` independence group as the $n_s/r/A_s$ rows (anti double-counting: never an independent hit); kill: robust $\alpha_s < -5 \times 10^{-3}$ or $> +3 \times 10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch. Probe: `experiments/tfpt-discovery/alpha_s_running_probe.py`.

- **μ-distortion (prediction of record, data-limited).** Chluba window integral (calibrated +20% to the full computation): sharp branch $N_\star$=51.4 (A_s=1.76 × 10^{-9}) $\mu \approx 1.6 \times 10^{-8}$, profiled record $N_\star$=56.1 (A_s=2.10 × 10^{-9}) $\mu \approx 2.0 \times 10^{-8}$, band 1.5–2.3 × 10^{-8}; the 23% branch split ($\Delta\mu = 3.0 \times 10^{-9}$) gives a 3σ $A_s/N_\star$ branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class; the PIXIE baseline $|\mu| < 9 \times 10^{-8}$ does NOT reach — factor 4–7 above the band; FIRAS $|\mu| < 9 \times 10^{-5}$). Honest S15 typing: the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is STRUCTURALLY BLIND in the μ window — window attenuation $|F(\omega)| = 0.033$, $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$ even at Voyage 2050 — the theory itself predicts where it is invisible, so NO separate comb-search row is opened. Shares the `Nstar_branch` alternative group with the two A_s rows (one branch question, several readings). Kill: robust $\mu < 0.9 \times 10^{-8}$ or $> 4 \times 10^{-8}$. Probe: `experiments/tfpt-discovery/mu_distortion_probe.py`.
- **Three parked analog beds (`parked_analog`; honest typing: they validate E8 edge/spectrum physics or the comb DETECTOR, never the TFPT axioms).** (i) Bosonic E8 phase / thermal Hall: $\kappa_{xy}/T = 8 \times (\pi^2 k_B^2/3h)$ with no charge Hall (seam edge c_-=8, `v450/v451/v489`); no engineered material yet (α-$RuCl_3$ c_-=1/2 contested: ultraclean half-quantized plateau, npj QM 2025, vs phonon-dominated κ_{xy}, PRX 12, 021025). (ii) E8 mass-ladder validation bed ($CoNb_2O_6/BaCo_2V_2O_8$): all 8 one-particle masses published (Zou et al. 2021 PRL 127, 077201, NMR+INS; Amelin et al. 2020 PRB 102, 104431, THz m_1–m_6; Coldea et al. 2010 Science m_2/m_1=φ) — blind Cartan-PF ladder recovery on the published spectra (8/8 masses + rejection of scrambled controls) as detector validation; confirms Zamolodchikov E8, not TFPT. (iii) Efimov DSI tuning: $s_0 = \pi/(6\ln(3/2)) = 1.2914 \Leftrightarrow \lambda = (3/2)^6 = 11.391$ at $M/m = 8.67$ (2 resonant pairs); best candidate $^{52}Cr_2$–^6Li (M/m=8.635, s_0=1.2890, λ=11.44, +0.44% vs $(3/2)^6$, gap deviation +0.18%); alternatives $^{39}K_2/^{41}K_2$–^6Li (λ = 11.59/11.20, needs double Feshbach control), $^{176}Yb_2$–^{23}Na (+5.9%); solver validated on the Naidon–Endo anchors (homonuclear 22.69, Cs_2Li 4.877); first end-to-end HARDWARE injection of the comb detector at tunable $\lambda \approx 11.39$; kill condition honestly "none on TFPT axioms — detector validation only". Probe: `experiments/tfpt-discovery/efimov_s0_massratio_probe.py`.
- Scorecard regenerated (`experiments/build_evidence_scorecard.py`): 120 rows — 48 consistent, 9 tension, 25 null, 34 data-limited, 4 parked; website mirror (`EXPERIMENTS_AUDIT` in `predictions.ts`) synchronised. Experiments level only; the verification suite, ledger and papers are untouched.

2026-07-20 (III) (`v491` — $g_{\rm car}$=5 as the partition COROLLARY of the four marks: 4 into exactly 3 positive parts is uniquely $\{1,1,2\}$, sharpening `v53`; the residual is the weight-typing postulate — `AX.P2.01` stays a declared axiom; *no gate closes, no marker moves*)

- **`v491_p2_partition_corollary.py` (`P2.PARTITION.01`).** Sympy-exact: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is UNIQUE — $\{1,1,2\}$ — so $e = (e_1, e_2, e_3) = (4, 5, 2)$ and hence $g_{\rm car} = e_2 = 5$, $|\mathbb{Z}_2| = e_3 = 2$, $N_{\rm fam} = e_2 - e_3 = 3$ are *corollaries*, not inputs (consistency fixed point $(2^4-1)/5 = 3 = \operatorname{rank} H^1$). **[E]** negative controls, each assumption load-bearing: drop positivity → 4 solutions, e_2 ambiguous $\{0,3,4,5\}$; sum 3 → e_2=3; sum 5 → not unique; 2/4 parts → $e_2 \in \{3,4\}/\{6\}$; Mehta–Seshadri RATIONAL weights → $e_2 = \frac{16}{3}$ resp. $\frac{44}{9} \neq 5$ (integrality = the Birkhoff–Grothendieck splitting-type typing, not MS weights). **[E]** the `v53` sharpening: `v53` pins $(1,1,2)$ with ALL THREE targets $(4,5,2)$; the partition needs only e_1=4. **[E]** sum rule: $4 \times \frac{N-1}{2} = 4$ iff N=3; the `v115` route $|a|_1 = |\mu_4| \operatorname{tr}(A_0) = 4$ agrees. Independence chain: #marks = 4 from `v216` (P1-side, no $g_{\rm car}$ input); $\operatorname{rank} H^1 = 3$ pure topology (`v137/v228`); $|a|_1 = 4$ $g_{\rm car}$-

free (`v115`). [**C**] honest residual: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality/positivity h^0=0/sum "given (U)"); `AX.P2.01` stays a declared axiom, and the $N_{\rm fam}$ direction reversal vs `tfpt_constants` is typed as an over-determination check (never double-counted). Integrated in `introduction` (anchor half-forced paragraph), `tfpt_1` ($(5,3)$ keybox), `tfpt_2` (H2 winbox) and `origin_theory` ($(5,3)$ keybox); ledger row `P2.PARTITION.01`; three exact Wolfram mirrors added to `tfpt_readouts_extension.wl` (counted with the next verified engine run, engine currently unactivated); website mirror (DAG node `P2`). Exact (sympy).

2026-07-20 (II) (`v490` — the T^2 spin-structure parity census: $\mu_{\rm pre}$=4 witnessed on the lattice, the 16-fold-way bridge θ_v=1 on measured numbers, and the honest TEE non-discriminator fence; *no gate closes, no marker moves*)

- **`v490_seam_parity_census.py` (`SEAM.MU.01`).** The T^2 ground-state fermion parity of the `v367` $p{+}ip$ collar, resolved per spin structure (AA, AP, PA, PP) and determined TWICE — Read–Green TRIM product AND real-space Majorana Pfaffian (Schur), relative to the trivial M=3 phase, with the twisted real-space spectra matching the Bloch bands to 9×10^{-15}. [**E**] per layer $(+,+,+,-)$ (the ν=1 Ising signature, TRIM $\equiv$ Pfaffian in every sector); [**E**] the 16-layer carrier has parity^{16} = + in ALL FOUR sectors $\Rightarrow$ gauged torus GSD = 4 = $\mu(SO(16)_1) = |\det \mathrm{Cartan}(D_8)|$ — the lattice ground-state witness of "$\mu_{\rm pre}$=4 before condensation" (step 1 of the `v378` KLM chain); [**E**] the 16-fold-way bridge on MEASURED numbers: $\theta_v = e^{2\pi i\nu/16} = 1$ exactly at $\nu = 2c_- = 16$ (`v489`), rivals $\nu = 1,2,8$ not holomorphically condensable; [**E**] honest fence: the fermionic Kitaev–Preskill TEE is identically zero for BOTH M=1 and M=3 (γ_f=0.000001/0.000000) — it does NOT discriminate $(E_8)_1$ vs $SO(16)_1$. [**C**] the KLM step $\mu: 4 \to 4/2^2 = 1$ stays the cited theorem (the condensed model is interacting); the numerics pin only the premise package $\{c_-{=}8,\ \mu_{\rm pre}{=}4,\ \theta_v{=}1\}$. [**O**] `SEAM.EQUIV.01` stays [**O**] (continuum existence, MMST `v336`). Integrated in `tfpt_research_contracts` (new S3-stack paragraph); ledger row `SEAM.MU.01`; website mirror (DAG node `qft`). Python (numpy + scipy Schur + sympy exact θ_v).

2026-07-20 (`v489` — the chiral central charge c_-=8 measured DIRECTLY from the bulk ground state via the Kim–Shehab–Kim modular commutator: the first ground-state witness for premise (i) of `SEAM.EQUIV.01`, independent of the Chern route (`v367`) and the KLM count (`v378`); *no gate closes, no marker moves*)

- **`v489_seam_modular_commutator.py` (`SEAM.CCC.01`).** The Kim–Shehab–Kim modular commutator $J(A,B,C) = i\,\mathrm{Tr}(\rho_{ABC}[K_{AB},K_{BC}]) = (\pi/3)\,c_-$ (PRL 128 (2022); `arXiv:2110.06932/2110.10400`) is evaluated on the EXACT BdG ground state of the `v367` $p{+}ip$ seam collar model, entirely from the Peschel Majorana covariance ($W_X = -2\,\mathrm{artanh}\,\Gamma_X$, $J = \frac{1}{4}\mathrm{Tr}([W_{AB}, W_{BC}]\Gamma_{ABC})$), on a disk about the torus centre cut into three 120° wedges. [**E**] ground-state purity eig$(i\Gamma) = \pm1$ to 6×10^{-15}; [**E**] per-layer $c_- = 0.499998$ at L=32 (convergence 0.499793 at L=16 $\to$ 0.500000 at L=48, deviation -1.8×10^{-7}); [**E**] controls: trivial M=3 $\to$ 0.000000, orientation flip M=-1 $\to$ -0.499986; [**E**] layer additivity exact (1.4×10^{-14}); [**E**] the 16-layer carrier lands on $c_-{=}8{=}g_{\rm car}{+}N_{\rm fam}$ (direct 7.9967 at L=16, additivity route 8.0000); [**E**] clip invariance 5×10^{-8}. The FIRST ground-state witness for premise (i) of `SEAM.EQUIV.01` (c_-=8) that is INDEPENDENT of both the Chern-number route (`v367`) and the counting route (`v378`) — read from entanglement data alone. [**O**] honest fence: does NOT close `SEAM.EQUIV.01`; the continuum scaling limit (MMST, `v336`) and the cited KLM condensation step are untouched. Integrated in `tfpt_research_contracts` (new S3-stack paragraph) and `tfpt_1` (the `SEAM.EQUIV.01` package paragraph); ledger row `SEAM.CCC.01`; website mirrors (DAG node `qft`, papers/glossary/FAQ inventories). Numerical (numpy), Python-only by nature.

2026-07-19 (editorial — positioning section *TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram* added to `introduction` + website mirror; *no result, no status change*)

- **`introduction.tex` — new section "TFPT in the light of algorithmic physics".** Positions the compiler against the Zuse–Schmidhuber–Wolfram algorithmic-physics lineage: three shared instincts (minimum description length as the selection principle, deterministic resource-bounded generation, compressibility as the aesthetic) and three deliberate divergences (one *forced* program vs. an ensemble of all universes; an algebraic Lie/lattice/VOA description language vs. Turing bit-strings; and falsifiability — dated, kill-tested predictions — as the load-bearing difference). Explicitly typed as *positioning*: it introduces no formula and moves no status marker; its only machine-checked hooks are already in the suite (the grammar-internal null model and the icosahedral/McKay bedrock).
- **Website mirror.** A matching curated section was added to the `introduction` entry in `website/lib/papers.ts`.

2026-07-15 (III) (v488 — the $3/5$ Clebsch door ENUMERATED AND CLOSED: no minimal Pati–Salam/$SO(10)$ Majorana channel can produce $5/3$ — the v482 escape is decided dead, the v481 window candidate stands as the honest endpoint; *a clean negative result, no gate closes*)

- **`v488_majorana_clebsch_door.py` (`FLAV.NUSCALE.04`).** v482 named ONE escape for the integer seesaw rung $M_R = c_3^3 \bar{M}_{\rm Pl}$: a third-generation Majorana factor $r \approx 1.6696$, 0.18% from $g_{\rm car}/N_{\rm fam} = 5/3$ — declined for lack of mechanism. The mechanism search is now EXECUTED: **[E]** NO RENORMALISABLE CHANNEL — $\mathrm{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$ and the $\overline{\mathbf{126}}$ is NOT in the E_8 hull $\{1, 10, 16, 45\}$ (v247), so M_R only via the d=5 operator $(\mathbf{16}_F\overline{\mathbf{16}}_H)^2/\Lambda$; channels $\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}$ with only singlet + $\mathbf{45}$ admissible. **[E]** CHARGE TABLE: $(B{-}L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$, $\mathrm{Tr}_{16}\, Y^2 = \frac{10}{3}$ (v485). **[E]** THE $\{2,3\}$-SMOOTH THEOREM (the door closes): every nonzero ν^c channel weight $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$ is $\{2,3\}$-smooth, and the multiplicative closure of $\{2,3\}$-smooth rationals never contains $\frac{5}{3}$ (no factor 5); the UNIQUE natural 5 of the embedding, $k_Y = \frac{5}{3}$ (Ginsparg, v470), is a full-multiplet trace whose direction has $Y(\nu^c) = 0$ EXACTLY — structurally decoupled from the Majorana operator. **[E]** GENERATION-BLINDNESS: Clebsches are family-space scalars; $\mathrm{diag}(1, 1, \frac{3}{5})$ cannot arise from group theory by construction. **[O]** compiler-flavor inventory typed and declined: exact $\frac{3}{5}$'s only in wrong slots (K lepton gen2/gen1; $R + L$ lepton gen3/gen2; never gen3 vs a degenerate base). NET: the rung$+5/3$ rescue is DEAD as a mechanism candidate; the v481 one-parameter window candidate (M_R free inside the PS window, the $\Sigma m_\nu = 0.059$ eV floor as kill test) stands. Contract provenance: `experiments/theory-contracts/clebsch01_majorana_35_door.py`. Integrated in `tfpt_2_standard_model` (new paragraph closing the seesaw arc); ledger row `FLAV.NUSCALE.04` + update note on `.03`; Python-only (Fraction/sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (II) (v487 — the lazy split FORCED by the resummed clock: the walk's one-step spectrum is the complete v124 clock ladder below the wall, deck parity = rung index, and the two comb tones are one bend apart — the v486 [O] "split selection" closes onto established structure; *no gate closes, no marker moves*)

- **`v487_transfer_clock_rungs.py` (`HYP.REWRITE.03`).** v486 left the split selection open ("$(\frac{1}{2}, \frac{1}{6})$ vs uniform $(\frac{1}{3}, \frac{1}{3})$ is selected by the physical spectrum, not yet derived"; named candidate: the vague "sheet-symmetric choice"). Closed: **[E]** RUNG READING — the lazy rule's one-step spectrum is EXACTLY the complete resummed-clock ladder below the wall (v124: $\lambda_n^{1/p_2} = 1 - n/N_{\rm fam}$, pole at $n{=}N_{\rm fam}$): $\mathrm{eig}(M) = \{1, \frac{2}{3}, \frac{1}{3}\} = \{1 - n/3 : n = 0, 1, 2\}$, while v327's uniform rule puts its second pair mode ON the wall ($0 = 1 - 3/3$) — the v486 degeneracy re-read as "odd channel collapsed to the wall". [**E**] DECK PARITY = RUNG INDEX: $[B, \sigma] = 0$; the σ-even pair mode carries rung n=1 (survival $\frac{2}{3}$), the σ-odd mode rung n=2 (survival $\frac{1}{3}$) — each clock step flips the deck parity; this is the structural home of the v486 parity assignment (ω_1 even-channel, ω_2 odd-channel) that `experiments/frb-parity-comb` tests. [**E**] FORCING: demanding the faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + $\mathbb{Z}_2$ equivariance + non-negative rates forces UNIQUELY both the split $(\frac{1}{2}, \frac{1}{6})$ AND the rung assignment even→1, odd→2 (the swapped assignment needs $h = -\frac{1}{6} < 0$); the "sheet-symmetric choice" candidate is superseded by this proof. [**E**] BEND COROLLARY: ω_1/ω_2 = rate(2)/rate(1) = $\log_{3/2} 3 = 2.70951\ldots$ — the parity-comb tone ratio IS the `v107`/`v124` bend of the pulsar-clock search. [**O**] what remains open is exactly what was open before v486: the arity $\{2, 3\}$ (anchor input) and the clock's semiclassical derivation (R1); no independent choice remains at the walk level. Contract provenance: `experiments/theory-contracts/lazy01_clock_rung_forcing.py`. Integrated in `origin_theory` (new paragraph after the v486 block); ledger row `HYP.REWRITE.03` + update note on `.02`; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-15 (`v486` — generator hunt: the FULL transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ from ONE forced local rule — the unique lazy $\mathbb{Z}_2$-pair walk (stay, hop, leak) = $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$**; the previously generator-less $(1/3)^6$ mode gets its origin; `v327`'s uniform rule superseded as full-spectrum generator — *no gate closes, no marker moves*)**

- **`v486_transfer_full_rule.py` (`HYP.REWRITE.02`).** The verified seam transfer spectrum is $\{1, (2/3)^6, (1/3)^6\}$ (`v54`/`v221`), but the only local generator in the suite (`v327`) produces $\{1, \frac{2}{3}, 0\}$ — its zero mode persists under every power, so no iterate reaches the third mode. [**E**] UNIQUENESS: for the symmetric 3-channel rule $M(s, h) = [[1, 0, 0], [0, s, h], [0, h, s]]$ (one absorbing family channel + $\mathbb{Z}_2$ pair) the survival spectrum is $\{s{+}h, s{-}h\}$; demanding the PHYSICAL pair $\{\frac{2}{3}, \frac{1}{3}\}$ forces UNIQUELY (stay, hop, leak) = $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/|\mathbb{Z}_2|, 1/(|\mathbb{Z}_2| N_{\mathrm{fam}}), 1/N_{\mathrm{fam}})$ — the lazy $\mathbb{Z}_2$-pair walk, every rate an atom expression (honest direction: spectrum → rates). [**E**] SIX-HAND GENERATION: over the order-6 $= 2N_{\mathrm{fam}} = |R^+(A_3)|$ hand, $\mathrm{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$ exactly — both decay gaps now have one recursive generator: $6 \ln \frac{3}{2}$ (recovery, `v76`) and $6 \ln 3$ (subdominant, previously generator-less). [**E**] SUPERSESSION: the uniform rule's zero mode is power-persistent; its λ_2 arity story ($\frac{2}{3} = |\mathbb{Z}_2|/N_{\mathrm{fam}}$) is unchanged. [**O**] the arity $\{2, 3\}$ stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay = $1/|\mathbb{Z}_2|$ as the sheet-symmetric choice). Discovery provenance: `experiments/tfpt-discovery/generator_hunt_full_transfer_rule.py` (plus a typed anchor-bilinear coverage inventory — known hits $a^T K a$=41, Koide 53, and the recorded-only $\mathbf{1}^T Q a$=16 — and the lepton-exponent observation $(5, 3, 2)$ = K-row = atom triple; no unforced claims, `v354`/`v355` discipline). Integrated in `origin_theory` (new paragraph after the `v327` rewrite block); ledger rows `HYP.REWRITE.02` + update note on `.01`; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

2026-07-14 (IV) (`v485` — the merged analytic target's computable side COMPLETED: the diagonal self-energy is renormalised to ZERO exactly at the KMS seam scale, the mark contact calculus resums in closed form (BFK route), and the 48/41 multiplicities are ONE state set under two response weights — the residual collapses onto the `SEAM.EQUIV.01` face alone; *no gate closes, no marker moves*)

- **`v485_contact_diagonal_closed.py` (`SEAM.CONTACT.UNIT.02`).** The v484 merge left one analytic step: the diagonal (self-energy) zeta-renormalisation at the marks + the multiplicity matching. Settled at the computable level: [**E**] DIAGONAL ZERO AT THE KMS SCALE — the renormalised coincident-point Green function of $|D|$ on circumference ℓ is $G_{\mathrm{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$

EXACTLY (symbolic limit + 30-digit checks); it vanishes iff $\ell = 2\pi = 1/(4c_3)$, the `v239` KMS seam circumference — the seam scale is the unique zero-self-energy scheme point, and the log is the pure SCALE channel (matching the structure: $F_{U(1)}$ carries exactly ONE log term, the transport $8b_1c_3^6\ln(1/\varphi_{\rm seam})$, while the $\varphi_0^{\rm ret}$ split is log-free). **[E]** CLOSED-FORM RESUMMATION (the BFK/gluing route already in the suite, `v151`): with the diagonal zero, the finite-rank mark determinant resums exactly — $\det(I-uC) = (1-4u)(1+2u)^2$, $u = \varepsilon c_3 \ln 2$, all orders, the linear term ABSENT precisely because the diagonal vanishes ($\operatorname{Tr} C = 0$); the series matches `v484`'s $\operatorname{Tr}(C^k) = 4^k + 2(-2)^k$ term by term. **[E]** WEIGHT DICHOTOMY: the SAME $48 = 3 \times 16$ admissible states carry BOTH multiplicities — counted flat they give the $\varphi_0^{\rm ret}$ puncture count $48 = \Omega_{\rm adm}$; Y^2/loop/GUT-weighted they give the alpha budget $41 = 10b_1 = 40 + 1$ ($\operatorname{Tr}_{16} Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg $(3/5)(41/6) = 41/10$, `v470`) — one state set, two response weights, no new number. **[C]**/**[O]** CONSEQUENCE: with `v484` EVERY finite/computable piece of the merged `ALPHA.QUILLEN.EXACT.01` + `HYP.PHI0.PUNCTURE.01` target is proven; the single remaining **[O]** is the identification of the abstract seam ζ-det with this glued mark determinant — a face of `SEAM.EQUIV.01`, exactly the `v382` typing, now substantiated computationally. Integrated in `tfpt_1_architecture_e8` (eighth step of the ALPHA.QUILLEN arc) and `origin_theory`; ledger rows `SEAM.CONTACT.UNIT.02` + update notes on `.01` and both targets; Python-only by nature, flagged in the Wolfram README.

2026-07-14 (III) (`v484` — the two open analytic targets MERGE: the shared "c_3 per boundary insertion" rule of `ALPHA.QUILLEN.EXACT.01` and `HYP.PHI0.PUNCTURE.01` is the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/|\mu_4|$, derived on the seam circle for the finite cycle sector — *no gate closes, no marker moves*)

- **`v484_seam_contact_unit.py` (`SEAM.CONTACT.UNIT.01`).** The v483 round left the two analytic targets sharing one rule ("weight c_3 per boundary contact insertion": the $F_{U(1)}$ ladder $\{0,3,6\}$, the $\varphi_0^{\rm ret}$ puncture $\{1,4\}$). This module shows the shared rule is NOT a new unknown: **[E]** the unit chain $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$ — one *bare* boundary insertion divided by the orbit order $|\mu_4|$=4 is exactly c_3; this is the v239 KMS seam unit and the v58/v73 one-sided Gauss–Bonnet in a single identity. **[E]** the mechanism is DERIVED on the actual seam circle for the finite cycle sector: the bare Green function $G(d) = -\frac{1}{\pi}\ln|2\sin(d/2)|$ (polylog identity, 40 digits) takes INTEGER multiples of $c_3 \ln 2$ at the μ_4 separations ($G(\pi/2) = -4c_3\ln 2$, $G(\pi) = -8c_3 \ln 2$; mark matrix $-4c_3 \ln 2 \cdot \operatorname{circ}(0,1,2,1)$, spectrum $\{4,-2,0,-2\}$), so the μ_4-invariant (orbit-averaged) insertion carries $-(c_3 \ln 2)C$ exactly and the log-det contact expansion is exactly graded in c_3 per insertion ($\operatorname{Tr} C^k = 4^k + 2(-2)^k$). **[E]** the grammar is suite-wide: the Λ prefactor $\frac{3}{4\pi^2} = (8\pi)^2 d_{\rm top} = 48c_3^2$ (`v60`) carries the SAME multiplicity $\Omega_{\rm adm}$=48 as the $\varphi_0^{\rm ret}$ puncture at TWO insertions; the $g-2$ seam loop divides by the same $2\pi = 1/(4c_3)$. **[C]**/**[O]** THE MERGE: the two tracked **[O]** targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ($48 = \Omega_{\rm adm}$, $41 = 10b_1$); the finite half is proven, no target coefficient is claimed (the integers are forced by the square configuration, `v354`/`v355`-clean). Integrated in `tfpt_1_architecture_e8` (seventh step of the ALPHA.QUILLEN arc) and `origin_theory` ($\varphi_0^{\rm ret}$ split paragraph); ledger rows `SEAM.CONTACT.UNIT.01` + update notes on both targets; Python-only by nature (mpmath polylog), exact identities flagged in the Wolfram README.

2026-07-14 (II) (v482–v483 — the deep-dive round: the ν-scale decision computation EXECUTED (the integer c_3^3-rung excluded unstructured, the $5/3$-proximate structure escape recorded and declined) and the $\varphi_0^{\rm ret}$ puncture target HALVED (the geometric side proven exact: all twisted pillowcase heat traces are t-independent rationals, so the π^{-4} must sit in the per-mark coupling weight) — *no gate closes, no marker moves*)

- **`v482_seesaw_rung_decision.py` (`FLAV.NUSCALE.03`).** The decision computation NAMED by `v481` is EXECUTED in bracketed (assumption-robust) form: the integer rung $M_R = c_3^3\bar{M}_{\rm Pl}$ needs a rescue factor $\times 1.670$ in m_3 (0.0301 vs 0.0503 eV), while $m_t(\overline{\rm MS}) \pm 3$ GeV ($> 3\sigma$), $\alpha_3^{-1} \mp 0.10$ ($> 3\sigma$), a $\pm 10\%$ band on the ADKLR κ run-down and a PS-leg Yukawa-beta envelope $\times[0.5, 1.5]$ move it by at most $\times 1.10$ singly and $\times 1.165$ ALL-favourably combined — the *unstructured* rung is EXCLUDED conditional on the carrier premise $y_\nu{=}y_t$ (the `v481` 1-loop miss is not an RG artifact; no exact higher-loop coefficients trusted, pure envelope argument). [**C**] The only escape is a third-generation Majorana STRUCTURE factor $r \sim 1.67$; its post-hoc 0.18% proximity to $g_{\rm car}/N_{\rm fam} = 5/3$ is recorded and DECLINED per `v354`/`v355` (no forcing mechanism; the $t_0{\sim}30$ discipline of `v478`). [**O**] the `v481` one-parameter window candidate (NO floor $\Sigma m_\nu{=}0.059$ eV) never used the integer rung and STANDS. Integrated in `tfpt_2_standard_model` (ν-scale paragraph) and `tfpt_5_redteam`; ledger rows `FLAV.NUSCALE.03` + `.02` update note; Python-only by nature.
- **`v483_pillowcase_exact_traces.py` (`HYP.PHI0.GEOM.01`).** The GEOMETRIC side of the $\varphi_0^{\rm ret}$ puncture target (`HYP.PHI0.PUNCTURE.01`/`v408`) is EXACT: every twisted/orbifold heat trace of the flat $\tau{=}i$ pillowcase is a t-INDEPENDENT RATIONAL — ${\rm Tr}(\gamma e^{t\Delta}) = 1$ for $\gamma = \sigma, \rho, \rho^3$ at every tested t (position-space Gaussian image sums $< 10^{-9}$; mode-space proof: only $n{=}0$ is γ-diagonal), matching the Atiyah–Bott fixed-point counts $4{\times}\frac{1}{4}$ (det $= 4{=}|\mu_4|$) and $2{\times}\frac{1}{2}$ (det $= 2{=}|\mathbb{Z}_2|$); the pillowcase contact term is EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark and the clock-equivariant trace exactly 1, for all t. THEREFORE the π^{-4} in $d_{\rm top} = 48c_3^4 = \frac{3}{256\pi^4}$ cannot come from flat orbifold geometry at ANY order: it must sit in the per-insertion COUPLING normalisation ($4{=}|\mu_4|$ marks of weight c_3 give c_3^4, multiplicity $\Omega_{\rm adm}{=}48$; the bare $1/(2\pi)$-propagator 4-cycle differs by the exact rational $\frac{3}{16}$). The target NARROWS from "derive the term" to ONE rule — "per-mark insertion weight $= c_3$" (the same residual class as the EM-Ward c_3-ladder, `EM.WARD.02`) — and stays [**O**]. Complements the experiments-side Donnelly probe (honest negative). Integrated in `origin_theory` ($\varphi_0^{\rm ret}$ split paragraph); ledger rows `HYP.PHI0.GEOM.01` + `HYP.PHI0.PUNCTURE.01` update note; Python-only by nature (quadrature verification of exact statements).

2026-07-14 (website + figure refresh — the `/verification` journey views become *live* HTML (the stale timeline/residual-chain images replaced by interactive components wired to the in-browser reproducer), the prediction surface's empirical-audit wall of text restructured into dated, expandable rounds, and the generated timeline/residual-chain figures extended to the current suite — *presentation only, no content or marker moves*)

- **Website `/verification`: live journey views.** The static `script_timeline.png` and `residual_chain.png` images (stale at the `v303`–`v407` figure state, with alt text and captions still describing the old eleven-phase/`v237` state) are replaced by two interactive components, `SuiteTimeline.tsx` (17 phases, `v1`–`v481`) and `ResidualChain.tsx` (the reduction chain through the certification round to the bedrock): every phase/step lists representative `vN` scripts that run live in the browser via the existing Pyodide reproducer, exactly like the dependency graph. Stale counts fixed ("$\sim$295-script journey", "260 scripts" $\to$ the generated suite total). The dependency graph itself was verified current (`v479`–`v481` present; all 288 referenced scripts exist in the registry and the website mirror).

- **Prediction surface: the empirical-audit narrative restructured.** The single ∼13 000-character audit paragraph in `lib/predictions.ts` is split into twelve dated, verdict-badged rounds (`AuditRound`: date, title, verdict, two-sentence summary, full unabridged record on expand) plus a domain-chip row and the anti-double-counting note — same content, readable surface. Long-token overflow fixed (`break-words`) in the dependency-graph detail panel and the universal-gap-lab residual cards; the over-long `ALPHA.QUILLEN` residual note tightened to its current state (`v470`/`v472`, residual unchanged).
- **Generated figures + captions synced.** `make_figures.py`: the script timeline gains the `v408`–`v481` certification-round phase (title now ∼475 scripts) and the residual chain gains the `v458`–`v480` certification step (Kronheimer `v479`, four-interval `v480`, crossed-product `v469`) with sub-text wrapping fixed; figures regenerated (PDF + website PNG). The stale figure captions updated: `introduction` (timeline caption now spans the full arc), `tfpt_research_contracts` (residual-chain caption now states the `SEAM.EQUIV.01` bedrock and certification round; QFT-skeleton caption now names the one **[O]** residual instead of the outdated "two open items"). No numbers, statuses or claims change anywhere.
- **KaTeX macro fix (broken formula rendering).** Formulas copied from the document set that use the shared TeX macros (`\gcar`, `\Nfam`, `\cthree`, . . .) rendered as red KaTeX errors on the site (e.g. the φ_0^{tree} icosahedral-combinatorics formula on the Origin-Theory page). A shared macro map (`lib/katex-macros.ts`, mirroring the `changelog.tex` preamble) is now passed to every KaTeX render; an automated scan of all 19 site routes finds zero `katex-error` nodes.
- **Readability pass (text walls clamped, content unchanged).** A reusable `Expandable` component clamps long mirror prose with a fade + "Show more" toggle — applied to the paper-section bodies and front-box bullet lists (`PaperSection.tsx`, the home set pages and `/papers/[slug]`), the dependency-graph node summaries and the prediction-card descriptions. Nothing is deleted; the full text expands in place.
- **Hylaean ecosystem card corrected.** The `hylaean.ai` card spoke of "Field AI" with structure tags that did not describe the project. It now speaks consistently of *Hylaean* and is grounded in the project documentation (`_documentation/core-physics/big-picture.md`) and the site's own status log: one cognitive substrate behaving like a physical material, knowledge = attractor valleys, question = boundary condition, answers committed only on physical landing, capability claims gate-checked against a machine-checked claim registry (honest negatives published) — borrowing TFPT's structural discipline, not its physics predictions. The Agentic-Commerce card reference updated accordingly.
- **Live-site correctness pass (counts + reading order).** Sitewide audit against the deployed page found stale counts and fixed them: the document set is stated everywhere as *10 documents = 1 reading guide + 5 numbered papers + 4 companions* (the safeguards companion was missing from the papers/downloads headers, the JSON-LD collection and the `tfpt_docset.tex` box, which still said "three companions" next to "four"); the stale "23 falsifiable predictions" claims (site metadata, OG image, `llms.txt`) now state the live count of the prediction surface (25 test surfaces, sourced from `predictions.length` where possible), while the interactive compiler page is honestly labelled as its own *23-readout prediction board* (the standalone app's board genuinely has 23 entries; the intro film keeps its spoken "twenty-three"); and the `/orientation` reading-order lists are fixed to the canonical numbering (Red Team = Doc 5 added, Appendix H = 6, Origin = 7, contracts = 8, safeguards = 9 — the old list mislabelled 6/7 and omitted three documents). Presentation/consistency only; no physics content changes.

2026-07-14 (`v479`–`v481` — the bird's-eye "overlooked tools" round: R3's graph→geometry bridge discharged to Kronheimer 1989; the four μ_4 marks realised as the exactly solvable *four-interval* multilocal modular geometry with $\omega \circ \rho = \omega$ manifest; and the absolute ν-scale collapsed to a one-parameter, PS-window-bracketed candidate under the carrier normalisation $y_\nu{=}y_t$ — *no gate closes, no marker moves*)

- **`v479_kronheimer_quiver_bridge.py` (`SEAM.KRONHEIMER.01`).** The R3 residual of the $\det K{=}1$ keystone (`v344`: "the rewrite attractor graph IS the du Val singularity") is an ESTABLISHED THEOREM — Kronheimer's ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685) — with all finite input data already compiler outputs, machine-checked **[E]**: the Kac marks $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ are the null vector of the affine-E_8 Cartan matrix AND the Perron vector $A\delta{=}2\delta$ of the rewrite attractor (`v312`); the hyper-Kähler quotient dimension count lands exactly on 4 (the $\mathbb{C}^2/\Gamma$ geometry; $\sum_{\text{edges}} d_i d_j{=}\sum d_i^2{=}120$ forced by the null identity); the gauge datum is $|2I|{=}120{=}|\mu_4|h(E_8)$, $h(E_8){=}30{=}2 \cdot 3 \cdot 5$; the resolution carries the unimodular $-E_8$ form ($\det{=}1$, SNF = identity — the Poincaré link) with $3 \cdot 8{=}24$ deformation parameters. **[C]** the theorems are cited, not re-proved (import class of NPW26/MMST/AGT); **[O]** the residual premise *transforms* to "the raw seam supplies ALE hyper-Kähler data asymptotic to the $(2, 3, 5)$ cone" — the same single arrow as `SEAM.EQUIV.01` (`v346` L2). Integrated in `origin_theory` (R3 keybox + new Kronheimer paragraph) and `tfpt_research_contracts` (Kummer/du Val); ledger rows `SEAM.KRONHEIMER.01` + `SEAM.DETK.01` update note; Wolfram mirror added to the extension file (+2 exact checks, counted with the next verified engine run — the local engine is currently unactivated; the identical statements are sympy-exact in `v479`). LEAN: the finite input data are kernel-checked in `TfptCarrier/KronheimerMarks.lean` (marks = null/Perron vector over $\mathbb{Z}$, $\sum \delta_i^2{=}120{=}4 \cdot 30$, $30{=}2 \cdot 3 \cdot 5$, dimension count = 4; lake build clean, `#print axioms` = the three standard kernel axioms only).
- **`v480_multilocal_four_interval.py` (`QGEO.MULTILOCAL.01`).** The seam circle minus its four marks IS the $n{=}4$ multi-interval region of a chiral free fermion — the class whose multi-interval modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu; Rehren–Tedesco multilocal fermions; KLM: holomorphic ⇔ trivial multi-interval inclusion, $\mu{=}1$ — the same one bit as $\det K{=}1$); until now the suite used only the single-ball CHM boost (`v358`). On the exactly solvable antiperiodic critical ring: **[E]** the quarter-turn clock is a signed permutation with $\rho^4{=}{-1}$ EXACTLY (the fermionic μ_4 clock is the order-8 double-cover lift — the same $\mathbb{Z}_2$-lift pattern as $2I$ over I); **[E]** its primitive-8th-root character projectors decouple the 4-interval covariance at 10^{-15}, each sector a single-interval quarter-ring kernel with boundary twist (10^{-14} spectra); **[E]** $\rho C \rho^{-1}{=}C$ at 10^{-14}, so $\omega \circ \rho{=}\omega$ holds *manifestly* at the 4-interval state level, and a one-site displacement re-couples the sectors at 10^{-2} (negative control); **[E]** the cross-interval modular coupling peaks on the conjugate-point diagonal (the Casini–Huerta mixing shadow). **[C]** continuum theory cited; **[O]** the raw-seam premise (`QGEO.SYM.01`/`SEAM.EQUIV.01`) unchanged. Integrated in `tfpt_research_contracts` (marks keybox + rigidity contract + v478 leg A), `origin_theory` (BW paragraph), `tfpt_4_frontier` (v358 companion note); Python-only by nature.
- **`v481_seesaw_carrier_ladder.py` (`FLAV.NUSCALE.02`, CANDIDATE class, genre `v467`/`v468`).** `FLAV.NUSCALE.01`'s $y_\nu{=}1$ probe was not the carrier normalisation: one $SO(10)$ **16** per family + the minimal Yukawa sector ($\mathbf{10}_H$ / PS $(1, 2, 2)$) forces $y_\nu{=}y_t$ at the matching scale (named premise). With explicit 1-loop RG (gauge/y_t/λ up; ADKLR Weinberg-operator running down, hep-ph/0108005) the (y_ν, M_R) trade-off collapses to M_R alone: the observed $m_3{=}0.0503$ eV demands $M_R{=}9.3{\times}10^{13}$ GeV — INSIDE the PS window $[4.2{\times}10^{13}, 2.4{\times}10^{15}]$ (`v249`) at $\log_{c_3}(\bar{M}_{\text{Pl}}/M_R){=}3.15$ (the $y_\nu{=}1$ inversion gives the structureless 2.58). **[X]** honesty gate: the integer rung $M_R{=}c_3^3\bar{M}_{\text{Pl}}$ predicts $m_3{=}0.030$ eV — 40% low at 1-loop (κ run-down 0.794) — the ladder pin is DECLINED per `v354`/`v355`; the decision computation is named

(2-loop SM + explicit PS thresholds). What survives: a *one*-parameter, window-bracketed **[O]** candidate with the falsifiable band $m_3 \in [0.002, 0.115]$ eV and the NO floor Σm_ν=0.059 eV; frozen record untouched. Integrated in `tfpt_2_standard_model` (new paragraph after `v272`), `introduction` status card (third named candidate), `tfpt_5_redteam`; ledger rows `FLAV.NUSCALE.02` + `FLAV.NUSCALE.01` update note; Python-only by nature.

- *Honest negative (experiments-only, no promotion):* the Donnelly/Atiyah–Bott equivariant heat-kernel route for the ϕ_0 puncture target (`HYP.PHI0.PUNCTURE.01`) types the KIND of term (order-4 equivariant contact datum; the exact weights $\sum_{k=1}^{n-1} \frac{1}{4\sin^2(\pi k/n)} = \frac{n^2-1}{12}$ verified) but does NOT force the value $48c_3^4$ (pure rationals vs π^{-4}) — recorded in `experiments/theory-contracts/phi0_donnelly_equivariant_probe.py`; the target stays **[O]**.

2026-07-13 (v478 — first computable steps on the two remaining legs of the entropic-action bridge: the state-side modular data flows to the CHM/Bisognano–Wichmann geometric form (Calabrese–Cardy $c_{\rm est}\to 1$, CHM parabola, even bands exactly zero) — the bridge's modular side meets TFPT's Einstein-derivation input; and the measure condition reduced to one exact KMS time $t_0 = \ln\frac{72}{17} + 9\ln(8\pi) = 30.461$ with the $h(E_8)$=30 near-miss explicitly declined; both legs stay [O], *no status moves*)

- **`v478_entropic_continuum_legs.py` (`GRAV.ENTROPIC.LEGS.01`).** *Leg A (AP2 continuum, exhibited):* on the exactly solvable critical chain (infinite-volume sinc kernel, 50-digit precision) the compressed interval state has $S(L)=(c/3)\ln L+k$ with $c_{\rm est}$=1.00089/1.00038/1.00021 (monotone $\to$ the Dirac c=1=2$\times\frac{1}{2}$, the `v450` reading; gapped control $\sim 6\times10^{-8}$, area law); the Peschel modular Hamiltonian K=$\ln((1-C_A)/C_A)$ organises on the CHM weight $\beta(x)=x(L-x)/L$ (corr 0.87$\to$0.99, shrinking plateau drift; constants deliberately not asserted), the even-range bands vanish identically (particle–hole, $\sim 10^{-36}$) and the odd tail is 1.8% with parabolic envelope — the state-side Δ_Σ (the reading forced by `v476`) flows to the CHM/BW geometric form, exactly the object TFPT's Einstein derivation consumes (`v323`/`v358`): the bridge's modular side and the gravity side meet in the continuum limit; a lattice exhibit, NOT a continuum proof — AP2 stays **[O]**. *Leg B (the measure):* **[E]** single-scale corollary — $d\mu$=$\delta(t-t_0)dt$ gives μ_2/μ_1^2=e^{t_0} exactly, so the `v477` one-moment condition forces t_0=$\ln\frac{72}{17}$+$9\ln(8\pi)$=30.461 (one dimensionless KMS time, closed form); the near-miss to $h(E_8)$=30 (difference 0.461) is explicitly DECLINED per the `v354`/`v355` no-free-pattern rule — deriving the measure from the entropic structure stays OPEN. Keybox extended in `tfpt_research_contracts`; high-precision numerical module, Python-only by nature (Wolfram count unchanged, 378/378; flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.LEGS.01`; both legs stay **[O]**, no marker moves.

2026-07-13 (v477 — the scale-flow representation of the entropic action: the surviving R^2 route typed as *one moment condition* $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$ with the exact closure $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$ — the kill-test gap is a scale-measure datum, satisfied by the same KMS moment TFPT already fixes (v36); zero new dials, *no status moves*)

- **`v477_entropic_scaleflow.py` (`GRAV.ENTROPIC.SCALEFLOW.01`).** **[E]** The Frullani identity upgrades to a scale-flow representation: $-\ln a = \int \frac{dt}{t}(e^{-ta} - e^{-t}) = 2\int \frac{d\chi}{\chi}(e^{-a/\chi^2} - e^{-1/\chi^2})$ (symbolic + 10^{-20} quadrature) — Bianconi's log action IS the *flat* scale-integral of relative heat-kernel actions, while TFPT's $S_{{\rm rel},\chi}$ evaluates *one* KMS scale with a weight: the bridge between the two action principles is a choice of SCALE MEASURE. **[E]** Moment dictionary: a weight $w(t)\,dt/t$ gives $3\beta R\,\mu_1 + \frac{17}{24}\beta^2R^2\mu_2$; the flat weight has μ_1=μ_2=1 and reproduces the `v475` raw coefficients identically. **[E]** The ONE moment condition: demanding m^2=$c_3^7\bar{M}_{\rm Pl}^2$ forces exactly $\mu_2/\mu_1^2{=}\frac{72}{17}(8\pi)^9$ — the `v475` kill-test number IS a moment ratio — and the closure identity $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}{=}c_3^7$ holds identically: the "13 orders" are a scale-measure datum, not a bridge mismatch. **[E]** v36 consistency: $f_0{=}6(4\pi)^2/c_3^7 \Leftrightarrow M^2/\bar{M}_{\rm Pl}^2{=}c_3^7$ — one KMS moment, two parametrisations, zero new dials. **[C]** A consistency statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein + Λ level and the scale-flow form, TFPT supplies the measure). Negative side-record: an exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain — kept out of the suite, logged in the research notes. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors both exact blocks (378/378); ledger row `GRAV.ENTROPIC.SCALEFLOW.01`; the R^2 gate and the `v359` typing stay **[O]**, no marker moves.

2026-07-13 (`v476` — the AP2 compression conjecture of the entropic-action bridge made *well-posed* on an exactly solvable model: the literal operator-side reading is ill-posed on a pure bulk, the state-side reading is forced, the mismatch is exactly quadratic in the cross-cut correlations and gap-suppressed; AP2 stays [O], *no status moves*)

- **`v476_entropic_compression_toy.py` (`GRAV.ENTROPIC.COMPRESS.01`).** The last open work package of the `v473` round, exercised on a gapped staggered-mass chain ($N{=}16$). The seam situation is reproduced (pure bulk: the ground-state covariance is a projection to 10^{-15}; thermal boundary: $\mathrm{spec}(PCP) \subset (0,1)$, Peschel $K{=}\ln((1{-}C_A)/C_A)$ finite, $S{>}0$ — `v155` in the bridge context). **The reading is decided:** on the pure bulk $f(x){=}x/(1{-}x)$ is singular on $\mathrm{spec}(C){=}\{0,1\}$, so the literal operator-side reading $Pf(C)P$ of the `v473` conjecture is *ill-posed* while the state-side $f(PCP)$ is finite — the compression identity can only mean "build Δ_Σ from the *compressed* relative metric", which matches Bianconi's own local-metric construction (AP2 retyped, not weakened). **[E]** Block algebra (symbolic): $[C^2]_{AA}{-}(C_{AA})^2{=}BB^t$, $[C^3]_{AA}{-}(C_{AA})^3{=}ABB^t{+}BB^tA{+}BDB^t$ — first order in the cross block vanishes identically, so for any analytic f the two readings agree to *second* order in the cross-cut correlations: the modular (entanglement) content lives exactly in that quadratic correction. Gap control at the bounded (covariance) level: the Bianconi-shaped bulk identity $f(C_T){=}e^{-h/T}$ holds to 10^{-11} relative; the mismatch obeys the quadratic law $d \le x^2$ ($d/x^2 \in [0.4, 1.0]$) and falls with the gap in the gapped regime, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\rm eff}{>}0$, `v76`/`v337`). Keybox extended in `tfpt_research_contracts`; numerical module, Python-only by nature (Wolfram count unchanged, 376/376; the block identity flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.COMPRESS.01`; AP2 (the continuum seam statement) stays **[O]**, no marker moves.

2026-07-13 (`v475` — the R^2 kill test of the entropic-action bridge, *executed*: exact tensorial factors on the maximally symmetric background, raw entropic scalaron $m^2 = \frac{4608\pi^2}{17}\bar{M}_{\rm Pl}^2$ ($\approx 51.7\,\bar{M}_{\rm Pl}$, trans-Planckian) $\Rightarrow$ the light-trace-mode reading is dead, KMS renormalisation the only surviving R^2 route; plus the explicit Lorentzian-positivity witness; *no status moves*)

- **`v475_entropic_scalaron_gate.py` (`GRAV.ENTROPIC.SCALARON.01`).** Work package 5 of the `v473` bridge. **[E]** On a maximally symmetric background the relative curvature operator $\widetilde{\mathcal{R}}\widetilde{g}^{-1}$ has eigenvalues *exactly* $\{R;\ R/4{\times}4;\ R/6{\times}6\}$ (Bianconi's Appendix-B flattening verified symbolically, incl. the 6×6 two-form block), giving the exact vacuum action $\mathcal{L}_{\rm B}(R) = 3\beta R + \frac{17}{24}\beta^2R^2 + O(R^3)$ — the 3 re-derives her Eq. (45) (= the `v473` pin), $\frac{17}{24}{=}\frac{1}{2}(1{+}\frac{1}{4}{+}\frac{1}{6})$ is the exact tensorial R^2 factor. **[E]** With the pinned $\beta'{=}c_3/6$ both mass routes (Starobinsky matching and $f'/3f''$) agree: the raw entropic scalaron is $m_{\rm raw}^2{=}\frac{4608\pi^2}{17}\bar{M}_{\rm Pl}^2$, i.e. $m_{\rm raw}{\approx}51.7\,\bar{M}_{\rm Pl}{\approx}1.3{\times}10^{20}$ GeV — TRANS-PLANCKIAN, not light; `v473`'s interpretation 2 (the TFPT scalaron as the light $\mathcal{G}$-trace mode) is *dead in naive form*; a viable mechanism must supply exactly $\frac{72}{17}(8\pi)^9\approx1.69\times10^{13}$ in mass^2 (the kill-test number now WITH tensorial factors). **[E]** Domain: first log pole at $R{=}1/\beta{=}384\pi^2\bar{M}_{\rm Pl}^2$ (trans-Planckian) — the obstruction is the mass, not the domain. **[E]** Lorentzian-positivity witness: a timelike gradient gives the Gg^{-1} eigenvalue $1-\alpha v^2\le 0$ for $v^2\ge 1/\alpha$ (ln undefined), spacelike/Euclidean stay positive — the `v473` **[C]** caveat exhibited, the OS route sturdier. **[C]** Verdict: interpretation 1 survives (Einstein + Λ level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving R^2 route. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors all three blocks (376/376); ledger row `GRAV.ENTROPIC.SCALARON.01`; no gate closes, no marker moves.

2026-07-13 (`v474` — the operator level of the entropic-action bridge: the D_5 Clifford/spinor structure exhibited on the carrier Fock space, the Hodge fold identified as the $5{\to}\bar{5}$ conjugation, and the `v473` Q-target *decided* — supports exactly $\{|\mathbb{Z}_2|,\operatorname{rank}E_8,2^{g_{\rm car}}\}$, the naive pair-block reading killed; *no status moves*)

- **`v474_entropic_hodge_carrier.py` (`GRAV.ENTROPIC.HODGE.01`).** Work packages 1 and 4-algebra of the `v473` bridge, machine-checked on the finite carrier Fock space $\Lambda^\bullet\mathbb{C}^5$ (dim $2^{g_{\rm car}}{=}32$). **[E]** CAR⇒Clifford: the ten Γ's from $(a_k^\dagger,a_k)$ satisfy $\{\Gamma_a,\Gamma_b\}{=}2\delta_{ab}$ exactly (all 100 pairs, integer matrices); the 45 bilinears $[\Gamma_a,\Gamma_b]/4$ commute with $\gamma{=}(-1)^N$, span exactly $45{=}\dim\mathfrak{so}(10)$ and leak nothing from the 16-dim even subspace — the carrier half-spinor *is* the even exterior algebra *with* its spinor structure (the operator level of `v2/v44/v197`). **[E]** the Hodge–Dirac symbol: $c(\xi){=}a^\dagger(\xi){-}a(\xi)$ satisfies $c(\xi)^2{=}-|\xi|^2\,\mathrm{Id}$ for symbolic ξ — Bianconi's $D{=}d{+}\delta$ is Clifford multiplication at symbol level. **[E]** the fold is the $5{\to}\bar{5}$ conjugation: the traceless $\mathfrak{u}(5)$ weights of Λ^4 are the negatives of those of Λ^1, so her fiber 1+5+10 becomes the GUT half-spinor $16{=}1{+}\bar{5}{+}10$ (the SM carrier decomposition). **[E]** honest obstruction: $\Omega^{\le2}$ is *not* Clifford-invariant ($\Lambda^2{\to}\Lambda^3$ leak) — the bridge must go through the fold, never through degree truncation (a recorded constraint on the AP2 compression). **[E]** the Q-target enumeration (the `v473` **[C]** conjecture decided at the algebraic level): $\operatorname{Tr}Q^2{=}d\,k^2c_3^4{=}32c_3^4$ with integer k has exactly the supports $(d,k){=}(2,4),(8,2),(32,1)$ $=$ $\{|\mathbb{Z}_2|,\ g_{\rm car}{+}N_{\rm fam}{=}\operatorname{rank}E_8,\ 2^{g_{\rm car}}{=}\dim\Lambda^\bullet\}$ — minimal uniform solution $q{=}c_3^2{=}1/(64\pi^2)$ per Fock mode, bonus identity $\operatorname{Tr}Q^2/\delta_{\rm top}{=}32/48{=}2/3{=}\dim\Lambda^\bullet/\Omega_{\rm adm}$; the two-form block 10 (the problem-note "pair sector" conjecture) and the fiber 16 need irrational multipliers $\sqrt{16/5}$, $\sqrt{2}$ — *killed* in uniform form. **[C]** the physical choice among the three admissible supports; **[O]** unchanged (AP2 compression + the R^2 mechanism). Keybox extended in `tfpt_research_contracts` (Research Contract 2, the `v473` keybox); Wolfram mirrors the enumeration and the weight-conjugation identity (373/373); ledger row `GRAV.ENTROPIC.HODGE.01`; no gate closes, no marker moves.

2026-07-13 (`v473` — the entropic-action bridge: Bianconi's *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); `arXiv:2408.14391`) quantified as an *external candidate* for the missing action level of the parameter-free Einstein equation; her free constant pinned $\beta'_{\rm B}=c_3/6=1/(48\pi)$, the R^2 gap $3(8\pi)^9$ pre-registered as kill test; *no status moves*, the `v359` equation-of-state typing stays [O])

- **`v473_entropic_action_bridge.py` (`GRAV.ENTROPIC.ACTION.01`).** TFPT's classical field equation $G_{ab}+\Lambda g_{ab}=c_3^{-1}T_{ab}$ is derived thermodynamically (`v358/v359`) and honestly typed "equation of state, not a from-action quantisation" **[O]**. Bianconi's entropic action $S_{\rm B}=-\operatorname{Tr}_F\ln(\widetilde{G}\widetilde{g}^{-1})$ (the quantum relative entropy between the spacetime metric and the matter-induced metric, Araki-related via $\Delta^{1/2}=\widetilde{G}\widetilde{g}^{-1}$) is a local covariant action whose variation gives Einstein + an emergent nonnegative Λ at low coupling — an explicit *external candidate* for exactly that missing variational layer. Machine-checked: **[E]** the carrier Hodge count (her $\Omega^{0,1,2}$ fiber on the five-slot carrier $\mathbb{C}^5$ counts $1{+}5{+}10{=}16{=}2^{g_{\rm car}-1}{=}\dim S^+_{D_5}$, while on 4d spacetime it is only 11 — the 16 *requires* the carrier); **[E]** the one quantitative pin — matching her weak-coupling $3\beta R$ to $1/(16\pi G)=c_3/(2G)$ fixes her free constant *exactly*,

$\beta'_{\mathrm{B}} = c_3/6 = 1/(48\pi)$; **[E]** her $\Lambda_{\mathcal{G}} = \frac{1}{2\beta}\operatorname{Tr}(\mathcal{G} - I - \ln\mathcal{G})$ is nonnegative and quadratic at the fixed point, and with $\varepsilon_\star = e^{-\alpha^{-1}}Q$ it reproduces the closed `v60` branch $(3/4\pi^2)e^{-2\alpha^{-1}}$ with the exact target $\operatorname{Tr} Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\delta_{\mathrm{top}}$ ([**C**] until Q is constructed); **[E]** the R^2 kill test — the raw log expansion sits *exactly* $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$ below the TFPT Starobinsky coefficient $1/(12c_3^7)$, so a direct identification is *false* without a mechanism (pre-registered); **[C]** the compression conjecture $P_\Sigma(\widetilde{G}\widetilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$ (the bridge proper) stays open, with Lorentzian positivity flagged (TFPT's OS route is the sturdier construction). New keybox in `tfpt_research_contracts` (Research Contract 2, after the `v359` keybox); residual pointers updated in `introduction/tfpt_4_frontier`/status card. Wolfram mirrors the three exact blocks (371/371); ledger row `GRAV.ENTROPIC.ACTION.01`; *no* gate closes, *no* marker moves, zero TFPT knobs added.

2026-07-03 (surfacing round — three already-computed unique readouts promoted to the public prediction surfaces: the axion–photon coefficient $g_{a\gamma\gamma} = -4c_3$, the free-seam Higgs band $m_H \approx 129$–$134\,\mathrm{GeV}$, and the never-run BH HFQPO $\times 1.5$ ladder-tooth search; no new vN, no status moves)

- **Uniqueness scan follow-up (no new results — presentation only).** A structure scan for "what does TFPT predict that nobody else pins?" found three concrete, falsifiable readouts that were fully computed and ledgered but missing from the public prediction surfaces; they are now surfaced, with their existing typing unchanged. (i) **Axion–photon coefficient**: the determinant-line anomaly coefficient $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$, $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$ (`v207`, the **[E]** leg of `GRAV.ASYMP.01`; the **[C]** caveat — coefficient, not a parameter-free $g_{a\gamma\gamma}$ in GeV^{-1} — stated verbatim) joins the axion card in `website/lib/predictions.ts` and the frontier dark-matter mirror in `papers.ts`; the TeX source (`tfpt_4_frontier`) already carried it. (ii) **Higgs free-seam band**: the double-criticality prediction $\lambda(\bar{M}_{\mathrm{Pl}}) = \beta_\lambda(\bar{M}_{\mathrm{Pl}}) = 0 \Rightarrow m_H = 133.5\,\mathrm{GeV}$ (band 129–134, m_t/α_s-sensitive; measured 125.25 sits a few GeV below — the known slight metastability, Buttazzo NNLO $\lambda(M_{\mathrm{Pl}}) = -0.0143 \pm 0.0057$, 2.5σ) becomes a prediction card ([**C**]; `v166`, `HIGGS.FREESEAM.01`, kill: settled-(m_t, α_s) RGE pull off the surface at $> 5\sigma$). (iii) **BH HFQPO ladder tooth**: the one discriminating HFQPO test — a third tooth at $\nu_3 = (3/2)\nu_u$ (661.5/414/252/363 Hz), integer harmonics forbidden — that *no published search ever targeted* although the teeth lie in the searched RXTE band (`experiments/hfqpo-ladder`, preregistered): added as an **[O]** search-target bullet to the `tfpt_horizon_readouts` "Search targets (not claims)" gapbox (firewall language verbatim: non-canonical mapping, selection null 18.5%, J1859+226 $+9.2\sigma$, GR resonance favored; even a tooth hit would be [**C**]) and as an **[O]** prediction card. Website mirrors updated in the same change (`predictions.ts` 23 $\to$ 25 cards, `papers.ts` frontier + horizon sections incl. the new search-targets mirror section); scorecard, ledger and all markers untouched.

2026-07-03 (v472 — the determinant line over the $U(1)$-twist moduli of the collar model carries curvature = the inflow level: the v470 bridge lemma exhibited at the finite/model level; ALPHA.QUILLEN.EXACT.01 stays [O]. Plus: bird's-eye rounds — electron-sector/Cabibbo whitespot probes, Calderón UV-universality, parameter-free flat-budget closure (sandbox); the flat-budget closure enters the evidence scorecard)

- **`v472_quillen_detline_moduli.py` (`ALPHA.QUILLEN.DETLINE.01`).** `ALPHA.QUILLEN.EXACT.01` names the α^3 level as the curvature of the *determinant line over the* $U(1)$ *seam moduli* (Quillen 1985 / Bismut–Freed CMP 106 (1986) curvature; Dai–Freed JMP 35 (1994) section); `v470` computed the bulk Chern invariant only over the Bloch Brillouin zone (the translation-invariant shortcut). `v472` computes the Quillen-shaped object itself: the same collar Hamiltonian ($h(k) = \sin k_x\,\sigma_x + \sin k_y\,\sigma_y + (M - \cos k_x - \cos k_y)\,\sigma_z$, `v367`/`v470`) in real space on an $L \times L$

torus with *twisted* boundary conditions — the twist torus (θ_x, θ_y) *is* the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the *determinant line of the occupied frame* over that torus: $\oint = 2\pi \cdot 1$ exactly (integer to 10^{-9}) at $M{=}1$ for $L{=}4$ AND $L{=}6$ (size-independent); controls: trivial collar $M{=}3 \to 0$, orientation flip $M{=}{-1} \to -1$; the twist-moduli integer equals the Bloch-BZ integer for all three M (Niu–Thouless–Wu PRB 31 (1985), exhibited on the S3 collar model); the Fermi gap (= 2.0) stays open over the whole twist torus, so the Dai–Freed section has no zero. **Reading [C]:** det-line holonomy over the $U(1)$ moduli = inflow level $k_0 = |C| = 1$ — the `v470` bridge lemma "$\delta \log \det_\zeta(\text{seam})$ = the inflow response" holds *verbatim at the finite/model level*, where $\log \det$ of the occupied frame replaces $\log \det_\zeta$. **Not closed [O]:** the *continuum* ζ-determinant identification on the abstract seam (= the `SEAM.EQUIV.01` face); `ALPHA.QUILLEN.EXACT.01` stays **[O]**; α^{-1} stays **[E]**. Structurally the same manoeuvre as `v471` for `SEAM.THEOREM.01`'s replica side. 7/7 checks; suite green through `v472` (466 modules); numerical (dense eigh + FHS twist grid), Python-only by nature. Papers updated: `tfpt_1` (sixth honest step in the EM closure), `tfpt_research_contracts` (residual certification), `tfpt_safeguards` (α-form lever), root `README`, `zenodo_description.html`. Exploration original: `experiments/tfpt-discovery/quillen_detline_twist_moduli.py` (same day).

- **Bird's-eye whitespot rounds (sandbox, `experiments/tfpt-discovery/`, nothing else promoted).** (i) `electron_sector_cabibbo_probe.py`: the α metrology triangle (TFPT α^{-1} = 137.0359992168 vs Rb-2020 $+0.99\sigma$ / CODATA22 $+1.90\sigma$ / a_e-route $+3.39\sigma$ / Cs-2018 $+6.33\sigma$ — TFPT takes the Rb side of the 5.5σ metrology split; new honest pressure point recorded); the `v204` seam-vertex universality catch (a flavor-universal reading of $a_\mu^{\text{seam}} = 2.879 \times 10^{-9}$ is excluded by the electron at $\sim 2 \times 10^4\sigma$; linear scaling at $\sim 10^2\sigma$; quadratic survives — the **[C]** bridge must scale at least quadratically in m_ℓ); the Cabibbo-angle anomaly as a third dissolution watchdog ($\lambda_C = 0.2243762$ exact sits $+0.08\sigma$ on the PDG-2026 rescaled kaon average, between $K_{\ell 3}$ and $K_{\mu 2}$; exact unitarity puts $V_{ud} = 0.97450$ — superallowed is the outlier at $+2.6\sigma$, all neutron/pion routes agree). (ii) `kernel_calderon_uv_universality.py`: the raw-collar Calderón/DtN datum is UV-universal (mode-by-mode first-order convergence to the exact Bessel symbol $mI'_\nu(mR)/I_\nu(mR)$, Richardson residual 3×10^{-4}; N_θ-independent; $|k|$ principal symbol; μ_4 block structure at machine precision; both spec-T masses; 3-fold mesh-defect negative control) — the operator equality demanded by `QGEO.KERNEL.01` has a well-defined discretization-independent left-hand side (gate stays **[O]**). (iii) `flat_budget_closure.py` + scorecard row: see next bullet.
- **Flat-budget closure enters the evidence scorecard (experiments-only, pattern candidate).** The zero-dial closure $\{\Omega_b = \varphi_0^{\text{ret}}(1 - \frac{1}{4\pi})$ [frozen], $\Omega_c = \frac{2}{7}(1 - \frac{1}{4\pi})$ [the 2026-07-02 numerology-flagged Nebenbefund], $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ [frozen], $\Sigma m_\nu = 0.0588\,\text{eV}$ [`v468` route]$\}$ + spatial flatness determines the whole flat ΛCDM budget: $H_0 = 67.15\,\text{km/s/Mpc}$ (-0.39σ Planck18, -5.67σ SH0ES22 — the parameter-free Planck side of the Hubble tension), $\Omega_m = 0.3133$ (-0.27σ), $t_0 = 13.86\,\text{Gyr}$; honest stresses recorded: ω_b -2.02σ (BBN-only -0.73σ), ω_c -1.19σ, DESI-DR2 H_0 -2.16σ (one correlated low-h direction). Scorecard row `cosmo / flat budget closure` (`consistent`, composite: shares the `phi0_seed` group (and the ρ_Λ leg the α engine), Ω_c leg in `alternative_group=axion_branch` vs the axion-spine DM reading — never double-counted); nothing promoted to `verification/`.

2026-07-02 (`v471` — the Seam–Horizon replica chain exercised numerically on the discretized collar: the kernel-identification premise of `SEAM.THEOREM.01` discharged at the finite/model level; residual retypes to the continuum MMST leg + the anchor; gate stays [O]. Plus: the GW ringdown-echo channel hardened — 10 loudest events incl. GW250114, redshift-corrected templates, stacked search and a 12-variant signature battery, all null)

- **`v471_seam_horizon_replica.py` (`SEAM.EHMODEL.04`).** The residual isolated by `v150`/`v151`/`v152` — "the RP seam kernel *is* this BFK Calderón datum" — is exercised numerically on a discretized seam collar with the seam's *own* kernel and *real* replica sheets: the matched conical deficit has slope $2C(\gamma)$ on deficit cones (0.01–0.05%) and replica sheets n=2 (0.8%), n=3 (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the ζ-scale μ_{lat} drifts — the `v152` anchor exhibited on data, with *no* canonical $\ln(m/\mu)$=3/4 emerging (recorded honestly); the BFK/Schur split is exact ($\sim 10^{-16}$) with the Calderón factor conically clean *measured on the kernel* (pure cut-edge law, tip term 0, power-checked) and the halves carrying $2C_{\text{cone}}$ via Kac doubling; the `v302` transfer masses $6\ln(3/2)$, $6\ln 3$ sit on the calibrated EH line ($\leq$ 0.05%); and the Perron/attractor mode (m=0) is IR-divergent — **the recovery gap Δ=$6\ln(3/2)$ is what makes the induced-gravity coefficient finite** (the same gap that makes the attractor unique makes $1/G$ finite). `SEAM.THEOREM.01` stays **[O]**; its residual retypes to the *continuum* scaling limit (the MMST class — the same single residual as `SEAM.EQUIV.01`, `v336`) plus the one declared anchor. 16/16 checks; suite green through `v471` (465 modules); numerical (sparse lattice determinants), Python-only by nature. Papers updated: `origin_theory` (new collar paragraph in the Seam–Horizon gapbox + residual retyping), `tfpt_research_contracts` (R3 premise), `tfpt_4_frontier` (3/4 audit note), root `README`, `verification/README`, `zenodo_description.html`. Exploration original: `experiments/theory-contracts/seam_horizon_replica.py` (17/17, same day).
- **GW ringdown-echo channel hardened (`experiments/gw-ringdown-echo`, scorecard row `data_limited`→`consistent`).** (i) Strain coverage extended to the *10 loudest* ringdown events incl. GW250114 (network SNR 78.6, O4b; new `fetch_strain_4096.py` crops the 4096 s GWOSC archive segments). (ii) **Redshift correction**: the GWTC catalogue reports source-frame masses, the observed ringdown is at $f_0/(1{+}z)$ — all templates moved to detector-frame $M(1{+}z)$ (before the fix GW190521, z=0.56, was filtered $\sim$1.6$\times$ too high); all stages re-run. (iii) Stage 1b *stacked* search over 23 detector streams: stacked p=0.262, kernel-consistent streams 0/23 — STACKED NULL. (iv) Stage 1c *signature battery* (Bonferroni $\times$12): ratio semantics (amplitude $(2/3)^6$ vs *energy reading* $(2/3)^3$ vs step 2/3) $\times$ μ_4 per-bounce phases $\{0, \pi/2, \pi, 3\pi/2\}$ (the boundary-birefringence analogue — path birefringence from parity violation c_{-}=8 cancels in the inter-echo ratio; only a per-reflection phase does not) $\times$ lags 0.5–350 ms (ECO through Planckian window), with joint $(2,2,0)$+$(2,2,1)$ subtraction and a template-agnosticism control: `NO_VARIANT_ECHO` on all 10 events — the null is robust against alternative signature readings. Search-target firewall unchanged: upper bound, no detection, no tension claim.

2026-07-02 (`v295` robustness patch — external-review follow-up: the numeric a_2-kernel tolerance must dominate its own finite-difference floor)

- **`v295_flataway_a2_exact.py` (`FLATAWAY.A2.01`).** Alessandro's Git-baseline audit (commit `8cb8ad0`) hit a platform-marginal failure in check 3: the exact divided-difference a_2 kernel is compared against an ε=10^{-3} second finite difference whose *own* truncation error is $O(\varepsilon^2) \sim 10^{-6}$, yet the tolerance was 10^{-6} — on his machine k=3 landed at 1.1×10^{-6} (ours: 2.9×10^{-7}). Fix: tolerance widened to 5×10^{-6} so it *dominates* the control's error floor, with the reasoning recorded in the check text; the load-bearing anchor remains the *closed form* at machine precision (`v296`), the numeric witness stays as the independent cross-check. A numerical-robustness patch, no claim or status change; ledger row updated; suite green (ALL CHECKS PASSED through `v470`).

2026-07-02 (v469/v470 — the closure-route round on the two named targets: the SEAM.EQUIV.01 extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the α^3 level = the *computed* bulk Chern invariant + the seam F-normalisation = the embedding index k_Y=5/3; both targets stay [O])

- **v469_seam_crossedproduct_route.py (SEAM.EQUIV.CROSSEDPRODUCT.01).** The 128-spinor extension leg of SEAM.EQUIV.01 no longer rests on 2024/2025 preprints: **[E]** the $SO(16)_1$ spinor has h_s=$N/16$=16/16=1$\in\mathbb{Z}$, so its statistics phase is +1 — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1\rtimes\{1,s\}$ is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B)$=4/2^2=1 $\Rightarrow$ holomorphic $\Rightarrow$ $(E_8)_1$ by the v463 pin. The AGT/AMT lattice-VOA route (v459) is *demoted to an independent second witness*. **[E]** the index-4 μ_4 glue runs on the same integer: $h(J^k)$=$\{1,1,1\}$ for k=1, 2, 3 (the v125 isotropy $q(k(1,1))$=k^2 at net level), μ=16/4^2=1, same endpoint. **[E]** the realisation input is reduced from model fiat to *invariants* (R1$'$): quasi-free **[C]** (v155/v160) + gap **[E]** (v302) + class D + c_-=8 **[E]** (v456, from P1), with the invariants *computed* — per-copy FHS Chern $|C|$=1 vs 0 (control), ν=16$\times|C|$=16, c_-=ν/2=8 — and $\nu\equiv0\pmod{16}$ is exactly Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of c_- is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). **[O]** *not closed*: the cited theorems stay cited; SEAM.EQUIV.01 stays **[O]**.
- **Lean hardening.** TfptCarrier/SeamResidualAxiom.lean gains the *parallel* peer-reviewed derivation seamResidualClosed$'$: the invariant-level axiom collar_invariants (R1$'$) + *one* combined 1995–2001 package crossedproduct_route_theorem; the new joints are kernel facts with *no* axioms — lr_locality_integer (h_s=16/16$\in\mathbb{Z}$), glue_locality_integers (μ_4 $h(J^k)\in\mathbb{Z}$), klm_holomorphic (4/2^2=1, 16/4^2=1), sixteenfold_pin (ν=16$\equiv$0 mod 16, c_-=8), bundled as crossedproduct_arithmetic ([propext] only); #print axioms seamResidualClosed$'$ = exactly four; the original seamResidualClosed (AGT/AMT witness) is kept unchanged as the independent second route; lake build OK (3285 jobs).
- **v470_alpha_inflow_level.py (ALPHA.QUILLEN.INFLOW.01).** Both residuals left on ALPHA.QUILLEN.EXACT.01 after v433–v435 are upgraded. **[E]** target re-verified (α^{-1}=137.0359992168; Quillen split <10^{-35}). **[C]** *the level is computed, not minimal*: the a^3 coefficient k_0=1 *equals* the FHS bulk Chern invariant $|C|$=1 of the *same* p+ip collar that realises S3 (v367/v461) — "why exactly 1" is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016) η=CS part of $\delta\log\det$; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), *replacing* v435's minimality assumption. **[E]** $k_Y=\mathrm{tr}(Y^2)/\mathrm{tr}(T_3^2)$=(5/6)/(1/2)=5/3 (Ginsparg, Phys. Lett. B197 (1987)): $(3/5)\times\frac{41}{6}=\frac{41}{10}=b_1$ — the "GUT 3/5" *is* the inverse embedding index; $\frac{41}{6}=\frac{20}{3}+\frac{1}{6}$ (carrier decomposition). **[C]** the seam F-normalisation (the one **[C]** of v434) is *retyped*: once the seam net is $(E_8)_1$ level 1, $\langle JJ\rangle=k/(z-w)^2$ is fixed by k_Y — level-1 current-algebra rigidity, zero independent content, a face of SEAM.EQUIV.01 (per the v382 typing). **[E]** *structural unification*: one invertible phase, two quantised responses — c_-=8 (gravitational, feeds SEAM.EQUIV.01/S3) and C=1 ($U(1)$, feeds EM1) — so closing R1$'$ feeds *both* named targets. **[O]** *not closed*: the bridge lemma "$\delta\log\det_\zeta$(seam)= the inflow response" stays the named cited step; ALPHA.QUILLEN.EXACT.01 stays **[O]**; α^{-1} stays **[E]**.
- **Integration.** Both registered in run_all.py + script_registry.csv (cluster

frontier); ledger rows SEAM.EQUIV.CROSSEDPRODUCT.01 + ALPHA.QUILLEN.INFLOW.01; FORM.SEAM.RESIDUAL.01 updated (the seamResidualClosed′ route); the exact joints Wolfram-mirrored (365 → 368: the LR/glue locality integers, the KLM μ-arithmetic, the sixteen-fold pin, the k_Y=5/3/b_1 identities); cited in tfpt_1 (α fixed point + EM-Ward residual prose), tfpt_research_contracts (seam residual + Lean section), introduction (closure ledger), tfpt_5_redteam (Target A residual wording) and the README/website status surfaces; exploration provenance experiments/tfpt-discovery/closure_route_*.py (2026-07-02 note in experiments/next.txt). Suite ALL CHECKS PASSED through v470. Curated version bumped to **v5.4** (Zenodo deposit updated).

2026-07-02 (v467/v468 — two named post-hoc [O] flavor candidates, record unchanged: the three $\sim 2\sigma$ mixing tensions as *one* third-generation $-\varphi_0$ pattern, and the splitting ratio $\Delta m^2_{21}/\Delta m^2_{31} = |J_{\rm PMNS}|$)

- **v467_thirdgen_mixing_pattern.py (FLAV.THIRDGEN.PATTERN.01).** The suite's three $\sim 2\sigma$ mixing tensions ($\sin^2\theta_{13}$ $+2.0\sigma$, $\sin^2\theta_{23}$ $+1.76\sigma$, $|V_{cb}|$ $+1.79\sigma$; v307) are exactly the three *third-generation* channels and share *one* relative deviation $-5.36\% \pm 1.67\%$ — non-zero at 3.2σ, -0.03σ from $-\varphi_0$. [**E**] vocabulary identities: λ_C^2=$\varphi_0(1-\varphi_0)$ (the frozen Cabibbo); $\sin^2\theta_{23}$=$(1-\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23}$=$\varphi_0$; the suppression size φ_0=5.3% sits inside the seam-gapped band $(2/3)^6$=8.8% (v393). [**N**] the ladder-base forms $\lambda_C^2 e^{-5/6}$=0.021880 / $(1-\varphi_0)/2$=0.473414 / $\lambda_C^2/(1+\lambda_C)$=0.041119 drop the joint four-observable χ^2 from 10.4 to 0.11 with *zero* new parameters; $|V_{ub}|$ (no bare φ_0) and the 1–2 channels stay untouched and correct (dressing λ_C would fail at -24σ — scope checked, not chosen); Daya Bay alone already prefers the pattern form ($\Delta\chi^2$=4.3). [**C**] **honest:** post-hoc (5/16 equally-simple dressings fix the two PMNS tensions; $(1-\varphi_0)/(1-c_3)$/resolvent degenerate <1.4%). [**O**] no seam-transport derivation (the Bernoulli-variance reading λ_Y^2=Var[Bernoulli(φ_0)] is an exact identity but only an interpretation) — the frozen record (v84/REG.FREEZE.01) is *unchanged.* [**X**] pre-registered, symmetric: θ_{13} stable at 0.0231 kills the pattern, at $\sim$0.0219 (sub-1%) kills the record's bare-φ_0 reading, exclusive $|V_{cb}|\sim$0.0395 kills both — JUNO-era reactors + Belle II decide. Python-only.
- **v468_dm2_ratio_jarlskog.py (FLAV.DM2RATIO.01).** The splitting ratio $\Delta m^2_{21}/\Delta m^2_{31}$ was previously *unpredicted*; the frozen PMNS matrix (v9/v268/v270, δ=$4\pi/3$) has one canonical derived CP invariant, and [**N**] $|J_{\rm PMNS}|$=0.029653 meets the NuFIT 6.0 measured ratio 0.029805 ± 0.000792 at -0.19σ (equivalently m_2/m_3=$\sqrt{|J|}$=0.17220 vs 0.17264(229)) — zero dials. [**N**] the informal tfpt_2 table heuristic $m_2/m_3 \sim \pi\varphi_0$=0.16704 (a "$\sim$" entry, never frozen) is now -2.44σ and is *superseded as the comparator.* [**C**] spectrum consistency: Σm_ν=$m_3(1+\sqrt{|J|})$=0.0588 eV reproduces the documented 0.0586 eV (v272); no absolute number is added. [**O**] no mechanism (the $\mu\tau$ texture gives θ_{13}=0 at leading order). [**X**] JUNO shrinks the window $\sim 4\times$; adopting the v467 pattern would shift $|J| \to 0.028849$ (-1.21σ today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.
- **Integration.** Both registered in run_all.py and script_registry.csv (cluster neutrinos); ledger rows FLAV.THIRDGEN.PATTERN.01 and FLAV.DM2RATIO.01 (both [**O**] candidates); cited in tfpt_2 (§ Neutrinos: pressure-point block + new candidate paragraphs, the m_2/m_3 table row retyped to the $|J|$ comparator, the octant prose kept honest — record does not select, the candidate would), introduction (closure ledger), tfpt_4_frontier (live kill-test board) and tfpt_5_redteam (seed-hyperplane caption); v328's "single outlier" wording scoped to the seed observables; exploration provenance experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py (2026-07-02 note in experiments/next.txt); Python-only (flagged in the Wolfram README).

2026-07-01 (v466 — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ back-solves the same axiom seed to -0.11%, extending the empirical over-determination to *five* sectors)

- **`v466_seed_leptonmass_channel.py` (`SEED.LEPTONMASS.01`).** The seed over-determination of `v306`/`v465` used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ is a sixth independent readout of the *one* axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$. **[E]** plumbing: the axiom φ_0 reproduces the frozen $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$ to 10^{-9}. **[N] new sector (headline):** back-solving φ_0 from the measured *pole* ratio $m_e/m_\mu = 0.004836$ (PDG 2024) gives $\varphi_0 = 0.053113$, only -0.11% from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. **[C] transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6 = 8.78\%$, `v393`) but largely cancels for m_e/m_μ, so the 0.11% residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. **[N] extends:** the six-channel cluster (`v306`'s five + this) has $\chi^2/\mathrm{dof} < 1$ and the error-weighted mean stays at the axiom to $<0.1\%$; with the corroborating heavy-quark m_t/m_b (-0.23%) it is a seven-channel, five-sector agreement. **[N] power:** a data↔formula shuffle explodes the cluster χ^2 by $>100\times$, and the scheme-ambiguous $m_c/m_s = (34/47)/\varphi_0$ (FLAG 11–14) is deliberately not counted. **[C] honest:** theoretically still the *one* seed (`v305` multiplicity 1) — an *empirical* strengthening, not theoretical independence; φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster `registry`); ledger row `SEED.LEPTONMASS.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside `v305`/`v306`/`v465`); Python-only (flagged in the Wolfram README).

2026-07-01 (v465 — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within 0.01σ, the θ_{13}-independent externally-runnable slice of v306)

- **`v465_seed_crosssector_joint.py` (`SEED.CROSSSECTOR.01`).** The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation β (a CMB EB/TB polarisation observable) and the Cabibbo angle $\lambda_C = |V_{us}|$ (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$. **[E]** plumbing: the axiom φ_0 reproduces the frozen λ_C, $\sin^2\theta_{12}$, β to 10^{-9}. **[N] cross-sector (headline):** back-solving φ_0 *independently* from β (CMB; ACT DR6 + Planck PR4 combined) and from λ_C (quark; PDG 2024) gives the same seed within $0.01\sigma_{\mathrm{comb}}$. **[N] one-parameter fit:** the error-weighted mean of the three back-solved φ_0 reproduces the axiom to 0.06%; the joint χ^2 at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\mathrm{dof} < 1$, combined pull 0.08σ. **[N] θ_{13} exclusion declared:** adding $\sin^2\theta_{13}$ raises χ^2 by ~ 4 (the documented $\sim 2\sigma$ pressure point, `v328`/`v338`), so this slice is the honest θ_{13}-independent statement, not a hidden dilution. **[N] power:** a data↔formula shuffle explodes the cluster χ^2 by $>5\times10^4$. A consistency / out-of-sample statistic (a subset of `v306`); φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster `registry`); ledger row `SEED.CROSSSECTOR.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside `v305`/`v306`) and in the `origin_theory` birefringence dependency chain; Python-only (flagged in the Wolfram README).

2026-06-30 (v463/v464 + Lean + Wolfram — the c=8 holomorphic-uniqueness pin and the one-particle realisation rigidity: the `SEAM.EQUIV.01` residual reduced to *entirely certification* on both halves)

- **The bird's-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (`v458` MMST + `v459` AGT/AMT); what was implicit was (B) *why* a c=8 holomorphic limit must be $(E_8)_1$, and (A) the *realisation* input R_1 that the abstract seam IS the concrete quasi-free collar. `v463` closes B; `v464` reduces A to its one-particle data.
- **`v463_seam_c8_holomorphic_uniqueness.py` (`SEAM.EQUIV.UNIQ.01`).** The c=8 holomorphic-uniqueness pin. [**E**] c=8 is NOT unique at level 1: $c = \dim\mathfrak{g}/(1+h^\vee)$=8 has THREE simple solutions A_8 (dim 80), D_8=$\mathfrak{so}(16)$ (dim 120), E_8 (dim 248), so the det K=4 $SO(16)$ rival is a genuine same-c competitor. [**E**] HOLOMORPHY FORCES dim V_1=248: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}$=E_4/η^8, q^1 coefficient 248 (the 1-dim candidate space, vacuum a_0=1). [**E**] dim 248+ level-1 c=8 $\Rightarrow$ E_8 uniquely (248/(1+30)=8); tower $\{1, 248, 4124, 34752, 213126\}$=$E_4/\eta^8$=$j^{1/3}$. [**C**] Dong–Mason/Schellekens: the holomorphic c=8 VOA is unique = V_{E_8} (even unimodular rank-8 lattice unique = E_8, Mordell–Witt) — so the identification half is classification-forced.
- **`v464_seam_oneparticle_rigidity.py` (`SEAM.EQUIV.ONEPARTICLE.01`).** The one-particle rigidity attacking R_1. The seam is quasi-free (`v113`/`v160`/`v161`), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol P. [**E**] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with P= projection onto K<0 (P^2=P to $< 10^{-12}$), K fixed by gap + chirality + reflection (`v426`). [**E**] ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window ($\max|P_{2N}-P_N|\sim 1/N\to 0$). [**E**] the limit is c=1 Dirac: entanglement $S(L)$=$(c/3)\ln L$ gives $c\to 1$, 16 copies $\Rightarrow$ c_-=8 (ties `v450`). [**C**] Shale–Stinespring makes the Bogoliubov map implementable, so R_1 is upgraded from a bald axiom to "the unique quasi-free realisation, modulo the named CAR functor". Python-only (numpy).
- **Lean (`SeamResidualAxiom.lean`).** Adds the `v463` kernel facts `c8_three_candidates` ($A_8/D_8/E_8$ each dim =$8(1+h^\vee)$) and `holomorphy_selects_e8` (forced dim V_1=248=$E_8 \neq 80, 120$) to `residual_arithmetic` (still `[propext]`-only); `v464` is numerical. `#print axioms seamResidualClosed` still lists exactly FOUR axioms.
- **Wolfram + integration.** `v463` mirrored (the three candidates, the forced 248, the E_4/η^8 tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)–(xxxi) the `SEAM.EQUIV.01` residual is now *entirely certification* — a named, hypothesis-audited package (MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet `SEAM.EQUIV.01` stays [**O**] because it still rests on cited continuum-existence theorems we do not re-prove.

2026-06-30 (v461/v462 + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the `SEAM.EQUIV.01` residual reduced to ONE realisation axiom + ONE combined cited theorem)

- **`v461_seam_strict_locality.py` (`SEAM.S3.LOCALITY.01`).** Sharpens `v460`'s verdict from a *cited* statement to an *exhibited* topological obstruction on the same `v367` p+ip collar. [**E**] CHERN INTEGER: the FHS Chern is C=+1 (chiral M=1), 0 (trivial M=3), −1 (opposite mass). [**E**] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $|C|$=1 (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [**E**] OBSTRUCTION INTEGERS TIED: winding = $|C|$=1$\neq$0 and c_-=8=$g_{\rm car}$+$N_{\rm fam}$= rank $E_8\neq 0$ (the same 8 as c_3's one-sided count, `v456`). [**E**] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [**C**][**O**] Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0, C=0; since C=1$\neq$0 NONE exists, so `v424` sub-step (i)'s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). `SEAM.EQUIV.01` stays [**O**]. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.

- **`v462_seam_spinor_continuum.py` (`SEAM.EQUIV.SPINOR.01`).** The constructive companion of the `v458` (MMST audit) / `v459` (lattice-VOA) residual: the 128-spinor extension $SO(16)_1{\to}(E_8)_1$ exhibited three ways. [**E**] $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/E_8 identity $\theta_2^8{+}\theta_3^8{+}\theta_4^8{=}2E_4$ gives $\chi_{(E_8)_1}{=}E_4/\eta^8{=}\chi_o^{SO16}{+}\chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. [**E**] THE 128 SPINOR WEIGHT-1 STATES FILL 120→248: χ_o gives 120, χ_s gives 128, summing to the exact tower $\{1, 248, 4124, 34752\}$ (248=120+128). [**E**] FINITE 16-MAJORANA FOCK: $C(16,2)$=120 NS bilinear currents and $2^{16/2}$=256 Ramond ground states = 128 spinor + 128 cospinor. [**E**] FINITE-L → CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives $c_-{\to}8$ and the lowest scaled NS gap $x_1{\to}1$ with $|x_1{-}1|{\sim}1/L^2$ (L=64..1024). [**C**][**O**] AGT/AMT (`v459`) supply the rigorous extension; `SEAM.EQUIV.01` existence (`v336`) stays [**O**]. Mixed exact (θ identity + tower + 120/128/256, Wolfram-mirrored) + numerical (finite-L, Python-only).
- **Lean (`TfptCarrier/SeamResidualAxiom.lean`, `FORM.SEAM.RESIDUAL.01`).** The two former cited axioms `mmst_existence` ∘ `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate `ChiralCFT_c8`), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (`v461`: winding = $|C|$=1$\neq$0, $c_-\neq$0) and `fock_sectors` (`v462`: $C(16,2)$=120, 2^8=256, spinor 128, 120+128=248); `residual_arithmetic` (now including both) depends only on `[propext]`. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram.** `tfpt_readouts_extension.wl` adds 3 exact mirrors (total 363/363): the `v461` obstruction integer-logic and the `v462` θ-identity + 120+128=248 sector / Fock counts.

2026-06-30 (`v460` — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the `SEAM.EQUIV.01` sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the μ_4 clock)

- **`v460_seam_s3_lto.py` (`SEAM.S3.LTO.01`).** NPW26 (`arXiv:2605.10693`) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a $\mathbb{Z}_2$ reflection; `v424` reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by `v426` and the continuum lift of BW by the uniform-in-N tower `v455`. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit `v367` $p{+}ip$ collar. [**E**] FLAT-BAND PARENT: the flattened Bloch $Q{=}h/|h|$ has $Q^2{=}I$ (flat ±1 bands) and $P{=}(I{-}Q)/2$ is an exact projector EQUAL to `v367`'s occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a k-grid). [**E**] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $|P(n,0)|$ decays rapidly ($\xi{\approx}1.14$ at M=1, exp R^2=0.995, $|P(20,0)|{<}10^{-6}$) and ξ grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law $|R|^{-2.1}$ (R^2=0.999) — locality needs the gap. [**E**] $\mathbb{Z}_2$ REFLECTION + μ_4 INHERITED:

the flattened modular Hamiltonian $K_{\text{flat}}=\text{sign}(K)$ inherits $\Theta K_{\text{flat}}\Theta=-K_{\text{flat}}$ ($u_\Theta=J$) and $[\rho, K_{\text{flat}}]=0$ EXACTLY (sign is odd; ρ commutes with any $f(K)$), with side-even and off-μ_4 neg controls firing. **[C][O]** the chiral $c_-=8\neq 0$ (FHS Chern $|C|=1$) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`; $C\neq 0$ forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances `v424` sub-step (i); `SEAM.EQUIV.01` stays **[O]** (the continuum theorem MMST `v336` is the backbone). Python-only (numpy); registered in `run_all.py`, the registry, the ledger and cited in `tfpt_research_contracts` (route (xxvii) of the convergent attacks).

2026-06-30 (v454–v459 + Lean + Wolfram — the post-F "G-block": the lattice Sugawara current algebra, a uniform-in-N Tomita–Takesaki tower, chirality forced by P_1, an $(E_8)_1$ character kill test, the exact MMST citation audit and the complementary lattice-VOA route)

- **`v454_seam_edge_virasoro.py` (`SEAM.EQUIV.EDGE.VIRASORO.01`).** Tests the cited MMST "Koo–Saleur lattice generators → continuum Virasoro with the correct c" fact on our own free-fermion edge (G2). **[E]** the Affleck–Cardy Casimir energy fits $c_{\text{Dirac}}=1.000$ ($\frac{1}{2}$ per Majorana, $L=64..1024$); **[E]** the lattice $U(1)$ current closes its affine level exactly ($\langle 0|J_n J_{-n}|0\rangle=n$, $n=1..8$; a gapped control vanishes); **[E]** the level-1 Sugawara $c=\dim G/(1+h^\vee)=8$ for *both* $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31). **[C]** $16\cdot\frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$ is the level-1 current-algebra c_-; `SEAM.EQUIV.01` stays **[O]**. Sugawara Wolfram-mirrored.
- **`v455_seam_bw_uniform.py` (`SEAM.EQUIV.BW.UNIFORM.01`).** A uniform-in-N Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition $u_\Theta=J$ — the inductive-limit companion to `v426`, advancing `v424` sub-step (i) (G1). **[E]** $\|\Theta K\Theta+K\|=0$ exactly at every N ($N=8..128$); **[E]** an N-independent gap $6\log\frac{3}{2}$; **[E]** $[\rho, K]=0$ exactly at every N; **[E]** an N-independent neg-control margin $2\|K\|$. **[C][O]** the inductive limit inherits $u_\Theta=J$; `SEAM.EQUIV.01` stays **[O]**. Python-only.
- **`v456_seam_chirality_from_c3.py` (`SEAM.S3.FROM-P1.01`).** The edge chirality ($c_-\neq 0$, the S3 phase) is *forced* by the one-sidedness that defines $c_3=1/(8\pi)$ (P1) (G6). **[E]** c_3's "8" is the one-sided count $8=|\mathbb{Z}_2|\cdot(\int K/\pi)=2\cdot 4$; **[E]** a reflection sends the Chern integer $C\to -C$ ($+1\to -1$ on the $p+ip$ model); **[E]** a two-sided boundary forces $C=-C=0$. **[C]** the one-sided seam (no reflection) forces $C\neq 0$, magnitude $c_-=8=\text{rank}\,E_8$ — S3 is a consequence of P_1; `SEAM.EQUIV.01` existence stays **[O]**. The arithmetic is Wolfram-mirrored.
- **`v457_seam_character_killtest.py` (`SEAM.EQUIV.KILLTEST.01`).** An explicit character/sector kill test of $(E_8)_1$ against $SO(16)_1$ (G7). **[E]** the $(E_8)_1$ character $\chi=E_4/\eta^8$ with tower degeneracies $\{1, 248, 4124, 34752, \dots\}$; **[E]** the weight-1 count $248=120+128$ (not the free 120); **[E]** the sector count $|\det\text{Cartan}\,E_8|=1$ (not $|\det\text{Cartan}\,D_8|=4$). **[X]** a measured 120, four sectors, or a wrong coefficient would *falsify* $(E_8)_1$; the data give $248/1/\{1, 248, 4124\}$ (passed). `SEAM.EQUIV.01` stays **[O]**. Wolfram-mirrored.
- **`v458_seam_mmst_citation_audit.py` (`SEAM.EQUIV.MMST.AUDIT.01`).** The exact hypothesis-by-hypothesis citation audit of MMST (`arXiv:2107.13834`) against the collar (G3, the "exakter Zitat-Abgleich"). **[C]** H1 CAR class, H2 massless limit, **[E]** H3 per-copy $c=\frac{1}{2}$ (Thm 4.11), **[C]** H4 decoupled additivity to $c=8$, H5 the WZW range $8\leq 8\leq 16$ with the 120 bilinear currents all match; **[O]** the single residual H6 is the 128-spinor extension $SO(16)_1\to(E_8)_1$ ($\det K\;4\to 1$), outside MMST's bilinear method. `SEAM.EQUIV.01` stays **[O]**, its open part narrowed to this one holomorphic extension.
- **`v459_seam_lattice_voa_route.py` (`SEAM.EQUIV.LATTICEVOA.01`).** The second, independent route to $(E_8)_1$ supplying exactly the residual of `v458` (G5). **[E]** E_8 even unimodular ($\det=1$); **[E]** the weight-1 space $248=8+240$; **[E]** the 240 roots split 112 integer +

128 half-integer (the spinor, 248=120+128). [**C**] the lattice net A_{E_8} (Adamo–Giorgetti–Tanimoto, `arXiv:2506.01008`; OS `arXiv:2407.18222`) constructs the holomorphic extension; [**O**] the two routes leave one shared realisation input (collar = A_{E_8}, tied to P_1 by `v456`). `SEAM.EQUIV.01` stays [**O**]. Wolfram-mirrored.

- **Lean (`FORM.SEAM.RESIDUAL.01`, `TfptCarrier/SeamResidualAxiom.lean`).** Kernel-checks the whole G-block arithmetic (the one-sided 8=$|\mathbb{Z}_2|\int K/\pi$, the reflection $C\to -C$, c_-=8, the MMST range 8≤8≤16, decoupled additivity, 248=8+240=120+128, 240=112+128, $\det K$ 1 vs 4) by `decide`, reducing the `SEAM.EQUIV.01` residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus two cited external theorems (MMST `v458`, AGT `v459`); `lake build` OK.
- **Wolfram + integration.** Six exact mirrors added to `tfpt_readouts_extension.wl` (Sugawara c=8 for 120/15 and 248/31; the one-sided $8=2\cdot 4$ and reflection $C{=}-C \Rightarrow 0$; the $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ with 248=120+128 and $|\det \mathrm{Cartan}|$ 1 vs 4; the root split 112+128 and 248=8+240) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

2026-06-30 (v449–v453 + Lean + Wolfram — the edge CFT named end-to-end: uniform-in-N convergence, three independent reads of c_-=8, the Ising operator tower, the $(E_8)_1$ modular data, and μ_4-symmetry derived from the four marks)

- **`v449_seam_mmst_uniform.py` (`SEAM.EQUIV.MMST.UNIFORM.01`).** Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (`v444`/`v448`) to the *uniformity* the theorem actually asserts, on the same `v367`/`v368` collar. [**E**] cross-ratio A $Q(1,2,4,7)/12 \to 2+\sqrt{3}$, the deviation shrinking monotonically; [**E**] an *independent* cross-ratio B $Q(1,3,5,9)/12 \to$ its own value 2, so convergence is not a tuned point; [**E**] a uniform $1/N$ rate $|Q(N)-Q_\infty|\cdot N \le 4$ for *both* observables at every $N_x \in \{36, 48, 60\}$ (one N-independent constant); [**E**] the bulk control is off-scale. [**C**] the MMST uniform-convergence hypothesis is empirically satisfied; `SEAM.EQUIV.01` stays [**O**].
- **`v450_seam_edge_entanglement.py` (`SEAM.EQUIV.EDGE.ENTANGLEMENT.01`).** A *third* independent reading of the edge central charge (after the correlator `v444` and the topology `v447`). [**E**] the Calabrese–Cardy entanglement law gives c_{Dirac}=1.000 for the critical free-fermion chain; [**E**] so c per Majorana = 1/2; [**E**] a gapped control saturates (c=0). [**C**] bulk–edge: $c_-{=}N_{\mathrm{Maj}}\cdot\frac{1}{2}{=}16\cdot\frac{1}{2}{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$. `SEAM.EQUIV.01` stays [**O**].
- **`v451_seam_edge_cardy_tower.py` (`SEAM.EQUIV.EDGE.CARDY.01`).** Names the edge CFT by its *operator content.* [**E**] the Cardy finite-size spectrum gives the spin dimension h_σ=1/16 (the periodic–antiperiodic ground-energy split) and [**E**] the energy dimension h_ϵ=1/2 (the lowest NS pair excitation), both [**E**] L-independent (conformal $1/L$ scaling). [**C**] $\{c, h_\sigma, h_\epsilon\}{=}\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$ is the chiral Ising (free-Majorana) minimal model $M(3,4)$; 16 copies give c_-=8. `SEAM.EQUIV.01` stays [**O**].
- **`v452_seam_e8_modular.py` (`SEAM.EQUIV.E8.MODULAR.01`).** The $(E_8)_1$ torus modular signature, extending `v156`'s q-expansion to the transformation data. [**E**] the character $\chi{=}E_4/\eta^8$ is S-invariant $\chi(-1/\tau){=}\chi(\tau)$ (one primary, S=[1]); [**E**] T-phase $\chi(\tau{+}1)/\chi(\tau){=}e^{-2\pi i/3}{=}e^{-2\pi i c/24}$ (c=8); [**E**] leading power $\chi\, q^{1/3} \to 1$ so $q^{-c/24}{=}q^{-1/3} \Rightarrow c{=}8$; [**E**] holomorphic $c{=}8 \equiv 0 \bmod 8$ with T-phase order $24/\gcd(8,24){=}3$. [**C**] the $(E_8)_1$ identity is now pinned on the torus *and* the edge. `SEAM.EQUIV.01` stays [**O**].
- **`v453_seam_mu4_from_marks.py` (`SEAM.RIGIDITY.MU4FROMMARKS.01`).** Folds the last structural premise of the rigidity chain (`v446`'s residual `QGEO.SYM.01`) into the existing realisation premise. [**E**] the four Gauss–Bonnet marks *are* μ_4= roots of $z^4{-}1$; [**E**] the order-4 clock $z \to iz$ is a 4-cycle permuting them (the `v445` clock); [**E**] it preserves the cross-ratio λ=2 (in the D_4 stabiliser, `v168`); [**E**] the form basis $\omega_k{=}z^{k-1}dz/(z^4{-}1) \to i^k\omega_k$ is μ_4-graded

(weights $(1, 2, 3) = A_3$ exponents). **[C]** so `QGEO.SYM.01` follows from marks= μ_4 + the existing `QGEO.REALIZE.01` — no new premise; the chain (`v453`→`v446`→`v445`) closes modulo realisation. `SEAM.EQUIV.01` stays **[O]**.

- **Lean: `TfptCarrier/SeamEdgeChern.lean` (`FORM.SEAM.EDGE.CHERN.01`).** Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers $(+1/0/-1)$, the bulk–edge channel count $|C|=1$, $N_{\rm Maj}=2^{g_{\rm car}-1}=16$, the assembly $2c_-=N_{\rm Maj}\cdot|C|$ (so $c_-=8$), $c_-=8=g_{\rm car}+N_{\rm fam}=\text{rank}\, E_8$, holomorphic $c\equiv 0 \bmod 8$ and the order-3 modular T-phase — all by kernel `decide` (`edge_kernel_facts`, axioms `propext` only). `lake build` OK, `#print axioms` clean.
- **Wolfram mirror.** `tfpt_readouts_extension.wl` gains nine exact checks (`v452` modular data ×4, `v450` entanglement assembly, `v451` Ising Kac weights, `v453` marks= μ_4 ×3): 345 → 354 checks, all passing (`GATE.WOLFRAM.02`). E_4 is evaluated via its q-series since `EisensteinE` stays symbolic for complex τ.
- **Papers.** The adversarial audit (`tfpt_5_redteam`, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge $c_-=8$ from three independent observables (correlator `v444`, topology `v447`, entanglement `v450`), the Ising operator tower (`v451`), the $(E_8)_1$ torus modular data (`v452`), and the μ_4 premise derived from the marks (`v453`) — while the only residual that stays **[O]** remains the cited continuum-existence theorem (`v336`).

2026-06-30 (v446/v447/v448 + Lean + Wolfram — deriving clock-invariance from μ_4-symmetry, an independent topological reading of the edge $c_-=8$, and pinning the one external MMST fact at four-point order)

- **`v446_seam_clock_invariance.py` (`SEAM.RIGIDITY.CLOCKINV.01`).** Closes `v445`'s single residual one notch further: the operator condition $[\rho,\Lambda_\Sigma]=0$ (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. **[E]** the OS canonical transfer $\Lambda=C(1)C(0)^{-1}=e^{-h}$ (fixed by RP+reflection, `v54`) of a μ_4-invariant RP covariance satisfies $[\rho,\Lambda]=0$ (swept N=4..32); **[E]** a μ_4-breaking covariance gives $[\rho,\Lambda]\neq 0$ (iff-sharp neg control); **[E]** Λ is a positive contraction (spectrum in $(0,1]$); **[E]** the free modular Hamiltonian $K=\log((1-C)C^{-1})$ (`v258`) of the invariant state has $[\rho,K]=0$ (Tomita–Takesaki), broken control $\neq 0$. **[C]** so with `v445` (commute ⇔ block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal Λ_Σ; the residual reduces from an operator commutation to "the seam RP data is μ_4-symmetric" (`QGEO.SYM.01`). `SEAM.EQUIV.01` stays **[O]**.
- **`v447_seam_edge_chern.py` (`SEAM.EQUIV.EDGE.CHERN.01`).** An *independent* (topological) reading of the edge central charge, corroborating `v444`'s correlator route from a different observable. **[E]** the M=1 collar has integer Chern number C=+1 (Fukui–Hatsugai–Suzuki link-variable method, no fit); **[E]** C=0 in the trivial phase and $C=-1$ at $M=-1$ (opposite chirality); **[E]** the open strip carries exactly $|C|=1$ chiral edge branch per boundary; **[E]** the Li–Haldane entanglement spectrum has in-gap modes at 1/2 only in the topological phase. **[C]** bulk–edge: $c_-=N_{\rm Maj}\cdot\frac{1}{2}\cdot|C|=16\cdot\frac{1}{2}=8=g_{\rm car}+N_{\rm fam}$ — the same $c_-=8$ as `v444`. `SEAM.EQUIV.01` stays **[O]**.
- **`v448_seam_edge_fourpoint.py` (`SEAM.EQUIV.EDGE.FOURPOINT.01`).** Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. **[E]** the ground-state correlation matrix is an exact projector $G^2=G$ ($\sim 10^{-13}$), so the state is quasi-free (the MMST CAR hypothesis holds); **[E]** the edge four-point cross-ratio $Q=[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement; **[E]** Q converges to the free-fermion conformal cross-ratio $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3}\approx 3.732$; **[E]** the bulk control is off-scale (exponential). **[C]** so the one external fact is supported at *both* two-point (`v444`) and four-point order. `SEAM.EQUIV.01` stays **[O]**.

- **Lean: `TfptCarrier/SeamRigidityForcing.lean` (`FORM.SEAM.RIGIDITY.FORCING.01`).** Kernel-hardens `v445` (and `v442`'s uniformity): the entrywise forcing iff clock4 a=clock4 $b \Leftrightarrow a=b$ (i.e. $i^a=i^b \Leftrightarrow a \equiv b \bmod 4$, residue-only hence uniform in N), the four distinct eigenphases, the exact commutant dimension $\sum n_s^2=64<256$ and the order discriminator (128>64) are all proved by kernel `decide` (`forcing_kernel_facts`, no domain axioms); only the operator-level lift to the full L^2 is the named premise. `lake build` OK, `#print axioms` clean.
- **Wolfram mirror.** `tfpt_readouts_extension.wl` gains three exact `v445` checks (entrywise forcing for all $a, b \in 0..31$; commutant dimension 64<256; order-2 commutant 128>64 swept N=4..64): 342 → 345 checks, all passing (`GATE.WOLFRAM.02`).
- **Papers.** The adversarial audit (`tfpt_5_redteam`, Target A) records the five-direction narrowing of the `SEAM.EQUIV.01` keystone (edge scaling `v444`, forcing converse `v445` + Lean, derived clock-invariance `v446`, independent topological $c_-=8$ `v447`, four-point pinning `v448`) explicitly as "not promoted": the only residual that stays **[O]** is still the cited continuum-existence theorem (`v336`). The research-contracts note lists the same as "eleven convergent attacks".

2026-06-30 (`v445` — the rigidity FORCING converse: commuting with the order-4 clock is *exactly* μ_4-block-diagonality, upgrading the rigidity step from "permit" to "force")

- **`v445_seam_rigidity_forcing.py` (`SEAM.RIGIDITY.FORCING.01`).** The rigidity route (`v398`/`v442`) showed a μ_4-block-diagonal K *commutes* with the order-4 clock $\rho=\mathrm{diag}(i^n)$. This module proves the *converse* and the exact structure, isolating the single remaining physical premise. **[E]** FORCING (entrywise iff): $[\rho, K]=0$ iff $K_{ij}=0$ whenever $i \not\equiv j \bmod 4$, verified entrywise and swept N=4..128 — commuting *forces* μ_4-block-diagonality, it does not merely permit it. **[E]** EXACT COMMUTANT DIMENSION: $\dim\{K : [\rho, K]=0\}=\sum_s n_s^2$ (the ad_ρ nullspace), a proper subspace of the full N^2 algebra (N=16: 64<256) — a genuine quantitative restriction. **[E]** OFF-BLOCK ⇔ NON-COMMUTING: every off-block matrix unit has $\|[\rho, E_{ij}]\| \geq \sqrt{2}>0$ (zero slack). **[E]** THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant (N=16: 64 vs 128) — the four Gauss–Bonnet marks (order $4=|\mu_4|=h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ ($\det=1$; `v281`/`v154`/`v351`). **[E]** NONDEGENERACY: exactly 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all $N \geq 4$. **[C]** SYNTHESIS: commuting with the order-4 μ_4 clock is *equivalent* to μ_4-block-diagonality, so RP-definability (`v54`/`v398`) + clock-invariance *force* the block-diagonal Λ_Σ — `v442`'s "permit" upgraded to "force". **[O]** the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = `QGEO.SYM.01`, `v323`/`v335`); `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays **[O]**. Python-only (numpy; det facts Wolfram-mirrored via `v89`/`v281`/`v422`). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

2026-06-30 (Lean kernel-hardening of two of the seven `SEAM.EQUIV.01` routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake` build OK; #print axioms clean. Tracked as two new FORM.* ledger rows; \veri-context in tfpt_research_contracts; Lean manifest refreshed.

- **FORM.SEAM.BW.HSMI.01 (TfptCarrier/SeamStandardPair.lean) — standard-pair relations, kernel-checked (hardens v438).** The four Borchers/Wiesbrock relations on the centred 5-mode ladder over $\mathbb{Z}$ are proved by kernel decide (axioms propext / Classical.choice / Quot.sound only, no domain axiom): $[K, P]{=}P$, $J^2{=}1$, $JKJ{=}{-}K$, $JPJ{=}P_+$ (and $\mathrm{tr}\,K{=}0$). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of SEAM.EQUIV.01 to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **FORM.SEAM.APPLICABILITY.01 (TfptCarrier/SeamApplicabilityLedger.lean) — the audit count, kernel-checked (hardens v441).** The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (total_count=12, internal_count=11, external_count=1), plus mmst_range ($8 \leq 8 \leq 16$) and npw_outside_bucket ($\det K{=}1 \neq 4$) — ALL with NO axioms (pure kernel decide). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root TfptCarrier.lean, axiom-audited in AxiomCheck.lean, and signature-locked in AuditContract.lean; full lake build succeeds. No new mathematics — the same facts v438/v441 verify numerically are now kernel-proved; SEAM.EQUIV.01 stays [**O**].

2026-06-30 (the chiral-edge scaling motor v444 — the edge half of the one SEAM.EQUIV.01 continuum residual: the lattice edge already carries the conformal data of a chiral $c{=}1/2$ Majorana CFT and converges under refinement, grounding the EDGE the way v439 grounded the BULK; the keystone stays [O])

- **Context.** v439 grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. v444 attacks exactly that residual on the SAME explicit v367/v368 $p{+}ip$ collar. vN-registered, runs in run_all.py (suite ends ALL CHECKS PASSED), \veri-cited in the body of tfpt_research_contracts (the G_{net} contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **v444_seam_edge_scaling.py (SEAM.EQUIV.CONTINUUM.EDGE.01) — chiral-edge motor.** The lattice edge carries the kinematic and conformal data of a chiral $c{=}1/2$ Majorana CFT: [**E**] a linear, chiral in-gap dispersion (velocity $v{\approx}0.98$, $R^2{=}0.9999$, with *opposite* per-edge velocities — one chiral Majorana per edge); [**E**] an *algebraic* equal-time edge two-point $\sim r^{-\eta}$ with $\eta{\approx}1$, versus the *exponential* bulk row and the exponential trivial-phase ($M{=}3$, no edge) control; [**E**] a Cauchy-convergent scaling-collapse under refinement ($\eta{=}1.001{\pm}0.007$, successive profile spread $0.028{\to}0.013$ across $N_x{=}36..60$) — the numerical hallmark of an existing scaling limit. [**C**] $\eta{=}1 \Leftrightarrow$ one chiral Majorana ($c{=}1/2$) per edge, so the 16-copy carrier has a chiral $c_-{=}8$ $(E_8)_1$ edge CFT ($c_-{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$). [**O**] the rigorous uniform-convergence theorem (cited MMST, v336) is not supplied.
- **Net.** With v439 (bulk) and v444 (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; SEAM.EQUIV.01 stays [**O**]. Ledger row, script_registry.csv index and the generated surfaces (verification.tex, ScriptIndex.tsx) updated in the same change; manifest refreshed last.

2026-06-30 (six convergent attacks on the one `SEAM.EQUIV.01` continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the β-homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)

- **Context.** The keystone `SEAM.EQUIV.01` is "closed modulo a cited theorem": its one open input is the continuum existence of the seam's chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are `vN`-registered, run in `run_all.py` (suite ends `ALL CHECKS PASSED`), and are `\veri`-cited in the body of `tfpt_research_contracts` (the $G_{\rm net}$ contract). Python-only (no new exact identity to Wolfram-mirror).
- **`v438_seam_hsmi_borchers.py` (`SEAM.EQUIV.BW.HSMI.01`) — intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation $P \geq 0$, boost K with $[K, P]{=}P$, reflection J with $JKJ{=}{-}K$, $JPJ{=}P_{-}$), so by Borchers' theorem the modular flow is geometric *intrinsically* — *without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). **[E]** ladder algebra exact, $sl(2, \mathbb{R})$ positivity (min eig > 0); **[C]** synthesis; **[O]** the continuum standard pair.
- **`v439_seam_lieb_robinson.py` (`SEAM.EQUIV.CONTINUUM.LR.01`) — continuum motor.** The explicit gapped $p{+}ip$ collar (`v367`) has a uniform-in-N gap, exponential clustering with N-independent ξ, and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. **[E]**/**[C]**, **[O]** the edge CFT.
- **`v440_seam_lto_rp_beta.py` (`SEAM.EQUIV.LTORP.BETA.01`) — β-homotopy.** The (LTO-RP)/BW data ($\Theta K \Theta{=}{-}K$, $[\rho, K]{=}0$) are β-independent, and along the whole path $C_\beta{=}1/(1{+}e^{\beta K})$ from the NPW26-proven *trace* state ($\beta{=}0$) to the seam KMS state ($\beta{=}1$) every covariance is reflection-related, μ_4-invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of `v426` to a path argument for sub-step (ii) of `v424`.
- **`v441_seam_applicability_ledger.py` (`SEAM.EQUIV.APPLICABILITY.01`) — audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the `v215` circularity), not an input. The keystone's external dependence is pinned to that one analytic fact.
- **`v442_seam_rigidity_uniform.py` (`SEAM.RIGIDITY.UNIFORM.01`) — uniform rigidity.** The band-limited μ_4-character rigidity of `v398` is uniform in N: exact 4-sector commutation to $N{=}256$ (not merely the $N \leq 64$ sweep), a uniform gap, and an N-independent clock-breaking lower bound $\|[\rho, V]\|{=}2$ — so it is not a small-N artifact (the full-L^2 forcing stays **[O]**).
- **`v443_seam_e8_discriminator.py` (`SEAM.E8.DISCRIMINATOR.01`) — kill tests.** Mode D of the closure-mode classification (`v347`) as concrete *falsifiable* discriminators against the nearest $c{=}8$ rival $SO(16)_1$: torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots; $c{=}8$ alone cannot decide (`v277`), only $\det K{=}1$ (the genus-1 statement, `v90`) selects E_8. A seam with GSD>1 would falsify $(E_8)_1$. **[X]**/**[C]**.
- **Status discipline.** `SEAM.EQUIV.01` stays **[O]** throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

2026-06-30 (shadow-export coherence — the `tfpt5-shadow` mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5_shadow.zip` the independent surfaces reproduced (`build.sh notes` → `11 ok`; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh audit` crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **`make_docs_map.py` is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh notes/gen` leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **`audit_sync.py` is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh audit` therefore runs to a clean `AUDIT OK` on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` *on the filtered tree* before the mirror is published. The manifest check catches the earlier `v235-v298` symptom at the source: a module referenced by `run_all.py/script_registry.csv/manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says `ALL CHECKS PASSED` while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or `vN` module changed — this is an export/infrastructure fix only.

2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated `/llms.txt`, explicit AI-crawler `robots.txt`, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)

- **New `/llms.txt` context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`, `version.ts`) so it never drifts. It carries the one-line summary, the "claims / does not claim" split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz "topological fixed point theory" of mathematics.
- **`robots.txt`** now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, ...) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries ("derivation of the fine-structure constant", "Standard Model from first principles", "where does 137 come from") and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root `README.md` and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the README gains a citation block with the archived deposit DOI `10.5281/zenodo.20846087`.

- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module `v437` (`E8.DEGREE.JOINT.01`) — a consolidation of `v6/v66/v355/v431` that derives more from the E_8 Casimir degrees *without* crossing the `v354/v355` anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
  - **[E] all of E_8 is fixed by its degree multiset.** From $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$ alone: rank $= \#\deg = 8$, $h = \max\deg = 30$, #positive roots $= \sum \exp = 120$, #roots $= 240$, $\dim E_8 = 248$, $|W(E_8)| = \prod \deg = 696{,}729{,}600$, $\sum \deg = 128 = \dim S^+$.
  - **[E] the joint quadratic (the new lock).** The two TFPT structural integers $(g_{\rm car}, N_{\rm fam})$ are the *unique* root pair of $x^2 - (\mathrm{rank}\,E_8)\,x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$: sum of roots $= \mathrm{rank}\,E_8 = 8$ (the rank-fill, `v6`), product $= h/2 = 15 = g_{\rm car}N_{\rm fam}$ ($2g_{\rm car}N_{\rm fam} = h$, `v431`); $\{3, 5\}$ is the unique positive-integer factor pair of 15 summing to 8. So $g_{\rm car} = 5$ and $N_{\rm fam} = 3$ are forced *together* by two E_8 degree-invariants, not chosen one at a time. The three primary-readout degrees $\{2, 8, 30\}$ are exactly $\{\min, \mathrm{rank}, \max\}$.
  - **[E] discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their `v431` family homes; the `v354/v355` line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`E8.DEGREE.JOINT.01`); cited in the `tfpt_5` degree-audit winbox beside `v431`; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339 → 342). Exact integer/lattice; no new physical number, no status change.

2026-06-29 (`v436` — hardening the unconditional "is it numerology?" floor to an assumption-minimal counting floor; plus a literature note grounding the α^3 Chern level)

- **New module `v436` (`OVERDET.FLOOR.02`) — it *hardens* `OVERDET.FLOOR.01` (`v432`) so the numerology verdict no longer rests on any subjective chance assignment.** `v432`'s nominal $\sim 10^{-10}$ multiplies the counting census by hand-chosen plausibilities (input-"8" independence 0.1^3, seed 0.012, kaon 0.15) a skeptic can contest; `v436` shows the conclusion survives without them.
  - **[E] assumption-minimal floor.** The α cubic-class census (`v100` layer D) is *pure counting*: of $N = 94500$ complexity-matched $F_{U(1)}$ variants only *one* lands in the 4×10^{-8} CODATA window — the TFPT instance, reproducing $\alpha^{-1} = 137.0359992$ — so $p_\alpha \le 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$ with *no* subjective probability, prior or multi-observable grammar.
  - **[E] monotone concession ladder.** $-\log_{10} p$ from generous to adversarial: full scorecard+α ~ 30.7 → scorecard-only ~ 25.8 → α-census-only ~ 4.98; each rung uses strictly less evidence and the "not chance ($> 4\sigma$)" verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of `v432`'s subjective factors ($\sim 1.8 \times 10^{-6}$) leaves the floor at 4.40σ — they only sharpen it.
  - **[C] honest 5σ gap.** The conventional 5σ line ($p = 5.7 \times 10^{-7}$) is crossed unconditionally by the census times exactly *one* further independent factor ≤ 0.054 (the input-"8" fourfold over-determination, or one foreign witness) — stated, not hidden. **[C]** Firewall unchanged: bounds only "is it numerology?", not the physics bridge (`v187`).
- **Literature note (research log, not a status change).** A `next.txt` entry grounds the α^3 Chern-level reading of `v435` in named theorems — Dai–Freed (*η-invariants and determinant lines*, the exponentiated η-invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the α^2 Calderón side) and Redlich/Stora–Zumino (the cubic $\mathrm{Tr}\,F^3$ anomaly descends to a Chern–Simons level, quantised to $\mathbb{Z}$). It records that

the `EM1` proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of `SEAM.EQUIV.01` — and remains **[O]** external math (no fabricated closure).

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`OVERDET.FLOOR.02`); cited in the `tfpt_safeguards` "unconditional floor" keybox beside `v432`; mirrored on the website. Python-only; no new physical number, no status change.

2026-06-29 (v435 — a fourth honest step on the α^{-1} Quillen target: a π-power test isolates the cubic α^3 as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)

- **New module v435 (`ALPHA.QUILLEN.PROGRESS.04`) — it attacks the *single* remaining [O] left after v434 (residual (2), the cubic α^3 Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target `ALPHA.QUILLEN.EXACT.01` (`v382`) stays **[O]**; $\alpha^{-1} = 137.0359992$ stays **[E]**.
  - **[E]** **π-power / metric-independence test.** The three EM-closure coefficients carry π-powers exactly $\{0, 3, 6\}$, equal to their c_3-orders: Maxwell $k_0 = 1 \to \pi^0$, Calderón $-2c_3^3 = -1/(256\pi^3) \to \pi^3$, transport $-8b_1c_3^6 = -41/(327680\pi^6) \to \pi^6$. So the Maxwell a^3 is the *unique* π^0/c_3^0 (metric-*independent*) term. A heat-kernel boundary coefficient always carries the $(4\pi)^{-d/2}$ measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level (a^3) + *two* heat-kernel boundary terms, confirming `v342`/`v434`'s "a^3 is a level, not a Seeley–DeWitt curvature integral" from a *test* rather than a claim.
  - **[C]** **conditional level quantisation.** *If* the a^3 term is the seam $U(1)$ Chern–Simons/η boundary level (the `EM1` reading, `v48`), large-gauge invariance quantises the level to $\mathbb{Z}$, so $k_0 = 1$ is the *unit* (minimal) level — an integer, not a tunable real. This converts "the coefficient happens to be 1" into "the coefficient is an integer level and 1 is the unit".
  - **[O]** **not closed:** the `EM1` lemma itself — the from-first-principles Chern–Simons boundary proof that $\partial_\tau \log Z_{\mathrm{Maxwell}} = a^3$ — stays external math (type A, `v384`); the seam F-normalisation stays the one **[C]** (residuals 1 & 3, `v434`); α^{-1} stays **[E]**.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.04`); cited in the `tfpt_1` EM-closure subsection beside `v433`/`v434`; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-28 (v434 — a third honest step on the α^{-1} Quillen target: b_1 is the a_4 heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])

- **New module v434 (`ALPHA.QUILLEN.PROGRESS.03`) — it settles the status of the three residuals named after v433 and shows they are *not* three independent open problems.** The target `ALPHA.QUILLEN.EXACT.01` (`v382`) stays **[O]**; $\alpha^{-1} = 137.0359992$ stays **[E]**.
  - **[E]** **b_1 *is* the $U(1)$ a_4 heat-kernel coefficient (the $\beta = a_4$ theorem).** The one-loop β function of a gauge coupling *is* the a_4 (Seeley–DeWitt a_2 in d=4) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ($\frac{2}{3}$ per Weyl fermion, $\frac{1}{3}$ per complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM) and $(3/5) \cdot$ that $= 41/10 = b_1$ (GUT). So the b_1 in the counterterm $8b_1c_3^6$ is the a_4 coefficient of the carrier $U(1)_Y$ content (the same number as the β, `v159`/`v246`), the *same kind* of heat-kernel object as the α^2 Calderón a_4 (`v342`) — not a fitted multiplicity.
  - **[E]** **the geometry factors off:** $8b_1c_3^6 = b_1 \cdot (\mathrm{rank}\, E_8) \cdot c_3^6$, with $\mathrm{rank}\, E_8 = 8$ the χ=2 seam boundary count (`v216`) and c_3^6 the six boundary insertions (the π-power test, `v342`).

- [**C**] **residual (1) $\equiv$ residual (3).** With b_1 and the c_3-powers heat-kernel-forced, the only freedom left in the exact seam a_4 value is how the gauge curvature couples to the boundary measure — the seam F-normalisation. So "the actual seam $a_4 = 8b_1c_3^6$" and "the seam F-normalisation" are *one* [**C**] obligation, not two.
- [**E**] **residual (2) located (not closed):** the determinant-line variation factors $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$; a^2 is the Gilkey gauge-curvature (Quillen/Chern) density (`v342`), so the leading $a^3 = a \cdot a^2$ is the *first moment* of the Chern density — a topological level at boundary order c_3^0, not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays [**O**].
- [**E**] **net reduction:** the EM-Ward residual is *one* [**C**] (residuals 1 & 3) + *one* [**O**] (residual 2), not three independent problems — a genuine narrowing of the open surface. [**O**] `ALPHA.QUILLEN.EXACT.01` is *not* closed (the α^3 Chern level and the from-first-principles proof stay external, type A, `v384`).

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.03`); cited in the `tfpt_1` EM-closure subsection beside `v433`; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-27 (`v433` — a second honest step on the α^{-1} Quillen target: a solvable 4D model reaches the a_4 order, connecting `v391` and `v342`)

- **New module `v433` (`ALPHA.QUILLEN.PROGRESS.02`) — a second honest step on the external target `ALPHA.QUILLEN.EXACT.01` (`v382`); it *grounds and connects*, it does *not* close it.** The target stays [**O**] (external math, type A, `v384`); the value $\alpha^{-1} = 137.0359992$ stays [**E**]. The earlier honest attempt `v391` gave a solvable model on a 1D circle whose ζ-determinant variation reached only the low Seeley–DeWitt orders a_0, a_1; separately, `v342` grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient a_4 ($30/360 = \frac{1}{12}$, $\Omega^2/12$). This module supplies the missing link between the two.
  - [**E**] **a solvable 4D model reaches the a_4 order:** the flat operator $\Delta = -\partial^2 + m^2$ on a 4-torus of volume V has heat trace $\operatorname{Tr} e^{-t\Delta} = V/(4\pi)^2\, t^{-2} e^{-tm^2}$, with Gilkey coefficients $a_0 = V/(4\pi)^2$, $a_2 = -Vm^2/(4\pi)^2$ and $a_4 = Vm^4/(2(4\pi)^2)$; the t^0 (conformal-anomaly) order a_4 is *non-zero* — exactly the order `v391`'s 1D circle toy could not reach.
  - [**E**] **it is the order `v342` uses:** so `v391` ("a ζ-determinant variation *is* closed heat-kernel data") and `v342` ("the seam term *is* the Gilkey a_4") are unified — the variation is heat-kernel data *at the a_4 order*, not only the low orders v391 reached.
  - [**E**] **the measure is c_3-built:** the universal 4D heat-kernel measure $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$ in the seam's one-sided 2D-boundary normalisation $c_3 = \frac{1}{2}(4\pi)^{-1}$; and the `v382` split is re-verified at $\alpha^{-1} = 137.0359992$.
  - [**O**] **not closed:** computing the *actual* seam $U(1)$ ζ-determinant a_4 coefficient ($= 8b_1c_3^6$) on the μ_4 seam moduli, and the origin of the cubic α^3 Maxwell/Chern moment, stay open (external math, type A, `v384`); the exact seam F-normalisation fixing the coefficient *value* stays [**C**] (`v342`). A second honest step that grounds and connects, it does not close.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.02`); cited in the `tfpt_1` EM-closure subsection beside `v382`/`v391`; mirrored on the website (the residual-gap lab). Python-only (like `v391`); no new physical number.

2026-06-27 (`v431`, `v432` — the "unmapped" E_8 degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module `v431` (`E8.DEGREE.LADDER.01`) — the reverse-audit "unmapped" E_8 Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (`v354`/`v355`) and its sheet/deck complement (`v430`): $\deg(E_8) = 6 \cdot \mathrm{spine}\{2,3,4,5\} \sqcup (\{2\} \cup \text{det-ladder}\{8,14,20\})$. **[E]** the two-family split is *forced* by $h(E_8) = 30 = 2 \cdot 3 \cdot 5$ — the exponents are the $\varphi(30) = 8$ totatives of 30, hence coprime to 6, hence $\equiv \pm 1 \bmod 6$, so the degrees occupy only the residue classes $\{0,2\} \bmod 6$. **[E]** the $6k$ family $\{12,18,24,30\}/6 = \{2,3,4,5\}$ is the `v91` spine; the $6k{+}2$ family $\{8,14,20\} = (\det R, \det C, \det L)$ is the winding line $\det M(s,0) = 6s+8$ of the flavor diamond (`v135`). **[E]** "$18 = 6N_{\mathrm{fam}}$" is not a holdout (spine family), and the clean split is special to E_8 among the simply-laced exceptionals ($E_6/E_7/D_5$ fail). **[C]** **honest `v355` line:** the arithmetic decomposition is exact, but the *functorial* claim "$6k{+}2$ Casimir stratum = flavor sheet operators" needs a representation-theoretic map and stays **[C]** ($12, 24$ admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the `v354` overhead to zero orphan degrees and closes no gate.
- **New module `v432` (`OVERDET.FLOOR.01`) — the unconditional improbability floor and the L1↔L2 bridge.** The complement of the conditional null model (`v100`) and the multiply-vs-compress framework (`v427`/`v428`): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. **[E]** compression removed (`v428`), the genuinely multiplicative pieces are the input "8" forced four independent ways plus the foreign witness $\alpha^{-1} \sim 137$. **[C]** multiplying only across disjoint pieces (input over-determination × the α census 1/94500 × the φ_0^{ret}-seed cluster counted once × one downstream witness) gives a floor $\leq 10^{-6}$ across a grid of generous assignments, nominal $\sim 10^{-10}$. **[E]** that floor sits ~ 20 orders *above* the `v100` conditional $10^{-30.7}$; the gap is exactly the evidential value of proving the forms forced, so the α form-derivation (`ALPHA.QUILLEN.EXACT.01`, `v382`) and this floor are the *same* lever. **[C]** **honest scope:** bounds only the "is it numerology?" question, not the physics bridge (firewall `v187`); a conservative floor, not a posterior.
- *Surfaces synced in the same change:* `v431`/`v432` added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); `v431` mirrored in `wolfram/tfpt_readouts_extension.wl` (`v432` is a conservative bound, Python-only).

2026-06-27 (`v430` — the seam's "other side" is forced-disjoint from E_8's unmapped Casimir degrees)

- **New module `v430` (`E8.OTHERSIDE.AUDIT.01`) — the sheet/deck complement of the reverse audit (`v354`/`v355`).** The reverse audit found five E_8 Casimir degrees $\{12,14,18,20,24\}$ with no primary readout ("hull overhead"); the double cover has an *other side* (the one-sided $\mathbb{Z}_2$ collar / conjugate half-spinor S^-), so the sharp adversarial follow-up is *"is the leftover where the second sheet lives?"* The answer is a disciplined **negative**. **[E]** the deck is the degree-2 invariant: the one-sided cover contributes $|\mathbb{Z}_2| = 2$ (the $\frac{1}{2}$ in $c_3 = 1/(|\mathbb{Z}_2|\,4\pi) = 1/(8\pi)$, `v58`/`v216`), and $2 = \min\deg(E_8)$ is the quadratic/metric — one of the *matched* primary readouts $\{2,8,30\}$, never one of the unmapped degrees. **[E]** the two sheets are the 128-spinor: $\dim S^+ = \dim S^- = 16$, the spinor block $128 = \mathrm{rank}\,E_8 \cdot \dim S^+ = 2^{\mathrm{rank}-1} = \sum \deg$, the spinor half of $248 = 120 + 128$; S^- enters 248 as the conjugate $(\overline{16}, \overline{4})$ ($128 = 2 \cdot 64$), not a spare singlet. **[E]** forced-disjoint: the sheet/deck set $\{2,16,32,128\}$ has empty intersection with $\{12,14,18,20,24\}$, its only degree contact the matched 2. **[O]** each unmapped degree, "explained" from sheet atoms, needs an unforced coefficient and admits ≥ 2 readings (the `v355` discriminator), so there is no forced sheet→unmapped-degree identity.

- **A structural negative, not a status change.** It consolidates the S^--dark-matter downgrade (`v227`) and the no-spare-singlet WIMP no-go of the frontier E_8 scan (`tfpt_4_frontier`): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live "other side" question — whether the 128 phase/glue channel hosts a dark sector (`v227`) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of `tfpt_5_redteam`, the magnitude/phase keybox of `tfpt_3_e8_audit_bootstrap`, the WIMP-no-go passage of `tfpt_4_frontier`; the `E8` node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

2026-06-26 (`v429` — the 'unmapped' golden E_8 structure is the geometry of the one external input θ_i)

- **New module `v429` (`DM.AXION.PENTAGON.01`) — a geometric re-reading answering "why does so much idle E_8 structure exist?" for the one case where the idle structure has a job.** The reverse audit (`v354`) had argued the golden ratio φ is *unmapped* (it appears only internally — the affine-E_8 attractor angle `v312`, the icosian lattice `v348` — and in no physical readout), and used exactly that to conclude φ cannot be numerology. **[E]** this sharpens that statement: because $N_{\text{fam}}=g_{\text{car}}-2$, the axion misalignment *spine* angle (`v211`, `DM.AXION.SPINE.01`) is $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=(g_{\text{car}}-2)\pi/g_{\text{car}}$, which is *exactly* the interior angle of the regular g_{car}-gon — for $g_{\text{car}}=5$ the pentagon, $\theta_i=3\pi/5=108°$. **[E]** its cosine is golden, $\cos(3\pi/5)=(1-\sqrt{5})/4=-1/(2\varphi)$ (partner $2\cos(2\pi/5)=1/\varphi=\varphi-1$, $\varphi=2\cos(\pi/5)$). **[E]** the golden character is *unique* to $g_{\text{car}}=5$: among the small regular n-gons only $n=5$ has an irrational interior-angle cosine ($n=3,4,6$ give $1/2, 0, -1/2$), so θ_i is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts — φ is the unmapped $g_{\text{car}}=5$ signature (`v354`/`v313`) and $\theta_i=3\pi/5$ is the one external cosmological input (`v211`) — and *refines* `v354` from "φ in no physical readout" (it is the unmapped $g_{\text{car}}=5$ signature, `v313`) to "φ touches *only* the **[C]** spine misalignment angle" (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop $\theta_i\approx170°$ (whose cosine is not a clean pentagon value), consistent with — not replacing — the finite-T solver that decides the branch (`v185`/`v211`). It does *not* derive θ_i, does *not* upgrade `DM.AXION.SPINE.01` (stays **[C]**) and closes no gate.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the dark-matter section of `tfpt_4_frontier`; the Wolfram extension/README/ledger counts (`GATE.WOLFRAM.02`) and the website verification count; manifests refreshed.

2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix `tex-artefacts/tfpt_references.tex`** ("External works built upon"), `\input` into the six math-heavy documents (`tfpt_1`, `tfpt_2`, `tfpt_4`, `tfpt_horizon_readouts`, `origin_theory`, `tfpt_research_contracts`). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto;

Adamo–Giorgetti–Tanimoto `2506.01008/2508.07109`; Naaijkens–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the `make_manifest.py` TEX list. No claim or status change — a provenance/citation completeness pass.

2026-06-25 (External AQFT citations + the four-fold μ_4 identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open `SEAM.EQUIV.01` continuum/realisation legs (`v336`) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (`arXiv:2407.18222`, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (`arXiv:2506.01008`, 2025) — "the lattice is a conformal net" is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (`arXiv:2508.07109`, 2025), which defuses the "Bisognano–Wichmann presupposes the covariance it should produce" circularity (`v215`) on the intrinsic-BW residual (`v424`). Added to `v336` docstring/registry, the `SEAM.EQUIV.CONTINUUM.01` ledger `external_data`, and `tfpt_research_contracts`.
- **The four-fold μ_4 identity made explicit (`v422`).** `v422` already proved that the seam deck/divisor μ_4, the carrier Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$, the discriminant $\mathrm{disc}(D_5)=\mathrm{disc}(A_3)=\mathbb{Z}/4$ and the $(E_8)_1$ simple-current glue are ONE cyclic $\mathbb{Z}/4$ (with the cyclic-vs-Klein negative control). It is now named the *four-fold μ_4 identity* and typed, per `v428`, as a *compression* (one object read four ways), NOT four independent witnesses – in `v422`, the registry, the `SEAM.GALOIS.NET.01` ledger row and `origin_theory`. (No new module: a separate `v429` would have duplicated `v422` and violated the very compression discipline.)

2026-06-25 (`v428` — honest self-correction of the over-determination map)

- **`v428` (`OVERDET.WITNESS.RECLASS.01`) — applying `v427`'s own multiply-vs-compress test to `v427` itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses "disjoint grammars that *multiply*" overstates independence. **[E]** by the Brieskorn classification (`v236`) the $(2,3,5)$ singularity is the ONE generator behind the order-30 clock (Milnor $\mu=(2-1)(3-1)(5-1)=8=\mathrm{rank}\,E_8$, $30=2{\cdot}3{\cdot}5=h(E_8)$, $\varphi(30)=8$), so each witness is a facet of that same object (primes $2,3,5$ of 30, the clock 30, or E_8): all seven collapse to one generator $\Rightarrow$ *compression*, not seven multiplications. **[E]** what genuinely multiplies is the input forced four independent ways — the "8" in $c_3=1/(8\pi)$ from $\mathrm{rank}\,E_8$, $h(D_5)=2(5-1)$, $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}\approx 137$. **[C]** refines and partly walks back `v427`, consistent with `v236/v305/v313`.
- **Surfaces corrected (same change).** `v427`'s prose, the ledger row (`OVERDET.WITNESS.MAP.01`, now noting the refinement), `tfpt_safeguards` Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (`papers.ts`, `Safeguards.tsx`, `release.ts`, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack "one richly over-determined input + a foreign readout".

2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures** (`make_figures.py`, in `manifest.sha256`): the layered defense, the α^{-1} uniqueness scan (one $F_{U(1)}$ variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs $\leq 5/13$).

- **`tfpt_safeguards` expanded.** Layer 5 now states the null model explicitly ($\prod p_i$=$10^{-25.8}$, 2×200k pseudo-theories ≤ 5/13, $\varphi_0^{\rm ret}$ ± 10% → 6/13, shuffle 1.2) and the α uniqueness (≤ $10^{-30.7}$, ∼102 bits, `v100`), with the data watchdog (`v307`) and kill board (`v321`); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (`components/Safeguards.tsx`) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-`v425`: the flow-time is *not* native and does *not* reduce to one scale — `v99` already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading t=$N_{\rm fam}$=3 ⇒ m_τ=1776.94 MeV, a Belle II kill test. No new module.

2026-06-25 (Three closure-strategy modules `v425`–`v427` + the Safeguards paper)

- **`v425` (`DYN.TRANSFER.UNIVERSAL.01`) — the single-flow reduction of $F_{\rm transfer}$.** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, `v238`/`v240`, gap $6\ln(3/2)$) restricted to each sector. **[E]** the `v99` Koide generator linearised at q=2 has rate $-\Delta$, time-1 contraction $(2/3)^6$; Boltzmann washout is that contraction (CP phase $4\pi/3$ native, `v320`); QCD rate b_3= − 7. **[C]** only the anchors ($v_{\rm geo}$, C_p, M_R) stay external; $F_{\rm relic}$'s cosmological IC θ_i is the one wall. A reduction (extends `v383` static→dynamic), not internalisation; the firewall holds.
- **`v426` (`SEAM.EQUIV.LTORP.KMS.01`) — the invertible β=1-KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). **[E]** a gapped reflection-symmetric μ_4-even collar at β=1 KMS satisfies $\Theta K\Theta^{-1}$=$-K$ (u_Θ=J) and $[\rho, K]$=0, so (LTO-RP) holds by direct Tomita–Takesaki; $|\det \mathrm{Cartan}(E_8)|$=1 (no anyons), state not a trace; neg controls have teeth. Advances `v424` sub-step (ii); `SEAM.EQUIV.01` stays **[O]** (continuum sub-step (i) remains).
- **`v427` (`OVERDET.WITNESS.MAP.01`) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. **[E]** seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss $N(3{+}2i)$=13, Eisenstein $N(3{+}2\omega)$=7, cyclotomy $N(3{+}2\zeta_5)$=55, Galois $|(\mathbb{Z}/5)^\times|$=4, $|\det \mathrm{Cartan}(E_8)|$=1, Pascal $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$=11 ($2^4$=16), Coxeter $h(E_8)$=30=2·3·5 with $\varphi(30)$=8 — and the anchor a=(1, 1, 2) is the compression generator. Extends `v305`.
- **New paper `tfpt_safeguards` (companion S).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (`v427`), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in `build.sh`, `make_docs_map.py`, `make_manifest.py` and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** `run_all.py` (421 modules), the registry, the ledger, `tfpt_research_contracts` (the $F_{\rm transfer}$ and keystone sections), `tfpt_safeguards`; manifests refreshed.

2026-06-25 (The (LTO-RP) reduction of the keystone `SEAM.EQUIV.01`: `v424`)

- **New module `v424` (`SEAM.EQUIV.LTORP.01`).** An honest MMST-style reduction of the one open keystone — `SEAM.EQUIV.01`'s intrinsic Bisognano–Wichmann lemma (the `v308`/`v309` residual) — to the recent commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`, [NPW26]), with the central mapping *corrected*. [**E**] the μ_4 clock is a modular *symmetry* ($[\rho, K]$=0, `v309/v199`), not the flow — σ^ψ=ρ is type-wrong (a one-parameter group versus a finite order-4 element). [**E**] NPW26's axiom (LTO-RP) means u_Θ=J (reflection = modular conjugation) = the intrinsic BW condition $\theta K\theta$=$-K$, *distinct* from $\omega\circ\rho$=ω: a positive character-block-diagonal K has $[\rho, K]$=0 yet never $\theta K\theta$=$-K$, so clock-invariance does *not* imply (LTO-RP) (refuting that identification). [**O**] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code $|\det K|$=4 via a trace; Levin–Wen anyonic), while the seam is invertible ($|\det \mathrm{Cartan}(E_8)|$=1, no anyons) and a β=1 KMS state (not a trace) — in neither proved bucket. [**C**] verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; `SEAM.EQUIV.01` stays [**O**]. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via `v89/v281`).

- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 (`arXiv:2605.10693`), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf (`arXiv:2203.07421`).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from `QG.AMB.01`: `v423`)

- **New module `v423` (`QGAMB.NODEP.01`).** Sharpens the `v369/v384` prose redundancy ("`QG.AMB.01` is a [**C**] redundancy, not dynamics") into a machine-checked DAG+script fact, told honestly. [**E**] gap-decoupling margin $\Delta_{\text{eff}} = 6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$ ($31 = 2^{g_{\text{car}}} - 1$), susceptibility $\chi = 729/665$ finite (negative control gap $\to 0 \Rightarrow \chi \to \infty$); [**E**] `QG.AMB.01` is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while α^{-1}, the mixing angles and cosmology are each *computed* by a compiler script, `v3/v9/v268/v7`); [**E**] the one transitive link is architectural — of the frozen readouts only θ_{13} references it, via the 4d-QFT contract spine (`PS.DIRAC`$\to$`CONTRACT.QFT4D`$\to$`QG.G2`$\to$`GATE.METRIC`), a coherence edge, not a numerical input. [**C**] verdict: `QG.AMB.01` is gap-decoupled certification — `v369` as a DAG/script fact; honest scope — this does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via `v337`).
- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open `SEAM.EQUIV.01` premise is the *holomorphic* $|\det K|$=1 point that selects $(E_8)_1$ — the central charge c=8 alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample (`v277/v281`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

2026-06-25 (The Galois$\leftrightarrow$Net bridge: `v422`)

- **New module `v422` (`SEAM.GALOIS.NET.01`).** The seam $\mu_4 = \mathrm{Gal}(\mathbb{Q}(\zeta_5))$ is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of `v419` to G_{net}/`SEAM.EQUIV.01`. [**E**] $\mathrm{disc}(A_3) = \mathrm{disc}(D_5) = \mathbb{Z}/4$ (one Smith invariant factor 4 = cyclic, $\det 4$); D_n disc is $\mathbb{Z}/4$ for n odd, $\mathbb{Z}_2\times\mathbb{Z}_2$ for n even, so the carrier D_5 (rank 5 = g_{car}, odd) is cyclic while D_8 (rank 8, even) is Klein; [**E**] $(\mathbb{Z}/5)^\times = \langle 2\rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); [**E**] in $\mathbb{Z}_4\times\mathbb{Z}_4$ ($16 = \dim S^+$) the glue $(1,1)$ is order 4 with Lagrangian quotient $16/4^2 = 1 = (E_8)_1$ (`v89/v154`); [**E**] negative control:

the Klein / order-2 $\langle(2,2)\rangle$ gives $16/2^2 = 4 = |\mathrm{disc}(D_8)| = SO(16)$ ($\det 4$), NOT E_8 ($\det 1$) — cyclic $\mathbb{Z}/4 \to E_8$, Klein$\to D_8$. **[C]** the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck μ_4 (`v419`), $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$ and the $(E_8)_1$ glue — the Galois gearbox $(\mathbb{Z}/30)^\times$ IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored ($324 \to 327$).

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts (327/327) and the website verification count; manifests refreshed.

2026-06-25 (F_pole external: the Koide source→pole transfer is one-loop EM: `v420`)

- **New module `v420` (`FR.POLE.QED.01`).** The honest external-physics follow-up to `v93` and the clock No-Go: the Koide source→pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. **[E]** Koide Q is rescaling-invariant ($Q(c\,m) = Q(m)$), so flavor-universal running leaves it fixed and the source→pole shift is purely flavor-split; **[C]** the tree deficit $\frac{2}{3} - Q_{\mathrm{src}} = 0.00220$ is one-loop-EM-sized ($= \varphi_0/24$ to 0.6%, $= \alpha/\pi$ to $\sim$5%), the PDG pole sits at 2/3 (the IR attractor); **[C]** a 1-loop QED pole$\leftrightarrow\overline{\mathrm{MS}}$ per-lepton-log estimate shifts Q by EM-loop order toward 2/3 (the `v82` direction); **[O]** the exact flow time $t \approx 2.84$ (`v93`) is scheme/scale-dependent external QED, only sized. **[C]** verdict: F_{pole} is a one-loop EM effect — the lens forces the rate $(2/3)^6$ (`v327`), external EM supplies the flow; `F_pole` stays **[C]**, firewalled (`v187`), so F_{transfer} is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like `v93`/`v371`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

2026-06-25 (The seam μ_4 is the carrier's Galois group: `v419` + figure)

- **New module `v419` (`SEAM.GALOIS.01`).** A positive resolution of the cyclic/Galois asymmetry (`v409`, `RES.COXETER.SYMMETRY.01`): why the seam μ_4 sits on the Galois side, not on the cyclic ζ_{30} dial. **[E]** $h(E_8) = 30$ is squarefree, so $\mathbb{Z}/30$ has element orders $\{1,2,3,5,6,10,15,30\}$ — *no* order 4, so the seam μ_4 cannot be a rotation hand and is forced onto the Galois side; **[E]** by CRT $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$ (order $\varphi(30) = 8 = \mathrm{rank}\,E_8$), and the μ_4 ($\mathbb{Z}/4$) factor IS $(\mathbb{Z}/5)^\times$ (the carrier prime 5), the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family 3 = CP conjugation, `v316`); **[E]** so the seam $\mu_4 = \mathrm{Gal}(\mathbb{Q}(\zeta_5))$, realised on the carrier clock C_5 (`v418`) by the Frobenius $\zeta_5 \mapsto \zeta_5^2$ — the explicit order-4 operator G with $GC_5G^{-1} = C_5^2$, $G^4 = I$, $G^2C_5G^{-2} = C_5^{-1}$. **[C]** this is the positive content of `v409`: the 2^2=4 is the order of $(\mathbb{Z}/5)^\times$, and "no contraction rate" is because the seam is an *automorphism*, not a hand; so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (cyclic h=30, Galois rank =8). Cited in `origin_theory`; Wolfram-mirrored ($321 \to 324$).
- **New figure `coxeter_galois.pdf`.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ($\mu_4 = \mathrm{Aut}(\zeta_5)$ the seam, $\mathbb{Z}_2 = \mathrm{Aut}(\zeta_3)$ the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: `v418`)

- **New module `v418` (`ARITH.NORMTRIPLE.01`).** The carrier split $(3,2)$ read in the three atom-rings gives $(7,13,55)$ =(scalaron, Δ_Q, quark numerator), and the carrier-5 clock missing from `v417` is found as the $4{\times}4$ Φ_5-companion C_5. **[E]** $C_5^5{=}I$, $\mathrm{tr}\,C_5{=}{-1}$, $\det C_5{=}1$ (the four primitive 5th roots), with the golden ratio as its real-part data ($2\cos 72°{=}1/\varphi$, $2\cos 144°{=}{-\varphi}$; output-side, `v313/v349`; dim 4, not 3 — this refines `v417`'s gap from "no operator" to "no $3{\times}3$ operator"); **[E]** $N_{\mathbb{Z}[\zeta_5]}(3{+}2\zeta_5){=}\det(3I{+}2C_5){=}55{=}5{\cdot}11{=}g_{\mathrm{car}}\|\,\mathrm{Pl}(K)\|_1$, the c_u/c_d numerator (`v410/v411`); **[E]** the triple $\det(3I{+}2\,\mathrm{Comp}(\Phi_n)) = (7,13,55)$ for $n{=}3,4,5$ over $\mathbb{Z}[\omega],\mathbb{Z}[i],\mathbb{Z}[\zeta_5]$ (atoms $3,2,5$, `v416`); **[E]** NEG control: the anchor $(5,4)\to(21,41,461)$ reaches named values only in the ω- and i-rings ($21{=}N_\omega(5,4)$, $41{=}10b_1$, `v222`) but 461 (prime) in the carrier ring, so the carrier ring separates $(3,2)\to 55$ from $(5,4)\to 461$. **[C]** honest: $(3,2)$ is the *forced* carrier split (`v14`), not the unique clean one — a scan finds $\{(2,1)\to(3,5,11),(3,2)\to(7,13,55)\}$, a small ladder. The sharper discriminator is *cross-sector*: only $(3,2)$ spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while $(2,1)$ is all compiler-core. Cited in `origin_theory`; Wolfram-mirrored ($318\to321$).

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, a triple paragraph in `origin_theory`, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

2026-06-25 (The Eisenstein/CP operator + the order-30 clock: `v417`)

- **New module `v417` (`DIAMOND.EISEN.01`).** The hexagonal dual of `v415`: the $\tau{=}\omega$ flavor side put on operators and unified with the seam into one order-30 Coxeter clock. **[E]** the family rotation $P=(1\,2\,3)$ is the order-3 Eisenstein deck ($P^2{+}P{+}I=\mathrm{ONES}$, so $\omega^2{+}\omega{+}1{=}0$ on the plane $\perp(1,1,1)$), the order-3 dual of $J^2{=}{-I}$; **[E]** it realises the hex CM norm as $(3I{+}2P)(3I{+}2P^2)=7I{+}6\,\mathrm{ONES}$, $\mathrm{Spec}\{7,7,25\}$ ($7{=}N_{\mathbb{Z}[\omega]}(3{+}2\omega){=}$ scalaron, $25{=}g_{\mathrm{car}}^2$) — the dual of `v415`'s $\{13,13,9\}$; **[E]** the CP clock $W{=}{-P^2}$ (ρ of `v233`) has $W^2{=}P$, $W^3{=}{-I}$, $\mathrm{Spec}\{-1,\zeta_6,\zeta_6^{-1}\}$, giving $\delta_{\mathrm{CKM,lead}}{=}\arg\zeta_6{=}\pi/3$ and $\delta_{\mathrm{PMNS}}{=}\arg\zeta_6^4{=}4\pi/3$ (`v316/v320`); **[E]** all flavor clocks are powers of ζ_{30} ($\zeta_{30}^{15}{=}{-1}$, $\zeta_{30}^{10}{=}\omega$, $\zeta_{30}^{6}{=}\zeta_5$, $\zeta_{30}^{5}{=}\zeta_6$) while the seam $\mu_4{=}i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30){=}8{=}\mathrm{rank}\,E_8$ — so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants ($h{=}30$ and rank $=8$). **[C]** honest gap: the carrier μ_5 has no $3{\times}3$ operator (the golden φ is an output-side trace, `v313/v349`); negatives: no U,V-diamond operator carries ζ_6, $[D,P]\neq 0$ (no μ_{12}), $N(3{+}2\zeta_6){=}19{\neq}7$. Cited in `tfpt_2`; Wolfram-mirrored ($315\to318$).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Sheet-Diamond keybox of `tfpt_2`, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

2026-06-25 (The Gaussian operator + the atom trichotomy: `v415`, `v416`)

- **New module `v415` (`DIAMOND.GAUSS.01`).** The square CM-norm dictionary of `v222/v230` is *operator-realised*, and the μ_4 deck becomes an intrinsic diamond object. **[E]** the integer commutator $J:=\frac{1}{3}[U,V]=\left(\begin{smallmatrix}-1&1&0\\-2&1&0\\-3&1&0\end{smallmatrix}\right)$ has $\mathrm{Spec}(J)=\{i,-i,0\}$, $J^2=-I$ on the rank-2 image and $J^3+J=0$ — a μ_4 quarter-turn born from the binary V and ternary U compilers (`v410/v95`); **[E]** $3I+2J$ and $5I+4J$ have spectra $\{3{+}2i,3{-}2i,3\}$, $\{5{+}4i,5{-}4i,5\}$, so the Gaussian integers $3{+}2i$ (norm $13{=}\Delta_Q$) and $5{+}4i$ (norm $41{=}10b_1$) are *literal eigenvalues*; **[E]** $(aI{+}bJ)(aI{-}bJ)$ reads $(a^2{+}b^2)$ on the μ_4-plane and the real part^2 on the kernel ($\{13,13,9\}$, $\{41,41,25\}$); **[E]** the intrinsic order-4 deck $D=-I+J-J^2$ ($D^4=I$, $\chi_D=(x{+}1)(x^2{+}1)$) has χ_2-line $\ker[U,V]=a-\mathbf{1}$. **[C]** the geometric-deck identification sharpens (does not close) the `v140` $\mathbb{Z}_3$ residue. Cited in `tfpt_2`; Wolfram-mirrored.
- **New module `v416` (`ARITH.TRICHOTOMY.01`).** The atoms $\{2,3,5\}$ are ramified/inert/split in the two exceptional CM rings. **[E]** in $\mathbb{Z}[i]$ (square/seam) $2{=}|\mathbb{Z}_2|$ ramifies, $3{=}N_{\mathrm{fam}}$ is inert, $5{=}g_{\mathrm{car}}$ *splits* as $(2{+}i)(2{-}i)$; **[E]** in $\mathbb{Z}[\omega]$ (hex/flavor) $3{=}N_{\mathrm{fam}}$ ramifies, $2,5$ are inert; **[E]** the ramified prime of each ring is its own atom; **[E]** the three quadratic facets $\mathbb{Q}(i),\mathbb{Q}(\sqrt{-3}),\mathbb{Q}(\sqrt{5})$

of $K=\mathbb{Q}(i,\sqrt{-3},\sqrt{5})$ (`v403`) ramify over exactly one atom each, product $2\cdot3\cdot5=30=h(E_8)$, two imaginary (CM/dynamic) + one real (RM/static). **[C]** the seam/flavor/golden reading. Cited in `origin_theory`; Wolfram-mirrored.

- **Wolfram extension** 308 → 315 + **a pre-existing fix.** Added the seven exact checks for `v415`/`v416` and corrected a one-character sign bug in the `v412` `CharacteristicPolynomial` check (Wolfram returns $-\chi$ for odd rank), so `tfpt_readouts_extension.wl` now runs genuinely clean at 315/315 (`GATE.WOLFRAM.02`); README and ledger counts updated.
- **Surfaces synced.** `run_all.py`, `script_registry.csv` and `status_ledger.csv` updated; a new keybox paragraph in the Sheet-Diamond section of `tfpt_2` and a trichotomy sentence in `origin_theory`; manifests refreshed.

2026-06-24 (The sheet generator V as a binary internal compiler: `v410`–`v414`)

- **New module `v410` (`SHEET.GEN.BINARY.01`).** The rank-2 sheet axis $V=Q\,\mathrm{diag}(0,1,1)$ of the centered cross (`v95`/`v218`) is a binary internal compiler: its powers print the carrier spine. **[E]** the closed form $V^n=\left(\begin{smallmatrix}0&2^{n-1}&0\\0&2^n&0\\0&2^{n+1}-2&1\end{smallmatrix}\right)$ (proved by the inductive step $F(n)V{=}F(n{+}1)$), so $V^n\mathbf{1}=(2^{n-1},2^n,2^{n+1}-1)=(1,2,3),(2,4,7),(4,8,15),(8,16,31)$; **[E]** the four bilinear families $\mathbf{1}^TV^n\mathbf{1}=7\cdot2^{n-1}{-}1$, $\mathbf{1}^TV^na=7\cdot2^{n-1}$, $a^TV^na=11\cdot2^{n-1}$, $a^TV^n\mathbf{1}=11\cdot2^{n-1}{-}2$ give $(6,7,9,11)$ at $n{=}1$ and $\mathbf{1}^TV^2\mathbf{1}{=}13{=}\Delta_Q$, $\mathbf{1}^TV^3\mathbf{1}{=}27{=}\mathbf{1}^TRa$, $\mathbf{1}^TV^4\mathbf{1}{=}55$, $\mathbf{1}^TV^4a{=}56{=}\dim\mathbf{56}_{E_7}$; **[C]** the binary spine is FORCED by $\mathrm{Spec}(V){=}\{0,1,2\}$, so the carrier/Lie matches ($\mathrm{rank}\,E_8{=}8$, $\dim S^+{=}16$, $248/8{=}31$) and the theta cross-link ($\sigma_3(2){=}9{=}a^TV\mathbf{1}$, $\sigma_3(3){=}28{=}\mathbf{1}^TV^3a{=}\det(I{+}R)$, $\sigma_3(5){=}126$) are audit-typed (the `v95` G_2-center typing).
- **New module `v411` (`FLAV.UD.VPOWER.01`).** The quark ratio is a pure V-power readout, $c_u/c_d=(\mathbf{1}^TV^4\mathbf{1})/((a^TV\mathbf{1})(\mathbf{1}^TV^2\mathbf{1}))=55/(9\cdot13)=55/117$, identical to the established $g_{\mathrm{car}}\|\mathrm{Pl}(K)\|_1/(N_{\mathrm{fam}}^2\Delta_Q)=5\cdot11/(9\cdot13)$ (`v94`). **[E]** the identity; **[C]** an exact *re-encoding* (no new physical input) — the value and the u/d identification stay conditional, coupled to the (U_{wall}) readout rigidity.
- **New module `v412` (`DIAMOND.SHEET.SOURCE.J.01`).** Names the operator on the $\mathbb{Z}_2$ wall ($t{=}{-}2$, $\det{=}2$, `v135`), $J{=}M(1,-2){=}C{-}2V$, completing the sheet axis $J,K,C,F{=}M(1,-2..1)$. **[E]** $\chi_J=\lambda^3-6\lambda^2+3\lambda-2\Rightarrow(\mathrm{tr},e_2,\det){=}(6,3,2)$; $a^TJa{=}30{=}h(E_8)$, $\det(I{+}J){=}12{=}\dim\mathfrak{g}_{\mathrm{SM}}$ and $\det(2I{+}J){=}40{=}|R(D_5)|$; **[C]** the Lie readings audit-typed like the G_2/F_4 shadows.
- **New module `v413` (`DIAMOND.SHEET.CALCULUS.01`).** On $M_t{=}C{+}tV$ the elementary-symmetric coefficients encode the atoms as difference orders: $e_1{=}3t{+}12$, $e_2{=}2t^2{+}15t{+}25$, $e_3{=}4t^2{+}14t{+}14$ give $\Delta e_1{=}3{=}N_{\mathrm{fam}}$, $\Delta^2e_2{=}4{=}|\mu_4|$, $\Delta^2e_3{=}8{=}\mathrm{rank}\,E_8$ (the e_3 curvature is the `v218`/`v224` sheet quadratic; the full triple is new); **[E]** the anchor energy $a^TM_ta{=}52{+}11t$ ($30{\to}41{\to}52{\to}63$, step $11{=}a^TVa$); **[C]** atom/Lie readings audit-typed.
- **New module `v414` (`DIAMOND.CENTER.RESOLVENT.01`).** The center $C{=}M(1,0)$ is a resolvent portal: **[E]** $\det C{=}14{=}\dim G_2$, $\det(I{+}C){=}52{=}\dim F_4$, $\det(2I{+}C){=}120{=}|R^+(E_8)|$, the exceptional shell ladder $G_2{\to}F_4{\to}E_8^+$ as $\det(C{+}kI)$, $k{=}0,1,2$ (with $\mathrm{tr}\,C{=}12{=}\chi_{E_8}(i)$, $\sum C{=}31{=}2^{g_{\mathrm{car}}}{-}1$, `v95`); **[C]** the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five `\veri` citations) and the E_8 theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror 303 → 308.

2026-06-24 (Two new tracked targets + axion-branch consistency: `v408`, `v409`)

- **New module `v408` (`HYP.PHI0.PUNCTURE.01`).** Names the φ_0 puncture term $48c_3^4$ as the order-$|\mu_4|{=}4$ local boundary heat-kernel (Seeley–DeWitt) contact term of the μ_4 puncture divisor, summed over $\Omega_{\text{adm}}=N_{\text{fam}}\dim S^{+}=48$ admissible states — the analytic complement of the icosahedral tree term (`v396`). [**E**] the exact split $\varphi_0=1/(6\pi)+48c_3^4$, the puncture count, the closed form $3/(256\pi^4)$ and the hypergraph negative (reused from `v396/ARCH.QUAD.01`); [**O**] the from-first-principles heat-kernel proof; [**C**] a FACE of the seam boundary analysis (the c_3 Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.

- **New module v409 (`RES.COXETER.SYMMETRY.01` closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*: $\sigma=\operatorname{diag}(1,-1,-1)$ has spectrum $\{\pm1\}$ and projector $(I+\sigma)/2$ spectrum $\{0,1\}$, none in $(0,1)$, so prime-2 carries no contraction rate, whereas the genuine rates $2/3$ and $(1/3)^6$ lie in $(0,1)$ and carry prime-2 only as a factor. [**E**] no prime-2-only attractor exists; [**C**] dynamics begins only after coupling to family-3 or carrier-5 ($|\mu_4|=2^2$ the Galois bridge). Resolves the `v394` open question's falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row `RES.COXETER.SYMMETRY.01` retyped [**O**] $\to$ [**E**]/[**C**].
- **Axion branch consistency (`tfpt_4_frontier`, website `predictions.ts`, `v185`).** The dark-matter section's opening + keybox are realigned with the finite-T result already stated later in the same section: the robust dominant-DM branch is the *spine* angle $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=3\pi/5=108°$ (untuned, in band), while the seam hilltop $\theta_i\approx170°$ over-produces ($\sim 5.4\times$) and is disfavoured; the misalignment angle is retyped from a closed [**E**] input to a [**C**] branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (`FORM.COXETER.PRIME2.01`).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ($r^2=1$) is ±1, a projector eigenvalue ($r^2=r$) is 0 or 1, neither lies in the open interval $(0,1)$, while $2/3$ does — so *no prime-2-only attractor exists.* Built clean (`lake build`); the new theorems are axiom-pure (`#print axioms` = `propext`, `Classical.choice`, `Quot.sound` only) and pass the `AuditCheck`/`AuditContract` signature locks. This lifts `RES.COXETER.SYMMETRY.01` to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant $11=\|Pl(K)\|_1=\det(\mathbf{1}|a|\sigma)$ is already typed honestly as an [**E**] identity with "why 11" the lone [**O**] residue (`v407`, `tfpt_1`); the repo carried no overstatement, so no edit was required.

2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: `GATE.UWALL.06-09` was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain `v114`–`v117` keeps `.06-.09` (as asserted by `v123_inventory_update.py` and `run_all.py`); the older exterior-leg chain `v42`–`v45` is renumbered to `GATE.UWALL.15-.18` (internal `depends_on` carried along). No claim content changed.
- **Website.** `papers.ts` document count 8 → 9 (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E→A–F* with a new Target F section + contribution bullet (the abstract already carried F); `StatusMatrix.tsx` *Doc 7→Doc 8* for the research contracts and the residual interfaces re-typed v_{geo} [**O**], G_{net} [**C**], F_{transfer} [**C**]; `objections` *Paper 1→Doc 1.*
- **Papers (marker/wording alignment, no canonical status change).** `tfpt_1` Strong CP [**E**]→[**E**]/[**C**] (given RP) and θ_{12} "genuinely open"→"conditionally derived"; `tfpt_2` Starobinsky A_s [**E**]→[**E**]/[**C**] (M exact, A_s a band over $N_\star\in[50,60]$) and the Part-II abstract "opens the one remaining gap"→"closes the last structural gap"; `tfpt_research_contracts` the QFT4D "nonperturbative measure stays open" line and the "only U_{point} stays open" headline reconciled with the [**C**] redundancy / v_{geo} reduction stated alongside; `origin_theory` class C re-typed to the ambient-measure [**C**] redundancy; `introduction` (U_{wall}) → v_{geo}.

Forbidden old-paper attributions ("Papers 1–7", "Papers 2/6", "Paper-7", "Paper-1") replaced by current document / neutral wording (the `tfpt_3` cascade bridge stays exempt).

- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model ($v_{\rm geo}$ **[O]**; $G_{\rm net}$/`SEAM.EQUIV.01`, $F_{\rm transfer}$, $N_\star$, `QG.AMB.01` **[C]**; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated `status_heatmap` (ledger-driven) and extended the narrative figures `script_timeline/residual_chain/qft_skeleton` through the `v303`–`v407` arc (typed solvers, parameter-free local Einstein equation `v359`, `QG.AMB.01` **[C]** redundancy, seam closed modulo cited theorems); the `qft_skeleton` "Stelle ghost" open box is retyped (resolved truncation artefact). Added `qft_skeleton/qft_unification` to the manifest figure list.

2026-06-24 (The $\tau{=}\omega$ dual keystone + the flavor residuals it homes: `v405`–`v407`)

- **[E]**/**[C]**/**[O]** **`v405_seam_equiv_omega.py`** (`SEAM.EQUIV.02`): the *dual keystone* at $\tau{=}\omega$ – the family/flavor sector as the order-3 Eisenstein/A_2 face of E_8, dual to the seam = $(E_8)_1$ at $\tau{=}i$ (`v404`). A named contract (genre of `v286/v398`), NOT a closure. **[E]** $\chi_{E_8}(\omega) = 0$ (the family CM point degenerates E_8) vs $\chi_{E_8}(i) = 12$; **[E]** A_2 is the $\tau{=}\omega$ Eisenstein lattice (Cartan det $= 3 = N_{\rm fam}$, $h = 3 = N_{\rm fam}$); **[E]** R lives in $E_6{\times}A_2$ ($248 = 78 + 8 + 2{\cdot}27{\cdot}3$, $\det R = 8 = \dim A_2 = \operatorname{rank} E_8$); **[E]** $\delta_{\rm PMNS} = 4\pi/3$ is the $\tau{=}\omega$ CM phase; **[E]** the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck $= N_{\rm fam} = \det Q$. **[C]** the family sector is the order-3 Eisenstein/A_2 deck; the two keystones sit at the only two elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$. **[O]** residual = the canonical $\mathbb{Z}_3$ deck map (`v140`) + the $v_{\rm geo}$-class scale. Cited in `origin_theory`; Python-only.
- **[E]**/**[C]**/**[O]** **`v406_pmns_phase_origin.py`** (`FLAV.PMNS.CLOSE.01`): the PMNS dynamics closed *modulo* the $\tau{=}\omega$ keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one $v_{\rm geo}$-class unit, so PMNS has *zero free flavor dials beyond* $v_{\rm geo}$. **[E]** the three angles (`v9/v268`) $\to$ unitary $U_{\rm PMNS}$ (`v270`); **[C]** $\delta_{\rm PMNS} = 4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase (`v405`), upgrading `v270`'s **[O]** "assembled input" to a keystone consequence; **[C]** $J_{\rm PMNS} = -0.0297$ derived; **[E]** the absolute scale = one seesaw ratio = $v_{\rm geo}$-class (`v272/v153`). **[O]** residual = the seesaw operator realisation (M_R, `v184`) + the Σm_ν floor kill test. Cited in `tfpt_2_standard_model`; Python-only.
- **[E]**/**[C]**/**[O]** **`v407_dn_pairings_omega.py`** (`FLAV.SELECTOR.CLOSE.01`): the R4′ selector residue folds into the $\tau{=}\omega$ keystone – the three (d, n) pairings ARE the $\tau{=}\omega$ family-slice atoms. **[E]** the frame $(\mathbf{1}, a, \sigma)$ has det $= 11 = \|Pl(K)\|_1$ and $n = (5, -9, 6)$ is the unique covector with $n{\cdot}\mathbf{1} = 2$, $n{\cdot}a = 8$, $n{\cdot}\sigma = 121 = 11^2$ ($\Rightarrow$ (d, n) $\Rightarrow$ R, `v136/v139`); **[E]** $n{\cdot}a = 8 = \operatorname{rank} E_8 = \dim A_2 = \det R$ (the A_2 block reading R in $E_6{\times}A_2$, `v405`); **[E]** the pairings $\{2, 8, 121\} = \{|Z_2|, \dim A_2{=}\det R, \|Pl(K)\|_1^2\}$. **[C]** deriving (d, n) folds into `SEAM.EQUIV.02` (d anchor-derived, `v139`); **[O]** the lone residue "why $11{=}\|Pl(K)\|_1$". Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (The arithmetic hull + the E_8-character CM duality: `v403`, `v404`)

- **[E]**/**[C]** **`v403_facet_compositum.py`** (`ARITH.HULL.01`): the three single-prime number-field facets of the order-30 clock (`v390/v394`, used so far only one at a time) *compose* to one field $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$. **[E]** the squarefree parts $-1, -3, 5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so $[K : \mathbb{Q}] = 2^3 = 8 = \operatorname{rank} E_8 = g_{\rm car}{+}N_{\rm fam}$ with Galois group $(\mathbb{Z}/2)^3$ (order 8); **[E]** exactly 7 quadratic subfields $\{-1, -3, 5, 3, -5, -15, 15\}$, splitting 4 imaginary $= |\mu_4|$ (CM/dynamic) + 3 real $= N_{\rm fam}$ (RM/static); **[E]** ramified *only* over $\{2, 3, 5\}$ = the atoms = the primes of $h(E_8) = 30$; **[E]** $2^{\omega(h)} = $ rank is E_8-special (the E_7 control $h = 18$ gives $2^2 = 4 \neq 7$). **[C]** K is the abelian $(\mathbb{Z}/2)^3$ *arithmetic* hull dual to the E_8 *lattice* hull; the 7 quadratic subfields are a candidate for the 7 E_8 audit slices. A synthesis composing the established facets, no new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. `v390`/`v394`).

- **[E]**/**[C]** **`v404_e8_char_cm_duality.py`** (`E8.CMDUAL.01`): the $(E_8)_1$ vacuum character $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$ read at *both* CM points, extending `v282` ($\chi_{E_8}(i) = 12$) to the second elliptic point. **[E]** $\chi^3_{E_8} = j$ (q-series); **[E]** SEAM $\tau{=}i$ ($\mathbb{Q}(i)$, order 4): cross-ratio 2 $\Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$ (E_8 present, `v214`/`v282`); **[E]** FAMILY $\tau{=}\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic λ ($\lambda^2{-}\lambda{+}1 = 0$) $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$ (E_4 vanishes, `v220`/`v267`); **[E]** $\tau{=}i, \tau{=}\omega$ are the *only* elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$, so the values are forced. **[C]** the seam/flavor rigidity duality: the seam is rigid where E_8 is present ($\chi = 12$, the keystone `SEAM.EQUIV.01` closes) while the flavor/CP sector is the soft residual where E_8 degenerates ($\chi = 0$: U_{wall}, v_{geo}, the CP texture), a candidate dual keystone at $\tau{=}\omega$. Cited in `origin_theory`; core values $j = 1728/j = 0$ already mirrored (`v214`/`v220`/`v267`/`v282`), Python-only.

2026-06-24 (The five TOE/QFT closure contracts `v398`–`v402` + a ledger rescope)

- **[E]**/**[O]** **`v398_seam_state_rigidity.py`** (`SEAM.RIGIDITY.01`, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (`v177`/`v199`/`v308`/`v309`/`v335`) and *hardening* the finite-block evidence to a multi-mode sweep. **[E]** RP-definability hinge (OS canonical transfer $T{=}e^{-H}$ from RP + reflection alone, `v54`, quasi-free `v155`, no μ_4 normal form); **[E]** band-limited character-block-diagonal core $[\rho, |k|]{=}0$ swept $N{=}4..64$ (beyond the 3-dim H^1); **[E]** holomorphy $\Rightarrow (E_8)_1$ ($c{=}8$, $|\det \mathrm{Cartan}|{=}1$ vs 4, index $4{=}|\mu_4|$). **[O]** the one residual: $[\rho, \Lambda_\Sigma]{=}0$ on the full L^2 (intrinsic Bisognano–Wichmann). NOT a closure of `SEAM.EQUIV.01` (continuum = cited MMST `v336`).

- **[E]**/**[C]**/**[O]** **`v399_gravity_complete.py`** (`GRAVITY.COMPLETE.01`, Paper B): the explicit verdict *Gravity-complete = Boundary-complete + Ambient-redundancy*, combining the parameter-free local equation (`v358`/`v359`) with the ambient redundancy (`v369`). **[E]** the five gravitational readout classes (Newton 8π, Λ, scalaron c_3^7, horizon $1/|\mu_4|$, recovery $(2/3)^6$) are all boundary/seam/gap, so $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$; **[O]** residual = `SEAM.EQUIV.01` + intrinsic BW (no new gate).

- **[E]**/**[O]** **`v400_grav_nonlocal_action.py`** (`GRAV.NONLOCAL.01`, Paper C): the nonlocal spectral-gravity action with the local $R{+}R^2$ as its IR projection (consolidating `v304`/`v370`/`v380`/`v386`, adding the IR-matching check). **[E]** entire $a{=}e^u \Rightarrow$ one pole (ghost-free), every truncation carries a Stelle zero (monotone, `v380`), and the IR order reproduces $R{+}R^2$ with the atom-fixed scale $M^2_{\text{scal}}/\bar{M}^2_{\text{Pl}}{=}c_3^7$ (`v36`/`v253`). **[O]** the entire-a assumption + perturbative-only.

- **[E]**/**[O]** **`v401_metrology_closure.py`** (`METROLOGY.CLOSURE.01`, Paper D): the final input set $\{a, \pi, v_{\text{geo}}\}$ certified — 0 dimensionless dials + 1 unit (consolidating `v153`/`v274`/`v364`/`v384`). **[E]** dimensional bookkeeping mass/$1G/\Lambda/U_{\text{point}}/(m/\mu)$= number $\times v_{\text{geo}}^{1,2,4,1,0}$; **[E]** the G-vs-Λ over-determination 0.11% (`v274`); **[O]** v_{geo} theorem-forbidden ($\mathrm{TOE}_{\text{strict}}{=}\mathrm{TOE}_{\text{dimensionless}}{+}[v_{\text{geo}}]$).

- **[E]**/**[C]** **`v402_ftransfer_suite.py`** (`FTRANSFER.SUITE.01`, Paper E): the four frontier sectors as ONE functor F_{transfer}, the composition $F_{\text{obs}} \circ F_{\text{thr}} \circ F_{\text{RG}}$ (consolidating the `FR.*.SOLVE` solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays **[C]**/**[O]**, only the algebraic source **[E]**; never re-pressed into the compiler, `v187`).

- **Ledger rescope (repo hygiene):** `GATE.METRIC.01` no longer reads "full covariant metric-sector field equation open" — the *local* covariant equation is closed parameter-free (`GRAV.NONLINEAR.01`/`v359`); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (`QG.AMB.01`, a **[C]** redundancy `v369`), removing the stale contradiction with the status card.

2026-06-24 (φ_0 leading term from icosahedral combinatorics: `v396`)

- **[E]**/**[C]** **`v396_phi0_icosahedral.py`** (`HYP.PHI0.01`): the tree-level seed $\varphi_0^{\rm tree} = 1/(6\pi)$ is exactly $F/(4h\pi) = F/(|\,{\rm Aut}\,|\pi) = (F/(g_{\rm car}N_{\rm fam}))c_3$ on the icosahedral hypergraph ($F = 20$ triangular hyperedges, $h = 30$, $|\,{\rm Aut}\,| = 120$); $F/h = |Z_2|/N_{\rm fam} = 2/3$ (the same survival ratio as `v327`). **[E]** Gauss–Bonnet consistent with $c_3 = 1/(|Z_2|4\pi)$ (`v216`/`v342`). **[E]** honest negative: the puncture $48c_3^4$ is still analytic, not a hypergraph fraction. **[C]** $\varphi_0^{\rm tree}$ is a combinatorial reinterpretation, not a new independent injection; the puncture stays **[O]**.

2026-06-24 (Coupled hypergraph rewrite: `v395` unifies carrier $\times$ family in one local rule)

- **[E]** **`v395_hypergraph_coupled.py`** (`HYP.COUPLED.01`): one coupled local rewrite on a 9×3 labelled grid (affine-E_8 nodes $\times$ family channels) unifies `v299` (carrier growth), `v327` (branching rule M) and `v324` (fiber product). Micro-step = network lazy diffusion on each cusp column + M on each node's family vector ($T_{\rm micro} = T_{\rm net} \otimes M$); joint attractor marks$\otimes e_0$; one clock hand (6 family steps) gives recovery $(2/3)^6$; `v324`'s $T_{\rm net} \otimes T_{\rm cusp}$ emerges as the cusp-readout basis. **[O]** the 3-arm seed (P_2) and analytic φ_0 remain irreducible (`v312`).

2026-06-24 (A falsifiable clock probe of the frontier: `v397` how external is the "external physics"?)

- **`v397` (`FR.CLOCK.PROBE.01`) — do the "external" frontier scheme-numbers land on the $\{2, 3, 5\}$ clock?** A falsifiable, anti-numerology probe (strict No-Free-Pattern, `v100`: "on the clock" means an *exact* match, not a wide-band fit). **[E]** the η_B decuple $A_\Lambda = 10 = |Z_2|g_{\rm car}$ is ON the clock (`v212`); **[E]** the Koide rate $(2/3)^6 = (|Z_2|/N_{\rm fam})^6$ is ON the clock (`v303`); **[E]** the axion spine angle $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5$ is ON the clock (`v211`/`v373`); **[E]** the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value — within its $\pm 5.3\%$ band several simple values fit $(2^{3/2}, 17/6, 14/5)$, so by No-Free-Pattern it is declared EXTERNAL (the firewall `v187`/`v374` is correct there). **[C]** verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible "external physics" residue shrinks to essentially the proton's non-perturbative QCD number C_p. Cited in `tfpt_4_frontier`; a falsifiable probe whose answer could have been "none on the clock"; Python-only.

2026-06-24 (Clockwork coherence + an open question: `v394` primes = atoms = facets, and `RES.COXETER.SYMMETRY.01`)

- **`v394` (`COXETER.ATOMS.01`) — the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets.** A bird's-eye **[E]** re-reading tying `v390` (the prime facets) to the anchor $a = (1, 1, 2)$ and `v319` (the 5×6 clock). **[E]** $a = (1, 1, 2)$ has $e_3{=}2{=}|Z_2|$, $p_0{=}3{=}N_{\rm fam}$, $e_2{=}5{=}g_{\rm car}$, product $2{\cdot}3{\cdot}5 = 30 = h(E_8)$ — the clock's primes *are* the anchor's atoms. **[E]** atoms = facet fields (`v390`): $2{\to}\mathbb{Q}(i)$, $3{\to}\mathbb{Q}(\sqrt{-3})$, $5{\to}\mathbb{Q}(\sqrt{5})$, each ramified at its prime. **[E]** the QG susceptibility is pure $\{2, 3\}$: $\chi = 1/(1{-}(2/3)^6) = 729/665 = 3^6/(3^6{-}2^6)$, $(2/3)^6 = 2^6/3^6$, exponent $6 = 2N_{\rm fam}$. **[E]** both dynamic rates carry prime-2: $2/3 = |Z_2|/N_{\rm fam}$ and $(\varphi{+}2)/4 = (\varphi{+}2)/|\mu_4|$ with $|\mu_4|{=}4{=}2^2$. **[E]** $30 = 5{\times}6 = g_{\rm car}(2N_{\rm fam})$ (`v319`). **[C]** the prime-2 seam is the universal dynamical engine; prime-3 (family) and prime-5 (carrier golden) the two attractors.

- **`RES.COXETER.SYMMETRY.01` — a new registered open research question.** Is the prime-2-as-common-factor reading *forced* (the seam involution provably in every sector's gap, no independent prime-2 attractor), or is there a residual asymmetry? Resolving it would either tighten `v383`'s "every sector is the same gapped operator" to "the same *seam-driven* operator" (with the falsifiable corollary that no prime-2-only attractor exists), or reveal a missing dynamical slot. Tracked in `tfpt_research_contracts`; a question, not a claim. Cited in `origin_theory`; Python-only.

2026-06-24 (Correction error-bars: `v393` the typed budget as actual first-correction magnitudes)

- **`v393` (`CORRECTIONS.NUMERIC.01`) — the typed correction budget (`v388`) as actual numbers.** The short-term, testable harvest of `v388`: instead of merely typing each prediction's correction, this gives the magnitude, so a parameter-free central value can be quoted as "value ± computed first correction". **[E]** FIXED-POINT band (the φ_0-derived $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\rm rad}$) = the explicit φ_0 puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; $= \theta_{12}$'s ε first-correction, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$); **[E]** SEAM-GAPPED band (Koide $F_{\rm pole}$, recovery) $= (2/3)^6 \approx 8.78\%$, the QG-decoupling bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (`v337`); **[E]** EXACT-IDENTITY band = 0 structurally ($\det R = 8$, $N_\Phi = 1$, $\theta_{\rm eff} = 0$); **[E]** EXTERNAL-RATE band = external physics, quoted (m_p/m_e ±7.5% `v374`; n_s/r the $N_\star \in [50, 60]$ band; η_B 1.07× `v212`), fenced by `v187`. **[C]** the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in `origin_theory`, mirrored on the website prediction surface; a harvest of `v388`, no new number; Python-only.

2026-06-24 (Attacking the last open lemma: `v392` the scaling-limit central charge *converges*)

- **`v392` (`SEAM.S3.SCALINGLIMIT.01`) — an honest attack on `SEAM.EQUIV.01`'s one open lemma (continuum existence, `v336`).** It does *not* close `v336` (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable `v376` did not test: that the finite-size data *converge*. **[E]** the Calabrese–Cardy entanglement slope $c_1(L)$ at $L = 66, 130, 258, 386, 514$ is a monotone Cauchy-like sequence approaching 1 ($|c_1(L) - 1|$ strictly decreasing, $4.6 \times 10^{-4} \to 7 \times 10^{-6}$); **[E]** a $1/L$ Richardson fit extrapolates to $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\rm car} + N_{\rm fam}$ is reached *in* the limit; **[E]** the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). **[O]** convergent data are evidence, not a proof of existence (the cited MMST theorem, `v336`), and do not pin $(E_8)_1$ over the same-c $SO(16)_1$ (the holomorphy discriminator, `v377`/`v378`). Strengthens the residual, does *not* close it. Cited in `tfpt_research_contracts`; Python-only.

2026-06-24 (Lean 4: `FORM.SPECTRALGAP.01` extended to the finite-dimensional Perron–Frobenius case)

- **`FORM.SPECTRALGAP.01` — the multi-dimensional case, reduced to the scalar core.** The Lean module `TfptCarrier/SpectralGapAttractor.lean` now also proves the *finite-dimensional* gapped-operator statement (`lake build` OK, no `sorry`): **[E]** `component_tendsto` (a non-dominant eigencomponent $|r| < 1$ decays, $r^n c \to 0$); **[E]** `deviation_tendsto` (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); **[E]** `deviation_unique` (two starts give deviation vectors both → 0, so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); **[E]** `flavor_deviation_tendsto` (a concrete instance on the cusp-transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, `v56`/`v82`/`v383`). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in `origin_theory`.

2026-06-24 (Website: an animated "Universal Gap Lab" visualising `v383`–`v391`)

- **Website — the `UniversalGapLab` animation on `/verification`.** A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction $\mathrm{iter}_n = x_\star + r^n(x_0 - x_\star)$ racing to its attractor at $r = (2/3)^6$ (seam-gapped, `v303`/`v387`/`v388`) or the golden $(\varphi + 2)/4$ (`v312`), with a gapless $r = 1$ track that never forgets its start (the Lean theorem `FORM.SPECTRALGAP.01`) — the rate *is* the first-correction size (`v388`); (B) the entire-form-factor graviton ratio e^{-u} falling faster than any power, ghost-free and UV-soft (`v386`/`v389`); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms (`v384`). Mirror-only: every number is grounded in a module; no theory change.

2026-06-24 (Honest attempt at an external target: `v391` reduces it, not a closure)

- **`v391` (`ALPHA.QUILLEN.PROGRESS.01`) — an honest attempt at the external target `ALPHA.QUILLEN.EXACT.01` (`v382`, door 5).** It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: `ALPHA.QUILLEN.EXACT.01` stays **[O]** (external math, type A, `v384`). **[E]** a solvable model $\det_\zeta(-\partial_\tau^2 + m^2)$ on a circle of circumference β equals $(2\sinh(\beta m/2))^2$, whose m-variation $\beta\coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \ldots$ is a closed heat-kernel/Laurent series (the m^3 coefficient $-1/360$ verified). **[E]** so a ζ-determinant's coupling-variation is structured heat-kernel data, and `v382`'s counterterm $8b_1c_3^6$ (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the `v382` split is re-verified at $\alpha^{-1} = 137.0359992$. **[E]** the reduction: the open step now reduces to the named sub-target "the seam $U(1)$ operator's Seeley–DeWitt coefficient $= 8b_1c_3^6$". **[O]** *not closed*: the actual seam-operator ζ-determinant proof stays open, and `SEAM.EQUIV.01` continuum existence (MMST, `v336`) stays the other external **[O]**; the value α^{-1} stays **[E]**. An attempt that reduces, does not close. Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (Lean 4: `FORM.SPECTRALGAP.01` formalises the scalar core of the universal spectral-gap principle)

- **`FORM.SPECTRALGAP.01` — a Lean 4 proof of the gap ⇒ unique-attractor ⇒ parameter-free core (door 4).** New module `TfptCarrier/SpectralGapAttractor.lean` (`lake build` OK, no `sorry`, only the three standard kernel axioms), formalising the meta-theorem of `v303`/`v383`/`v387` at the one-dimensional reduction — the affine gap contraction $x \mapsto x_\star + r(x - x_\star)$ with multiplier $r = \lambda_2/\lambda_1$. **[E]** `iter_sub`: the closed form $\mathrm{iter}_n - x_\star = r^n(x_0 - x_\star)$ (general n); **[E]** `iter_tendsto`: $0 \le r < 1$ (the gap) ⇒ $\mathrm{iter}_n \to x_\star$ (the unique attractor); **[E]** `starts_agree`: two starts differ by $r^n(a - b) \to 0$, so the attractor is independent of the initial state (parameter-freeness); **[E]** `no_gap_no_forget`: $r = 1$ (no gap) leaves $\mathrm{iter}_n - x_\star = x_0 - x_\star$ constant — the initial condition is then a free parameter, so the gap is exactly what removes it; **[E]** the seam rate $(2/3)^6$ and the golden $(\varphi + 2)/4 = (5 + \sqrt{5})/8$ both lie in $[0, 1)$. Honest scope: the scalar reduction (the contraction core `v303`/`v383` iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in `origin_theory`; ledger row `FORM.SPECTRALGAP.01`.

2026-06-24 (Completing the Coxeter clock: `v390` the prime-2 (Gaussian / $\mathbb{Z}/4$-seam) facet of $30 = 2{\cdot}3{\cdot}5$)

- **`v390` (`COXETER.PRIME2.01`) — the prime-2 facet of the order-30 Coxeter clock.** `v383`/`v387` named two number-field facets of the clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ — the golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (prime-5, carrier) and $(2/3)^6 \in \mathbb{Q}$ (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field $\mathbb{Q}(i)$, i.e. the CM point $\tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ *square* seam deck (`v282`/`v288`/`v333`). **[E]** $h(E_8) = 30$ factors into exactly three primes; one quadratic facet field per prime, each ramified at its prime:

disc $\mathbb{Q}(i) = -4$ (2 ramifies), disc $\mathbb{Q}(\sqrt{-3}) = -3$ (3 ramifies), disc $\mathbb{Q}(\sqrt{5}) = 5$ (5 ramifies); the ramified primes are $\{2, 3, 5\}$ with product $30 = h(E_8)$. **[E]** the prime-2 (Gaussian) facet is the $\mathbb{Z}/4$ seam deck: min. poly $x^2 + 1$, $j(i) = 1728 = 12^3$, $\tau = i$ fixed by the modular S of order $4 = 2^2$, $|\mu_4| = 4 = 2^2$ — the static, even, seam-orientation facet. **[C]** so the three primes of the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set $\{2, 3, 5\}$ is forced by $h(E_8) = 30$, not chosen. Cited in `origin_theory`; a synthesis, no new number; Python-only.

2026-06-24 (Graviton loop finiteness: `v389` the entire form factor is UV-finite at the loop level by power counting)

- **`v389` (`GRAV.LOOP.FINITE.01`) — the entire-form-factor graviton is UV-finite at the *loop* level.** The honest next step past `v386` (which established the tree amplitude), extending `v304`/`v370`/`v380`/`v386` from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. **[E]** super-polynomial falloff: $p^k e^{-p^2/M^2} \to 0$ for every degree k. **[E]** one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$, while the GR $\int p^3/p^2\,dp$ diverges quadratically. **[E]** the superficial degree $\to -\infty$ at every loop order L (the form factor multiplies the integrand by $e^{-(\#\text{internal})p^2/M^2}$; dressed cutoff-ratio $\to 1$ vs GR $\to 4$). **[E]** the regulator is the seam scale $M = M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}}$ (`v253`/`v259`), not a dial. **[C][O]** loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is `v386`; not claimed solved. Cited in `tfpt_4_frontier`; Python-only.

2026-06-24 (Typed correction budget: `v388` the gap correction is keyed to the mechanism class, not a uniform band)

- **`v388` (`CORRECTIONS.BUDGET.01`) — the gap-driven correction (`v387`) is a *typed* budget.** A machine-checked answer to “does the $(\lambda_2/\lambda_1)^n$ correction apply to *all* predictions?” — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator’s leading eigenvector, it is defined only where a subleading λ_2 exists, so the prediction surface splits into four classes. **[E]** *seam-gapped* (Koide F_{pole} `v303`, recovery `v221`, the QG bound capped by $\chi = 729/665$ `v337`): rate $(2/3)^6$; the discrete compiler: golden $(\varphi + 2)/4$ (`v312`). **[E]** *exact identity* ($\det R = 8$, $N_\Phi = 1$, $\theta_{\text{eff}} = 0$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6}$): no λ_2, so the correction is 0 structurally and a band would contradict the **[E]** status. **[E]** *external-rate* (η_B washout, axion freeze, m_p/m_e RG): the gapped *shape* with an external rate — only F_{pole} has the seam rate (`v303`), firewalled (`v187`). **[E]** *fixed-point*: the texture is already explicit (e.g. $\sin^2\theta_{12}$, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$). **[E]** the Koide first correction (door 2): F_{pole} is the one seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (`v371`), the source quotient 0.33% below 2/3 — the rate is the suppression *scale*, not the residual. **[C]** so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of `v387` (as `v384` is of `v383`); cited in `origin_theory`; Python-only.

2026-06-24 (Harvesting the QFT/gravity machinery: `v385` the UV-branch kill-test surface, `v386` the graviton-exchange amplitude, `v387` the gap-driven correction calculus)

- **`v385` (`UVBRANCH.KILLTEST.01`) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (`v381`) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy.* **[E]** the carrier content is the SM with no new states, so the SM-non-unification (`v246`) is now all-order-robust; **[X]** the plain-SM 4D-GUT stays killed ($\sin^2\theta_W = 3/8$ vs 0.231); **[C]** the optional carrier-PS branch sits at $M_{PS} = \kappa M_{\rm scal} \approx 3 \times 10^{13}$ GeV ($\kappa \in [1.0, 1.2]$, `v253`); [**E**] minimal PS gauge bosons are $SU(4)$ leptoquarks that mediate rare LFV ($K_L \to \mu e$), *not* $p \to e^+\pi^0$, so the branch clears the binding bound by $M_{PS}/M_{\rm bound} \sim 10^7$ – proton-safe, with *no* fake τ_p window (`v265`). Cited in `tfpt_5_redteam`; Python-only.

- **`v386` (`GRAV.AMPLITUDE.01`) — the graviton-exchange amplitude is finite and UV-soft.** Extends `v304`/`v370`/`v380` from the propagator to the actual tree amplitude. [**E**] the dressed spin-2 propagator $e^{-p^2/M^2}/p^2$ has its only pole at $p^2 = 0$ (residue $1 > 0$, ghost-free), is exponentially softened in the UV (the ratio e^{-u} falls faster than any power) and reduces to GR + a calculable $-p^2/M^2$ correction at low energy ($M = M_{\rm scal}$, `v253`); [**C**] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering $A \sim (T{\cdot}P_2{\cdot}T')e^{-p^2/M^2}/p^2$ is an explicit finite computation (perturbative-only; `QG.AMB.01` stays a [**C**] redundancy, `v369`). Cited in `tfpt_4_frontier`; Python-only.
- **`v387` (`CORRECTIONS.GAP.01`) — the gap-driven correction calculus.** The practical harvest of `v383`: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction ($\text{correction}_n \sim (\lambda_2/\lambda_1)^n$). [**E**] flavor, horizon recovery and the QG bound all share the $(2/3)^6$ rate (the family-3 facet; QG susceptibility $\chi = 729/665$), while the discrete compiler transient decays at the golden $(\varphi + 2)/4$ (the carrier-5 facet) — the two number-field facets of `v314`/`v383`. [**C**] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in `origin_theory`; Python-only.

2026-06-23 (Bird's-eye synthesis: two new structural patterns — `v383` the universal spectral-gap principle and `v384` the residual is certification, not construction)

- **`v383` (`DYNAMICS.UNIVERSAL.01`) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird's-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). `v303` named this only for $F_{\rm transfer}$; this module extends it to the sectors made parameter-free *after* it. [**E**] three gaps are computed — the flavor transfer $\{1, (2/3)^6, (1/3)^6\}$ with gap $6\ln(3/2)$ (`v56`/`v82`); the affine-E_8 adjacency with Perron eigenvalue 2 (Kac marks) and subleading $2\cos(\pi/5) = \varphi$, gap $2 - \varphi = 1/\varphi^2$ (`v312`/`v313`); the decoupling margin $\Delta_{\rm eff} \approx 1.648$ (`v76`/`v337`). [**E**] the two number-field facets (`v314`): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$. [**C**] the other sectors are the same structure — gravity (δS=0 equilibrium, `v358`/`v359`), the QG measure (Gaussian saddle, `v365`), the boundary QFT (holomorphic c=8 RG fixed point, `v157`/`v158`), the all-order S-matrix (adiabatic limit, `v381`), recovery (Petz fixed point, `v221`), τ=i (`v333`). [**C**] the meta-theorem: gapped $\Rightarrow$ unique attractor $\Rightarrow$ parameter-free, theory-wide. Cited in `origin_theory`; a synthesis (no new number), Python-only.
- **`v384` (`RESIDUAL.CERTIFICATION.01`) — the residual is certification, not construction.** After `v369`/`v381`/`v382` the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [**E**] (A) external math proof (`SEAM.EQUIV.01` continuum existence = the cited MMST/Adamo theorem `v336`; `ALPHA.QUILLEN.EXACT.01` `v382`; `QG.AMB.01` the constructive measure, a [**C**] redundancy `v369`), [**E**] (B) theorem-forbidden ($v_{\rm geo}$, the No-Unit theorem `v153`/`v364`), [**E**] (C) external physics ($F_{\rm transfer}$, `v371`–`v374`, firewalled `v187`). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), and the count of open *internal* TFPT mechanisms is **zero**. [**C**] convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in `tfpt_research_contracts`; an audit (no new number), Python-only.

2026-06-23 (External-review closure: the two genuine residual targets named — `v381` the all-order Epstein–Glaser/BRST contract for S_{pert}, and `v382` the Quillen determinant-line variation as a tracked target)

- **`v381` (`QFT4D.EG.ALLORDER.01`) — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix S_{pert} (the previous documents had the S_{pert} skeleton and one-loop, `v269`/`v271`/`v273`/`v278`, but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [**E**] power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order $\omega \leq 4$ and the counterterm space is finite at every order; [**E**] BRST nilpotency $s^2{=}0$ verified exactly via the Jacobi identity for $su(2)$ and $su(3)$ (the carrier weak+colour content); [**E**] the seam gap $\Delta = 6 \log \frac{3}{2} > 0$ (`v302`) gives the IR-safe adiabatic limit. [**C**] the two heavy legs are imported rigorous theorems — all-order T_n existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the R^2/Weyl^2 gravity sub-sector is the entire-form-factor resummation (`v304`/`v370`/`v380`), and S_{pert} is not the ambient measure `QG.AMB.01` (a [**C**] redundancy, `v369`). NET [**C**], prerequisites [**E**]; not [**E**] and not a path integral. Cited in `tfpt_5_redteam` and `tfpt_2`; Python-only.
- **`v382` (`ALPHA.QUILLEN.EXACT.01`) — the Quillen determinant-line variation, named as a target.** Per the same review, the "why *this* $F_{U(1)}$ functional" obligation (previously tracked only as `EM.WARD.01`'s residual, `v341`/`v342`) is elevated to its own citable target: $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log \varphi_{\text{seam}}) = 0$ on the determinant line over the μ_4 seam moduli. [**E**] the Quillen split holds at the $\alpha^{-1}{=}137.0359992$ root with every coefficient a named atom (index $8b_1$, heat-kernel $c_3^{3,6}$, discriminant $q(D_5){+}q(A_3){=}2$); [**E**] the value stays closed (`v3`); [**O**] the from-first-principles exactness proof is the target. [**C**] it is a *face* of `SEAM.EQUIV.01` (`v336`) — the seam $U(1)$ determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored `v341` identities). Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)

- **`v369` (`QGAMB.REDUNDANCY.01`) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, `v332`/`v334`), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/`QG.AMB.01`) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. [**E**] holomorphy $\Rightarrow$ $\mathrm{DHR}((E_8)_1) = \mathrm{Vec}$: $\#\text{primaries} = |\det \mathrm{Cartan}| = 1$ for E_8 vs 4 for $SO(16)$ $\Rightarrow$ no nontrivial superselection charge a bulk could carry invisibly; [**E**] no torus ground-state degeneracy ($|\det K| = 1$); [**E**] finite Petz recovery rate $(2/3)^6$ (`v221`); [**E**] the gap-decoupling margin $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (`v76`/`v337`). [**C**] with the Bisognano–Wichmann reconstruction map (`v258`/`v309`/`v323`) the redundancy statement $\mathcal{O}^{\text{bulk}}_{\text{phys}}/\mathrm{Diff} \cong \mathcal{A}^{\mu_4}_{\Sigma}$ is well-posed. [**O**] residual: the two consumed premises — `SEAM.EQUIV.01` (the S3 lattice input, `v366`) and the BW intrinsicality of the raw collar (`v329`); it does *not* construct `QG.AMB.01` but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Status propagation: the v365–v380 + FORM.SEAM.MMST.01 status carried into the remaining papers, the README and the Zenodo description)

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, `introduction`) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1`–`tfpt_5`, `tfpt_research_contracts`, the root `README.md`, the Zenodo description and `website/lib/papers.ts`: `SEAM.EQUIV.01` is *closed modulo a cited theorem* (lattice `v367`/`v368` + the S_3 stack `v376`–`v379` + Lean `FORM.SEAM.MMST.01`; residual **[O]** = the cited MMST/OS existence only); `QG.AMB.01` is a **[C]** *redundancy* (`v369`/`v379`), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, `v304`/`v370`/`v380`); and the frontier transfers are typed runnable solvers with kill tests (`v371`–`v374`) + the observatory CI (`v375`). No new claim — a wording propagation so every surface matches the ledger; `AUDIT OK`.

2026-06-23 (v380 — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)

- **`v380` (`GRAV.KMS.HESSIAN.01`).** Upgrades `v304`/`v370` from the *assumption* "the resummed graviton form factor is entire" to a derived statement. **[E]** the seam KMS cutoff $f(u)=e^{-u}$ (`v259`) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u)=e^u$ entire and nowhere zero (only the $p^2=0$ pole). **[E]** each finite Seeley–DeWitt truncation is a Taylor partial sum of a and carries a Stelle-ghost zero ($T_1=1+u \to u=-1$; $T_2 \to -1\pm i$); the entire a has none. **[E]** the modulus of the nearest truncation-zero grows monotonically with the order $(1, 1.41, \ldots, 3.07$ for $n=1..8) \to \infty$ — the ghost decouples, "entire" = "do not truncate". **[C]** $\Rightarrow$ perturbative graviton unitarity; **[O]** the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

2026-06-23 (Lean: the S3 continuum leg formalised as "closed modulo a cited theorem" — FORM.SEAM.MMST.01)

- **`FORM.SEAM.MMST.01` — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms): $D = 16$, rank $= c = 8$, the range $8 \leq 8 \leq 16$, $c_- = 8 \neq 0$, and $\det K = 1$ vs $SO(16)$'s 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion $(E_8)_1$ follows by a well-typed composition whose residual (`#print axioms`) is exactly those two published theorems plus the carrier gap — no hidden assumption, no `sorry`. Mirrors the S3 closure stack `v376`–`v379`; so `SEAM.EQUIV.01` is now "closed modulo a cited theorem", the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge $c=8$ from the lattice (v376), the $(E_8)_1$ character (v377), the genus-1 torus count $= 1$ (v378), and reflection positivity (v379); only the abstract continuum existence (cited MMST) stays external)

- **`v376` (`SEAM.S3.CENTRALCHARGE.01`).** The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives $c = 1.0000$ per complex mode (Peschel correlation-matrix EE; the $L=4m+2$ sizes avoid the half-filling zero-mode), so the 16-Majorana ($= 8$ complex) collar has $c = 8 = g_{\text{car}}+N_{\text{fam}}$, chiral ($c_-=8$, `v367`/`v368`). Numerical $c=8$ from the lattice. Python (numpy).

- `v377` (`SEAM.S3.E8CHARACTER.01`). The exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ gives $c = 8$ and 248 level-1 currents (one primary); the μ_4/GSO promotion $248 = 240 + 8 = 120 + 128$ ($240 = 16{\cdot}5{\cdot}3$) pins $(E_8)_1$ over the same-c rival $SO(16)_1$ (120 currents, 4 primaries). Python (sympy q-series).
- `v378` (`SEAM.S3.MODULAR.01`). The genus-1 discriminator: KLM condensation μ(carrier)=16 $\to$ $\mu(E_8)$=16/4^2=1, $|\det \mathrm{Cartan}(E_8)|$=1 vs D_8=4, and the single character's modular T-eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 (holomorphic = $(E_8)_1$) — the "hard part" `v344` isolated. Python (sympy).
- `v379` (`SEAM.S3.RP.01`). Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With c=8 + $(E_8)_1$ in hand, the OS inputs are complete and C7/`QG.AMB.01` is then discharged by the redundancy theorem `v369`. Python (numpy).
- **Net:** S3 is the master key — on the explicit lattice model the target net $(E_8)_1$ is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem `v336`), which lives outside the computational suite. Cited in `tfpt_research_contracts`.

2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (`v375`))

- `v375` (`OBSERVATORY.REGISTRY.01`). A CI over `freeze_file.csv`: [**E**] every prediction carries a kill criterion (falsifiability complete, no orphan claims); [**E**] the headline values re-derive from $\{\varphi_0^{\mathrm{ret}}\}$ ($\sin^2\theta_{12}$=1/3$-\varphi_0^{\mathrm{ret}}$/2, $\sin^2\theta_{13}$=$\varphi_0^{\mathrm{ret}}e^{-5/6}$, β_{rad}=$\varphi_0^{\mathrm{ret}}/(4\pi)$); [**C**] a live scorecard records θ_{12} vs the JUNO first run (0.28σ, compatible), θ_{13} vs NuFIT 6.0 NO (2.0σ, the flagged most-tensioned core prediction), β_{rad} vs ACT DR6 (0.37σ, a hint), $r < 0.036$ (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in `tfpt_5_redteam`; Python (stdlib).

2026-06-23 (Track 3 — the F_{transfer} functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source$\to$pole (`v371`), η_B Boltzmann (`v372`), axion finite-T relic (`v373`), m_p/m_e uncertainty budget (`v374`))

- `v371` (`FR.POLE.SOLVE.01`). A clean *negative*: standard QED/EW running cannot be the Koide source$\to$pole transfer (it pushes Q above 2/3; the required ϵ is negative while QED's is positive), so the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (`v183`/`v82`). [**C**], plain running excluded as the kill. Python (numpy+scipy).
- `v372` (`FR.BOLTZMANN.SOLVE.01`). The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen M_1=$M_{\mathrm{scal}}(\varphi_0^{\mathrm{ret}})^2/A_\Lambda$ (`v212`) gives η_B within a factor $\sim$1.1 of 6.1×10^{-10} with no M_R dial; replaces the `v326` toy washout. [**C**]/[**X**].
- `v373` (`FR.RELIC.SOLVE.01`). The finite-T axion misalignment ODE decides the branch: the frozen spine angle $3\pi/5$ gives $\Omega_a h^2$$\sim$0.12 (in [0.08, 0.16], lands on Ω_{DM} untuned) while the hilltop 170° over-produces ($\sim$5.4$\times$); replaces the `v326` toy. [**C**]/[**X**](lattice $\chi(T)$ input).
- `v374` (`FR.QCD.BUDGET.01`). The m_p/m_e uncertainty budget: with $b_3 = -7$ [**E**] and the two declared externals ($\alpha_s(M_Z)$, C_p, $\sim$5.3% each), the central $\sim$1824 (`v262`) carries a propagated $\pm$7.5% band [$\sim$ 1690, $\sim$ 1960] CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to `v262`; [**C**].

2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana ($p+ip$) lattice model with numerical Chern $|C|=1$ realises the $c_-=8$ edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))

- `v367` (`SEAM.S3.LATTICE.01`) — **Track 1.** The abstract S3 residual of `v356/v366` ("the collar realised as a genuine lattice chiral free-fermion invertible phase") is turned into a RUNNABLE model. [**E**] the $p+ip$ Bloch model is gapped at $M=1$ with a numerically computed Fukui–Hatsugai–Suzuki Chern number $|C|=1$ ($C=0$ trivially at $M=3$); [**E**] $N_{\mathrm{Maj}}=\dim S^+=16$ copies give $c_-=16/2=8=g_{\mathrm{car}}+N_{\mathrm{fam}}$, with the edge K-matrix $SO(16)_1$ $(\det 4) \to (E_8)_1$ $(\det 1)$ by the order-4 μ_4 clock (`v351/v369`). [**C**] the μ_4 condensation (`v154/v351`); [**O**] the continuum scaling-limit existence (MMST, `v336`) stays open. Cited in `tfpt_research_contracts`; Python (numpy + sympy).
- `v368` (`SEAM.S3.INFLOW.01`) — **Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the `v367` model: [**E**] a finite strip has an edge-localised in-gap chiral branch ($\min_{k_x}|E|\sim 0$) in the topological phase and a clean gap in the trivial phase (neg control); [**E**] bulk-edge correspondence gives one chiral Majorana per edge ($c_-=8$). [**C**] so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the `v336/v351` "does the edge exist?" residual on the lattice. [**O**] the continuum scaling limit stays open. Cited in `tfpt_research_contracts`; Python (numpy).

2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))

- `v370` (`GRAV.SPIN2.UNITARITY.01`) — **Track 2.** Extends `v304` (the scalar $1/(p^2(p^2+M^2))$ pole algebra) to the actual tensor spin structure. [**E**] the Barnes–Rivers projectors are complete ($\operatorname{tr} P_2=(d-2)(d+1)/2$, $\operatorname{tr} P_1=d-1$, $\operatorname{tr} P_{0s}=\operatorname{tr} P_{0w}=1$ sum to $d(d+1)/2$; $d=4$ split $5+3+1+1=10$), the EH propagator is $P_2/p^2-\frac{1}{d-2}P_{0s}/p^2$, and the Stelle ghost is a *spin-2* state. [**E**] the entire KMS form factor e^{p^2/M^2} (`v304/v259`) makes the spin-2 sector ghost-free; [**C**] the wrong-sign spin-0 conformal mode (`v332`) is cured separately by the GHP contour (`v334`). [**C**] so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on `v304`'s entire-analyticity assumption). [**O**] residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. [**O**] DECLINE: $\operatorname{tr} P_2=5$ is dimension counting, not g_{car} (`v354/v355`). Cited in `tfpt_5_redteam` (Target F); Python-only.

2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST's free-fermion class so the scaling limit is $(E_8)_1$ (v366))

- `v365` (`QGAMB.SADDLE.01`) — **Direction 3.** With the parameter-free saddle (`v358/v359`) the QG *measure* (gate C7/`QG.AMB.01`) organises as a one-loop Gaussian fluctuation determinant. [**E**] the fluctuation stiffness is $1/c_3=8\pi$ (no free dial, so the one-loop problem is now well-posed); [**E**] the fluctuation operator is gapped $M\geq\Delta_{\mathrm{eff}}=6\log\frac{3}{2}-\frac{31}{4\pi^2}\approx 1.648>0$ (`v76/v337`) $\Rightarrow$ mode-by-mode convergence, finite model log-det $\operatorname{Tr}'\log M=6\log\frac{9}{2}$; [**C**] the conformal mode converges on the GHP contour (`v334`); [**E**] the admissible projective limit exists ($\chi=729/665$, `v330`) with 0 new dials (`v364`). [**O**] residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from "diffuse full QG" to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in `tfpt_research_contracts`; Python-only.

- `v366 (SEAM.MMST.INCLASS.01)` — **Direction 7.** The seam collar is verified IN MMST's free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: **[E]** $D{=}\dim S^{+}{=}16$ Majoranas, $c{=}g_{\rm car}{+}N_{\rm fam}{=}8$, $\operatorname{rank} E_8{=}8$ so rank $\le c \le D$ is $8 \le 8 \le 16$ (in range); **[E]** gapped ($\Delta{=}6\log\frac{3}{2}{>}0$, derived `v302`); **[C]** quasi-free CAR class (`v155/v160`); **[E]** chiral ($c_{-}{=}D/2{=}8 \neq 0$). **[C]** the MMST scaling-limit theorem then applies, and **[E]** the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K{=}1$ vs $SO(16)$ $\det K{=}4$, `v351`). **[O]** residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, `v297/v356`). Reduces the continuum side of `SEAM.EQUIV.01` to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (`v358/v359`) propagated across ALL stale paper and website surfaces; new `v359` keybox; homepage gravity animation)

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as "exact only in the $R + R^2$ regime" or the metric-sector field equation as "open". These were reconciled with the canonical post-`v358/v359` status: the *local* covariant field equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (`QG.AMB.01`/C7), and the single unit $v_{\rm geo}$. Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New `v359` keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume $\Rightarrow$ Einstein tensor, Lovelock $\Rightarrow$ conservation an output); the `v358` residual corrected from "linear$\to$nonlinear" (stale — `v359` closed it) to the equation-of-state + C7 + $v_{\rm geo}$ residual, including the generated script index (`script_registry.csv`).
- **Website.** The `GravityEmergence` animation (three origins of c_3 converging to $1/(8\pi)$; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.

2026-06-23 (Direction 6, $v_{\rm geo}$ sharpened: after the parameter-free gravity, the *gravity* sector adds NO new dimensionful input either, so $v_{\rm geo}$ is the SINGLE dimensionful input of the complete theory — final tally $\{v_{\rm geo}, \pi\}$, ZERO dimensionless dials: `v364`)

- `v364 (VGEO.SHARPEN.01)` — **Direction 6.** The No-Unit Theorem (`v153`) already made $v_{\rm geo}$ a forced metrology primitive. **[E]** The SHARPENING after the parameter-free gravity (`v358/v359/v361`): the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\rm geo}^2$ and $\Lambda \sim$ (dimensionless) $v_{\rm geo}^4$ reduce to $v_{\rm geo}$ times atoms — the gravity sector adds no new scale. **[E]** FINAL TALLY: every dimensionful quantity = (a dimensionless ratio)$\times$(a power of $v_{\rm geo}$); TFPT's free content is $\{v_{\rm geo}, \pi\}$ with ZERO dimensionless dials, a $\sim$26 $\to$ 1 reduction vs the SM. **[O]** $v_{\rm geo}$ stays irreducibly primitive (No-Unit; $1/G$ UV-sensitive, Sakharov `v68`). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier's gravitational anomaly is forced c_-=8, and the matter vacuum-energy backreaction is finite (→ the forced Λ from α, not $M_{\rm Pl}^4$) — the loop closes; no new SM-quantum-number read-out (declined): `v361`)

- `v361 (GRAV.BACKREACT.01)` — **Direction 5.** With the parameter-free equation (`v359`) and the explicit J3 flux (`v358`), the backreaction of TFPT's carrier matter is computed. **[E]** the carrier's gravitational anomaly is forced: $c_- = 16/2 = 8 = g_{\rm car}+N_{\rm fam}$ (the chiral central charge of the 16-Majorana content). **[E]** the backreaction is FINITE → Λ from α: the matter vacuum energy through $G_{ab} = c_3^{-1}T_{ab}$ is gap-suppressed (Decoupling Thm `v337`) to $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (`v60`), the 123 orders $= 119.03+3.92$ — *not* the $M_{\rm Pl}^4$ catastrophe; the loop (`v359`+`v60` + the carrier vacuum) closes. **[E]** J3 explicit: $\delta\langle K_B\rangle = (8\pi^2R^4/15)\delta\langle T_{00}\rangle$, carrier T_{ab} = the SM's (`v159`). **[C]** the SM trace anomaly is reproduced (content = SM). **[O]** DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond c_-=8 and Λ — per `v354/v355` declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known R^2 scalaron — already falsifiable via n_s/r; a *new* forced near-term deviation does NOT survive the discriminator and is declined: `v360`)

- `v360 (GRAV.GAPCORR.01)` — **Direction 2, run honestly.** Does the seam transfer gap $\{1,(2/3)^6,(1/3)^6\}$ yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known R^2 scalaron? With the `v354/v355` discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. **[E]** the leading higher-curvature term IS the R^2 scalaron (spectral-action a_4=$R^2/72$, `v36`; $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^7$, exponent 7=$g_{\rm car}$+$N_{\rm fam}$−1) — the calculable beyond-Einstein correction, already in TFPT. **[C]** it is the entanglement-equilibrium NEXT order (the Wald first law → $f(R)$=R+R^2, `v28/v359`). **[E]** it is ALREADY falsifiable via n_s=1−2/$N_\star$, r=12/$N_\star^2$ (live CMB-S4 kill test). **[E]** the gap Δ=$6\ln\frac{3}{2}$ is a dimensionless rate, so a distinct gap-induced curvature^2 term sits at $\xi\, v_{\rm geo}$ ∼ the seam/Planck scale (not near-term observable). **[O]** DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent 7=g+N−1 and the gap exponent 6=2$N_{\rm fam}$ are different; c_3^7 and $(2/3)^6$ unrelated) — per `v354/v355` declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives $G_{ab}+\Lambda g_{ab} = c_3^{-1}T_{ab}$ with BOTH coefficients TFPT-fixed ($8\pi = 1/c_3$ from `v358`, Λ from α via `v60`); B6 closed at the local level: `v359`)

- `v359 (GRAV.NONLINEAR.01)` — **Direction 1, the full covariant equation.** Extends `v358` from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. **[C][E]** Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}) — trace $g^{ab}G_{ab} = (1-d/2)R = -R$ in d=4, the structure that carries Λ. **[E]** Lovelock: G_{ab} is the unique symmetric divergence-free second-order metric tensor, so $\nabla^a(G_{ab}+\Lambda g_{ab}) = 0$ forces $\nabla^a T_{ab} = 0$ — conservation is an output. **[E]** coefficient 1: $8\pi G \to 8\pi = 1/c_3$ fixed (`v358`). **[E]** coefficient 2: Jacobson leaves Λ free, TFPT fixes it — $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ (`v60`); so BOTH coefficients are TFPT-fixed, not free integration constants. **[C]** all timelike directions → the covariant tensor equation; the matter side is the assembled CHM ball modular flux (`v358/v323`). NET: the original B6 "full covariant Einstein-side field equation" is now PARAMETER-FREE at the local level. **[O]** residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/`QG.AMB.01`, separate, gap-decoupled); v_{geo} the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (303/303). (`next-plan.md` Direction 4 folded in: the Λ here matches the α-readout.)

2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law $\delta S = \delta\langle K\rangle$ with TFPT's atoms gives the LINEARISED Einstein equation with G from c_3; the thermodynamic and geometric origins of c_3 coincide via $|\mu_4| = |\mathbb{Z}_2|\chi = 4$; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: `v358`)

- **`v358` (`GRAV.ENTROPY.EQUILIBRIUM.01`) — the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. "first law of entanglement ⇒ Einstein equation" with TFPT's atoms. **[E]** PARAMETER-FREE COEFFICIENT: $\delta S = \delta\langle K\rangle$ gives $G_{ab} = c_3^{-1} T_{ab}$ with $c_3^{-1} = 8\pi$ fixed; all four c_3-roles consistent (Unruh $1/(2\pi)$=$4c_3$, EH $1/(16\pi)$=$c_3/2$, Einstein 8π=$1/c_3$, Bekenstein $1/4$=$1/|\mu_4|$) — no free dimensionless Newton dial. **[E]** THERMO = GEO COINCIDENCE (new): the first law's coefficient $2\pi/\eta$ equals the geometric $|\mathbb{Z}_2|2\pi\chi$ iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (`v216`) — the thermodynamic and geometric origins of c_3 are ONE. **[C]** J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, `v323`) gives the matter flux $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\delta\langle T_{00}\rangle$. **[E]** ENTROPY DENSITY ATOM-FIXED: $1/4$=$1/|\mu_4|$ (`v57`) + central charge $c = g_{\text{car}}$+$N_{\text{fam}} = 8$ (form `v59`). **[O]** residual: linear→full non-linear (the original B6) + the v_{geo} unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (302/302).

2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of `SEAM.EQUIV.01` theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by `v40` (harmonic metric = finite linear algebra, not a PDE): `v356`)

- **`v356` (`SEAM.EQUIV.CONTINUUM.03`) — Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped (Δ=$6\ln(3/2)$>0, `v302`) quasi-free 16-Majorana collar (`v155`/`v160`) → **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, `v301`) → **S3** the *nontrivial* invertible phase (c_-=8≠0) → **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk–edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly reduces the `v336`/`v351` "does the collar have a chiral edge?" residual → **S5** its chiral-CFT scaling limit (MMST, `v336`; range rank $\le c \le D$, $8 \le 8 \le 16$) → **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (`v351`/`v297`). **[E]** the only non-theorem link is S3, and **[C]** S3 is the seam being *one-sided* (chiral, c_-≠0) — the *same* one-sidedness that makes c_3=$1/(8\pi)$ a one-sided Gauss–Bonnet ($|\mathbb{Z}_2|\oint K$, `v58`). **[O]** the one remaining input is "the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase" (the `v297` Flat-Away). NET: `SEAM.EQUIV.01` reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.
- **Direction B — found ALREADY closed by `v40`.** The U-wall "transcendental Hitchin solve" (`v31`) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta–Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs Φ=0) and $\det R$=8 is read off — not a PDE. Only the absolute amplitude U_{point} (an anchor) remains. A redundant new B module was built, found to duplicate/contradict `v40`, and *deleted* (no padding). Honest outcome: B needs no new work.

2026-06-22 (bandwidth search of the unmapped E_8 region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ($\sum \deg = 128 = \dim S^+$, $\prod \deg = |W(E_8)|$, exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling `v66` and `v354`: `v355`)

- **`v355` (`E8.UNMAPPED.BANDWIDTH.01`) — the next step after `v354`, run honestly.** The obvious follow-up ("search those five degrees for hits we overlooked") is itself the numerology temptation `v354` warns about: a rich object like E_8 returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about E_8), never a coincidence. [**E**] FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number): $\sum \deg = 128 = \dim S^+$, the spinor half of $248 = 120 + 128$ (theorem: $\sum \deg = \#\text{pos roots}+\text{rank}= 120 + 8$) – the invariant budget of E_8 equals the fermion content the carrier reads; $\sum \exp = 120 = \#\text{pos roots}$; $\prod \deg = |W(E_8)| = 696{,}729{,}600$; exponents = the $\varphi(30)$=8 totatives of 30 (`v223`), so the unmapped degrees' exponents are part of the forced set; $\max \deg = 30 = h(E_8)$. [**C**] SUGGESTIVE, not a theorem: 12=$h(E_6)$, 18=$h(E_7)$, 30=$h(E_8)$ are all degrees of E_8 (the $2T/2O/2I$ McKay ladder, `v219`), but "deg(E_8) contains its subalgebras' Coxeter numbers" is not a general theorem – flagged, not claimed. [**O**] DECLINED (the numerology temptation): the per-degree atom matches 14=$\dim G_2$, 20=$\det L$, 24=$|W(A_3)|$, 12=$|R(A_3)|$, 18=$p_4(a)$ (`v66` already flagged $12, 18$ as fittable) each admit ≥ 2 readings, and the lone extra prime in $|W(E_8)| = 2^{14}3^5 5^2\, 7$ "=" the scalaron exponent is one of many readings of 7 – all declined. [**E**] RECONCILES `v66` (the 8 degrees coincide with the load-bearing integers) and `v354` (only $2, 8, 30$ are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens `v354`: the five are "no PRIMARY readout / hull overhead", not literally "unmapped" – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (`301/301`).

2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of E_8's 8 Casimir invariants only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: `v354`)

- **`v354` (`E8.REVERSE.AUDIT.01`) — the reverse complement of `v305`.** TFPT's forward discipline (`v305`) is one-directional: every physical READOUT must map onto an E_8 structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much E_8 structure maps onto NOTHING, and is the recurring golden ratio numerology? [**E**] THE REVERSE AUDIT: of E_8's eight Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ (exponents +1), exactly THREE feed a readout — 2 (metric), 8 (rank $\to c_3$=$1/(8\pi)$), 30 (Coxeter $\to g_{\text{car}}$=5 and the order-30 cycle) — and FIVE carry no primary readout $\{12, 14, 18, 20, 24\}$ (they coincide with structural atoms per `v66`, but unforced – reconciled in `v355`); so 3/8 are primary readouts, 5/8 hull overhead. [**E**] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, h=30, $\det K$=1, $D_5{\oplus}A_3$, the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of E_8 untouched, the anti-numerology signature being precisely the large unused remainder. [**E**] THE GOLDEN RATIO IS UNMAPPED $\Rightarrow$ NOT NUMEROLOGY: $\varphi = 2\cos(\pi/5)$ appears only in internal structure (the affine-E_8 attractor eigenvalue `v312`, the icosian lattice `v348`) and in NO physical readout; numerology means a number FITTED to an observation, φ is fitted to nothing — it is the algebraic signature of the 5-fold (h=30=$2 \cdot 3 \cdot 5$), the carrier announcing it is icosahedral. [**O**] the 5/8 unmapped is the HULL OVERHEAD (E_8 is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly, not hidden. Cited in `tfpt_5_redteam`; mirrored in Wolfram (`300/300`).

2026-06-22 (the bird's-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system; π is a fixed geometric constant, not a dial; the one residual is the continuum closure: `v353`)

- `v353` (`TFPT.SELFLOOP.01`) — **refining `v352`.** Stepping all the way back: TFPT is not a LINEAR theory (axioms → theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. [**E**] it is a loop: $g_{\rm car}$=5= max prime of the output $h(E_8)$=30, and the 8 in c_3= rank E_8 – the outputs fix the inputs (`v6`), with a unique fixed point (`v56`); "which is the axiom?" is the wrong question for a loop. [**E**] ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. [**E**] π is a FIXED geometric constant (the boundary 2-sphere, $\oint K$=4π), not a dial. [**C**] this refines `v352`'s "framework + π": π is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is "a unique self-consistent loop with zero free dials". [**O**] the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, `v336`/`v351`); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

2026-06-22 (two consequences: the c=8 which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework + π: `v351, v352`)

- `v351` (`SEAM.EQUIV.CONTINUUM.02`) — **the which-net ambiguity falls.** The long-standing "c=8 ambiguity" (E_8 det 1 vs $SO(16)$=D_8 det 4, both c=8, flagged open in `v277`/`v344`) is RESOLVED, non-circularly, by the seam's ORDER-4 μ_4 clock: the index-4 simple-current extension of $D_5 \oplus A_3$ is E_8 (`v154`), the index-2 would be $SO(16)$, so the order-4 clock (the four Gauss–Bonnet marks, `v216`) selects E_8 (full μ_4 condensation 16→4→1) not $SO(16)$ (partial $\mathbb{Z}_2$ 16→4). The bootstrap agrees ($h(E_8)$=30 max prime 5=$g_{\rm car}$ vs $h(D_8)$=14 max prime 7). [**E**] so det K=1 is a consequence, not an open input. [**O**] the residual of `SEAM.EQUIV.01` is then PURELY the chiral-edge / scaling-limit existence (`v336`), not which-net.
- `v352` (`TFPT.IRREDUCIBLE.01`) — **the inputs are not axioms.** The deepest answer to "the inputs are back-determined": BOTH P1 and P2 are bootstrap-forced fixed points. [**E**] $g_{\rm car}$=5 is forced three ways (`v6`); the 8 in c_3 four ways (rank E_8 = $h(D_5)$ = $\phi(30)$ = det R, `v54`); the E_8 hull is forced (unique even unimodular rank-8, `v83`/`v154`). [**E**] so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental π. [**O**] the framework is the META-AXIOM — not math-reducible, the foundational stance (like "spacetime is a manifold"), minimal, tested by its consequences not proved. TFPT's true input is ONE framework + π. Wolfram mirror +2 (297 → 299).

2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms – $g_{\rm car}$=5 IS the icosahedral 5 of $h(E_8)$=30, and the golden ratio is emergent from the μ_4-glue, not external: `v350`)

- `v350` (`SEAM.EQUIV.BOOTSTRAP.01`) — **correcting `v349`.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. [**E**] the inputs ARE bootstrap-forced (`v6`): $g_{\rm car}$=5 is over-determined three ways – rank-fill ($g_{\rm car}$+$N_{\rm fam}$=8), Coxeter-match ($g_{\rm car}$= max prime of $h(E_8)$=30), Pascal (2^4=$C(5,0)$+$C(5,1)$+$C(5,2)$). [**E**] so $g_{\rm car}$=5 IS the icosahedral 5-fold (the 5 in $h(E_8)$=30=2·3·5), not the D_5 rank; the atoms $(2,3,5)$ ARE the primes of the icosahedral Coxeter number 30. [**E**] the golden ratio is EMERGENT from the glue: D_5 (h=8)/A_3 (h=4) are non-golden alone, but $D_5 \oplus A_3 \to E_8$ (μ_4 index-4, `v154`) gives $h(E_8)$=30 (golden) – the seam's own μ_4 produces the 5-fold, so `v349` checked the un-glued pieces, the wrong object, and its pentagon-vs-count dichotomy was false. **[O]** the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (`v336`), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296 → 297).

2026-06-22 (the honest test – does the raw seam carry φ? – answer NO: the bare carrier gives $\sqrt{2}$, golden is the icosahedral input; the keystone reduces to "is g_{car}=5 a pentagon or a count?": `v349`)

- **`v349` (`SEAM.EQUIV.GOLDEN.01`).** The honest, computed test of the `v348` question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. **[E]** the golden ratio $2\cos(\pi/5)$ is the signature of a 5-fold rotation ($5 \mid h$); the carrier D_5 has h=8 (eigenvalue $\sqrt{2}$) and A_3 has h=4 – no 5-fold; the dynamical data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed φ_0) are rational/transcendental. **[E]** $5 \mid h$ holds only for A_4=$SU(5)$ (h=5), H_3 (h=10), E_8 (h=30) – the output/icosahedral side, so φ is the icosahedral INPUT. **[E]** φ alone would not suffice ($2I \neq C_5 \times \mu_4$, $20 \neq 120$). **[O]** the resolution: the keystone reduces to one question – is g_{car}=5 a PENTAGON (a 5-fold rotation ⇒ golden ⇒ icosahedral $2I = E_8$) or just a COUNT (the rank of $D_5 \Rightarrow \sqrt{2}$)? The "absurdly simple solution" is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of g_{car}=5, not derivable from g_{car}=5-as-a-count); there is no hidden φ. L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295 → 296). A NEGATIVE/clarification; prevents a false closure.

2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: `v348`)

- **`v348` (`SEAM.EQUIV.RIGID.01`).** Executes the rigidity route (mode B of `v347`) and lands on a strikingly simple statement. **[E]** rigidity closes the crystallographic part: the order-4 clock has the unique normal form $z \mapsto iz$ (Nielsen/Kerckhoff, `v180`), Troyanov gives the unique flat pillowcase metric (`v284`). **[E]** the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians = $2I$, golden-ratio quaternion norm) IS the rank-8 E_8 lattice. **[E]** the single bridge is the golden ratio: the crystallographic restriction allows only orders $\{1, 2, 3, 4, 6\}$ (the 5-fold absent), so $\varphi = 2\cos(\pi/5)$ is exactly what extends the crystallographic μ_4 to the non-crystallographic icosahedral $2I$ (the quasicrystal). **[O]** so the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes, μ_4+$\varphi \Rightarrow 2I \Rightarrow E_8$ and rigidity fixes the rest. **[O]** honest subtlety: TFPT's φ is currently E_8-side (`v312`/`v313`); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries φ non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294 → 295). An analysis/reduction; closes nothing.

2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat→spherical base locus, four modes, no free theorem – only rigidity or axiom P_3: `v347`)

- **`v347` (`SEAM.EQUIV.CLOSURE.01`).** A direct, honest answer to "is there any other way to fully solve the one open arrow L2?". **[E]** the precise locus: the seam gives the *flat* pillowcase base $S^2(2, 2, 2, 2)$ ($\chi_{\mathrm{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$, v216) while the $2I/E_8$ structure is the Seifert base over the *spherical* $S^2(2, 3, 5)$ ($\chi_{\mathrm{orb}} = 2 - (\frac{1}{2} + \frac{2}{3} + \frac{4}{5}) = \frac{1}{30}$), so L2 bridges two geometric types – why it needs the carrier $(2, 3, 5)$ input. **[E]** as a "physics-realises-geometry" statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom P_3 (then TFPT is complete relative to 3 axioms), D empirical. **[E]** the "no new state" discipline (v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [**E**] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [**O**] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293 → 294). An analysis/roadmap; closes nothing.

2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam ⇒ $\det K{=}1$ is a six-link chain, five theorems + one open orbifold realisation, the group forced by the $(2,3,5)$ atoms: `v346`)

- **`v346` (`SEAM.EQUIV.GEOM.01`).** The honest capstone of the `SEAM.EQUIV.01` investigation: it lays out the full path raw-seam ⇒ $\det K{=}1$ as six links and shows exactly one is open. [**E**]/[**C**] L1 raw seam → μ_4 + four marks (Gauss–Bonnet, `v216`); [**O**] L2 μ_4 + the $(2,3,5)$ atoms → the $2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE ONE OPEN ARROW = the geometric content of `SEAM.EQUIV.01`); [**E**] L3 $2I$ → affine E_8 McKay (`v219`); [**E**] L4 $2I$ → E_8 du Val (`v232`); [**E**] L5 $\mathbb{C}^2/2I$ → Poincaré link $H_1{=}0$ (`v345`); [**E**] L6 $H_1{=}0$ → $\det K{=}1$. [**E**] five of six links are theorems; [**E**] the group is FORCED by the seam's own atoms (the exceptional $SU(2)$ axis signatures $(2,3,3)/(2,3,4)/(2,3,5)$ → $2T/2O/2I$; $(2,3,5)$ unique to $2I$, and $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}){=}(2,3,5)$ ARE those axes); [**E**] everything downstream of L2 is then forced. [**O**] the irreducible open content is L2 alone ("the raw seam normal bundle is an ADE orbifold point $\mathbb{C}^2/\Gamma$") = `SEAM.EQUIV.01`; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

2026-06-22 (executing route R3: the $(2,3,5)$-rewrite attractor link computed to be the Poincaré homology sphere ($\det K{=}1$), the combinatorial core complete, only the geometric bridge open: `v345`)

- **`v345` (`SEAM.DETK.02`).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from `v344`) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has H_1 = coker(Cartan). [**E**] the Smith normal form of the E_8 Cartan matrix is the identity, so coker$(E_8) = 0$ – the E_8-graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8{=}1$); among ADE only E_8 does this ($A_n \to \mathbb{Z}/(n{+}1)$, $D_n \to \mathbb{Z}/4$, $E_6 \to \mathbb{Z}/3$, $E_7 \to \mathbb{Z}/2$). [**E**] $\pi_1 = 2I = SL(2,5)$ is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so $H_1 = 0$ and the link is THE Poincaré sphere, $|2I| = 120 = |\mu_4|\, h(E_8)$. So R3's combinatorial half is [**E**]-complete: the E_8 attractor link is forced to be the Poincaré sphere ($\det K{=}1$), uniquely among ADE. [**O**] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this E_8 du Val plumbing) = the `SEAM.EQUIV.01` content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292 → 293).

2026-06-22 (the one keystone bit $\det K{=}1$ mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: `v344`)

- **`v344` (`SEAM.DETK.01`).** A bird's-eye, full-spectrum map of `SEAM.EQUIV.01`'s only open residual, the holomorphy bit $\det K{=}1$. [**E**] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link $H_1{=}0$ / one 1-dim irrep / det Cartan = 1 / full μ_4 condensation / $\tau{=}i$ CM), unified by $|\det \text{Cartan}| = |H_1(\text{link})| = \#(\text{1-dim irreps})$ across ADE ($A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$) – only E_8 gives 1. [**E**] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (`v87`); $\det K$ is the torus degeneracy. [**E**] THREE genus-1 access routes: R1 continuum-modular (`v336`), R2 anyon-condensation (`v281`), R3 hypergraph-homotopy. [**C**] the HYPERGRAPH route is the most promising fresh angle: the $(2,3,5)$ rewrite attractor is the affine-E_8 graph (`v312`) and the E_8 du Val link is the Poincaré sphere (`v232`), so $\det K{=}1$ becomes a finite combinatorial statement (no continuum limit). [**O**] none closes it; the gap is the combinatorial-to-geometric bridge = the `SEAM.EQUIV.01` content. A synthesis/roadmap; closes nothing.

2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on `SEAM.EQUIV.01` + the decoupled QG contour: `v343`)

- **`v343` (`FOUR.ROUTES.01`).** A direct, honest answer to "which black-hole-birth route closes ALL problems and is 100% provable": *none individually* — but the four converge on the same two residuals as `v335`. [**E**] **(A)** the finite causal diamond reframes `QG.AMB` to a finite thermal trace (de Sitter $S_{\mathrm{dS}} = 32\pi^4 e^{2\alpha^{-1}}$ finite, $\chi = 729/665$) but inherits the GHP contour (`v334`, [**O**]); does not touch the seam. [**C**] **(B)** Carlip–Cardy gives a second route to existence+c (c=8 consistent, but the Carlip ambiguity and no holomorphy $\det K{=}1$) — inherits `SEAM.EQUIV.01`. [**E**] **(C)** the modular/thermal flow (`v239`) is well-defined but its geometric content is `QGEO.SYM.01`, a corollary of `SEAM.EQUIV.01` (`v335`) — a reduction. [**E**]/[**O**] **(D)** the unique attractor (`v56`) gives parameter-freeness but the cyclic dynamics is [**C**] interpretation. [**E**] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from `v335`); the horizon premise makes both more natural but eliminates neither — focus = `SEAM.EQUIV.01`. An analysis module; closes nothing, fabricates nothing.

2026-06-22 (the heat-kernel origin of the α terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the `EM.WARD.01` residual: `v342`)

- **`v342` (`EM.WARD.02`) — the heat-kernel sharpening of `EM.WARD.01`.** Derives the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [**C**] (conjectured) to [**E**] for the term structure; no new gate. [**E**] $c_3 = 1/(8\pi)$ *is* the boundary Gauss–Bonnet coefficient ($\oint_{S^2} K = 2\pi\chi = 4\pi$, χ=2, `v216`). [**E**] the α^2 (Calderón) term *is* the Gilkey gauge-curvature coefficient ($30\,\Omega^2/360 = \frac{1}{12}\Omega^2$, $\Omega{=}i\alpha F \Rightarrow -(\alpha^2/12)F^2$), so it is forced, not an ansatz. [**E**] the c_3-orders $\{0,3,6\}$ are an arithmetic ladder and $2c_3^3 = 1/(256\pi^3)$ carries π^3 (three boundary insertions). [**C**] the exact coefficient needs the seam F-normalisation. [**O**] the cubic α^3 Maxwell moment (a topological/Chern level) + the full "this is the seam $U(1)$ ζ-determinant" identification stay the `EM.WARD.01` residual, tied to `SEAM.EQUIV.01`. α^{-1} stays [**E**] (`v3`). Wolfram mirror +1 (290 $\to$ 291).

2026-06-22 (α as the stationarity of a $U(1)$ determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: `v341`)

- **`v341` (`ALPHA.QUILLEN.01`).** Answers the external review's sharpest valid point — "derive the *form* of α, not more decimals". The EM closure $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6 M \log(1/\varphi_{\mathrm{seam}}) = 0$ is rewritten as a *stationarity* of the boundary $U(1)$ determinant line: $[\alpha^3 - 2c_3^3\alpha^2]_{\delta \log\det\Delta_{U(1)}} = [8b_1c_3^6]_{\text{counterterm}} \cdot [\log(1/\varphi_{\mathrm{seam}})]_{\text{seam response}}$. [**E**] every ingredient is a NAMED object, not a knob: M=41 is a $U(1)$ index ($\frac{4}{5}M = 8b_1$, b_1=41/10 the SM $U(1)_Y$ beta, EM Ward `v48`); c_3=$1/(8\pi)$ the Gauss–Bonnet boundary coefficient (8= rank E_8); $-5/4 = -q(D_5)$ a discriminant form and φ_{base}=$1/(6\pi)$=$\frac{4}{3}c_3$ carries the other, $c_3/\varphi_{\mathrm{base}}$=3/4=$q(A_3)$; δ_{top}=$\Omega_{\mathrm{adm}}c_3^4$. The Quillen split is verified at the α^{-1}=137.0359992 root. **No new gate:** α^{-1} stays closed [**E**]; v341 adds no open item — it *sharpens* the pre-existing [**O**]/physical-origin obligation `EM.WARD.01` ("why *this* function", already conjectural). [**O**] the one still-open part *is* that residual: the from-first-principles AQFT proof that $F_{U(1)}$ is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 ($289 \to 290$).

2026-06-22 (A_s reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: `v340`)

- **`v340` (`AS.RECONCILE.01`).** Answers "why don't we have A_s / the old versions derived it" and "can a ratio define mass itself". **[E]** A_s *is* predicted and *dimensionless*: $A_s = N_\star^2(M_{\rm scal}/\bar{M}_{\rm Pl})^2/(24\pi^2) = N_\star^2 c_3^7/(24\pi^2)$ with $M_{\rm scal}/\bar{M}_{\rm Pl} = c_3^{7/2} = 1.26\times10^{-5}$ a pure ratio, so A_s pins the e-fold count $N_\star$, NOT $\bar{M}_{\rm Pl}$ — it cannot define an absolute mass (No-Unit, `v153`, stands). **[C]** the -11.4σ at the slow Higgs channel ($N_\star$=51.4) is the slow-vs-fast reheating dichotomy; the A_s-preferred $N_\star$=56.2 $\sim$ the instantaneous bound. **[C]** the archived TFPT v4.5/v4.6 reported a clean $A_s = 2.124\times10^{-9}$ using the algebraic $N_\star = 3/\varphi_0^{\rm ret} - 1 = 55.42$; at that $N_\star$ the current normalisation gives 2.047×10^{-9} (-1.9σ) — the old clean A_s is near-instantaneous reheating, not the slow channel. **[E]** nothing lost: the current theory replaced the posited algebraic $N_\star$ with one derived from reheating physics and exposed honestly that A_s requires fast (pre)heating. Record band $N_\star \in [50, 60]$ (`v84`) untouched.

2026-06-22 (the first-principles boundary, mapped: can $M_{\rm Planck}$ be computed? can $F_{\rm transfer}$ be first-principles? both NO, with the exact reason: `v339`)

- **`v339` (`FIRSTPRINCIPLES.BOUNDARY.01`).** An honest boundary audit answering two head-on questions and fabricating nothing. **[E]** *The Planck mass cannot be computed* — $v_{\rm geo}/M_{\rm Planck}$ is forbidden by theorem (No-Unit, `v153`): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, `v274`); every dimensionless *ratio* is derived, only the one unit is not. **[E]** $F_{\rm transfer}$ *cannot be made first-principles* either: for m_p/m_e the structure is in place (`v262`, carrier b_3=−7 + closed electron) but two inputs stay external. **[X]** $\alpha_s(M_Z)$ is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread ∼8.8 at 2×10^{16} GeV, `v246`). The lattice $C_p\sim2.83$ is parameter-free but lattice-only (`v35`); η_B/axion ride on cosmological initial conditions; Koide Q=2/3 is structural **[E]**. **[E]** The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.

2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the θ_{13} budget: `v336`, `v337`, `v338`, `FORM.CARTAN.DET.01`)

- **`v336` (`SEAM.EQUIV.CONTINUUM.01`) — the one open lemma, literature-anchored.** SEAM.EQUIV.01's only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. **[C]** continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8, CMP **398** (2023) 219) give, building on the MMST (Morinelli–Morsella–Stottmeister–Tanimoto) operator-algebraic (wavelet) renormalization framework (CMP 387 (2021) 299; PRL 127 (2021) 230601), the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur → Virasoro, correct c); WZW range rank $\le c \le D$, and $(E_8)_1$ (c=8, rank 8, D=16 Majoranas, $SO(16)_1 \to (E_8)_1$) is inside it. **[C]** OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. **[E]** target pinned by c=$g_{\rm car}$+$N_{\rm fam}$=8, $\det\mathrm{Cartan}(E_8)$=1. **[O]** narrowed residual = "the raw collar's massless scaling limit is the holomorphic $(E_8)_1$ net" ($\det K$=1), not a from-scratch construction.

- **v337 (DECOUPLING.THEOREM.01) — the Decoupling Theorem.** [**E**] every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$; finite susceptibility $\chi = 1/(1-(2/3)^6) = 729/665$ gap-suppresses the ambient contribution (margin $6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$, with $2\dim(E_8)c_3^2 = 31/(4\pi^2)$ exactly), so no readout depends on the ambient measure $\Rightarrow$ `QG.AMB.01` decoupled; negative control (gap closed $\Rightarrow \chi$ diverges) makes it non-vacuous. This is the theorem that makes the `QG.AMB.01` reclassification (`v335`) honest; consolidates `v76/v311/v330/v332`.
- **v338 (THETA13.BUDGET.01) — the θ_{13} budget, fenced.** The $+2\sigma$ pressure point (`v328`) quantified against *both* NuFIT 6.0 NO variants: $+2.0\sigma$ vs the no-SK central (0.02195) but $+1.7\sigma$ vs the with-SK best fit (0.02215 ± 0.00057), a $\sim 0.3\sigma$ systematic spread; [**E**] 0.023108 lies inside the with-SK 3σ band (upper 0.02388), an upper-edge pressure point not a falsification; [**X**] JUNO's sub-1% precision turns it into a $\sim 6\sigma$ kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01).** `CartanDeterminants.lean` extended with the explicit $D_8 = SO(16)$ Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator: D_8 is even and symmetric just like E_8 (same c=8) but $4 = \det \mathrm{Cartan}(D_8)$ is genuinely not a unit while $\det \mathrm{Cartan}(E_8)$ is — so unimodularity (one anyon) is the load-bearing selector. `lake build` green.

2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird's-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the "two open structural items" narrative. [**E**] **`QGEO.SYM.01` is a COROLLARY of `SEAM.EQUIV.01`** (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations $U(1)$ are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the μ_4 clock = rotation by $\pi/2$ ($(e^{i\pi/2})^4$=1) fixes it $\Rightarrow \omega \circ \rho$=$\omega$. So `SEAM.EQUIV.01` $\Rightarrow$ `QGEO.SYM.01` with NO extra premise (the v323 "rotation-invariant vacuum" is a net axiom) — the two bedrock items collapse to ONE. [**E**] **the one keystone theorem**: `SEAM.EQUIV.01` = a construction theorem (c=g_{car}+N_{fam}=8, $\det \mathrm{Cartan}(E_8)$=1, gap $6\ln\frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). [**E**] **`QG.AMB.01` decoupled, not a gate**: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v311/v330/v332) $\Rightarrow$ no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from "TFPT structural frontier" to "decoupled general QG problem". [**E**] **collapsed status**: open structural items $2 \to 1$; live residual = $v_{\mathrm{geo}} + G_{\mathrm{net}}$(=`SEAM.EQUIV.01`) + F_{transfer}. [**O**] the one residual = `SEAM.EQUIV.01`'s continuum OS reconstruction.
- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** `BWKeystone.lean` extended with the cited conformal-net vacuum axiom `conformal_net_vacuum_rotation_invariant`; with it, `StateInvariance` (=`QGEO.SYM.01`) follows from `SeamIsE8` alone, so `QGEO.SYM.01` is a COROLLARY of `SEAM.EQUIV.01` with no independent open premise (`#print axioms` confirms only the chain + cited BW/net axioms). `lake build` green (3362 jobs).
- **Status deep-sync.** The ledger (`QG.AMB.01` reclassified to "decoupled general QG; non-TFPT"; `QGEO.SYM.01` marked a corollary; new `SEAM.EQUIV.UNIFY.01` row), the canonical status card (`tfpt_status.tex`), `introduction` (status core + gate table), `origin_theory`, `tfpt_4_frontier`, `tfpt_5_redteam`, `tfpt_research_contracts`, `README`, `next.txt` and the website (`OpenGates`, FAQ, `papers.ts`) all moved to "one solvable theorem (`SEAM.EQUIV.01`) + one decoupled foreign problem (`QG.AMB.01`)". No falsifiable prediction changed; this is a status-framing sharpening backed by `v335` + the Lean corollary.

2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, `FORM.CARTAN.DET.01`)

- **`v333_cm_selection.py` (`QGEO.CMSELECT.01`, [E]/[O]).** The CM-selection mechanism for the last keystone residual: $\tau{=}i$ (the square) is the *unique* order-4 modulus — $S{=}\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$ has $S^4{=}I$ and fixes i, and an order-4 deck selects $j{=}1728$ (vs the order-3 hexagon $j{=}0$) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal $\pi/2$ spacing). So given the μ_4 deck, $\tau{=}i$ is forced; the residual sharpens from "why the square" to "the deck acts geometrically" (= `QGEO.SYM.01`). NOT a closure.

- **`v334_conformal_resolution.py` (`QGAMB.CONFORMAL.01`, [E]/[C]/[O]).** The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation $\phi \to i\phi$ flips the wrong-sign kinetic term, making the divergent conformal Gaussian $\int e^{+|c|\phi^2}$ into the convergent $\int e^{-|c|\phi^2}{=}\sqrt{\pi/|c|}$; the entire IDG form factor e^{p^2/M^2} dresses the propagator pole-free (the polynomial $R{+}R^2$ ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour's nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.

- **`CartanDeterminants.lean` extended (`FORM.CARTAN.DET.01`, [E]).** The Conway–Sloane lattice side: the E_8 Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ($|\det|{=}1$) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel `decide` / the integer-inverse argument. The remaining content (Witt's uniqueness: E_8 is the only even unimodular lattice in dim < 16) is the cited classification residual.

2026-06-22 (endgame round: the necessity of H reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem, $|\det E_8|{=}1$ and the μ_4 commutation grounded in Lean: v331, v332, `FORM.MU4COMM.01`)

- **`v331_necessity_of_H.py` (`QGEO.NECESS.01`, [E]/[O]).** The necessity companion to `v329`: a concrete Steklov (DtN) computation shows the seam $\Lambda{=}|k|{+}M_f$ commutes with the μ_4 clock *iff* the four marks sit at the square ($\tau{=}i$) — moving a mark off the square breaks $\mathbb{Z}_4$ and $[\rho, \Lambda] \neq 0$. The square has cross-ratio 2 ($j{=}1728$), the unique 4-config with an order-4 automorphism (`v214`/`v267`), so $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$ and "necessity of H" sharpens to: the raw dynamics places the four Gauss–Bonnet marks (`v216`) at the order-4 CM point. Residual = that dynamical selection.

- **`v332_qgamb_metric_sector.py` (`QGAMB.METRIC.01`, [E]/[C]/[O]).** The open metric sector of `QG.AMB.01` is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient $-(D{-}1)(D{-}2)/4$ ($= -\frac{3}{2}$ at $D{=}4$), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (`v330`/`v76`); the only route is the conditional IDG taming (`v304`), with the polynomial $R{+}R^2$ ghost as the control. Open, but pinned to one named obstruction.

- **`CartanDeterminants.lean` extended (`FORM.CARTAN.DET.01`, [E]).** The rank-8 holomorphy discriminator $|\det \mathrm{Cartan}(E_8)|{=}1$ (E_8 has one anyon, vs D_8's four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the 8×8 PRODUCT gives `IsUnit (det E8)` via $\det E_8 \cdot \det E_8^{-1}{=}\det 1{=}1$ — no 8! determinant, no `native_decide` (Mathlib itself leaves `E₈_det = 1` a `proof_wanted`).

- **`Mu4Commutation.lean` (`FORM.MU4COMM.01`, Lean 4, [E]).** The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers $\mathbb{Z}[i]$: $i^4{=}1$, $i^2{=}-1$, and (on concrete 6×6 matrices) an offset-4 coupling commutes with the μ_4 clock while an offset-2 coupling does not — all by kernel `decide`. Built `lake build` on Lean v4.29.1 + Mathlib.

2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, FORM.CARTAN.DET.01)

- **`v329_os_gap_reduction.py` (QGEO.OSGAP.01, [E]/[O]).** The one analytic input the v308 keystone chain still cited — "the OS-reconstructed bulk gap = the transfer gap" — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization $d\Gamma(-\log T)$, whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $= 6\ln\frac{3}{2}$, *independent of system size*. Together with $H \Rightarrow \omega \circ \rho{=}\omega$ and $H \Rightarrow$ SRE $\Rightarrow (E_8)_1$, this exhibits the rotation-invariant flat-pillowcase premise H as the *single sufficient premise* of both bedrock IDs; residual = the necessity of H on the raw collar.
- **`v330_qgamb_admissible_measure.py` (QGAMB.MEASURE.01, [E]/[O]).** The constructive step `v275`'s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ($C(n)/C(n{-}1) \to (2/3)^6$), with finite susceptibility $\chi = 729/665$ so $\mathrm{Var}(S_\Lambda)/\Lambda \to \chi$ converges $\Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit exists; OS reconstruction gives $H_{\mathrm{OS}} \geq 0$ with gap $6\ln\frac{3}{2}$. The full $\mathrm{QG}_{\mathrm{amb}}$ (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **`CartanDeterminants.lean` (FORM.CARTAN.DET.01, Lean 4, [E]).** The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom): $|\det\mathrm{Cartan}(A_3)|{=}4{=}|\mu_4|$ ($3{\times}3$, `det_fin_three`) and $|\det\mathrm{Cartan}(D_5)|{=}4$ ($5{\times}5$, `decide`), giving the carrier $|\det\mathrm{Gram}|{=}4\cdot 4{=}16$ (the `v281` anyon count). The rank-8 E_8/D_8 case stays the Nat discriminator — Mathlib's own `Data.Matrix.Cartan` leaves `E₈_det = 1` as a `proof_wanted`. Built `lake build` on Lean v4.29.1 + Mathlib.

2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the F_{transfer} solver suite, the cusp weight derived from a minimal rewrite, and the θ_{13} pressure point: FORM.QGEO.BW.01, v325, v326, v327, v328)

- **`BWKeystone.lean` (FORM.QGEO.BW.01, Lean 4, [E]/[O]).** The `v323` Bisognano–Wichmann linkage formalised: the nodes RotationInvariantVacuum $\to$ GeometricModularFlow $\to$ Mu4ModularSymmetry $\to$ StateInvariance (=`QGEO.SYM.01`) composed over the `v308` chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock `QGEO.SYM.01` reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator $(1 \mid 4) \wedge \neg(4 \mid 1)$ (BW geometric flow strictly stronger than `v309` clock invariance). Built `lake build` on Lean v4.29.1 + Mathlib.
- **`v325_pillowcase_keystone.py` (QGEO.KEYSTONE.01, [E]/[O]).** The keystone as ONE citable theorem: IF the raw seam state is the flat $\tau{=}i$ pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks, μ_4 deck, $\omega\circ\rho{=}\omega$, holomorphic $c{=}8 \Rightarrow (E_8)_1$, with the recovery gap as input); all pieces machine-checked (incl. the pillowcase $\chi_{\mathrm{orb}}{=}0$), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.
- **`v326_ftransfer_suite.py` (FR.SUITE.01, [E]/[C]).** The four F_{transfer} readouts productised into one runnable harness, each a gapped relaxation to a unique attractor: F_{pole} (Koide, seam

rate $(2/3)^6$), $F_{\text{Boltzmann}}$ (η_B, H-theorem), F_{relic} (axion adiabatic invariant), F_{QCD} (m_p/m_e, RG to the UV fixed point), plus the F_{RareKaon} bridge. Only F_{pole} carries the seam rate; the four external rates are honestly fenced (`v187/v303`).

- **`v327_hypergraph_rewrite.py` (`HYP.REWRITE.01`, [E]/[O]).** The cusp weight 2/3 is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum $\{1, 2/3, 0\}$, so $2/3{=}(N_{\text{fam}}{-}1)/N_{\text{fam}}{=}|\mathbb{Z}_2|/N_{\text{fam}}$ emerges from the rule arity and $(2/3)^6{=}64/729$ is the recovery gap; with a PROOF that 2/3 is not a root of the affine-E_8 charpoly ($p(2/3) = -3520/19683 \neq 0$, sharpening `v312`). `v324`'s injected datum is reduced to the anchor arity $\{2, 3\}$.
- **`v328_theta13_pressure.py` (`FLAV.TH13.PRESSURE.01`, [C]/[X]) + `v307` hardening.** The honest weak spot named and quantified: $\sin^2\theta_{13}{=}\varphi_0^{\text{ret}} e^{-5/6}{=}0.023108$ is $+2.0\sigma$ above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (`v268`), not relieved by any freeze-respecting correction (RG $< 1\%$, exponent 5/6 exact), pre-registered as the kill. The `v307` watchdog gains a data provenance/date check and the rare-kaon F_{transfer} bridge row.
- **Wolfram mirror** 285 → 288. `v325` ($\chi_{\text{orb}}{=}0$) and `v327` (rewrite spectrum $\{0, 2/3, 1\}$, $(2/3)^6{=}64/729$, and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; `GATE.WOLFRAM.02` and the Wolfram README updated to 288/288.

2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)

- **Keystone naming reconciled to `SEAM.EQUIV.01`.** Following `v323/QGEO.BW.01` (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now `SEAM.EQUIV.01` *everywhere*, with `QGEO.SYM.01` stated as its *downstream* conformal-deck face. The ledger row `QGEO.SYM.01` was updated (it no longer calls itself "the single structural residual of the whole theory"; it now depends on `SEAM.EQUIV.01/QGEO.BW.01`), and the superseded "`GATE.QGEO` open gate" / "two research gates (U_{wall}) and G_{metric}" framing was replaced by the current Rest $= v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ with the two structural items `SEAM.EQUIV.01` + `QG.AMB.01` (intro status card, `tfpt_4_frontier`, `origin_theory`, `tfpt_5_redteam`, `tfpt_research_contracts`, and the website).
- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger ("honest bottom line") and hard-reduction subsection; the `origin_theory` "residual inventory" (the five "Update/. . . /Terminal" passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ "What is still open?", the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label `5.0`→`v5.2` (intro, single-sourced via `\TFPTversion`); Lean build `1886`→`3357` jobs (`FORM.SEAMEQUIV.01`); README Python suite `v213/ 212`→`v324/ 322` modules; Wolfram extension `269/269 / 249/249`→`285/285` (README, website replication page, `zenodo_description.html`); residual "Paper-3" old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: `v202`, `v203`, `v305`, `v307`, `v321`)

- **Bug fix — $\delta_{\rm CKM}$ display value.** The predictions table (`tex-artefacts/predictions.tex`) showed $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96°$, but that formula evaluates to 68.65° (the canonical frozen `DELTA_CKM_RAD`= 1.198232 rad, used by `v88/v202/FLAV.CP.01` everywhere else). Corrected to 68.65°. An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **`v202_rare_kaon.py` — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ ($\sim 1.2\sigma$ above) to the full 2016–2024 combination $\mathrm{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$ (La Thuile 2026), on which the prediction 9.45×10^{-11} lands at $+0.08\sigma$. Framed honestly: a strong *consistency* hit for the closed CKM point, but an $F_{\rm transfer}$ bridge — *not* a unique TFPT-vs-SM discriminator (the SM $(8.6 \pm 0.4) \times 10^{-11}$ is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (`v307`, a new $F_{\rm transfer}$ bridge row), the forward board (`v321`, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from "a relation to the *measured* quark phase" to "a relation to the *leading* ($\pi/3$) component of the quark phase": the lock is to the structural $\pi/3$, not the full measured $\gamma = \delta_{\rm CKM}^{\rm lead} + 3\lambda_C^2 \approx 68.7°$, so $\delta_{\rm PMNS} = 240°$ (not 248.7°). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **`v305_witness_independence.py` — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule: $\phi = 2\cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$ is the *static* carrier ($\mathbb{Z}/5$) field, every *dynamic* rate is rational ($\mathbb{Q}$, the $\mathbb{Z}/6$ hand), so a dynamic rate explained by ϕ is a cross-field false friend (`v314/v315`).
- **`v203_eht_achromatic.py` — the μ_4 Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at $m \equiv 0 \bmod 4$ (the μ_4 fingerprint of the four marks, `v201`); a μ_4-orbit profile has ~ 0 off-mode power while a $\mathbb{Z}_2$ control leaks at $m = 2$.
- **Seed-hyperplane figure (Fig. ??).** A new figure visualises `v306`: every scorecard observable inverts *independently* to the one latent seed φ_0, the error-weighted mean matching the axiom to 0.04%, with $\sin^2\theta_{13}$ the $\sim 2\sigma$ pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: `v322`, `v323`, `v324`)

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks, 275 → 285, *all passing*): `v313` (the affine-E_8 characteristic polynomial with the golden factor $x^2{-}x{-}1$; the $(2,3,5)$ atoms as $2\cos(\pi/k)$; the 31/30 selection), `v314` (the rational kernel vs $\mathbb{Q}(\sqrt{5})$ dynamic-rate split), `v315` (30=5·6, $\varphi(30)$=8, the Galois group $\mu_4 \times \mathbb{Z}_2$, the $\sqrt{5}$ Gauss sum), `v316` (ρ=ζ_6=ζ_{30}^5, ρ^4= $-\rho$, Gal=$\mathbb{Z}_2$= CP conjugation), `v317` ($N_{\rm fam}$=3 as the μ_3 orbit, the 1+2 Galois split), `v318` ($\varphi_0^{\rm ret}$=(4/3)c_3+48c_3^4 a pure function of π) and `v320` (the CP lock $\delta_{\rm PMNS}$=240°). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.
- **`v322_cp_lock_sharpened.py` (`GALOIS.CP.SHARPEN.01`, [E]/[C]/[X]).** The `v320` CP-lock sharpened into a quantitative, pre-registered kill test. **[E]** the leading phase sits on

the six hexagonal nodes $\arg\rho^k = k\cdot 60°$, and *which* one is selected is the deck order: $\delta^{\rm lead}_{\rm PMNS}=|\mu_4|\,\delta^{\rm lead}_{\rm CKM}=4(\pi/3)$ (node 4); **[C]** the sub-leading correction is bounded by the quark analogue $3\lambda^2\approx 8.7°$, so the prediction is the band $240°\pm\sim 9°$; **[C]** against NuFIT 6.0 (NO, $\delta_{CP}=212^{+26}_{-41}°$) the leading value is $+1.08\sigma$, the band edge $+0.74\sigma$ – compatible today; **[X]** the nearest wrong node is 60° away, far beyond the budget, so DUNE/Hyper-K ($\sim$ 5–15°) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.

- **`v323_bw_geometric_modular.py` (`QGEO.BW.01`, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. **[E]** the μ_4 clock is a geometric rotation $\rho=\mathrm{diag}(i^n)=\exp(i\frac{\pi}{2}L)$, $\mu_4\subset U(1)_{\rm rot}$; **[E]** for a rotation-covariant covariance $C=f(L)$ the modular Hamiltonian $K=\log((1-C)/C)=g(L)$, so $[K,L]=0$ (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric, $[K,L]\neq 0$); **[C]** given the seam is the $(E_8)_1$ chiral net (`v308`), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ($\omega\circ\rho=\omega$) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; **[O]** residual: the raw collar vacuum is rotation-invariant (cleaner than `v309`, shared with `v308`). Cited in `origin_theory`; ledger `QGEO.BW.01`.
- **`v324_hypergraph_fiber.py` (`HYP.FIBER.01`, [E]/[C]).** The constructive `v312` follow-up: the *minimal* substrate that carries both the E_8 skeleton and the recovery gap is a *product.* **[E]** the carrier network $T_{\rm net}=(A+2I)/4$ (attractor = Kac marks = skeleton) fibred by the 3-node family cusp $T_{\rm cusp}=\mathrm{diag}((1-w)^{2N_{\rm fam}})$, $w\in\{0,\frac{1}{3},\frac{2}{3}\}$ (spectrum $\{1,(2/3)^6,(1/3)^6\}$); the fibred $T_{\rm net}\otimes T_{\rm cusp}$ has top eigenvalue 1 (eigenvector marks $\otimes$ $(w=0)$) and $(2/3)^6$ as a genuine eigenvalue (marks $\otimes$ $(w=\frac{1}{3})$) – the $N_{\rm fam}=3$ cusp is the minimal fibre and the substrate = carrier × family = the $30=g_{\rm car}(2N_{\rm fam})=5\times 6$ of `v315`; **[C]** honestly (`v312`) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of 2/3 from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: `v321`)

- **`v321_killtest_board.py` (`KILL.BOARD.01`, [E]).** The forward companion to `v307` (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO $\sin^2\theta_{12}=0.306747$ (fastest), #2 CMB-S4/LiteBIRD r (kill $r>0.01$), #3 DESI/Euclid w (kill $w\neq -1$), #4 DUNE/Hyper-K $\delta_{\rm PMNS}=240°$ (the new `v320` Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering+$m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2\theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β. A forward decision ledger (publicly-stated experimental reach), complementing `v307`; no new physics. Cited in `origin_theory`; ledger row `KILL.BOARD.01`.

2026-06-22 (Lean: the `SEAM.EQUIV.01` composition chain formalised — the keystone residual machine-pinned to the named cited steps: `FORM.SEAMEQUIV.01`)

- **`SeamEquivChain.lean` (`FORM.SEAMEQUIV.01`, Lean 4, [E]/[O]).** A Lean 4 mirror of the `v308` closing chain: the four nodes GapPositive → InvertibleSRE → HolomorphicC8 → SeamIsE8 assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Müger / Kawahigashi–Longo–Müger holomorphy; Conway–Sloane E_8 uniqueness) as named `axioms` and the carrier recovery gap (the OS step) as the single open input. **[E]** `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); **[E]** a `decide`-proven discriminator primariesE8 $=1\neq$ primariesD8 $=4$ (the $|\det\mathrm{Cartan}|$ that separates the holomorphic E_8 from the same-c rival $D_8=SO(16)$). Built end-to-end (`lake build`, 3357 jobs) on Lean v4.29.1 + Mathlib. **[O]** The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close `SEAM.EQUIV.01`. Ledger row `FORM.SEAMEQUIV.01`; mirrors `v308`.

2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta^{\text{lead}}_{\text{CKM}} + \pi$: `v320`)

- **`v320_galois_cp_relation.py` (`GALOIS.CP.PREDICT.01`, [E]/[C]/[X]).** The `v316` follow-up turned into a testable cross-prediction. **[E]** both leading CP phases are powers of the *one* hexagonal unit $\rho = \zeta_6$ of the family Galois factor: $\delta^{\text{lead}}_{\text{CKM}} = \arg\rho = \pi/3$ (60°), $\delta_{\text{PMNS}} = \arg\rho^4 = 4\pi/3$ (240°), and $\rho^4 = -\rho$ locks them as $\delta_{\text{PMNS}} = \delta^{\text{lead}}_{\text{CKM}} + \pi$ (the $\mathbb{Z}_2$ sheet flip); **[C]** this upgrades the previously assigned $\delta_{\text{PMNS}} = 240°$ to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); **[X]** kill test: a δ_{PMNS} robustly incompatible with 240° ($> 3\sigma$ at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation; 240° is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the `v313`–`v319` arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row `GALOIS.CP.PREDICT.01`.

2026-06-22 (the translation clock: the discrete↔dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands 5×6, read law-inclusive $(0..5)$ or live-only $(1..5)$: `v319`)

- **`v319_translation_clock.py` (`TRANSLATE.CLOCK.01`, [E]/[C]).** A synthesis of the `v313`–`v318` arc with `v124`/`v55`, answering the question "is the static↔dynamic translation a clock running $0, 1, 2, 3, 4, 5$ or $1, 2, 3, 4, 5$?". **[E]** the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (`v314`); $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ and $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (two coprime hands, `v315`). **[E]** the *dynamic* hand is $\mathbb{Z}/6 = 2N_{\text{fam}}$ (ticks $0, \ldots, 5$): the recovery exponent is exactly $6 = 2N_{\text{fam}}$, rate $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$, order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$, `v316`). **[E]** position 0 is the *conserved law* — the resummed clock rate$(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (`v124`) has rate$(0) = 0$ (the attractor, eigenvalue 1, `v56`), the live decaying modes from n=1, and the E_8 live phases (totatives of 30) start at 1 ($\{1, 7, \ldots, 29\}$), the 0 phase not a totative — so "$0, 1, 2, 3, 4, 5$" is law-inclusive, "$1, 2, 3, 4, 5$" live-only. **[E]** the *static* hand is $\mathbb{Z}/5 = g_{\text{car}}$ (ticks $1, \ldots, 5$): the golden field $\mathbb{Q}(\sqrt{5})$ ($\varphi = 2\cos(\pi/5)$), field-obstructed (no rational dynamic image, `v314`) — static-only; $\zeta_5 = \zeta_{30}^6$ carries no phase. **[E]** one full turn $= \operatorname{lcm}(5, 6) = 30 = h(E_8)$, rank $E_8 = 8 = \varphi(30)$ live phases pairing into $|\mu_4| = 4$ invariant planes (`v55`). **[C]** verdict: the discrete→dynamic translation *is* the order-30 clock = (static 5-ring) × (dynamic 6-ring), the dynamic hand carrying the rate (exponent $6 = 2N_{\text{fam}}$), the static hand the golden carrier (no rate), and "0 vs 1" = law-inclusive vs live-only; the arithmetic is **[E]**, the "it is one clock" reading **[C]**. Python-only (`sympy`), like the rest of the arc. New figure `figures/translation_clock.pdf` (concentric 5- and 6-hands). Cited in `origin_theory`; ledger row `TRANSLATE.CLOCK.01`.

2026-06-21 (capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30})$+ Galois $\mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\text{geo}}\}$ — zero dimensionless free parameters: `v318`)

- **`v318_arithmetic_capstone.py` (`ARITH.CAPSTONE.01`, [E]/[O]).** The synthesis of the `v313`–`v317` arc. **[E]** the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field $\mathbb{Q}(\zeta_{30})$ with Galois group $\mu_4 \times \mathbb{Z}_2$ (degree $\varphi(30) = 8 =$ rank E_8, $30 = |\mathbb{Z}_2| N_{\text{fam}} g_{\text{car}} = h(E_8)$), forced by the anchor atoms $\{2, 3, 5\} = e(a)$; **[E]** the magnitude seed $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}} c_3^4 = \frac{4}{3}c_3 + 48c_3^4$ ($c_3 = 1/(8\pi)$) is a pure function of π (no free parameters, anchor-derived coefficients), so the magnitudes ($\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$) are $\{a, \pi\}$-determined, not free; **[E]** free inventory: *zero* dimensionless free parameters, π the unique transcendental primitive (`ARCH.CORE.01`), v_{geo} the unique dimensionful input (No-Unit

Theorem) — complete input $\{a, \pi, v_{\mathrm{geo}}\}$. **[O]** honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (`QGEO.SYM.01`/`SEAM.EQUIV.01` stay **[O]**). Cited in `origin_theory`; ledger row `ARITH.CAPSTONE.01`.

2026-06-21 (are the 3 generations a μ_3/Galois orbit? Yes (μ_3), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)

- **`v317_galois_family.py` (`GALOIS.FAMILY.01`, [E]/[C]).** The `v316` follow-up to the generation structure. **[E]** the three cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (with $\mathrm{Spec}(Q_+) = \{1, 2, 3\}$, the A_3 exponents) are the cube roots of unity $\{1, \zeta_3, \zeta_3^2\}$ under $w \mapsto e^{2\pi i w}$, and the cyclic family symmetry $w \mapsto w+\frac{1}{3}$ is a *transitive* μ_3 orbit (the family IS the cube-root orbit); **[E]** the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at w=0); **[E]** the family Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$, = the `v316` CP conjugation) refines the orbit to 1+2 — fixing w=0 (rational) and swapping $w=\frac{1}{3} \leftrightarrow \frac{2}{3}$; **[E]** so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. **[C]** verdict: the family STRUCTURE is the same arithmetic as the CP phase (a μ_3 orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in `origin_theory`; ledger row `GALOIS.FAMILY.01`.

2026-06-21 (does the Galois $\mu_4 \times \mathbb{Z}_2$ organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois $\mathbb{Z}_2$ = CP conjugation: v316)

- **`v316_galois_readout.py` (`GALOIS.READOUT.01`, [E]/[C]).** The `v315` follow-up to phenomenology: does the Galois coupling $\mu_4 \times \mathbb{Z}_2$ reach the physics? **[E]** it does, through the *family* factor — the hexagonal CP unit $\rho = \zeta_6 = \zeta_{30}^5$ is exactly the order-6 $= 2N_{\mathrm{fam}}$ factor c^5 of `v315` (the carrier being $\zeta_5 = \zeta_{30}^6$); **[C]** the leading CP phases are its cyclotomic arguments $\delta_{\mathrm{CKM}}^{\mathrm{lead}} = \pi/3 = \arg\zeta_6$ and $\delta_{\mathrm{PMNS}} = 4\pi/3 = \arg\zeta_6^4$ (power $4 = |\mu_4|$), with $\zeta_6^4 = -\zeta_6$ the $\mathbb{Z}_2$ sheet flip and $\mu_6 = \mu_3 \times \mu_2$ (inherits `v231`/`v233`); **[E]** the family Galois $\mathrm{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$ ($\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$) *is* CP conjugation $\delta \mapsto -\delta$; **[E]** the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (μ_4 static), and the magnitudes ($\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$, transcendental φ_0) are not cyclotomic. **[C]** verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the `v314` field split. Cited in `origin_theory`; ledger row `GALOIS.READOUT.01`.

2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois = $\mu_4 \times \mathbb{Z}_2$: v315)

- **`v315_coxeter_coupling.py` (`COX.COUPLE.01`, [E]/[C]).** Answers the `v314` question — does $30 = g_{\mathrm{car}}(2N_{\mathrm{fam}}) = 5{\cdot}6$ couple the carrier (5-fold, golden, $\mathbb{Q}(\sqrt{5})$) and family ($6 = 2{\cdot}3$, rational) sectors, or just factorize? **[E]** the cyclic group *factorizes*: $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled, in line with the `v314` field obstruction); **[E]** the E_8 exponents = totatives(30) split as totatives(5)$\times$ totatives(6), $\varphi(30) = 4 \cdot 2 = 8 = \mathrm{rank}\ E_8$; **[E]** but the *field* couples: $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$ and the family $\mathbb{Q}(\zeta_3)$, degree 8 = rank E_8; **[E]** the Galois group is exactly $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$ (order 8, max element order 4), so the μ_4 seam clock *is* the Galois group of the carrier's golden field and $\mathbb{Z}_2$ that of the family field (refining `v223`); **[E]** $\sqrt{5} = \phi + 1/\phi$ is the quadratic Gauss sum in $\mathbb{Q}(\zeta_5)$. **[C]** verdict: $30 = 5 \cdot 6$ does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static$\leftrightarrow$dynamic bridge being *Galois*, not a dynamical map. Cited in `origin_theory`; ledger row `COX.COUPLE.01`.

2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$: v314)

- **`v314_rate_translation.py` (`RATE.TRANSLATE.01`, [E]/[C]).** Answers, exactly, whether the discrete kernel (the affine-E_8 network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate $(2/3)^6$) translate. [**E**] FIELD SPLIT: the adjacency spectrum is the rational part $\{0, \pm1, \pm2\} \subset \mathbb{Q}$ plus the golden part $\{\pm\phi, \pm1/\phi\} \subset \mathbb{Q}(\sqrt{5})$, while the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ is *entirely* rational; [**E**] the family/sheet rates DO translate (the rational angles $2\cos(\pi/2)=0 = |\mathbb{Z}_2|$ and $2\cos(\pi/3)=1 = N_{\text{fam}}$ build $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$, same field, same atoms), but [**E**] the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has NO rational dynamic image – a field obstruction (2/3 is neither an adjacency nor a lazy-update eigenvalue; heat-kernel $t = \Delta/(2-\phi) \approx 6.37$ not clean), so the carrier is *static-only*; [**E**] the only object holding both is the order-30 Coxeter element $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$, and the ϕ-quasicrystal $E_8 = H_4 + \phi H_4$ $(240 = 2 \cdot 120)$ carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in `origin_theory`; ledger row `RATE.TRANSLATE.01`.

2026-06-21 (the golden ratio made precise: the $(2,3,5)$ atoms are the affine-E_8 network spectral angles, $\phi = 2\cos(\pi/5)$ the g_{car}=5 signature: v313)

- **`v313_golden_atoms_spectral.py` (`GOLD.ATOMS.01`, [E]/[C]).** A follow-up to `v312`'s observation, made exact. [**E**] the affine-E_8 adjacency characteristic polynomial factors as $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$, so the golden minimal polynomial x^2-x-1 divides it and $\phi = 2\cos(\pi/5) = (1+\sqrt{5})/2$ is an *exact* eigenvalue; [**E**] the non-trivial eigenvalues are $2\cos(\pi/k)$ for exactly $k \in \{2,3,5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$ $(2\cos(\pi/2)=0,\ 2\cos(\pi/3)=1,\ 2\cos(\pi/5)=\phi,\ 2\cos(2\pi/5)=1/\phi)$, so the golden ratio is *specifically* the g_{car}=5 (5-fold) signature and g_{car}=5 gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the `v305` multiplicity); [**E**] the $(2,3,5)$ spherical excess $1/2+1/3+1/5 = 31/30 > 1$ (the condition selecting the finite $2I \Rightarrow E_8$) ties the `v63`/`v76` gap-margin numerator $31 = 2^{g_{\text{car}}}-1 = 248/8 = 1+h(E_8)$ to the Coxeter $30 = h(E_8)$ — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, `v305`); [**E**] honest null: ϕ is a structural lattice/network quantity, not a phenomenological readout. [**C**] direction: ϕ is the E_8-quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the `v312` question. Cited in `origin_theory`; ledger row `GOLD.ATOMS.01`.

2026-06-21 (TOE-roadmap solver round: the SEAM.EQUIV.01 keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: v308–v312)

- **`v308_seam_equiv_chain.py` (`SEAM.EQUIV.CHAIN.01`, [E]/[O]).** TOE attack point 1: the keystone "the raw seam-Calderón net is the holomorphic c=8 $(E_8)_1$ net" assembled into ONE well-typed logical chain gap>0 → invertible/SRE (Kitaev) → holomorphic c=8 (Müger/KLM) → $(E_8)_1$ (Conway–Sloane), with the exact discriminators $|\det \text{Cartan}(E_8)|=1$ vs $|\det \text{Cartan}(D_8{=}SO(16))|=4$, the carrier tower $4{\cdot}4=16 \to 4 \to 1$ and $c=g_{\text{car}}+N_{\text{fam}}=8$ all machine-checked; the input gap is the derived recovery gap $6\ln(3/2)>0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row `SEAM.EQUIV.CHAIN.01`.

- **`v309_modular_bedrock.py` (`QGEO.MODHAM.01`, [E]/[O]).** TOE attack point 2: the bedrock $\omega\circ\rho=\omega$ via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (`v216`) and the order-4 clock from cross-ratio 2 (`v214`), so μ_4 is derived; the quasi-free modular Hamiltonian $K = \log((1-C)/C)$ round-trips to the seam operator and $[\rho, K]=0 \Leftrightarrow \omega\circ\rho=\omega$ (a $\mathbb{Z}_2$ profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicality of the raw collar; not a closure of `QGEO.SYM.01`. Cited in `tfpt_research_contracts`; ledger row `QGEO.MODHAM.01`.

- **`v310_carrier_sm_anomaly.py` (`CAR.SM.ANOM.01`, [E]).** TOE attack point 3: the carrier half-spinor $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$ is exactly one anomaly-free Standard-Model generation

($SU(5) \subset SO(10)$: $10+\bar{5}+1$), with $\sum Y = \sum Y^3 = [SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$ (exact) and the one-loop $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1, so $N_{\rm fam}$=3 pins the RG. Closes the spectrum/anomaly/β sub-question; the interacting Epstein–Glaser S-matrix and the Yukawa sector stay **[C]**/**[O]**. Cited in `tfpt_research_contracts`; ledger row `CAR.SM.ANOM.01`.

- **`v311_gap_decoupled_measure.py` (`QGAMB.CLUSTER.01`, [E]/[C]/[O]).** TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ (correlation length $1/(6\ln(3/2))$, finite susceptibility 729/665), and the `v76` margin $6\ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. `QG.AMB.01` stays **[O]** (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row `QGAMB.CLUSTER.01`.
- **`v312_hypergraph_substrate.py` (`HYP.INJECT.01`, [E]/[O]).** TOE attack point 5: the hypergraph substrate sharpened — the affine-E_8 Kac marks are the Perron eigenvector ($A\,m$=$2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)$ = the golden ratio, but the recovery rate $(2/3)^6$ is *not* any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3$=$|\mathbb{Z}_2|/N_{\rm fam}$ and the analytic seed φ_0. The open question is precisely delimited, not closed. Cited in `introduction`; ledger row `HYP.INJECT.01`.

2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: `v305`–`v307`)

- **`v305_witness_independence.py` (`WITNESS.ECON.01`, [E]).** The *structural* anti-numerology firewall complementing the `v100` phenomenological null test, against the sharpest self-deception worry ("the same integer counted many times as if independent"). Every headline integer ($|\mu_4|$=4, $g_{\rm car}$=5, $|\mathbb{Z}_2|$=2, $N_{\rm fam}$=3, $\dim S^+$=16, $\operatorname{rank} E_8$=8, 240, 248, $|2I|$=120, $h(E_8)$=30, $|W(D_5)|$=1920, $\Omega_{\rm adm}$=48, the gap exponent 6) is a documented image of the single anchor a=$(1,1,2)$ with π the only transcendental primitive (13 atoms from 2 free primitives, 6.5× compression made explicit), so the recurring $2/3$=$|\mathbb{Z}_2|/N_{\rm fam}$ and $(2/3)^6$ are *one* ratio; the E_8 spine is overdetermined (≥ 3 structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by `v100`, never multiplied); the negative control shows α^{-1}'s "137" is foreign to the anchor family (an independent witness via the $F_{U(1)}$ cubic, `v3`). Cited in the `tfpt_5_redteam` body; ledger row `WITNESS.ECON.01`. Python-only.
- **`v306_seed_crossval.py` (`SEED.CROSSVAL.01`, [E]).** A leave-one-out cross-validation turning the "many observables, one seed φ_0" worry into a falsifiable out-of-sample test: back-solving φ_0 from each of the five seed-grammar observables ($\sin^2\theta_{12}$, $\sin^2\theta_{13}$, β, Ω_b, λ_C; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$ to 0.04%; leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2\theta_{13}$, $\sim 2\sigma$); a data↔formula shuffle explodes the χ^2 by $> 100\times$. Cited in `tfpt_5_redteam`; ledger row `SEED.CROSSVAL.01`. Python-only.
- **`v307_data_watchdog.py` (`DATA.WATCHDOG.01`, [E]).** The operational decision pipeline over the frozen registry (`v84`): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live n_σ against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core prediction $\sin^2\theta_{13}$ (2.0σ, the honest current pressure point); $\sin^2\theta_{12} = 0.306747$ is compatible (0.02σ, the JUNO falsifier), the $r \in [0.0033, 0.0048]$ band is below the BK18 limit and the kill, and w=-1 is flagged as the DESI dark-energy watchdog. Cited in `origin_theory`; ledger row `DATA.WATCHDOG.01`. Python-only.

- **Target-A holomorphy kill test (`freeze_file.csv`).** A new `seam_holomorphy` freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ($|\det K| > 1$, the same-c rival $SO(16)_1$) would break holomorphy and kill "seam net $= (E_8)_1$" — since holomorphic c=8 ⇔ E_8 is machine-proven (`v83`/`v143`) and $\det K$=1 ⇔ invertible/SRE (`v237`/`v301`/`v302`).

2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the "constancy of c" framing of `SEAM.EQUIV.01`: `v304`)

- **`v304_idg_nonlocal_ghost.py` (`QGAMB.IDG.01`, [C]/[O]).** A conditional, parameter-free narrowing of the red-team `rt_F`/`v278` Stelle-ghost attack on the perturbative gravity sector. [**E**] the local R+R^2/Weyl^2 four-derivative propagator $1/(p^2(p^2+M^2))$ carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [**E**] for an *entire* graviton form factor $a(p^2) = e^{p^2/M^2}$ (nowhere zero) the dressed propagator $1/(p^2a)$ has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [**C**] the spectral-action cutoff is the seam KMS weight $f(u) = e^{-u}$ (`v259`), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [**O**] It does *not* close `QG.AMB.01`: the trace regulator is not yet shown to resum to an entire $a(\Box)$, and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row `QGAMB.IDG.01`.
- **Double-pushout figure (`tfpt_1_architecture_e8`).** A new shared TikZ macro (`TFPTdoublepushout` in `tex-artefacts/tfpt_figures.tex`) draws the commuting square showing that the order-$|\mu_4|$=4 glue is *one* operation in two categories: the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ (`v1`) and the conformal-net / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$ (`v92`/`v281`/`v234`/`v277`/`v260`). Expository only; no new claim.
- **Framing (`introduction`, `tfpt_research_contracts`).** `SEAM.EQUIV.01` is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of c plays in relativity (cf. `v267`). Prose sharpening; no status change.

2026-06-19 (the discrete→dynamic lens on `F_transfer`: one shape, one seam-rate, `v303`)

- **`v303_ftransfer_dynamics.py`** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update → the E_8 marks, gap $6\ln(3/2)$) — is also the shape of the four `F_transfer` instances, or an unrelated bolt-on. It is the *same* shape: [**E**] F_pole (Koide) *is* the seam dynamics (Möbius multiplier F'=$(2/3)^6$=λ_2, the seam-clock eigenvalue ⇒ a unique attractor at the seam rate, `v82`/`v54`); [**C**] F_Boltzmann (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$, H-theorem); [**C**] F_relic (axion) freezes at an adiabatic fixed point; [**E**] F_QCD (m_p/m_e) is the 1-loop RG flow with the carrier b_3= -7 (asymptotic freedom, Gaussian UV attractor). [**O**] Verdict: one shape underlies all four; F_pole has the seam rate $(2/3)^6$ *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So `F_transfer` is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (`v187`), not reduced to the seam. [**E**] suite `v303` 5/5, `AUDIT OK`.

2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6\ln(3/2)$, `v302`)

- **`v302_seam_gap.py`** closes the chain on the single statement `v301` had left: "the quasi-free seam bulk is gapped". That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. [**E**] the frozen transfer spectrum is $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (`v160/v162`), a simple Perron eigenvalue 1 separated from the rest; **[E]** the gap is $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1-(2/3)^6 = 665/729$; $(2/3)^6$ the μ_4-deck transfer eigenvalue forced by the carrier); **[E]** with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (`v76`) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; **[C]** by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian. **[O]** **Bottom line:** with `v300/v301`, `SEAM.EQUIV.01`'s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the `v297` AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap $6\ln(3/2)$) — *no undischarged TFPT-internal assumption remains.* It stays **[O]** (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. **[E]** suite `v302` 5/5, `AUDIT OK`.

2026-06-19 (Route A's last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, `v301`)

- **`v301_route_a_invertible.py`** attacks the single remaining *unconditional* gap of `SEAM.EQUIV.01`. Since `v300` collapsed Route B into Route A's rationality, that gap is Route A's open hypothesis (`v297` LIT-A): "the raw quasi-free seam bulk is invertible (SRE)". A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev's periodic table, class D). **[E]** the carrier is 16 Majoranas, $c = g_{\mathrm{car}}+N_{\mathrm{fam}} = 8 = 16/2$ (`v148`); **[E]** $SO(16)_1 \supset (E_8)_1$ ($\dim\mathfrak{so}(16) = 120$ currents $+\,128$ spinor $= 248$, `v154`); **[C]** so a gapped quasi-free bulk is invertible automatically; **[E]** the invariant $|\det K| = \#$anyons is 1 for E_8 (invertible) vs 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce. **[O]** LIT-A's 'invertible bulk' now *follows* from 'gapped 16-Majorana quasi-free bulk' (`v148/v160`+the mass gap), so the residual changes from a topological-order statement to a spectral one. **[O]** **Bottom line:** with `v300`, the whole open content of `SEAM.EQUIV.01` reduces to the *single spectral statement* "the quasi-free seam bulk is gapped" — the topological-order obstruction (hidden anyons) is removed; the gap is not manufactured. **[E]** suite `v301` 6/6, `AUDIT OK`.

2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$, `v300`)

- **`v300_flataway_rigid.py`** attacks the single external fact the three Flat-Away routes (`v291/v293/v295`) left open, on two fronts. *Harden:* the flat seam was only a *soft* minimum ("the spectral mismatch grows with ε"); here the obstruction becomes a *discrete* on/off invariant. **[E]** the flat fingerprint — $\mathrm{spec}(|D_\theta|)$ is the integer ladder with every level $n \geq 1$ of multiplicity exactly 2; **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, + an $O(g^2)$ common shift), validated against the full Toeplitz operator; **[E]** any $g_k \neq 0$ lifts the resonant degeneracy ($2 \to 1{+}1$) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — strictly stronger than the "deviation grows" scan; **[E]** with the `v295` convexity, flat is isolated analytically *and* discretely. *Derive the pin:* **[C]** the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q-coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, i.e. the external pin *is* the $(E_8)_1$ rationality (Route A). **[O]** verdict: Flat-Away carries *zero* new open content beyond Route A — the two `SEAM.EQUIV.01` routes converge on one rationality fact. **[E]** suite `v300` 6/6, `AUDIT OK`.

2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2,3,r)$ seed reaches E_8 as the last finite point, v299)

- **`v299_hypergraph_0d_autonomous.py`** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on `v219/v298`): **[E]** the icosahedron is a 3-uniform hypergraph ($|\mathrm{Aut(graph)}| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$; the McKay group is the $SU(2)$ binary lift $2I$, a *different* order-120 group whose McKay graph is affine E_8; 30 edges $= h(E_8)$); **[E]** the $1 \to 4$ subdivision rewrite is equivariant (all 120 symmetries at every scale); **[E]** the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; **[E]** Smith ($\rho \le 2 \Leftrightarrow$ affine ADE) + Collatz–Wielandt ($\rho \le 2 \Leftrightarrow$ a local witness $Am \le 2m$); **[E]*** the autonomous rule (diffusion-generated witness + *local* gate $Am \le 2m$, Perron only as audit) runs $E_6 \to E_7 \to E_8 \to \hat{E}_8$; **[E]** the negative controls $\rho(A_n) = 2\cos\frac{\pi}{n+1}$ and $\rho(D_n) = 2\cos\frac{\pi}{2n-2}$ are < 2 for *all* n (closed form), so neither family ever reaches E_8. Honest type: **[O]** the $(2,3,\cdot)$ seed is the single irreducible input = P2 — *the rule reaches E_8 as the last finite point on this seed; it is not a derivation of P2.* **[E]** build green, `AUDIT OK`.

2026-06-18 (E8 as a network attractor — the Wolfram-program reading, v298)

- **`v298_e8_network_attractor.py`** is an honest audit of the "TFPT as a hypergraph/rewrite system" idea, on the `v219` McKay bedrock. **[E]** the affine-E_8 Kac marks $(1,2,3,4,5,6,4,2,3)$ are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4\,m$ on the $(2,3,5)$ McKay graph (converges from any positive start; $A\,m = 2m$, top eigenvalue 2); the spherical tower terminates at E_8 and the cascade is nested-subgraph coarse-graining. **[C]** the Wolfram reading "E_8 is the universal attractor of a minimal $(2,3,5)$ network" holds at the Coxeter-skeleton level. **[O]** a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives: φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. **[E]** build green, `AUDIT OK`.

2026-06-18 (a_2 closed form + Route A literature stack, v296–v297)

- **`v296_flataway_a2_closed.py` (`FLATAWAY.A2.CLOSED.01`)** evaluates the exact a_2 coefficient in *closed form*: the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1-e^{-4kt})$) to $W_k^{\mathrm{tail}} = t(1-e^{-4kt})/(8k(1-e^{-t}))$, leaving a finite $(4k-1)$-term middle; e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$. Manifestly positive, small-t law $W_k = t/2 + O(t^3)$ (the t^2 term cancels). Reproduces `v295` to machine precision.
- **`v297_route_a_literature_stack.py` (`SEAM.EQUIV.A.LIT.01`)** writes Route A's one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins) → LIT-B (Müger; Kawahigashi–Longo–Müger) → LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk $\Rightarrow (E_8)_1$", all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B's Flat-Away, and the stack is non-circular with Route B (the `v286` firewall holds). **[E]** build green, `AUDIT OK`.

2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **`v295_flataway_a2_exact.py` (`FLATAWAY.A2.01`)** upgrades the `v292` positive-definiteness from a numerical scan to a *proof*: the first variation of $\mathrm{Tr}\,e^{-t\Lambda}$ vanishes (zero-mean off-mark curvature, Gauss-Bonnet), and since $\Lambda \mapsto \mathrm{Tr}\,e^{-t\Lambda}$ is convex (Klein/Peierls), $\varepsilon{=}0$ is the *global* minimum — so $\Delta\mathrm{Tr} \ge 0$ for all deformations, $= 0$ iff $f = 0$. The exact second-order coefficient is the divided-difference kernel of e^{-tx} (Duhamel), validated to rel. err $< 10^{-6}$.
- **Lean `FORM.FLATAWAY.01`** (`SeamDeckClosure.lean` Part 5): the formal $\Delta{=}0 \Leftrightarrow$ flat step — a positive-weighted sum of squares is positive-definite (`heatDeviation_eq_zero_iff`); built and axiom-checked (only `propext`, `Classical.choice`, `Quot.sound`). So the heat route's

positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes a_2) remains. **[E]** build green, `AUDIT OK`.

2026-06-18 (Flat-Away attacked on all three routes, `v292`–`v294`)

- **`v292_flataway_heat_reduction.py` (`FLATAWAY.HEAT.01`)**: the heat route, made precise. The heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a *positive-definite* quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode, positive on all combinations), leading invariant $\|f\|_{L^2}$ — so Flat-Away reduces to the single fact "the seam fixes the a_2 heat coefficient".
- **`v293_flataway_spectral_hessian.py` (`FLATAWAY.SPEC.01`)**: the spectral route, upgraded from one direction to the *full* smooth-$\mathbb{Z}_4$ space. Mode $4k$ splits the degenerate Steklov level $|2k|$ at first order; the Hessian of the spectral mismatch over modes $\{4, 8, 12\}$ is positive-definite (eigenvalues $\{0.497, 0.500, 0.503\}$), so the flat seam is a strict isolated minimum — the "only one direction" objection is answered.
- **`v294_flataway_rp_energy.py` (`FLATAWAY.ENERGY.01`)**: the variational route. With the rigid mark divisor (4 cone angles π, Gauss-Bonnet 4π), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy $\int f^2$ is strictly convex (Hessian $2\pi I$) with a unique minimiser — Flat-Away reduces to "RP+gap realises the energy-minimiser". The three routes converge on one selection principle for the flat metric; closing any one closes Route B. **[E]** build green, `AUDIT OK`.

2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, `v290`–`v291`)

- **`v290_z4_smooth_curvature_adversary.py` (`SEAM.ADVERSARY.01`)** is the honest red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality: a smooth $\mathbb{Z}_4$ off-mark curvature $f = \varepsilon\cos(4\theta)$ **[E]** passes the `v288` commutator ($\|[\rho, \Lambda]\| = 0$, same as mark-local) yet **[E]** shifts the DtN spectrum ($\max|\Delta\lambda| = 0.155$) and the heat trace ($\Delta = 0.019$). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not $[\rho, \Lambda]$=0 — which is exactly why Flat-Away cannot be read off the commutator.
- **`v291_flataway_contract.py` (`FLATAWAY.RP.01`)** states Flat-Away as its own named mini-theorem and types three independent proof routes: **[E]** spectral-rigidity and **[E]** heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family — deviation zero only at ε=0) and **[C]** RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. **[E]** build green, `AUDIT OK`.

2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, `v289` / `v287`–`v288`)

- **`v289_marklocal_raw.py` (`MARKLOCAL.RAW.01`)** decomposes the one remaining Route-B residual ('why is the raw seam sub-principal term mark-local?') into a 5-lemma chain and isolates the *single* open analytic lemma: **[C]** L1 conic parametrix (b-calculus/Melrose, imported); **[O]** L2 **Flat-Away** — RP+gap+4 marks ⇒ curvature vanishes away from the marks; **[E]** L3 orbit-source (`v195`/`v216`), L4 Fourier-support on $4\mathbb{Z}$ (`v264`/`v288`), L5 full-operator $[\rho, \Lambda]$=0 (`v288`). 4/5 discharged — Route B reduces to exactly L2.
- **`v288` reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, `v264`; mark-locality from the RP seam Open Selection, `v289`) — so it can never be read as 'full-L^2 QGEO proved'.

- **v287 literature matrix**: L3 ('invertible bulk ⇒ single-sector') is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Müger/Kawahigashi–Longo–Müger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [**E**] build green, `AUDIT OK`.

2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **`v286_seam_equivalence_contract.py` (`SEAM.EQUIV.01`)** names the one open theorem — *the raw RP seam state is the holomorphic* $(E_8)_1$ *net at* $\tau=i$ — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional `v282`, 1 open Gral), an **import firewall** proving the two routes (`v276/v280/v284` vs `v277/v281/v285`) never import each other (no 'E_8 into the geometry and back' circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **`v287_free_rp_bulk_to_holomorphic_boundary.py` (`SEAM.EQUIV.A01`)**: Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible ⇒ μ=1 ⇒ holomorphic ⇒ $(E_8)_1$ is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk ⇒ single-sector boundary* (the AQFT form of $\det K$=1). 4/5 discharged.
- **`v288_full_l2_subprincipal_z4.py` (`SEAM.EQUIV.B01`)**: Route B (DtN) — *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \bmod 4$, so $\Lambda = |n| + M_f$ commutes with $\rho = \mathrm{diag}(i^n)$ on the full operator, $\|[\rho,\Lambda]\| \sim 2\times10^{-15}$; generic curvature → 4.44), lifting `v201/v284` to L^2. The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [**E**] build green, `AUDIT OK`.

2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **`v284_route_i_rp_uniqueness.py` (`QGEO.ROUTEI.01`)** turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [**E**] 5/6 lemmas are already discharged — L1 RP→DtN (`v194`), L2 marks→pillowcase (`v195/v216/v214`), L3 flat Steklov = $\sqrt{-\Delta_{\text{flat}}}$ (verified here), L4 covariance from DtN (`v258`), L5 μ_4-invariance (`v276`/Lean/`v280`); [**C**] Troyanov (χ_{orb}=0 ⇒ unique flat $\tau=i$) reduces the residual to one lemma. [**O**] the one open lemma: *the raw seam is the constant-curvature geometric state.*
- **`v285_route_ii_seam_condensation.py` (`QGAMB.ROUTEII.01`)** turns Route (ii) (seam-net condensation) into a 4-lemma chain: [**E**] 3/4 discharged (no abelian sector ⇔ $\det K$=1 ⇔ holomorphic ⇔ E_8; the tower 16→4→1), and **its open lemma coincides with Route (i)'s** — both are "the raw seam is the $(E_8)_1$-at-$\tau=i$ object" (`v282`), so proving *either* route closes *both.* [**O**] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau=i$. Build green, `AUDIT OK`.

2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at-$\tau=i$ unification, live scorecard, v280–v283)

- **`v280_pillowcase_steklov.py` (`QGEO.STEKLOV.01`)** runs the QGEO chain on the actual flat $\tau=i$ pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain $[\rho,\Delta]$=0 ⇒ $[\rho,H]$=0 ⇒ $[\rho,C]$=0 ⇒ $\omega\circ\rho$=ω is numerically exact — route-(i) evidence for `QGEO.OBLIG.01`; the premise stays [**O**].
- **`v281_holomorphy_modular_data.py` (`QGAMB.MODULAR.01`)**: #anyons = $|\det \mathrm{Gram}|$ (E8→1, SO(16)→4), the anyon-condensation tower 16 → 4 → 1, Gauss–Milgram → c=8; holomorphy ($\det K$=1) is the single Tier-B discriminator.

- **`v282_e8_tau_i_unification.py` (`QGAMB.UNIFY.01`)**: the candidate *simplification* — $\chi_{E_8}=j^{1/3}$ gives $\chi_{E_8}(i)=1728^{1/3}=12$, so $\tau=i$ is both the order-4 CM point (QGEO) and the $(E_8)_1$ modulus (QG.AMB): the two obligations are two faces of one object (the seam is $(E_8)_1$ at $\tau=i$), **obligation count** $2 \to 1$.
- **`v283_live_killtest_scorecard.py` (`PRED.LIVE.01`)**: the recent round's computable predictions vs current data — all consistent $< 2\sigma$, with two sharp near-term kill tests (Σm_ν floor 0.0586 eV; proton decay 1.55×10^{35} yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, `AUDIT OK`.

2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/`FORM.QGEO.03`)

- **`v279_qgeo_obligation_lemma.py` (`QGEO.OBLIG.01`) writes the bedrock as ONE precise constructive-QFT lemma.** Given the premise *the raw seam carries the flat $\tau=i$ pillowcase metric* (Troyanov-unique, $\Lambda_\Sigma=\sqrt{-\Delta_{\text{flat}}}$), then $\omega_\Sigma \circ \rho = \omega_\Sigma \to$ mark-locality $\to E_8$. **[E]** the reduction DAG from 'free RP seam' has exactly *one* open leaf — 'the raw seam state is the flat state'; **[E]** the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders), so the obligation is a state-identification, with **[C]** two proof routes (RP-state uniqueness, or Troyanov + OS).
- **`FORM.QGEO.03`: the all-orders step is now machine-proved in Lean.** `SeamDeckClosure.lean` gains `diagonal_commutesClock` and `flat_all_orders_clock`: any spectral function $g(\Delta_{\text{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho=\text{diag}(i^n)$ — so the `v276` all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full `lake build` green, 3356 jobs). The premise (the seam *is* flat) stays the **[O]** `QGEO.SYM.01` input. Build green, `AUDIT OK`.

2026-06-18 (4D QFT: the $S_{\text{pert}} \to S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **`v278_lsz_bridge_unitarity.py` (`QFT4D.SPERT.04`) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity.** **[E]** LSZ bridge: the EG T-products of S_{pert} (`v269`) *are* the OS Wick-rotated correlators (`v240`), so amputate+on-shell-residue gives $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$; **[C]** the gap $\Delta=6\log(3/2)>0$ gives Haag–Ruelle asymptotic states. **[E]** *one-loop optical theorem*: the bubble discontinuity $\operatorname{Im} B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2\operatorname{Im} M = \sum\int d\Pi|M|^2$ (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. **[O]** the gravity sector's Stelle ghost (`v269`/`rt_F`) is non-unitary $\to$ `QG.AMB.01`. Exact core mirrored in Wolfram (273/273). Build green, `AUDIT OK`.

2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\to (E_8)_1$ match, v277)

- **`v277_seam_calderon_e8_match.py` (`QGAMB.TIERB.01`) reduces Tier-B to ONE bit.** **[E]** the full $(E_8)_1$ matching target is computed and matches: $c=8$, $\det \text{Cartan}(E_8)=1$, the holomorphic character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ with a single primary and exactly $248=\dim E_8$ weight-1 currents (240 roots). **[E]** the discriminator is holomorphy — the $c=8$ counterexample $(D_8)=SO(16)_1$ has $\det\text{Cartan}=4$ (non-holomorphic), so the one condition is $|\det K|=1$. **[C]** the seam side matches via `v276` (flat $\tau=i$) + `v260` (Kummer/K3). **[O]** residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, `AUDIT OK`.

2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **`v276_qgeo_flat_closes_commutator.py` (`QGEO.SYM.03`) collapses the bedrock to ONE geometric postulate.** The state-level residual the whole bedrock rests on is the operator equation $[\rho, H]$=0. **[E]** if $H = f(\Delta_{\text{flat}})$ and the order-4 deck ρ ($z \mapsto iz$) is a flat-τ=i isometry, then $[\rho, \Delta]$=0 ⇒ $[\rho, H]$=0 *exactly*, to all orders (verified on the discretised square torus: ρ^4=I, $[\rho, \Delta]$=0, $[\rho, \sqrt{-\Delta}]$=0; generic non-isometry fails) — upgrading the leading-order `v198`/`v201` to the full H. **[C]** τ=i is forced by order-4 (j=1728; hexagonal j=0 is order-6, excluded). **[O]** `QGEO.SYM.01` therefore collapses to the single statement "the raw seam IS the flat τ=i pillowcase quasi-free state" — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, `AUDIT OK`.

2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, `rt_F`)

- **`redteam/rt_F_qft4d.py` (`REDTEAM.F.01`) stress-tests the just-published round** (the perturbative S_{pert} layer, the PMNS Jarlskog, the ν-scale, the over-determination, the `QG.AMB.01` roadmap). Survives (narrowed): two attacks land and are folded in. **(1)** the gravity R^2/Weyl^2 sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator $1/(p^2(p^2 + M^2))$ has a negative-residue massive spin-2 mode), so S_{pert} is unitary for the matter+gauge sector only — which is exactly why the non-perturbative `QG.AMB.01` is the real frontier (caveat added to `v269`). **(2)** the 0.11% anchor over-determination is conditional on the Λ-branch readout (`LAMBDA.BRANCH.01`, **[C]**), the gravity route being unconditional (caveat added to `v274`). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo.** Version `5.2 (rev 191)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20743347`, DOI `10.5281/zenodo.20743347`. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser S_{pert} skeleton `v269`, the concrete one-loop quartic `v271` and gauge β-coefficients `v273`), the complete complex PMNS matrix and leptonic CP strength `v270`, the honest absolute ν-mass-scale typing `v272`, the over-determination of the single mass anchor `v274`, and the `QG.AMB.01` obligation roadmap `v275`. Suite `273` modules (highest ID `v275`; v186/v226 skipped), Wolfram `271/271`, `run_all.py` green, `AUDIT OK`, `npm` build clean. The Zenodo description (`zenodo_description.html`) and the public website were synced in the same release.

2026-06-18 (scale + frontier round: ν-mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **`v272_nu_mass_scale.py` (`FLAV.NUSCALE.01`) types the absolute neutrino-mass scale honestly** (no fabricated Σm_ν — would contradict the `v184` firewall). **[E]** the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ (one UV input, like v_{geo}, `v153`); **[C]** the observed m_3=0.050 eV with the TFPT PS→GUT window $M_R \in [4.2{\times}10^{13}, 2.4{\times}10^{15}]$ GeV (`v249`) needs $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; **[O]** M_R is not a clean $\bar{M}_{\text{Pl}}$ power, so the absolute value stays open; **[X]** the NO floor Σm_ν=0.0586 eV is the cosmological kill test. Closes the PMNS bookkeeping (the last **[O]** is one seesaw ratio).
- **`v273_eg_gauge_running.py` (`QFT4D.SPERT.03`) derives the SM one-loop b_i via the EG gauge self-energy.** **[E]** the gauge 2-point is the same scaling-degree-4 extension as the quartic (`v271`), giving $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ from the carrier/SM content (the same

41 as the carrier algebra, `v159`); [**O**] two-loop + all-order stay open. Exact b_i mirrored in Wolfram.

- **`v274_scale_overdetermination.py` (`SCALE.OVERDET.01`) shows the single mass anchor is over-determined.** [**E**]/[**C**] two independent routes to $\bar{M}_{\rm Pl}$ agree — gravity $(8\pi G)^{-1/2} = 2.4353\times10^{18}$ GeV and cosmology (the compiler prediction $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4} = 2.24$ meV) $= 2.438\times10^{18}$ GeV, to 0.11%; the seam being a conformal c=8 CFT makes "no derivable absolute scale" a theorem, not a gap. Sharpens `v153`/`v78`.
- **`v275_qgamb_roadmap.py` (`QGAMB.ROADMAP.01`) gives the open gate `QG.AMB.01` a module.** [**E**] Tier A gap decoupling ($\Delta_{\rm eff} = 1.648 > 0$, `v36`/`v76`); [**E**]/[**C**] Tier B reduction to the holomorphic $(E_8)_1$ boundary net (`v77`); [**C**] the perturbative layer in hand (`v269`/`v271`/`v273`); [**O**] the nonperturbative ambient measure stays the one structural frontier.
- Suite `271/271` Wolfram, `run_all` green, `AUDIT OK`, build green.

2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, `v271`)

- **`v271_eg_oneloop_quartic.py` (`QFT4D.SPERT.02`) takes the `v269` $S_{\rm pert}$ skeleton from "exists" to an actual EG number** (no path integral). [**E**] the first renormalization is the order-2 product T_2; the massless propagator has scaling degree sd=2 in d=4, the one-loop bubble has sd=4=d, so ω=sd$-d$=0 $\Rightarrow$ exactly one logarithmic local counterterm ($C(\omega{+}d, d)$=1, finite, no infinite tower); the loop factor $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly — the *same* factor that normalises the spectral-action a_4 geometric quartic. [**C**] the ϕ^4 one-loop $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry $3 = s, t, u$ channels), EG initial condition $\lambda_\Phi(\mathrm{UV}) = 1/(16\pi^2)$. [**O**] order-2/one-loop only; the all-order EG construction and the nonperturbative measure `QG.AMB.01` stay open. Exact core mirrored in Wolfram (270/270). [**E**] build green, `AUDIT OK`.

2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, `v270`)

- **`v270_pmns_jarlskog_assembly.py` (`FLAV.PMNS.03`) assembles the four TFPT channels** ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$, $\sin^2\theta_{23}{=}\frac{1}{2}$, $\sin^2\theta_{13}{=}\varphi_0^{\rm ret}e^{-5/6}$, $\delta{=}4\pi/3$) **into one exactly-unitary complex PMNS matrix and derives the leptonic CP strength.** [**E**] $U = R_{23}U_{13}(\delta)R_{12}$ is unitary; [**C**] the Jarlskog $J_{\rm PMNS} = \Im(U_{e1}U_{\mu2}U^*_{e2}U^*_{\mu1}) = -0.02965$ ($J_{\max} = 0.03424$) is a *derived* number, the leptonic analogue of the CKM J (`v88`); [**C**] data: $J_{\max}$ vs NuFIT 5.2 0.0332 ($\sim 3\%$), J vs best fit -0.026, the negative CP-violating sign reproduced. [**O**] δ is the μ_6/triality phase (assembled, not from M_ν/D_F) and the absolute ν-mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [**E**] build green, `AUDIT OK`.

2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, `v269`)

- **`v269_spert_paqft_skeleton.py` (`QFT4D.SPERT.01`) builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [**E**] three distinct layers: $S_{\rm top}$ (2d DHR braiding $|M|$=1, statistics, `v243`), $S_{\rm pert}$ (4d Epstein–Glaser/BV S-matrix of the spectral action), $S_{\rm phys}$ (LSZ on the OS Wightman functions, `v240`). [**E**] the spectral-action a_4 interaction is power-counting renormalizable (all operators dim ≤ 4; 7 marginal + 3 relevant, a finite counterterm basis), so [**C**] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [**C**] the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit ($S_{\rm pert} \to S_{\rm phys}$ via LSZ). [**O**] perturbative only — the ambient measure `QG.AMB.01` stays open, the canonical 4d reading remains boundary-only (`v265`). [**E**] build green, `AUDIT OK`.

2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **`v268_theta13_carrier_trace.py` (`FLAV.TH13.01`) closes the θ_{13} exponent origin.** The value $\sin^2\theta_{13} = \varphi_0^{\rm ret} e^{-5/6} = 0.0231$ was already checked (`v16`); now the exponent is exact: **[E]** $\gamma = 5/6 = \mathrm{tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = \mathrm{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, roots of $6Y^2 - Y - 1 = 0$; complement $\frac{1}{6}$ the neutrino ratio). **[E]** own channel: the $\mu\tau$-breaking $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret}$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel — correcting the tempting (and wrong) "θ_{13} from M_ν" route flagged by the new script atlas. **[O]** the CP magnitude and the absolute ν-mass scale remain the open PMNS items. Exact core mirrored in Wolfram. **[E]** build green, `AUDIT OK`.

2026-06-18 (Infrastructure: a generated causal "script atlas" so the suite is navigable at a glance)

- **`verification/make_script_atlas.py` generates `verification/script_atlas.md`** — a single grouped, dependency-aware view fusing the existing single sources (`status_ledger.csv` for claim→status→dependencies→supersedes, `script_registry.csv/script_clusters.csv` for the theme grouping, `docs_map.csv` for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into `bash build.sh gen`. **[E]** build green, `AUDIT OK`.

2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for `QGEO.SYM.01`, v267)

- **`v267_qgeo_rigidity_minimal_axiom.py` (`QGEO.SYM.02`) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). **[E]** the *single* bare statement "the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks" ($= \omega \circ \rho = \omega$) forces *everything*: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 independent of a; cross-ratio 2 $\Leftrightarrow$ j=1728 (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism on y^2=x^3−x; and the unique flat pillowcase metric, mark-locality, $[\rho, \Lambda]$=0 and $\omega \circ \rho$=ω all follow (`v214/v201/v264/v198`). **[E]** neg controls: generic config $j \neq 1728$ ($\mathbb{Z}_2$ only), hexagonal j=0 ($\mathbb{Z}_6$, order 6) — only the square (order-4) realises the μ_4 clock. **[C]** second, independent reason: $\{1, i, -1, -i\}$ over $\mathbb{Q}$ with j=1728 (CM by $\mathbb{Z}[i]$), Belyi/dessins Galois-rigidity. **[O]** the bare order-4 symmetry stays the one foundational postulate (like c=const) — the *reduction* is closed (axiom ⇒ everything), the axiom is not. **[E]** build green, `AUDIT OK`.

2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, +$(15, 1, 1)$ survives, Hyper-K decisive, v266)

- **`v266_ps_threshold_proton.py` (`PS.PROTON.02/PS.THRESHOLD.02`) decides branch B of the `v265` fork.** The explicit two-step Pati–Salam thresholds give d=6 $\tau_p(p{\to}e^+\pi^0)$ (writes `ps_threshold_proton.json`): **[X]** the minimal 16-Higgs content (M_{GUT}∼$2.4{\times}10^{15}$) gives τ_p∼$4.4{\times}10^{33}$ yr < Super-K and is *killed* even at the high hadronic band; **[C]** the E_8-allowed $(15, 1, 1)$ [the single 45] raises M_{GUT}∼$5.9{\times}10^{15}$ ⇒ τ_p∼$1.6{\times}10^{35}$ yr, proton-safe *now*; **[X]** but that sits at the Hyper-K reach (∼$1.4{\times}10^{35}$ yr) — a *dated* kill test decisive within the decade. So branch B is promoted from "well-mannered candidate" to *content-selected (must be* +$(15, 1, 1)$*), Hyper-K-decisive UV branch*; **[O]** structurally marginal (only one 45 in the hull, `v247`) with an $O(3)$ τ_p band. Branch A (boundary-only) is unchanged and canonical. **[E]** build green, `AUDIT OK`.

2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, `v265`)

- **`v265_qft4d_fork_freeze.py` (`QFT4D.FORK.01`) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree** (`qft4d_fork_freeze.json`), no new physics. **A (canonical):** boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* (E_8 is the audit hull). [**X**] SM-only 4D-GUT unification is killed (1-loop: at the α_1=α_2 scale α_3^{-1} off by ∼5.6, spread ∼8.8 at 2×10^{16} GeV, re-deriving `v246`) — the canonical expectation, not a failure. **B (conditional):** carrier-native Pati–Salam is a falsifiable UV branch only with the E_8-allowed content $\{1, 10, 16, 45\}$ (no **126**, `v247`), $\sin^2\theta_W$=3/8 (`v248`), the $O(1)$ κ=M_{PS}/M_s band (`v249/v253`), a threshold model and a proton-decay kill test ([**X**], no fake τ_p window; carrier gauging stays the contract). [**E**] a prose guard (`QFT4D.NO_OVERCLAIM.01`) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows `QFT4D.BOUNDARY.DEFAULT.01`, `QFT4D.SMONLYGUT.01`, `PS.UVBRANCH.01`, `PS.THRESHOLD.01`, `PS.PROTON.KILL.01`. A fork-policy box added to `tfpt_research_contracts`; [**E**] build green, `AUDIT OK`.

2026-06-18 (Three open-residual advances: the F_{QCD} solver, M_ν from D_F, and the raw-seam⇒mark-locality reduction, `v262`–`v264`)

- **`v262_fqcd_mp_me.py` (`FR.MPME.02`) implements the one contract-only F_{transfer} pipeline.** A real two-loop α_s solver runs $\alpha_s(M_Z)$=0.1179 with n_f thresholds (carrier $b_3 = -7$, [**E**]) to $\alpha_s(2\,\mathrm{GeV})$∼0.30 and $\Lambda^{(3)}_{\overline{\mathrm{MS}}}$∼0.33 GeV; with the lattice bridge C_p (external $O(1)$) and the closed m_e it reproduces m_p/m_e∼1824 vs the measured 1836.15 within the ∼ 7% external budget. A [**C**] transfer reproduction (firewall: no compiler integer for 1836), advancing `FR.MPME.01` ([**O**]) to `FR.MPME.02` ([**C**]).
- **`v263_mnu_from_df.py` (`FLAV.PMNS.02`) makes M_ν a seesaw operator of D_F.** [**E**] $M_\nu = -m_D M_R^{-1} m_D^\top$ is symmetric, seesaw-suppressed and (for $\mu\tau$-symmetric inputs) $\mu\tau$-symmetric; [**C**] it reproduces θ_{23}=45° and $\sin^2\theta_{12}$=$\frac{1}{3}-\frac{\varphi_0^{\mathrm{ret}}}{2}$ (`v9`) under the seam misalignment; [**O**] θ_{13}=0 in the leading texture, the CP *magnitude* (240°) is the μ_6/triality phase (`v231/v233`), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.
- **`v264_raw_seam_marklocal.py` (`QGEO.ENERGY.03`) sharpens the bedrock.** Composing `v216`+`v201`+`v198` on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is $\mathbb{Z}_4$-invariant (Fourier on $4\mathbb{Z}$, FFT-verified), block-diagonal, and $[\rho, \Lambda]$=0 ⇒ $\omega \circ \rho$=ω on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=`QGEO.SYM.01`). A reduction/sharpening, [**O**] — it does *not* prove the seam must carry that metric (negative control: curvature off the μ_4 orbit breaks the chain). Paper citations added in `tfpt_4_frontier`, `tfpt_2_standard_model` and `tfpt_research_contracts`. [**E**] `AUDIT OK`, build green.

2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)

- **Five paper sections that predated the `v238`–`v261` round are brought up to date** (additive, no status marker moved). The `introduction` closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ closed modulo `QGEO.SYM.01` (`v258`–`v261`) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The `tfpt_2` “ambient closure” section now states that the cutoff of its $S_{\rm rel}$ *is* the seam KMS weight (f_2/f_0=1, `v259`) and $D_{\rm rel}$ the covariance induction (`v258`); the `tfpt_2` "(4) Constructive QFT closure" section adds the relative-object pointer; `tfpt_4_frontier`'s full-quantum-gravity section records that the cutoff moment f_0 is pinned by the seam state (`v255`/`v259`); and `origin_theory`'s honest-status keybox notes that the whole boundary QFT collapses onto the same `QGEO.SYM.01` postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [**E**] build green (`10 ok, 0 failed`), `AUDIT OK`.

2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The `v258`–`v261` closure is now reflected on every "what is open" surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary $G_{\rm net}$ entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise `QGEO.SYM.01` via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card ("the structural residual is one condition") carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [**E**] build green, `AUDIT OK`.

2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, `v258`–`v261`)

- **`v258_dirac_covariance_induction.py` (`PS.DIRAC.03`) closes the "posited Yukawa vs derived operator" gap of `v250`/`v252`.** The finite Dirac operator D_F is the modular/covariance *induction* of the boundary seam quasi-free (KMS, β=1) state: a Gaussian state is fixed by its covariance C (a positive contraction) with one-particle modular Hamiltonian $H = \log((1{-}C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1{+}e^H)^{-1}$. [**E**] the inversion identity holds exactly; [**E**] C_F is a positive contraction; [**E**] the induced operator is chirality-odd; [**E**] it is unitarily equivalent to the `v252` D_F (same mass spectrum); [**E**] the Majorana/seesaw m_D^2/M_R is an off-diagonal covariance entry. [**C**] $C_F = P_F C_\Sigma P_F$, so D_F is the Kasparov shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\rm car}]$ of the boundary geometry and the `v18` Yukawas are the readout of C_Σ.
- **`v259_modular_cutoff_kappa.py` (`PS.SPECACT.02`) fixes the last open spectral-action input.** The cutoff f is the seam's own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at β=1 (`v239`) and 2π=$1/(4c_3)$ (`v58`) fixes β, so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*. [**E**] its moments give $f_2/f_0 = f_4/f_2 = 1$ exactly; [**E**] a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$ (the choice is non-vacuous); [**E**] hence $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\rm grav}} = \sqrt{c_{PS}/c_{\rm grav}}$ loses its scheme factor. [**C**] the open [**O**] cutoff freedom of `v253`/`v255` becomes a finite trace ratio over the 96-dim H_F (the RG $\kappa{\approx}1.3$ pins $c_{PS}/c_{\rm grav}{\approx}1.7$).
- **`v260_k3_kummer_unification.py` (`ARCH.K3.01`) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z{\mapsto}{-z})$ (`v214`); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [**E**] the E_8 Cartan is even/det 1/positive-definite; [**E**] $H^2(\mathrm{K3},\mathbb{Z}) = U^3 \oplus E_8(-1)^2$ (rank 22, det -1, even, signature $(3,19)$); [**E**] the seam is $T^2/(z{\mapsto}{-z})$ with 4 marks (orbifold Euler char 0); [**E**] the carrier-16 are the $|A[2]|$=2^4=16 Kummer nodes = $\dim S^+$ (`v197`), 16=4×4; [**E**] honest note — the rank-16 Kummer lattice is *not* the $E_8(-1)^2$ summand. [**C**] so the `v1` "glue $D_5{\oplus}A_3{+}\mu_4{\Rightarrow}E_8$" is the lattice shadow of a single object; E_8 appears both locally (du Val $\mathbb{C}^2/2I$, `v232`/`v236`) and globally (H^2 summand). A unification, not a closure.

- **v261_modular_spectral_closure.py (QFT.MSC.01) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ reduced to *one* open premise. **[E]** the cross-consistency invariants agree: $4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap $6\log(3/2)$ (OS = recovery = Lindblad); **[E]** the three closures re-derived; **[E]** the ten-component manifest present. **[O]** the QFT layer reduces to `QGEO.SYM.01` (seam realisation), the ambient QG measure `QG.AMB.01` kept separate. **[C]** the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

2026-06-17 (Infrastructure: the changelog is now a public website page, generated from `changelog.tex` and audit-gated)

- **The canonical changelog now also drives a public `/changelog` website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf`/`\emph`/`\texttt`/`\vref`, the status markers **[E]**/**[C]**/**[O]**/**[X]**, inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. **[E]** build (`npm run build` green, `/changelog` prerendered static) and `bash build.sh audit` (`AUDIT OK`).

2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **v257_quasiorient_modpd.py (PS.NCG.ORIENT.02) corrects the slightly too-generous v256 typing.** The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit `v252` triple: **[E]** strict orientability fails ($\gamma \notin \mathrm{span}\{\lambda(a)\rho(b)\}$); **[E]** *quasi-orientability* holds (the left/right A-irrep boxes are disjoint by chirality: $L = H$ weak-doublet, $R = \mathbb{C}$ singlet, so $L^{\mathrm{LR}}(\mathcal{H}^{\mathrm{even}}, \mathcal{H}^{\mathrm{odd}}) = \{0\}$); **[E]** strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal-L/R-neutrino case); **[C]** modified Poincaré duality holds (CCM), Stephan's alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **v254_inner_fluctuations.py (PS.NCG.FLUCT.01) closes v250 #5 + the no-junk obligation.** The inner fluctuations $\Omega^1_D(A_F)$ of the finite Dirac operator are computed explicitly: **[E]** they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet $(1,2)_{1/2}$ — no junk, no triplet, no $(10,1,3)/\mathbf{126}$. **[E]** the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and **[E]** σ is an SM singlet $(1,1,0)$ with $B{-}L = 2$. **[C]** σ = scalaron.

- **`v255_spectral_action_expansion.py` (`PS.SPECACT.01`) closes the spectral-action expansion (`v252` #11).** The Seeley–DeWitt expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$: [**E**] structure ($a_0$ cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl^2 + R^2+Higgs-quartic, $N = 96$); [**E**] $b/a^2 = 1/3$ (top-dominated); [**E**] $\sin^2\theta_W = 3/8$; [**E**] the R^2 spin-0 mode = the Starobinsky scalaron; [**C**] Higgs quartic → ∼125 GeV with σ; [**C**] $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\rm grav}}$ explicit; [**O**] exact decimal needs the cutoff moments.
- **`v256_orientability_poincare.py` (`PS.NCG.ORIENT.01`) — honest status, no faked proof.** KO-dimension 6 forces the intersection form $\mathrm{Tr}(\gamma\,\pi(p)J\pi(q)J^{-1})$ to be *antisymmetric*: [**E**] skew form (verified), [**E**] rank 2, corank 1, null $\sim (1,1,-1)$ on the SM K-theory $\mathbb{Z}^3$; [**E**] the orientability local condition $[\gamma_F, \pi(a)] = 0$. [**C**] the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); [**O**] a self-contained machine proof needs the complete canonical bimodule.

2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and κ pinned to $O(1)$, `v252`–`v253`)

- **`v252_full_finite_triple.py` (`PS.DIRAC.02`) advances the Dirac contract `v250` from a 7-obligation contract to a concrete object discharging 5 to [E].** The full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ — H_F the particle⊕antiparticle doubling of three $SO(10)$ **16**-plets — is built explicitly and the NCG axioms are checked by direct computation: [**E**] reality ($J^2{=}{+}1$, $JD{=}DJ$), grading ($\gamma^2{=}1$, $[\gamma, a]{=}0$, $J\gamma{=}{-}\gamma J$), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to $\{\nu_R, e_R\}$ — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\rm light}{=}m_D^2/M_R$. [**C**] Poincaré duality; [**O**] orientability, no-junk, spectral-action expansion.
- **`v253_kappa_scalaron_ps.py` (`PS.KAPPA.01`) discharges residual 6(b) of `v249`.** κ in $M_{PS} = \kappa\,M_s$ ($M_s = c_3^{7/2}\bar{M} \approx 3.06{\times}10^{13}$ GeV, exact) is pinned by an RG scan (PDG errors × the two E_8-allowed contents) to a tight $O(1)$ interval (≈ 1.3). [**C**] heat-kernel origin: $\kappa = \sqrt{(f_2/f_0)\,c_{PS}/c_{\rm grav}}$ is a scale-free ratio of heat-kernel traces over the 96-dim H_F (Λ cancels). [**X**] the **126**-content drives κ out of the $O(1)$ window (content-selective). [**O**] the exact decimal needs f_2/f_0 fixed.

2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/σ — the CCvS Pati–Salam mechanism, `v251`)

- **`v251_first_order_sigma.py` (`PS.DIRAC.01`) advances the Dirac contract `v250` from a contract to an exhibited mechanism.** On a faithful lepton-block model ($H = H_F \oplus H_F^c$, $H_F = \mathbb{C}^3$), the real even structure is checked ($J^2{=}{+}1$, $\gamma^2{=}1$, $D\gamma{=}{-}\gamma D$, KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ($\nu_R\nu_R^c = \sigma$) block — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ. Honest correction to the review: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [**C**] σ = scalaron (`v249`); [**O**] the full 96-dim triple + spectral action.

2026-06-17 (Carrier-native Pati–Salam program: E_8 forbids the 126, PS algebra, unification kill-test, Dirac contract, `v247`–`v250`)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** `v247_e8_branching_no126.py` (`PS.E8BRANCH.01`): the E_8 hull branches under $SO(10){\times}SU(4)$ as $(45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4})$, so the $SO(10)$ content is exactly

$\{1, 10, 16, 45\}$ and the **126** is *absent* — the renormalisable 126_H is algebraically forbidden and 10+16+45 is E_8-supplied (only one 45 $\Rightarrow$ proton-marginal). `v248_ps_algebra.py` (`PS.ALGEBRA.01`): $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \to SU(2)_L \times SU(2)_R \times SU(4)_c$, exact hypercharges $Y = T_{3R} + (B-L)/2$, matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$ (colour $SU(4)_c \subset D_5$, distinct from the family A_3). `v249_ps_unification.py` (`PS.RGTEST.01`): the two-step PS$\to SO(10)$ unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content), $M_{PS}/M_s = 1.0$–1.7. `v250_dirac_triple_contract.py` (`CONTRACT.QFT4D.DIRAC.01`): the spectral-triple contract ($H_F = 3\times16$, the seven NCG axioms, σ =scalaron) that would turn "carrier is gauged" from a fork into a theorem. The root-level `qft.txt` carries the full analysis.

2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, `v246`)

- **`v246_unification_data_crosscheck.py` (`QFT4D.RGTEST.01`) — the first hard data confrontation of the spectral-action prediction.** TFPT's gauge-charged content *is* the Standard Model (`v159`) and the theory adds no new states; the E_8 cascade is an arithmetic spine (`v5`), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread $\sim 10^{13}$–10^{17} GeV (residual $\Delta\alpha^{-1} \approx 9$ at 2×10^{16} GeV), and at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at 1.7×10^{14} GeV — the three never meet. Typed [**X**]/[**O**]: with "no new state" there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (`figures/qft_skeleton.pdf`, `figures/qft_unification.pdf`) added via `make_figures.py` (PDF + website PNG) and embedded in `tfpt_research_contracts.tex`; a root-level `qft.txt` summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

2026-06-17 (Spectral-action matter content: the $SO(10)$ **16** = one anomaly-free SM generation, `v245`)

- **`v245_spectral_matter_content.py` (`QFT4D.MATTER.01`) discharges the computable core of the 4d contract `v244`.** The carrier half-spinor (the $SO(10)$ **16**) is verified, in exact rational arithmetic, to be one Standard-Model generation: $\mathbf{16} = SU(5)(\mathbf{10}+\bar{\mathbf{5}}+\mathbf{1})$, the SM multiplet dimensions sum to $16 = \dim S^+$ (15 Weyl fermions $+\,\nu_R$), the electric charges $Q = T_3 + Y$ are exactly the SM values, all four gauge anomalies cancel ($\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$), the spectral-action normalisation gives $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\,g_1$), and the Higgs is the $(1,2)_{1/2}$ inner fluctuation. So the matter-content + tree-relation half of `CONTRACT.QFT4D.01` is now [**E**]; only the seam$\to$Dirac realisation and the run-down couplings stay [**O**]. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, `v239`–`v244`)

- **Six emergent-QFT skeleton scripts (`v238`).** `v239_kms_thermal_time.py` (`DYN.KMS.01`): the modular flow is KMS at $\beta = 1$, and $2\pi = 1/(4c_3)$ fixes the modular temperature as $T_H = c_3/M$ — thermal time = horizon time. `v240_gns_os_reconstruction.py` (`DYN.GNS.01`): GNS reconstructs the Hilbert space from $(\mathcal{M}, \omega)$ and the OS Hamiltonian $H_{\mathrm{OS}} = -\log T = -L$ is positive with gap Δ (the `v64` mass gap). `v241_dhr_sectors.py` (`DHR.SECTORS.01`): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower $16 \to 4 \to 1$ (`v235`/`v237`). `v242_spin_statistics_modular.py`

(DHR.MODULAR.01): spin-statistics, the modular (S, T) data, Verlinde fusion, and Gauss–Milgram $c = 8$. `v243_haag_ruelle_braiding.py` (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the v238 mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is v36; the matter Lagrangian is the open [**O**] contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New `v238_modular_lindblad_dynamics.py` (DYN.SEMIGROUP.01, cluster registry).** A companion to the modular round (v196/v198) and the recovery code (v221/v64) that reads the *same* seam objects *dynamically* rather than statically. [**E**] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the μ_4 clock, $\rho\sigma_t\rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$, IS the v198 state-invariance — so the static commutator is the conservation law of modular time. [**E**] the recovery channel's generator $L = \log T$ is a doubly-stochastic/CPTP-classical Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time). [**E**] keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2) =$ the OS mass gap Δ (v64) — the static recovery bound (v221) and the dynamical mass gap (v64) are one number, one exponentiated. [**E**] the GKSL split $G = i\Lambda_\Sigma + L$ (imaginary + real≤ 0 spectrum). [**O**] the residual is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra = QGEO.SYM.01 (v198) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

2026-06-17 (Experiments: black-hole-cosmology signatures, the η_B Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (`experiments/`, all `data_limited`, not load-bearing).** From the `problem_b` black-hole-cosmology survey: (i) `ccbh-dark-energy` — the de Sitter seam interior (w_{in}=−1) forces the Croker–Weiner cosmological-coupling index $k = -3w_{\text{in}} = 3$, hence a population $w = -1$; vs. Farrah+2023 $k = 3.11 \pm 0.79$ (-0.14σ), a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI watchdog tests (`alternative_group w_de_eos`, never double-counted). (ii) `gravastar-compactness` — the Nariai Q_{geom}=3/8 equals the Jampolski–Rezzolla (2026) maximum horizonless compactness $\mathcal{C}$=3/8 (shared de Sitter endpoint 1/2); since 1/3<3/8<4/9<1/2 this is a light-trapping ECO, yielding an echo template (round-trip delay ∼0.7 ms at $62\,M_\odot$, amplitude $\leq (2/3)^6$) — a [**C**] structural echo, no $\mathcal{C}\leftrightarrow Q_{\text{geom}}$ map proven. (iii) `cosmic-handedness` — a parity watchdog (Shamir JADES ∼3.3σ spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- η_B **— full BDP Boltzmann ODE solve confirms the route at the frozen scale (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`; `tfpt_4_frontier`, [C]).** Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency $\kappa_f = 0.092$, validating the analytic fit) at the *frozen* heavy scale $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV with $\delta_{\text{CP}} = 4\pi/3$ gives $\eta_B = 6.5 \times 10^{-10}$ — a factor 1.07 from the observed 6.1×10^{-10}, with *no* free M_R dial. This moves the leptogenesis route from "solve pending" to a consistent [**C**] readout (the full *flavored* density-matrix solve is the next refinement); `tfpt_4_frontier` Route C synced.

- **The Petz recovery map and the rank-one baby universe, realised numerically (`experiments/recovery-channel`; `tfpt_horizon_readouts`, [C]; companion to `v221`).** The gapped transport contracts to a *rank-one* projector at exactly $\|T^n - P_\infty\| = (2/3)^{6n}$ — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected λ=1 (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification `v221` deferred is now verified at the channel level (`internal_consistency`, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.

- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (`experiments/gw-ringdown-echo`; `tfpt_horizon_readouts` search-targets).** The Stage-1 machinery (Kerr-ringdown subtraction $\to$ matched filter on the residual, ratio $(2/3)^6$ frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in ≥ 2 detectors**, consistent with the $(2/3)^6$ *upper* bound (a single loud H1 excess has $\hat{q} \approx 2$, far from $(2/3)^6$, so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.

- **Scorecard + website synced.** `experiments/evidence_scorecard.json` now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website `EXPERIMENTS_AUDIT` mirror and the η_B prediction entry are updated to match. These are standalone `experiments/` surfaces (not in the verification suite or ledger); the `tfpt_4_frontier/tfpt_horizon_readouts` and `papers.ts/predictions.ts` edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors `figures/`)

- **Red Team Target A re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, infrastructure).** The Python red-team surface still emitted the historical *three-residual* "A reduced, not closed" form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement "the seam–Calderón boundary net is holomorphic with $c = 8$" ($\Leftrightarrow$ the index-4 simple-current extension), with the index-4 $(D5)_1 \times (A3)_1 \to (E8)_1$ glue realised at Lie level (`v143`) and its odd sectors twisted/Ramond ($E_8 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$, `v148`), while the $q(A_3)$ normalisation collapses into the one declared anchor (`v152`). Two exact adversarial checks added ($120+128 = 248$; glue index $= |\mu_4| = N_{\mathrm{fam}}+1 = 4$); status stays **[C]**/**[O]** (reduced, not closed).

- **Shadow export now mirrors `figures/` but not `website/` (`scripts/export-shadow.sh`, `verification/make_script_index.py`, infrastructure).** `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh notes` runs on the subset without crashing.

2026-06-17 (CP phases as μ_6 powers — a reduction of red-team Target D)

- **Both CP phases are μ_6 powers of one hexagonal CM unit, split by the sheet (`v231_cp_mu6_phases`, `tfpt_5_redteam`, [E]/[C]).** Sharper than `v220`/`v225` (which only *located* the CP residual). With $\rho = e^{i\pi/3}$ the primitive 6th root (the j=0 Eisenstein CM unit): δ_{CKM} leading $= \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\mathrm{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the phase lattice), with $\rho^4 = -\rho$, so $\delta_{\mathrm{PMNS}} = \delta_{\mathrm{CKM,lead}} + \pi =$ (CM unit) $\times$ ($\mathbb{Z}_2$ sheet, $\rho^3 = -1$). The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ($\pm 21 \sin(\pi/3)$). A structural reduction of

Target D (removes one of its two phase inputs); the quark seam correction $3\lambda_C^2$ stays the `v88` misalignment, the deck stays $\mathbb{Z}/4$, and the physical CP derivation stays [**C**] (`CP.MU6.01`).

- **The seam as the E_8 Kleinian singularity (`v232_e8_kleinian_seam`, `origin_theory`, [E]/[C]).** The du Val side of the McKay correspondence (`v219`) gives the open seam-realisation premise `QGEO.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine-E_8 McKay graph of $2I$ leaves the finite E_8 Dynkin = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$; the eight curves = rank $E_8 = g_{\text{car}}+N_{\text{fam}}$ (a fourth reading of the 8), their negated intersection form is the E_8 Cartan (det 1, even, (-2)-curves), and the link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ perfect $\Rightarrow H_1{=}0$, $\pi_1{=}2I$ order 120). The raw seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes E_8; `TOPO.E8.01`, bedrock stays [**O**]).
- **CP is the universal family/triality phase, only sheet-split (`v233_cp_triality_phase`, `tfpt_5_redteam`, [E]/[C]).** One step past `v231` in the $\mu_6 = \mu_3(\text{family/triality})\times\mu_2(\text{sheet})$ factorisation: $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight). Since $\rho^4 = \rho^1\rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (`CP.TRIALITY.01`).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate (`v234_seam_holomorphy_selection`, `tfpt_research_contracts`, [E]/[O]).** The whole structural residual is *one* condition — "the seam carries no nontrivial abelian sector" — with three provably-equivalent faces, all forcing E_8: (i) holomorphy (μ-index 1; Target A), (ii) the seam link is a homology 3-sphere (Γ perfect $\Leftrightarrow 2I$; `v232`), (iii) exactly one 1-dim irrep (`v219`). All equal $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$, which is 1 only for E_8; and holomorphic $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice E_8. So P2, G_{net}, Target A and the Kleinian seam are one gate. The stated closing theorem ([**O**]): RP+gap+chirality $\Rightarrow$ quasi-free bulk (`v160`) $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$; the single residual analytic step is "free bulk $\Rightarrow$ holomorphic boundary" (`GATE.HOLO.01`).
- **The closing step in Chern–Simons language: holomorphic $\Leftrightarrow$ $\det K{=}1$ (`v235_seam_chern_simons`, `tfpt_research_contracts`, [E]/[O]).** A genuine reduction (not a closure) of `GATE.HOLO.01` into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer K-matrix with #anyons $= |\det K|$, edge $c = \text{signature}(K)$, and the edge net holomorphic $\Leftrightarrow |\det K|{=}1$. The TFPT extension tower (`v92`) *is* the condensation tower read by determinant: carrier $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic) — all rank 8, $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ — so holomorphic $= E_8 =$ the Kitaev E_8 quantum-Hall state. Holomorphy $\Leftrightarrow$ condensing the order-$|\mu_4|$ Lagrangian glue (det 16$\to$1, vs. a partial order-2 isotropic $\to 4{=}D_8$). The residual narrows to one integer condition "the free RP seam condenses the Lagrangian glue (det $=$1)" = the sheet/QGEO selection, now with holomorphy$\Leftrightarrow$ det 1 a theorem (`GATE.HOLO.02`).
- **The $(2,3,5)$ Brieskorn singularity is the one generator of the skeleton (`v236_brieskorn_capstone`, `origin_theory`, [E]).** Capstone of the icosahedral round: the singularity $x^2{+}y^3{+}z^5$, whose exponents are the atoms $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}){=}(2,3,5)$, has Milnor number $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}8{=}\operatorname{rank} E_8$ (a *fifth* origin of the "8"), Milnor monodromy = the order-30 E_8 Coxeter element (eigenvalues the primitive 30th roots = the E_8 exponents, charpoly Φ_{30}), Milnor lattice E_8, link the Poincaré sphere; the two clocks are facets of this one monodromy — the μ_3 triality (CP, `v233`) is h^{10} in $\langle h\rangle{=}\mathbb{Z}/30$, the μ_4 clock (`v223`) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$ (`TOPO.BRIESKORN.01`).
- **The closing step as a physical condition: no topological degeneracy $\Leftrightarrow$ $\det K{=}1$ $\Leftrightarrow$ the seam is SRE (`v237_seam_sre_closure`, `tfpt_research_contracts`, [E]/[O]).** Sharpens `GATE.HOLO.02`'s residual from definitional to falsifiable: the K-matrix ground-state

degeneracy on a genus-g surface is $|\det K|^g$ (torus: carrier 16, D_8 4, E_8 1), so "no topological degeneracy on any closed surface" $\Leftrightarrow$ $\det K{=}1$; among rank-8 even positive-definite K ($c{=}8{=}g_{\rm car}{+}N_{\rm fam}$) only E_8 is short-range-entangled (the Kitaev E_8 phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$); so the one open step is the physical statement "the free RP seam is SRE (no topological order)" (`GATE.HOLO.03`).

2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are $\{2,3,5\}$ (`v219_icosahedral_mckay`, `origin_theory`, [E]).** E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T{\to}\hat{E}_6$, $2O{\to}\hat{E}_7$, $2I{\to}\hat{E}_8$). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group $2I$ (element orders $\{1,2,3,4,5,6,10\}$ — it contains the $2,3,5$ rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are $\{1,2,2,3,3,4,4,5,6\}$, sum $30{=}h(E_8)$, sum of squares $120{=}|R^+(E_8)|$; the McKay adjacency from the natural 2-dim rep *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8, not a P2 proof (`MCKAY.E8.01`).
- **CM-norm duality: 41 (square) and 7 (hex) (`v222_cm_norm_duality`, `origin_theory`, [E]).** The square seam (Gaussian $\mathbb{Z}[i]$, $j{=}1728$) gives $N(g_{\rm car}{+}i|\mu_4|){=}5^2{+}4^2{=}41{=}10b_1$ (the EM index *is* the Gaussian norm of the carrier-glue vector) and $N(3{+}2i){=}13{=}\Delta_Q$; the hexagonal partner (Eisenstein $\mathbb{Z}[\omega]$, $j{=}0$) gives $N(3{+}2\omega){=}7{=}$ scalaron. The $(3,2)$ split generates $(5,6,7,13)$ by sum/product/Eisenstein/Gauss norm; rings rigid (`CMNORM.DUAL.01`).
- **The μ_4 clock is the order-4 character of the E_8 Coxeter cycle (`v223_coxeter_totative_clock`, `origin_theory`, [E]).** $(\mathbb{Z}/30)^\times$ are the E_8 exponents; the order-4 generator 7 ($\langle 7\rangle{=}\{1,7,13,19\}$) is a literal μ_4 clock permuting the four conjugate planes; only E_8 has $\varphi(h){=}$rank and $h{=}2{\cdot}3{\cdot}5$ (`COX.CLOCK.01`).
- **$248{=}120{+}128$ as a magnitude/phase channel typing (`v227_degree_exponent_channel_split`, `tfpt_5_redteam`, [E]).** Exponents sum $120{=}|R^+(E_8)|$ (magnitude channel), degrees sum $128{=}\mathrm{rank}\cdot\dim S^+$ (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong S^- reading; `E8.CHAN.01`).
- **P2 as the mode space of a degree-4 seam divisor (`v228_rr_index_gate`, `origin_theory`, [E]/[C]/[O]).** A Riemann–Roch index gate: $\deg D{=}4{=}|\mu_4|$ on $\mathbb{P}^1$ forces $h^0{=}5{=}g_{\rm car}$, $\mathrm{rank}\,H_1{=}3{=}N_{\rm fam}$, $h^0{+}H_1{=}8{=}\mathrm{rank}\,E_8$, $\Lambda^{\rm even}(\mathbb{C}^5){=}16{=}\dim S^+$ — bedrock language for P2, conditional on the seam producing the degree-4 divisor (`QGEO.RR.01`).
- **The seam as a finite recoverability code (`v221_seam_qecc`, `origin_theory`, [E]/[C]).** Code dimension $\dim S^+{=}16$; the gapped transport is a CPTP doubly-stochastic contraction with spectrum $\{1,(2/3)^6,(1/3)^6\}$, recovery bound $\|T^n\delta\|{\le}(2/3)^{6n}\|\delta\|$ (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading **[C]**, gated on the Seam–Horizon theorem (`QEC.SEAM.01`).
- **CP lives in the hexagonal phase fiber (`v220_cp_hexagonal_modulus`, `tfpt_5_redteam`, [E]/[C]).** The seam deck is the square modulus $j{=}1728$ ($\mathbb{Z}/4$); its partner is the hexagonal $j{=}0$ ($\mathbb{Z}/6$, $\arg\rho{=}\pi/3$). The frozen $\delta{=}\pi/3{+}3\lambda_C^2{=}68.654^\circ$ has leading term $\pi/3{=}\arg\rho$; the deck stays $\mathbb{Z}/4$, CP is the $\mathbb{Z}/6$ fiber (`CP.MOD.01`).
- **The dual normal frame (d,n) with CP as orientation (`v225_dual_normal_frame`, `tfpt_5_redteam`, [E]/[C]).** $d{=}a^\top R^{-1}{=}(-\frac{1}{2},-\frac{1}{2},1){=}-\frac{1}{2}$ Nariai; $n{=}(5,-9,6)$ reads det on the first-generation column; $\det(\mathbf{1},d,n){=}21{=}3{\cdot}7$; $\mathrm{Im}\det(\mathbf{1},d,\rho n){=}21\sin(\pi/3)$ — CP as orientation (`DUAL.FRAME.01`).

- F_{transfer} **as one path on the Sheet Diamond (`v224_diamond_ftransfer_path`, `tfpt_2_standard_model`, [E]/[C]).** $M(s,t)$=R+$Q\,\mathrm{diag}(s,t,t)$: K,C,F on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps $(1,8,10),(1,8,16)$; the Koide source→pole is the 1-D projection (`FTR.PATH.01`).
- **The charged leptons as an étale Frobenius algebra (`v229_lepton_frobenius_algebra`, `tfpt_2_standard_model`, [E]/[C]).** (c_e,c_μ,c_τ)=$(\frac{16}{7},\frac{4}{3},\frac{7}{6})$, ring closure c_ec_τ=$\frac{8}{3}$=$|\mathbb{Z}_2|c_\mu$, product $\frac{32}{9}$; A=$\mathbb{Q}[t]/(m)$ is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the μ_6 resolvent (`LEP.FROB.01`).
- **The center budget $(7,11,13)$ as three local norms (`v230_center_budget_norms`, `tfpt_2_standard_model`, [E]).** C=R+$Q\,\mathrm{diag}(1,0,0)$ row sums $(7,11,13)$: 7=$N_{\mathbb{Z}[\omega]}(3$+$2\omega)$ (hex), 13=$N_{\mathbb{Z}[i]}(3$+$2i)$ (square), 11=$\binom{4}{0}$+$\binom{4}{1}$+$\binom{4}{2}$ (QBL Fock count); the two CM rings and the boundary count meet in one operator center (`CENTER.NORM.01`).

2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (`v218_diamond_axis_geometry`, `tfpt_2_standard_model`, [E]).** Sharpens `v94`/`v95` with three exact **[E]** lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis $\det(C{+}xU) = 14{+}6x$ (slope $6{=}|R^+(A_3)|$) and quadratic on the sheet axis $\det(C{+}yV) = 14{+}14y{+}4y^2$ (2nd difference $8{=}\mathrm{rank}\,E_8$); the anchor block has $\det B(C{+}xU) = 2\det(C{+}xU)$ and sheet curvature $6{=}|R^+(A_3)|$ — two curvatures $(8,6) = (\mathrm{rank}\,E_8, |R^+(A_3)|)$. **(2) Plücker transfer ladder:** $\mathrm{Pl}(K) \to \mathrm{Pl}(C) \to \mathrm{Pl}(F)$ lifts in steps $(1,8,10) = (N_\Phi, \mathrm{rank}\,E_8, A_\Lambda)$ then $(1,8,16) = (N_\Phi, \mathrm{rank}\,E_8, \dim S^+)$ — the decuple then the full generation (identity **[E]**, source→pole reading **[C]**). **(3) Spectral ramification:** for Q,K,C,F the cubic discriminant factors as $q(r)^2\,\mathrm{Disc}(q)$ with squares $(1,3,4,6)$ and kernels $(13,48,65,105)$, F carrying $105 = 3{\cdot}5{\cdot}7$. The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ($28{=}2\dim G_2$, $52{=}\dim F_4$ stay audit-only, not spine, as in `v94`/`v95`); F as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger `DIAMOND.AXIS.01`, `DIAMOND.PLUCKER.01`, `DIAMOND.SPECTRAL.01`; Wolfram-mirrored (249/249).

2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed (`v216_marks_gauss_bonnet`, `tfpt_research_contracts`, [E]; ledger `QGEO.MARKS.03`).** The K1 lever of the `v215` kill-test ("exactly four marks") is *executed*: if the marks are $\mathbb{Z}_2$ branch points (deficit π) on the flat seam sphere ($\chi = 2$, the same χ as the 8 in c_3), Gauss–Bonnet forces $n\pi = 2\pi\chi = 4\pi$, so $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$. The closed Euclidean sphere 2-orbifolds are exactly $(2,3,6)$, $(2,4,4)$, $(3,3,3)$, $(2,2,2,2)$, and three independent criteria — all cone orders $= 2$ (the $|\mathbb{Z}_2|$ sheet branch), $\mathrm{rank}\,H^1 = \#\text{marks} - 1 = 3 = N_{\text{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase $(2,2,2,2)$. The numerical sibling `v217_marks_emergence_scan` ([C], ledger `QGEO.EMERGE.01`) confirms it on the actual DtN/state (number n and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap $(2/3)^6$). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays **[O]**. `v216` exact (Wolfram extension → 248/248); `v217` numerical (Python-only). Suite → 217 modules.

2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **`QGEO.SYM.01` as a kill-test, not a proof-promise (`v215_seam_deck_killtest`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.KILL.01`).** A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock $\omega \circ \rho = \omega$ is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (`v198/v199/v201`) gives $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ with $|k|$ commuting exactly, so the residual is the bounded sub-principal M_f (equivalently the raw Gaussian bulk form A is μ_4-deck-invariant). Levers: **K1** exactly four marks ($b_1 = \#\text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \#\text{marks} = 4 = |\mu_4|$); **K2** the square config (cross-ratio $2 \Rightarrow j = 1728$, $\text{Aut} = \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j = 0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$-invariant); **K4** transfer gap $(2/3)^6 = 64/729$. Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side **[E]** (regression guards), with the *decisive* kill deferred to `QGEO.REALIZE.01` **[O]** (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts $j = 1728$ and $(2/3)^6$ are Wolfram-mirrored via `v214/v54/v56`). Suite $\to$ 215 modules.

2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of `QGEO.SYM.01` (`v214_seam_pillowcase`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.PILLOW.01`).** The two separately-tracked open QGEO residuals — `QGEO.ISO.01` (`v180`: the carrier clock is an order-4 *isometry* of the seam metric) and `QGEO.SUBPRIN.01` (`v201`: the DtN sub-principal symbol is *mark-local*, the seam flat away from the μ_4 marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold $S^2(2,2,2,2)$ ($\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet $4\pi = 2\pi\chi$) — this *is* the `v201` "flat away from the marks". **(2)** New link: cross-ratio$(\mu_4) = 2$ (`v168`) $\Rightarrow j = 1728$ ($j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$; all six harmonic cross-ratios give 1728) $\Rightarrow$ the *square* modulus $\tau = i$, whose CM by $\mathbb{Z}[i]$ *derives* the order-4 clock $z \mapsto iz$ (explicit $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$); generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$. **(3)** Unification: "seam = uniformising flat pillowcase metric" implies *both* `v201` and `v180`, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close `QGEO.SYM.01`: the choice "RP seam metric = the constant-curvature representative" stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/j-invariant/CM arithmetic); Klein-J values mpmath-numerical. Suite $\to$ 214 modules.

2026-06-15 (the F_transfer functor contract CONTRACT.F.01)

- F_{transfer} **formalised as a typed functor with four axioms (`v213_ftransfer_functor`, `tfpt_research_contracts`, [C]).** The four frontier transfers are consolidated into ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$, each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)** μ_4-deck equivariance (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ is the deck transfer eigenvalue); **(2)** Plücker preservation ($53 = a^T(R+Q)\mathbf{1}$, $\text{Pl}(F) = (1, 22, 30)$, $\|\text{Pl}(K)\|_1 = 11$); **(3)** positivity/stochasticity (spec T positive, $\kappa_f \in (0, 1)$); **(4)** external modules explicit ($b_3 = -7$ a carrier output). The four instances are F_{pole} (`v183`), $F_{\text{Boltzmann}}$ (`v212`), F_{relic} (`v211`), F_{QCD} (`v164`). `CONTRACT.F.01` is the third research contract alongside (U_{wall}) and (G_{metric}), the structural complement of the `v187` typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger `CONTRACT.F.01`.

2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (`v211_axion_spine_angle`, `tfpt_4_frontier`, [C]).** A more *robust* misalignment angle than the determinant-line hilltop

$\theta_i \approx 170^\circ$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$): the central spine quotient gives $\theta_i = 3\pi/5 = 108^\circ$ exactly (no fit), the finite-T solver reaches $\Omega_{\rm DM}$ at $\approx 106^\circ$, and 108° sits in the *mild*-anharmonic regime (62° below the hill-top) — not exponentially sensitive. An alternative ansatz to $\theta_i = \pi(1 - \varphi_\Sigma)$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); **[X]** $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes it. A sharper scenario, not a derivation.

- **Leptogenesis scalaron-decuple branch (`v212_leptogenesis_decuple`, `tfpt_4_frontier`, [C]).** Both Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$: $M_1 = M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV and $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. M_1 uses *only* $\{M_{\rm scal}, \varphi_0^{\rm ret}, A_\Lambda\}$ — no hidden seesaw scale, cleaner than `v184`'s $M_R\varphi_0^4$ — and lands in the thermal window where $\eta_B \approx 1.2 \times 10^{-9}$ brackets the observed 6.1×10^{-10}. The coupling mechanism is posited not derived, so η_B stays **[C]**; **[X]** a precise Boltzmann/RGE solve outside the band falls the route, not the theory (`FR.ETAB.04`; Route A independent).

2026-06-15 (Lean: QGEO.COHOM.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (`CohomologyGrading.lean`, ledger `FORM.QGEO.03`, **[O]**). A new module machine-proves the COHOM node of the QGEO proof tree (`v177`; AUDIT: PASS, no `sorry/admit`, axioms `propext, Classical.choice, Quot.sound` only): the three $H^1(\mathbb{P} \setminus \mu_4)$ eigenforms $\omega_k = z^{k-1}/(z^4 - 1)$ (k=1, 2, 3) have pullback character $\rho^*\omega_k = i^k\omega_k$ under the clock $\rho : z \mapsto iz$ (each an exact rational-function identity, the denominator clock-invariant since i^4=1), so the grading is $(i, -1, -i)$ with weights $(1, 2, 3)$ = the A_3 exponents = $\mathrm{Spec}(Q_+)$, rank $3 = N_{\rm fam}$. The MODULE *parity* is also proved: the reflection $\sigma : z \mapsto 1/z$ satisfies $\sigma^*\omega_1 = \omega_3$, $\sigma^*\omega_2 = \omega_2$, $\sigma^*\omega_3 = \omega_1$ (swap $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed). Signature-locked in `AuditContract.lean`. The geometric COHOM/MODULE-parity nodes are now **[E]**; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay **[O]**.

2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (`MobiusUniformisation.lean`, ledger `FORM.QGEO.02`, **[O]**). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (`v177`; AUDIT: PASS, no `sorry/admit`, axioms `propext, Classical.choice, Quot.sound` only): the clock $\rho : z \mapsto iz$ has ρ^4=id and ρ^2≠id (order exactly 4); the reflection $\sigma : z \mapsto 1/z$ is an involution with $\sigma\rho\sigma = \rho^{-1}$ (so $\langle\rho, \sigma\rangle$ is the dihedral D_4 of order 8); a non-fixed orbit $\{a, ia, -a, -ia\}$ scales to $\mu_4 = \{1, i, -1, -i\}$; σ permutes μ_4 (fixes $1, -1$, swaps $i, -i$); and the multiplier classification "$z \mapsto \zeta z$ has order exactly 4 $\Leftrightarrow \zeta = \pm i$". Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation `QGEO.MARKS.01` stays **[O]**, *not* closed.

2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local $\Rightarrow \omega\circ\rho = \omega$ implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, **[O]**). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of `v201/v210` (AUDIT: PASS, no `sorry/admit`, no domain axioms — `#print axioms = propext, Classical.choice, Quot.sound` only): (i) the 4th-root character orthogonality $\sum_{j<4} \zeta^j$ = (if ζ=1 then 4 else 0) for ζ^4=1 (so a μ_4-mark sum is supported only on modes $\equiv 0 \bmod 4$); (ii) the clock generator $-i$ has order 4; (iii) `markLocal_blockDiagonal`: a mark-local Toeplitz symbol connects only equal clock-characters, $[\rho, M_f] = 0$; (iv) the premise is a typed `structure SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence $\omega \circ \rho = \omega$. So the *implication* is now **[E]** (formally proved); the *premise* (the physical seam being mark-local) stays **[O]** — the one fundamental seam-identification postulate, *not* closed.

2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN $\Rightarrow \omega \circ \rho = \omega$, certified on realistic Steklov profiles and at the state level (`v210_mark_local_dtn`, `tfpt_research_contracts`, [O]).** A numerical sharpening of `QGEO.SUBPRIN.01/v201`: a real von-Mises curvature bump summed over the four μ_4 marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = |D_\theta| + M_f$ whose off-(mod 4) Fourier support vanishes to $< 10^{-16}$, $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the quasi-free covariance $C = \frac{1}{2}(1 + \operatorname{sgn} H_1)$ is clock-invariant ($\rho C \rho^\dagger = C$ to $< 10^{-13}$), i.e. the actual $\omega \circ \rho = \omega$. Negative controls (single off-centre bump, $\mathbb{Z}_3$ source, four generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. This certifies the *implication*; the premise that the physical seam *is* mark-local stays **[O]** (it does *not* close `QGEO.SYM.01`; net existence and full-cone RP remain the **[E]** of `v175`). Ledger `QGEO.MARKLOCAL.01`. Python-only (the exact mark-sum identity is Wolfram-mirrored via `v201`).

2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon $g-2$ as a seam vertex (`v204_muon_seam_g2`, `tfpt_4_frontier`, [C]).** The carrier's second-order topological defect $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$ projected through the seam-loop phase gives $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\,\pi^9) \approx 2.879 \times 10^{-9}$ (0.81σ vs the dispersive Δa_μ; $\sim$1.5σ vs lattice/CMD-3). The value is exact **[E]**; the vertex identification is the **[C]** bridge. Ledger `FR.MUONG2.01`; Wolfram-mirrored.

- **An independent gravitational 3/4 (`v205_xi_threequarter`, `tfpt_4_frontier`, [E]/[C]).** $\xi = c_3/\varphi_{\text{tree}} = 3/4$ exactly (Einstein limit of $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$) — the same $3/4 = q(A_3) = \ln(m/\mu)$ as the seam-determinant replica (`v152`), an over-determination. Ledger `GRAV.XI.01`; Wolfram-mirrored.

- **Quantitative H_0 from the Λ branch (`v206_h0_lambda_branch`, `tfpt_4_frontier`, [C]).** $\delta_\Sigma = 48\,e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$, the $(8\pi)^2$ Planck-convention split, and $H_0 = 66.5$–67.1 km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger `GRAV.H0.01`.

- **Asymptotic Safety as a second QG route (`v207_asymptotic_safety`, `tfpt_4_frontier`, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms; $y^2 = 16c_3^2 = 1/(4\pi^2)$) — a plausibility cross-check, *not* load-bearing, does *not* close G_{metric}. Ledger `GRAV.ASYMP.01`.

- **Black-hole thermodynamics from the seam** (`v208_bh_thermodynamics`, `tfpt_horizon_readouts`, **[E]**/**[C]**). Modular $T_H = \kappa/(2\pi)$ (the $2\pi = 1/(4c_3)$ seam unit, ties to `v198`/`v199`) and the scalaron-corrected Wald entropy $S_W = (A/4G)(1 + R_h/3M_s^2)$ (exact from `v28`). Ledger `HOR.BHTHERMO.01`; Wolfram-mirrored.

- **The black hole as a seam fixed point** (`v209_bh_regular_core`, `tfpt_horizon_readouts`, **[C]**). No singularity ($\varphi \to \varphi_\star$ at the gapped attractor $(2/3)^6$); the maximal BH = the anchor (Nariai roots $(1, 1, -2)$); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger `HOR.BHCORE.01`.

2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (`tfpt_3`, [O]).** Added a paragraph to the "Möbius origin of π" section: $\text{Möb}(\hat{\mathbb{C}}) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$ (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock $z \mapsto iz$ is an elliptic $\mathrm{PSL}(2,\mathbb{C})$ rotation (`v180`) and the RP→OS causal cone is the same Lorentz structure — so "Möbius origin of π" and "one Lorentzian cone" are two faces of one boundary $\mathrm{PSL}(2,\mathbb{C})$. A foundation citation, not a new claim (π stays primitive).

2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3$ + retired small-angle reading)

- **Haloscope coupling tied to c_3 (`tfpt_4_frontier`, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ($y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$), so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and $\beta_{\mathrm{rad}} = \varphi_0/4\pi$ — a **[C]** structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$). The earlier small-angle $\theta_i = \varphi_0$ reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop $\theta_i = \pi(1 - \varphi_{\mathrm{seam}}) = 170.4°$, recorded as a retired reading and guarded by `v188` (a new sentinel forbids that stale axion mass from re-entering the active docs).

2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **`HOR.EHT.01` (`v203`, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts `v101`–`v104`/`v190`; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole, $\beta_{\mathrm{BH}}(r) = 16c_3^4\, Q_e^{\mathrm{eff}} Q_m^{\mathrm{eff}}/r^2$. The coupling $16c_3^4 = 1/(256\pi^4) = \delta_{\mathrm{top}}/3$ is *exact* (**[E]**, Wolfram-mirrored) — the *same* top-form coefficient $\delta_{\mathrm{top}} = 48c_3^4$ that fixes the α-kernel precision-zone correction, one row from the cosmic-birefringence seed $\beta_{\mathrm{rad}} = \varphi_0/4\pi$. TFPT fixes the $1/r^2$ shape, achromaticity and sign-flip (**[C]**); the amplitude Q_eQ_m is an MHD/GR weight, never a pixel prediction. **Dated kill test [X]**: the residual intercept must pass three nulls (frequency, spatial $1/r^2$, sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

2026-06-15 (archive integration #1: the rare kaon sector $K \to \pi\nu\bar{\nu}$ — v202)

- **`FR.RAREKAON.01` (`v202`, [C]).** The closed TFPT CKM point ($s_{12}{=}\lambda_C$, $s_{23}{=}\varphi_0/(1{+}\lambda_C){=}|V_{cb}|$, $s_{13}{=}\lambda_C^3/3$, $\delta{=}\pi/3{+}3\lambda_C^2{=}1.19823$ rad; `v18`/`v88`, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou $X_t, P_c, \kappa_\pm$) it gives $\mathrm{BR}(K^+ \to \pi^+\nu\bar{\nu}) = 9.45 \times 10^{-11}$ and $\mathrm{BR}(K_L \to \pi^0\nu\bar{\nu}) = 3.33 \times 10^{-11}$ — a **[C]** downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted $\mathrm{BR}(K_L) = 3.47 \times 10^{-11}$ from an $\Im(\lambda_t)/\lambda^5$ slip, corrected here). Grossman–Nir ratio $0.352 \ll 4.3$; NA62 $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ is $\sim 1.2\sigma$ above. **Dated kill test [X]**: a stable NA62 $\mathrm{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, `v187` guard (new `FR.RAREKAON.` family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **`QGEO.SUBPRIN.01` (`v201`, [E]/[O]).** The `v199` bedrock residual — "the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements" — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4-character-block-diagonal iff f is $\mathbb{Z}_4$-invariant (**[E]**; principal $|k|$ already commutes, `v198`); and a μ_4-mark sum $f = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$

has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for any profile g, hence is automatically $\mathbb{Z}_4$-invariant ([**E**]). So the μ_4-mark orbit (forced by `v195`) *forces* the sub-principal symbol block-diagonal; 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to "the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck, `v177`)", structurally definitional — the narrowest form yet of `QGEO.SYM.01`, [**O**]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (`experiments/ftransfer/axion_relic/relic_solver.py`, [C]).** The first real F_{transfer} pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest finding: at the *predicted* $\theta_i \approx 170°$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6$–2.3 (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120$–$130°$ or extra dilution) — this *refines* `v185`'s optimistic "can be all-DM" toward an over-production tension. The near-hilltop regime is O(1)-uncertain, so a full finite-T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the `v185` docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\text{seam}})$ only.
- **Full finite-T solve confirms it (`axion_relic/full_finiteT_solve.py`).** A converged nonlinear misalignment solve ($\theta'' + (3 + \mathrm{d}\ln H/\mathrm{d}N)\theta' + (m_a(T)/H)^2 \sin\theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i{=}1 \to \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170°$ — $\sim$5.5$\times$ above $\Omega_{\text{DM}} = 0.12$, reached only at $\theta_i \approx 106°$. The over-production is thus *confirmed* (not the optimistic "all-DM"); `tfpt_4_frontier`, `v185` docstring and the website updated accordingly. Stays [**C**]; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **`v199_seam_state_invariance.py` [E]/[O] (claim `QGEO.STATE.01`, work package A).** Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) [**E**] the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, `v117`), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [**E**] for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1+\operatorname{sgn} H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) [**E**] $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4-character classes $\{n{=}r \bmod 4\}$ (verified). (4) [**O**] the residual is sharpened to "the bounded sub-principal symbol has no off-character matrix elements" — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.
- **`v200_seam_variational_scan.py` [C]/[O] (claim `QGEO.VARI.02`, work package B).** The discretised variational test: with the clock angle free, $E_{\text{fail}}(\theta)$ has its unique minimum at $\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i =$ the $(1,2,3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4; the full operator-valued minimisation stays the `v199` residual [**O**]. Cited in `tfpt_research_contracts`.
- *Also (this pass, `experiments/` only, not the ledger): the F_{transfer} four-pipeline scaffold (`experiments/ftransfer/`) with one CONTRACT per interface, and confirmation that the FRB preregistration (`frb_tfpt_v1.yaml`) and its status semantics already exist.* Suite now 199 modules, highest ID `v200`; Wolfram extension 241/241.

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **`v198_modular_commutator_reduction.py` [O] (claim `QGEO.MODULAR.01`).** The decisive crack on the last residual. (1) [**E**] the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \mathrm{diag}(|n|)$ and the clock $\rho{:}z \mapsto iz = \mathrm{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [**E**] by Tomita–Takesaki the modular operator is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log\Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [**E**] the finite H^1 block is already μ_4-invariant (`v177`). So the entire residual collapses to the single *state-invariance* "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of `QGEO.SYM.01`, [**O**]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 197 modules, highest ID `v198`; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- **`v195_marks_lefschetz_character.py` [E]/[O] (claim `QGEO.MARKS.02`).** The four seam marks are *forced* (not posited): $\rho{:}z \mapsto iz$ fixes only $\{0, \infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\mathrm{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$) has no trivial component ⇒ no puncture at a fixed point, and $\mathrm{rank}\, H^1 = n-1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So Marks reduces from the geometric posit "D_4 marks" to the milder premise "H^1 carries the A_3-exponent rep" — less circular, still [**O**]. Cited in `tfpt_research_contracts`.
- **`v196_seam_energy_functional.py` [E]/[O] (claim `QGEO.VARI.01`).** The failure functional $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the `QGEO.SYM.01` conditions; on the finite H^1 block it vanishes at the μ_4 config [**E**] and is > 0 for any μ_4-breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the `v194` Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [**O**]. Cited in `tfpt_research_contracts`.
- **`v197_rr_carrier_clifford_d5.py` [E]/[C] (claim `ARCH.RRCAR.02`).** Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \to g_{\text{car}} \to D_5$. [**E**] arithmetic, [**C**] identification (as `v189`). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop "$\pi - 3\phi_0$" proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID `v197`; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **`v194_raw_seam_dtn_rp.py` [O] (claim `QGEO.DTN.01`).** The next hard lever — "is the raw RP-seam DtN definable from RP data alone?" (sub-claim 2 of `QGEO.ENERGY.02`) — was tackled, and the honest result is a *reduction*, not a closure. (1) [**E**] the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; `v177`). (2) [**E**] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, `v54`) on a quasi-free seam state (`v155`), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \setminus \mu_4$. (3) [**O**] the residual relocates: the full-L^2 commutation ⇔ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So `QGEO.SYM.01` reduces from a definitional postulate to "intrinsic BW geometric modular covariance" — one notch sharper, still [**O**]; the last structural fog point sharpens, it does *not* close to [**E**]. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 193 modules, highest ID `v194`.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **`v193_qgeo_energy_commutator.py` [O] (claim `QGEO.ENERGY.02`).** The non-circular sharpening of `QGEO.ENERGY.01`: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) [**E**] ρ is carrier-defined (Coxeter of $W(A_3)=S_4$, `v117`), not imported from the seam; (2) [**O**] the non-circularity hinge: Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [**E**] the commutator forces the $(1,2,3)$ grading on H^1 (`QGEO.COHOM.01`). Exact [**E**] rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (`v150`) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, `QGEO.SYM.01` becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID `v193`; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to "Constructive QFT closure of the admissible physical sector", and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the `v188` prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **`v189_riemann_roch_carrier.py` [E]/[C] (claim `ARCH.RRCAR.01`).** The pair $(g_{\text{car}}, N_{\text{fam}}) = (5,3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$ (already `QGEO.COHOM.01`) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$, matched by $\text{rank}(D_5) = 5$. The carrier-as-mode-space reading is [**C**] (geometric, does not displace the over-determined $g_{\text{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.
- **`v190_nariai_entropy_bound.py` [E] (claim `HOR.NARIAI.02`).** A variation-free one-line proof of the SdS entropy floor: $3(x^2{+}1) - 2\Phi_3(x) = (x{-}1)^2 \geq 0$, so $S_{\text{tot}} \geq \frac{2}{3} S_{dS}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **`v191_universal_branch_line.py` [C] (claim `HOR.BRANCHLINE.01`).** The affine map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2} m$ carries the SdS branch points $\pm 1/N_{\text{fam}}$ onto the flavor labels $\{2,5\}$. *Honestly typed [C]* — an exact *relabeling*, not a theorem (a decoy pair $\{3,4\}$ admits an equally exact map; the shared content is the known 2/3 ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **`v192_qgeo_energy_clock.py` [O] (claim `QGEO.ENERGY.01`).** An operator-checkable restatement of the bedrock: $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ (the clock preserves the DtN energy form). *Honestly typed:* it *relocates* `QGEO.SYM.01` to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID `v192` (v186 skipped).
- *All four are validated external-review proposals (`problem_b`); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version `5.1 (rev 120)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20687564`, DOI `10.5281/zenodo.20687564`. This version carries the F_transfer-firewall round (`v184`–`v188`): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger `v187` and prose `v188` guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-`QGEO.SYM.01` two-layer clarification. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Suite 187 modules (highest ID `v188`; v186 skipped). `run_all.py` green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- **`v184_etaB_anchored_boltzmann.py` [C] (claim `FR.ETAB.03`).** An honest test of the external-review η_B proposal ($M_1 = M_R\varphi_0^4$, $\tilde{m}_1 = m_3/A_\Lambda$). Firewall verdict: *sharper scenario, not a closure.* The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_\Lambda = 10$ atom, in the strong-washout window) [**C**]; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{\rm Pl}$ both miss it (c_3-/φ_0-powers off $\sim$75%/$\sim$40%), so $M_1 = M_R\varphi_0^4$ merely relocates the free input. η_B stays [**C**], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.
- **`v185_axion_relic_solver.py` [C] (claim `FR.DM.03`).** Honest misalignment estimate: the determinant-line axion $f_a = M_{\rm scal}/128 \approx 2.39\times10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) CAN be all dark matter, but the harmonic misalignment under-produces ($\sim$21% at $\theta \sim O(1)$), so the $\theta_i \approx 170°$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [**C**] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.
- **`v187_ftransfer_laws.py` [E] (claim `FR.TRANSFER.01`).** A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B, axion, m_p/m_e) are read from the ledger and required to carry a [**C**]/[**O**] marker — never a primitive [**E**] compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be [**E**], the physical prediction never is). Records the four interfaces $F_{\rm pole}/F_{\rm Boltzmann}/F_{\rm relic}/F_{\rm QCD}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **`v188_frontier_wording_guard.py` [E] (claim `AUDIT.WORDING.01`).** A prose sentinel (no new physics): `v187` protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [**E**]-typed Route B leptogenesis claim and its heavy-scale wording → v184 [**C**]; the old "open relic scale" dark-matter line → v185; the old "near, not exact" Koide line → v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [**C**]), the Koide and dark-matter summary lines, and added the P1/P2-vs-`QGEO.SYM.01` two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID `v188` (v186 was intentionally skipped as redundant — m_p/m_e is already typed [**O**] as a QCD matching contract, `QCD.LAMBDA.01`).
- *Review framing points (P1/P2 as projections of the seam-deck postulate, $v_{\rm geo}$ as unit gauge, $G_{\rm net}$ a corollary of `QGEO.SYM.01`, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed [**O**] as a QCD matching contract (`QCD.LAMBDA.01`), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **`v183_koide_f_corner_transfer.py` [E]/[C] (claim `FR.KOIDE.07`).** An external-review proposal, validated exactly: the Koide source→pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2\cdot 27) = \frac{53}{54}$. Here $\mathbf{1}^T Ra = 27$ is the $E_6\times A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78+8+2\cdot 27\cdot 3$) and $a^T(R+Q)\mathbf{1} = 53 = 52+1$, where $F = R+Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (`v80`). Hard negative control: $a^T M\mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades `FR.KOIDE.03` from "near miss + conjecture" to "operator-sourced source→pole transfer": the *factor* is [**E**] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the F corner) stays [**C**], an F_{transfer} special case. The prediction itself stays [**C**]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, exponent ratio $|\mathbb{Z}_2|$, "one clock, two geometries") is exactly `HOR.TRISECT.01/v103` — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **`v182_reviewer_residual_map.py` [E]/[C] (claim `REVIEW.MAP.01`).** A careful external review raised four concerns (A mapping $2D \to 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as `v175`–`v181`, onto the *two already-named* residuals $\{\texttt{QGEO.SYM.01}, F_{\text{transfer}}\}$: A (Plücker-11 = QBL boundary count `v174` + holographic reduction), B ($\equiv$ `QGEO.SYM.01`), D (Möbius forced by $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2,\mathbb{C})$ on the deck; the missing generator = F_{transfer}). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a "Rosetta" translation lexicon** (website `RosettaLexicon`): TFPT's compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, F_{transfer}, v_{geo}, `QGEO.SYM.01`) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website `TrustContract`): the frozen pre-data predictions ($\sin^2\theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \le 10^{-30.7}$) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate.** `QGEO.SYM.01` is now framed confidently as TFPT's *one fundamental postulate* (the role "c = const" plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website "what is open" card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded.** Three rows (`AX.P1.01`, `AX.P2.01`, `QGEO.ISO.01`) had *unquoted commas* in a status/external field (e.g. "a=(1,1,2)", "PSL(2,C)"), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.

- G_{net} **typed in three explicit levels** (website "What is open" card): (1) *algebra* [**E**] $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$; (2) *AQFT machinery* [**E**] (net existence + full-cone RP, CAR functor, `v175`); (3) *seam instantiation* [**O**] (`QGEO.SYM.01`). The target object's mathematics is closed; only the physical coupling of the raw seam is open.
- **`QGEO.SYM.01` now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than `CONF/ISO`, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\mathrm{fam}} \neq 3$); with the explicit conditional framing "*given* `QGEO.SYM.01`, the closure follows" rather than "the theory derives the seam geometry".
- **Claim-language detox.** "no free fundamental number" softened to "no free *dimensionless compiler dial* beyond the anchor and π (dimensionful scale + transfer physics stay typed)" (`origin_theory`, `introduction`, website); the site title softened from "the Standard Model Derived" to "the Standard-Model *Skeleton* Derived".
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [**E**], conditional physics [**C**], open premises [**O**], kill tests [**X**]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version `5.1 (rev 114)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20686284`, DOI `10.5281/zenodo.20686284`. This version carries the AQFT-closure-to-bedrock chain (`v175`–`v181`, the structural residual reduced to the single definitional premise `QGEO.SYM.01`), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The `README` and the Zenodo description were refreshed (181 modules), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain `v175`–`v181` as a descending staircase from "build a quantum-gravity measure" to the single definitional bedrock `QGEO.SYM.01`; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the $\sim$181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1,1,2)$ plus π (with $g_{\mathrm{car}} = 5$ an over-determined bootstrap fixed point — forced three ways, `v6`, Pascal-unique and Lean-formalised, `FORM.LADDER.01`) was already established (`ANCHOR.GEN.01/v23`, `ARCH.CORE.01/v53`) and stated in the papers/glossary, but the *axiom surfaces* still read as bare "declared inputs". Fixed: the ledger rows `AX.P1.01/AX.P2.01` now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\mathrm{car}} = e_2(a)$, bootstrap-forced) rather than "declared axiom". No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the `FIG` list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **`v181_clock_is_conformal_symmetry.py` [E]/[O] (claim `QGEO.SYM.01`).** The isometry premise `QGEO.ISO.01` (`v180`) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (`v117`), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification `QGEO.SYM.01`: "the seam's conformal structure *is* the μ_4-deck structure of the carrier clock". This is bedrock — a physical/definitional statement tying P2 to the seam's conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE(v175)`$\to \cdots \to$`SYM.01(v181)` ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the `introduction` "latest reductions" prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A $\times 2$, G_{net}), the `glossary` G_{net} entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **`v180_clock_is_mobius.py` [E]/[O] (claim `QGEO.ISO.01`).** Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, `v179`) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)$=4) is topologically $z \mapsto iz$; (3) *isometry⇒ conformal⇒Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2, \mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i, free orbit μ_4). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly **[O]**. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **`v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`).** Investigating the two residual premises left by `v178` shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays **[O]**). Marks's residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the same $\chi = 2$ that gives the "8" in c_3 (`v58/v73`); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (`v117`), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). Kernel then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann `v156`; $c = 8$ rigidity `v158`; the cusped-surface residual $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: "the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism". The two doors of `v177/v178` are one geometric realisation, left honestly [**O**]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- **`v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`).** An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [**E**] core of each and reduces each to *one sharper* premise. *Marks core* [**E**]: given a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1, i, -1, -i\} = \mu_4$ (*not* the fixed points), distinct iff $b_1 = 4 - 1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving a faithful D_4 of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [**E**]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (`v162`) and the unique residue-normalised basis (`v177`) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee–Uhlmann/`v156`; no drift `v158`). Net: neither obligation closed; both sharpened to milder premises with their finite cores [**E**]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- **`v177_seam_marking_kernel.py` [E]/[O] (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`).** A second external residual-class audit pointed out that the `v176` statement takes the marks/D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET and closes *four* nodes *constructively* (all validated symbolically before adoption): UNIFORM [**E**] — an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}, \mu_4)$ (strengthening `v168`'s cross-ratio to the full uniformisation); COHOM [**E**] — $\rho^*\omega_k = i^k\omega_k$, grading $(1, 2, 3) = A_3$ exponents; MODULE [**E**] — a *unique* μ_4-equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); NET [**E**] (`v154/v175`). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: `QGEO.MARKS.01` (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (the raw seam Calderón operator *is* the μ_4-equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via `v168/v137/v162/v154/v175`). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- **`v176_seam_collar_realisation.py` [E]/[O] (claim `QGEO.REALIZE.02`).** Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem:* given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible c=8 data, its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* **[E]**: geometry/uniformisation (`v168`), Calderón DtN symbol $|k|$ (`v156`), H^1 character = generation grading $(1, 2, 3)$ = A_3 exponents (`v137/v168`), μ_4-equivariant contraction with gap $(2/3)^6$ (`v162`), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all m (`v154/v175`). The whole structural residual thus reduces to the *one* open premise `QGEO.REALIZE.01`: that the physical seam collar's boundary *is* this $\mathbb{P} \setminus \mu_4$ — a constructive-geometry statement, left honestly **[O]**. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via `v168/v156/v162/v154/v175`). Suite now 176 modules. *Audit note:* the metrology residual the same review flags is already discharged — `v153` (No-Unit Theorem) is the metrology-line typing ($v_{\rm geo} \mapsto \lambda^{-1} v_{\rm geo}$, $1/G \mapsto v_{\rm geo}^2$, dimensionless data invariant), and the Koide $u \to \frac{53}{54} u$ source→pole transfer is already typed **[C]** (`FR.KOIDE.03`); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version `5.1 (rev 105)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20681451`, DOI `10.5281/zenodo.20681451`. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to **[E]**. The `README` and the Zenodo description were refreshed to the current state (175 modules, Wolfram $116/116 + 231/231$), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **`v175_net_existence_full_cone.py` [E]/[O] (claim `QFT.NETRP.01`).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m*: the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of spec(t), so all 2^{16} products lie in $[0, 1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every m* (not only $m \leq 6$, `v161`) to the one-particle contraction (Shale–Stinespring), discharging the `v160` scope premise. (2) *A_2 assembled + verified*: the $(E_8)_1$ character $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$ (level-1 = 248, extends the order-4 `v156` check by one order), the embedding $248 = 120 + 128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, `v83`); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, `v125`). (3) Both then need the *same* input — the seam realisation `QGEO.REALIZE.01` (finite half proven, `v168`) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly **[O]**. **Net:** net existence and full-cone RP become **[E]**; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174_seam_fock_readings.py [E] (claim QFT.FOCK.01).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\mathrm{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind `v161`. (2) *Witten index = the Plücker norm* 11: the four μ_4-corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL (deg ≤ 2) keeps $\sum_{k\leq 2}\binom{4}{k} = 11 = \dim S^+ - g_{\mathrm{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5+3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the `v157`/`v158` rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170_diamond_slice_compression.py [E] (claim E8.SLICE.01).** E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3,4,6,10)$ ($p_n = 2+2^n$) and Sheet-Diamond determinants $(3,4,8,14,20,32)$ summing to $81 = N_{\mathrm{fam}}^4$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7,11,13)+s(3,3,3)+t(1,2,3)$ follow; $78 = \dim E_6 = p_2(p_0+p_3)$. Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- **v171_os_moment_cluster.py [E]/[C] (claim QFT.OSMOMENT.01).** Atomic OS Moment Lemma: $\mathrm{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster $729/665$, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1 + h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\mathrm{eff}} = 6\log\frac{3}{2} - 31/(4\pi^2) > 0$. Cited in `tfpt_research_contracts` (G5).
- **v172_trace_anomaly_seed.py [E] (claim SEED.ANOMALY.01).** The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7 =$ scalaron exponent $48-41$ (confinement $\leftrightarrow$ inflation), and the Volkov–Wipf $-98/45 = -2\cdot 7^2/\dim(D_5)$ factorisation. Cited in `tfpt_2` (seed).
- **v173_pfaffian_cp_car.py [E] (claim FLAV.STRONGCP.02).** Strong CP $\theta_{\mathrm{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D, \Sigma\} = 0$ pairs the spectrum, γ_5-Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version `5.1 (rev 97)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20678568`, DOI `10.5281/zenodo.20678568`. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- **`v168_qgeo_rigidity.py` [E]/[C] (claims `QGEO.RIGID.01`, `QGEO.REALIZE.01`).** Hardens the *finite* half of the one remaining Tier-1 hinge `GATE.QGEO` (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \backslash \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4 - 1)$ $(k = 1, 2, 3)$ are μ_4-eigenforms with characters χ_1, χ_2, χ_3 of weights $(1, 2, 3) = \mathrm{A}_3$ exponents $= \mathrm{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \backslash \mu_4$ — the QGEO Rigidity Theorem (`QGEO.RIGID.01`, **[E]**). The seam-collar realisation stays the one premise (`QGEO.REALIZE.01`, **[C]**). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167's completeness claim).
- **`v169_etaB_boltzmann_interface.py` [C] (claim `FR.ETAB.02`).** Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron 28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240°$. Over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10}, untuned) — the route is *viable*. But M_1/washout are scenario inputs, so η_B stays **[C]**; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **`v166_higgs_free_seam.py` [E]/[C] (claim `HIGGS.FREESEAM.01`).** A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral c=8 fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\text{seam}}) = 0$ and $\beta_\lambda(M_{\text{seam}}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from $\sim$0.022 at one loop to $\sim$0.002). The double condition predicts $m_H \sim$ 129–134 GeV; the same condition at the scalaron scale gives $\sim$107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ, reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE F_{transfer} follow-ups: flavor scale-tagging v163, QCD scale v164)

- **`v163_rg_stability_flavor.py` [E]/[C] (claim `FLAV.RGSTAB.01`).** Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2-suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2}Y_e Y_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2/(16\pi^2)\ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$ and $|\Delta \sin^2\theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2-driven quark mass ratios ($m_t/m_b \sim 41$) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- **`v164_qcd_scale.py` [E]/[O] (claim `QCD.LAMBDA.01`).** The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-loop,

n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest "**[O]** — not forced" frontier ratio. Cited in `tfpt_4_frontier`.

- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β's, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the F_{transfer} gauge inputs — v159)

- **New module `v159_pyrate_gauge_crosscheck.py` [E]/[C] (claim `EM.BUDGET.02`).** The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$, $\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\text{car}}2^{g_{\text{car}}-2}+1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and PyR@TE — and the split $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of F_{transfer} is thus not a free knob.
- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F__transfer + irreducible core, `v167`)

- **New module `v167_inventory_post_closure.py` [E]/[C].** The terminal residual inventory (line `v105`→`v123`→`v167`), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the `v123` *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q-realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology b_1=N_{fam}=3); the EH step (R3) folds into v_{geo} (`v152`) and the ambient measure into G_{net} (`v76`). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem, No-Unit `v153`); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = F_{transfer}, one functor with four typed conditional interfaces (Koide, η_B, axion, m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A2 by assembly, R1 = GATE.QGEO, zero independent open gates, `v165`)

- **New module `v165_premise_a_closure.py` [C]/[O].** Pins the two residual gates of the `v160`–`v162` (A)-chain to their floor. A_2 **(net existence) closes by assembly** of established mathematics **[C]**: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic c=8 lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (`v125`), $\mu(B)$=16/16=1, 248=120+128, character E_4/η^8=$j^{1/3}$ (`v156`); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $|k|$ (`v156`/`v157`). **R_1 (seam clock) reduces to `GATE.QGEO` [C]**: a μ_4-equivariant flat gapped generator is *unique* (= $A_0^\star$, `v115`/`v116`), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification "seam boundary = $\mathbb{P} \setminus \mu_4$" = `GATE.QGEO` (cohomology established, `v137`, b_1=3=N_{fam}). **Final accounting:** by the No-Unit Theorem (`v153`) the irreducibles stay $\{\pi, v_{\text{geo}}\}$; premise (A) = (A_2 assembly) + (R_1=`GATE.QGEO`) + the $\{\pi, v_{\text{geo}}\}$ core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).

- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)

- **New module `v162_seam_transport_identification.py` [E]/[O].** Pushes the `v161` residual as far as it rigorously goes and shows it carries *no new open content.* The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced:** a μ_4-equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4| \operatorname{diag} X$ for $U = \operatorname{diag}(1, i, -i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights $\operatorname{spec}(A_0^\star)$=$\{0, \frac{1}{3}, \frac{2}{3}\}$ (`v115`) and the transfer eigenvalue $(1-\alpha)^{p_2}$ with p_2=6=$|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6 \log \frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (`v137`). (α) is the rank-8 Calderón polarisation of ω_k=$|k|$ (`v113`/`v156`). **Closure:** with `v160`+`v161`, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, `HOR.CLOCK`). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger `GATE.METRIC.18`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the cone reduces to one particle: premise A's scope premise discharged to a 16-dim statement, v161)

- **New module `v161_one_particle_reduction.py` [E]/[O].** *Discharges* the scope premise of `v160` ("the transport is gapped on the *full* Schwinger cone") via Bogoliubov second quantisation. A quasi-free transport is T_Σ=$\Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) $\operatorname{spec}(\Gamma(t)|_{\deg m})$ = products of m one-particle eigenvalues (compound-matrix theorem, all m=0..6); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by `v113` (one polarisation ⇔ one vacuum); (4) quasi-freeness is *forced* — the 120=$\dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and `v158` makes every non-Gaussian deformation irrelevant. **Consequence:** `v160`'s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger `GATE.METRIC.17`). Python-only (numpy + stdlib).

- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, `v160`)

- **New module `v160_seam_gaussianity_from_pf.py` [E]/[O].** Formalises the proposed closure of premise (A) ("the seam bulk is a reflection-positive free field") as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants $\kappa_{2n}=0$ — verified by the *signed* set-partition/Pfaffian moment$\leftrightarrow$cumulant inversion on a concrete pure Majorana covariance (the `v113` "one kernel = net" property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray $\Rightarrow$ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap>0, the value $(2/3)^6$ only sets the rate. **The honest residual:** `v56`'s gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement "the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of `v56`", which the suite does *not* yet establish (ledger `GATE.METRIC.16`). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.
- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (`v160` is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection` $\times$ `{YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as `2026-06-07 / 2026-06-08` sort by the later date). A manual prepend had left five `2026-06-12` entries above seven `2026-06-13` blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, `v158`)

- **The free chiral c=8 fixed point is provably stable (`v158`, exact dimension count).** The finite core of (A)'s "reaches-the-fixed-point" residual: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the h=1 currents are chiral (`v157`, not $(1,1)$-marginal), and the lowest interaction (quartic, h=2) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger `GATE.METRIC.15`). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the b_1=41/10 cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, `v157`)

- **The free-bulk premise becomes a fixed-point property (`v157`, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally*

$|k|$ (Lee–Uhlmann — order-1 ΨDO, principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic c=8 theory is *rigid* — every field has $\bar{h}$=0, so no marginal $(1,1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving μ=1 (248=$120_{\rm NS}$+$128_{\rm R}$). With $(E_8)_1$ unique (`v83`), premise (A) shrinks from "prove the seam is Gaussian" to "the gapped (Δ=$6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point" — freeness then forced by rigidity, not fine-tuned (ledger `GATE.METRIC.14`).

- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, `v156`)

- **The constructive route carried to its exact endpoint (`v156`).** "$B \cong (E_8)_1$" is no longer an assertion: the chiral seam net is built explicitly and the identification *checked.* (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5{+}3$; (c) the weight-1 currents are $248 = 120_{\rm NS} + 128_{\rm R}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q-series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual:* exact given (i) the free-bulk premise (DtN = $|k|$, `v155`) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger `GATE.METRIC.13`).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN = $|k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the 248=120+128 split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, `v155`)

- **$G_{\rm net}$'s last premise reduces to a single shared physical premise (`v155`, Python-only).** The seam-side identification of the Simple-Current Extension Theorem (`v154`) splits (`v110`) into (a) "the Calderón involution is sheet-odd" (the double-cover deck, structural) and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from "the seam bulk is a reflection-positive free field", the *same* premise already load-bearing in the area law (`v59`) and the EH mechanism (`v150`/`v151`). **The entire seam-side residual is therefore one physical premise**, shared by $G_{\rm net}$, the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger `GATE.METRIC.12`).

- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (`v155` is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, `v153`–`v154`)

- **$v_{\rm geo}$ re-typed: the No-Unit Theorem (`v153`).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. $v_{\rm geo}$ is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ($U_{\rm point} \sim v_{\rm geo}$, $1/G \sim v_{\rm geo}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay {one unit, π} (ledger `ANCHOR.VGEO.02`).

- **$G_{\rm net}$ stated as the central closing theorem (`v154`).** The Simple-Current Extension Theorem: $A{=}(D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L{=}\langle(1,1)\rangle$ has $[B{:}A]{=}|L|{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (`v89`/`v92`/`v125`/`v143`/`v148`); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).
- **Status surfaces re-typed** (review): the canonical status card now reads $v_{\rm geo}$ as a metrology *primitive*, $G_{\rm net}$ as the simple-current extension $\cong (E_8)_1$ ([**E**]/[**C**]), and $F_{\rm transfer}$ as one functor with four typed interfaces — "not structural gaps". The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\rm geo}\}$; one seam-net theorem; frontier = $F_{\rm transfer}$) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/`origin_theory`/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series "Paper N" cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series "Paper N" attributions and removed them: four literal "Paper 6" references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old "Paper 4" (OS/admissibility) and "Paper 5" (APS/Hodge) references in `tfpt_2`, the old "Paper 3" transport-kernel attributions in `tfpt_2`, and the internal "Paper 1/2/4" cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts $\leftrightarrow$ papers $\leftrightarrow$ ledger $\leftrightarrow$ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, `v152`)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (`v152`).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in $d{=}2$ that ratio *is* the induced Newton constant $1/G$ (`v68`). So $v_{\rm geo}$ (`v78`), $1/G$ (`v68`) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay {one scale, π}, and `SEAM.THEOREM.01`'s residual reduces to the kernel-identification premise alone. *Audit (not promoted):* the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/`origin_theory`/`next.txt`/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion} (rev {git count})` (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (`ZENODO_TOKEN` secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to `main`, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ∼270 files / ∼3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts ("Why the interpretation is internally consistent", "Dependency chain of the birefringence test", "The precise reformulation", "The three theses honestly graded", "The whole story", "One-line summary") become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (`v53`–`v56`, `v101`, `v105`, `v113`, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class [**O**]/[**E**].

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a handwritten, *stale* copy of the document-set card ("five short documents", missing `tfpt_5`) *and* the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box "clock" wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon "Scope/typing" marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document's `.aux`/`.toc`/`.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (`v151`).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghelea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ($C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the `v150` EH form *verbatim*, with all of it localised in the *local* half-space determinants — the "Calderón version" demanded by `v150` needs no separate derivation. `SEAM.THEOREM.01` narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays **[O]** (ledger `SEAM.EHMODEL.02`).

- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/`origin_theory/next.txt` moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, `v149`–`v150`)

- **R4′: both dual normals are anchor data (`v149`).** In the cusp frame the torsion normal pairs to $(6,3,5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ the dual normal pair is anchor data through and through. Equivalences to the $(2,8,121)$ frame data and the w_0-lift exact. R4′ residue = *one* discrete assignment (ledger `GATE.UWALL.14`).
- **R3: the mechanism exists at the gapped-model level (`v150`) — the first movement on `SEAM.THEOREM.01`.** The conical heat-trace deficit is exact and t-independent (Cheeger image sum $(N^2-1)/(12N)$); for a *gapped* operator the ζ-regularised replica variation is *finite and cutoff-independent*, $\Delta \log \det' = 2C(\gamma)\ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m/12\pi)\int\sqrt{g}R$. The `v73` normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays **[O]** (ledger `SEAM.EHMODEL.01`).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, `v145`–`v148`)

- **R4′: the pairing values are atom identities (`v145`).** Reading the `v139` lift structurally ($n = w_0(\sigma) + |\mu_4| e_3$, w_0 = the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\text{car}} = N^2_{\text{fam}}$ (4+5=9, $\sigma \perp w_0(a)$) and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2| g_{\text{car}} \|\text{Pl}(K)\|_1$. R4′ = derive *one* map (ledger `GATE.UWALL.13`).
- **R5: the Möbius D_4 realisation (`v146`).** The *full* dihedral symmetry $\langle z \mapsto iz,\ z \mapsto 1/z \rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity +1 on the self-conjugate line (= T_A in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger `FLAV.QGEO.07`).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is det′-clean (`v147`).** Variance ratio $(1-\alpha)$ (forced by norm = area) gives the Born-squared ratio $(1-\alpha)^{p_2}$, the ln series *is* the `v127` ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the `v144` obstruction is localised to the $\alpha = 0$ reference (ledger `HOR.CLOCK.13`).
- **R2: the NS/R sector census (`v148`).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ($248 = 120_{\text{NS}} + 128_{\text{R}}$); the one remaining net statement is irreducibly twisted-sector (ledger `GATE.METRIC.10`). *Correction note (same day):* the first version of `v148` graded the Ramond labels by the sum $q_D + q_A$ (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.

- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + `GATE.WOLFRAM.02` + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{\rm Pl}$ vs $M_{\rm Pl}$ — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.
- **Stale references cleaned.** "How each of the seven papers simplifies" → "How the legacy layers simplify (bridge map)" with the lead-in "(Paper 6)" reference replaced by the scalaron-reheating chain (`v86`); "(Paper~6/7)" in the `tfpt_3` bridge table reworded; the `tfpt_1` "single remaining geometric input (H2)" passage and the `tfpt_2` "single remaining input (U)" passage carry dated status updates pointing at the current decomposition (`v118`–`v122`, `v139`, `v141`, `v142`, `v75`).
- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, 2/3 motif and K_5 figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes **[E]**/**[C]**/**[O]** instead of the legacy $[A]/[I]/[L]/[B]/[P]/[R]$ badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + $c{=}8 \Leftrightarrow$ index-4 inclusion; `v83`/`v87`/`v89`, Lie-level realisation `v143`); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never "Paper 5–7"); the marker system on every website surface migrated to $[E]/[C]/[O]/[X]$ (fine types named in prose; the script index keeps quoting the scripts' internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the `v141`–`v144` outcomes (sheet question: deck derived; G_{net}: Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review's narrative reframings (anchor-algebra trilogy, "one seam clock" framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ *error* was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, `v141`–`v144`)

- **R5 closed at the discrete level (`v141`, deck selection theorem).** The `v140` $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A\Sigma$ (`v97`/`v98`) pairs the $Q_+{=}1$ cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2,3,1) = \mathrm{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \to -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading $= Q_+ \circ \rho$. `GATE.QGEO` keeps only its realisation premise — no discrete freedom left (ledger `FLAV.QGEO.06`).
- **R4′ narrowed: three pairings → two (`v142`, frame integrality).** Integer covectors in the $(\mathbf{1}, a, \sigma)$ dual frame form an index-11 sublattice ($x - 3y + z \equiv 0 \bmod 11$; the index *is* $\|\mathrm{Pl}(K)\|_1$);

with $(x, y) = (|\mathbb{Z}_2|, \operatorname{rank} E_8)$ the σ-pairing is forced $\equiv 0 \bmod 11$, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger `GATE.UWALL.12`).

- **R2 realised at the finite Lie level (`v143`, graded Frobenius).** The $\mathbb{Z}_4$-graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion $= \mathbb{C}[\mathbb{Z}_4]$) — the `v125` Q-system realised by the hull; residue = one conformal-net statement (ledger `GATE.METRIC.09`).
- **R1's det-ratio step derived in-family (`v144`).** e_2-rigidity of the traceless SdS cubic gives $r_b r_c = 1 - \Delta^2/3$ exactly; with the `v132` anomaly the non-zero-mode determinant ratio is $(1 - \Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the `v131` cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays **[C]** with the $6 \to \frac{14}{3}$ obstruction stated sharply (ledger `HOR.CLOCK.12`).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant→EH step (`SEAM.THEOREM.01`); the gate keeps its **[O]** typing untouched.
- **Wolfram mirror extended** to `v141`–`v144`: extension file now 211/211 (`GATE.WOLFRAM.02` and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the `vN` scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite ↔ `run_all.py` ↔ registry ↔ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri`/`\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the `vN` scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: `v17` and `v85` are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. `v91` (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` ("Three views of one spine": anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the `ARCH.SPINE.01` ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.

- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare `vN` ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[**E**] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): tfpt_3 from 39 to 27 boxes (e.g. "Purpose and discipline", the five "(i)–(v)" audit-lemma boxes, "Net", "The connection in one sentence", "The unification", "The question and the honest answer", "Net: two axioms"); tfpt_4 from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item "Honest status of . . . " keyboxes are kept — tfpt_4 is the status authority for the frontier.
- **Colour-marked module references.** `\vref` applied to the inline `vN` references in tfpt_3 and tfpt_4 (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in tfpt_4 (Koide flow-time) replaced by the 4-class [**C**]; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant [**E**]) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): tfpt_1 from 21 to 14 loud boxes (+3 `keyeq`); tfpt_2 from 52 to 43 (e.g. "In one sentence", "The one-paragraph version", the four "Sharpening" notes, "What the simplification means net", "The disciplined main statement", "What improves. . . ", "Scope and honesty"). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout tfpt_1 and tfpt_2 (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** tfpt_2's inflation-pivot table gains the `v86` scalaron-reheating point ($N_\star$=51.4 ⇒ n_s=0.9611, r=0.0045, A_s-disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed [**C**]. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: [**E**] exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), [**C**] conditional (physical / bridge / readout under named hypotheses), [**O**] open / axiom, and [**X**] kill test. `tex-artefacts/tfpt_style.tex` defines `\mE`/`\mC`/`\mO`/`\mX` and keeps the legacy `\tI ...\tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L]` → [**E**], `[N]/[P]` → [**E**]/[**C**]). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.

- **Status tables reconciled against the ledger.** The introduction's inflation/$N_\star$ rows are made consistent with `v86/COSMO.NSTAR.01`: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star$=51.4 (n_s=0.9611, r=0.0045), replacing the stale point values $N_\star$=55 (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed [**E**] → [**C**](conditional on M_{Star}=M_{scal} + reheating), matching the other two tables and the ledger `[P]` typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today` + version + revision). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document's title page and page footer carry author, date and `v5.2 (rev N)` automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.
- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. `v108`–`v113`, `v49/v71`) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction's callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and E_8 as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the "Reduction in one line" and the fermion-spectrum callouts now carry inline `\veri{}` citations (`v1`, `v6/v14/v23`, `v18/v20/v46`).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to `tex-artefacts/`.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`).

`changelog.tex` stays at the repo root because it is also a standalone `build.sh notes` deliverable. The `make_manifest.py` `TEX` list and the workflow rule were updated to the new paths.

- **Master script index outsourced (`tex-artefacts/verification.tex`).** The long `vN`→check table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction's eleven `keybox`es are reduced to the single headline "In one sentence" card; the explanatory keyboxes (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winbox`es become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winboxes`, one `gapbox`) plus the two shared orientation cards.
- **Document-count consistency.** The introduction's document map gained the missing `tfpt_5_redteam` row; "seven/eight companion documents" and the ledger "six documents" wording are corrected to the true count (nine active documents); `README.md` ("10 ok", 140 scripts) and the workflow rule ("9 docs") are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across tfpt_1, tfpt_2, tfpt_3 and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer's prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the 2/3 *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document's layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in tfpt_1, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types **[E]** (exact algebra), **[E]** (exact number), **[C]** (readout / standard physics in TFPT units) to the existing **[C]** (bridge) and **[X]** (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain **[E]** where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction's one-sentence claim is softened from "derives all constants" to "constructs a discrete compiler; physical readouts run through named, status-typed bridges"; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says "stays [P]" — ratios are **[E]** closed on the derived stratum, only the absolute scale stays **[O]**, with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\mathrm{ret}} = c_3 + 36c_3^4$ (leading c_3, fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO's first 59.1-day result (0.3092 ± 0.0087, compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. "Complete solutions of the five open questions" becomes "Closure status of the five former open questions".

- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ (historical $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\to v_{\text{geo}} \to G_{\text{net}}$); F_{transfer} is named as one missing functor with Koide/η_B/axion/m_p/m_e as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv$ `tfpt_N`). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four tcolorbox families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` ("TFPT status at a glance") gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \to 12 \to 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4$'$ selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form: $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q-system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from "find" to "identify".
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights (spec $A_0^\star$); $\det(\mathbf{1} - A_0^\star) = \frac{2}{9}$ = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$-graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law $\to$ zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\text{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).

- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1, 1, -2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s,t) = 3s(t{+}1)(t{+}2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8, 14, 20)$, $\det F = 32$, row-budget axes with $\mathrm{rowsum}(V) = \mathrm{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6, 8, 7)$); the $A_0^\star$ zero mode carries the spectral normal $\sigma = (2, -9, 5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1, 2, 3) = \mathrm{Spec}(Q_+)$ = the A_3 exponents and cusp weights = $\mathrm{spec}\, A_0^\star$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2, 8, 121)$ — the R4′ residue is three pairings; the $H^1 \to$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the `GATE.QGEO` residue collapses to one $\mathbb{Z}_3$ choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded "single remaining input" claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated ∼110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s,t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5, -9, 6)$; documentation pass ("The Sheet Diamond and the Winding Line" in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1, 1, -2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\mathrm{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\mathrm{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank $= c$) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- **U_{wall} made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue $A_0^\star$ (diag $= a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff$

$a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.

- **Second review validation + address pinning (v119–v121).** Anchor ratio triad $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$; the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and det 8 — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.